

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 425 — Past Questions & Materials
by Renance

Course: CSC 425
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 400

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Paper 1 of 5 — Csc 425

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF COMPUTER SCIENCE
2nd Semester: BSc (Computer Science Examination) 2023/2024 Session
CSC425 - MANAGEMENT SCIENCE Time Allowed: 2½ Hrs
Instruction: Attempt Questions 1, 2 AND Any Other

Question 1:

A small plant assembles personal computers via 9-interlinked stages according to the following precedence and process: (1-2 = 4), (1-3 = 12), (1-4 = 10), (2-5 = 6), (3-6 = 8), (4-6 = 10), (5-7 = 10), (6-7 = 0), (6-8 = 8), (7-8 = 10) and (8-9 = 6) respectively. Evaluate the following:

- Draw the network arrow diagram representing the above assembly line. 4mks
- Show workings of the Forward-Pass and Backward-Pass for the Assembly line. 8mks
- Compute the critical path and the assembly duration. 6mks
- Tabulate the Start, Finish and Float/Slack times for the Critical Path Method. 4mks
- Evaluate the total slack time for the project. 3mks

Question 2:

Given the matrix diagram below, resolve the transportation problem using: 25mks
The Vogel's Approximation Method

Plants	Destinations				Total
	1	2	3	4	
1	8	4	6	10	300
2	12	9	10	8	250
3	7	8	8	9	400
4	6	7	10	12	250
	300	400	200	300	1200

Question 3:

- Show the schematics sources of changes in a Project. 4mks
- Solve the Transportation Problem in Question 2 using North-West Corner Rule. 16mks

Question 4:

- Describe Project Management with a well-labelled Diagram. 4mks
- Solve the Transportation Problem in Question 2 using the Least-Cost Method. 16mks

Question 5:

- What are the responsibilities of a project manager? 5mks
- State seven (7) major reasons why do Projects fail. 7mks
- Show the Schematics of an Integrated Project Management. 8mks

OJUGO, Arnold Adimabua (PhD) - Examiner

Paper 2 of 5 — CSC 425

Level 400 • Second Semester • PastQuestion

Question 4:

A small plant

Personal computers via 9! er linked

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OJUGO, Arnold Adimabue (PRD) + Exemeriner

Paper 3 of 5 — CSC 425 Project management materials 2025...

Computer science 400level 2nd semester

Level 400 • Second Semester • Handout

INTRODUCTION TO PROJECT MANAGEMENT

OVERVIEW

The purpose of presentation is to provide leaders and team members of projects, committees or task forces with advanced techniques and practical skills for initiating, planning, tracking, controlling and evaluating any kind or size of project.

Introduction to Project Management

- ☐ On time
- ☐ On budget
- ☐ Meeting the goals that have been agreed upon

WHAT IS A PROJECT?

- A project is an activity that :
 - is temporary having a start and end date
 - is unique
 - brings about change
 - has unknown elements, which therefore create risk
- Generally projects are formed to solve a problem or take advantage of an opportunity.
- Business as usual activities can often be mistaken for projects.
- Generally it is the uniqueness of the activity that is the deciding factor – do we do this every year? If so, then it is not truly a project – although you can use project methods to get it done.

QUIZ – ARE THESE PROJECTS?

- ☐ Building a deck Yes No
- ☐ Implementing a new system Yes No

- ☐ Mowing the lawn Yes No
- ☐ Planning a wedding Yes No
- ☐ Planning a fundraiser Yes No
- ☐ Planning a student graduation Yes No

COMMON PROJECT TERMS

- Deliverables: Tangible ‘things’ that the project produces
- Milestones: Dates by which major activities are performed.
- Tasks: Also called Actions. Activities undertaken during the project
- Risks: Potential problems that may arise
- Issues: Risks that have happened
- Gantt Chart: A specific type of chart showing time and tasks. Usually created by a Project Management program like MS Project.
- Stakeholder: Any person or group of people who may be affected by your project

EXAMPLE: Building a deck

- Deliverables: A plan, a consent form, the deck
- Milestones: Plan drafted 1 Jun

Plan approved 15 Jun

Plan submitted 16 Jun

Plan approved 19 Jun

Materials purchased 16 Jul

Resources booked 16 Jul

Equipment identified 16 Jul

Deck constructed 20 Jul

Deck tested 24 Jul

Deck quality approved 24 Jul

“Deck warming” completed 28 Jul

tasks Subtasks

- Tasks:

Plan drafted Requirement gathered

Best practice researched

Draft 1 prepared

Distributed to stakeholders

Plan approved Feedback gathered

Amendments made

Final plan prepared

Distributed to stakeholders

Sign-off obtained

WORK-BREAKDOWN STRUCTURE

□ WBS

□ Hierarchy of tasks required to complete project

□ Each task is broken into smaller tasks that can be managed and estimated

□ Define task dependencies

■ Some tasks must begin at the same time, some must end at the same time and some cannot start until the other tasks have finished.

□ Estimate task durations and cost

□ May be inputted into project management software

- Final WBS plan is called baseline WBS

- Risks:

- Plan is not approved after first round of feedback

- Resources are not available at the required time

- Plan is not given consent

- For each of the above, you should have a contingency plan, or do some activity that may prevent it happening in the first place.

- Issues:

- If any of the above actually happens, then it becomes an issue to solve.

- Gantt Chart:

- Stakeholder:

- House owner, Builder, Council, ???

A successful Project Manager must simultaneously manage the four basic elements of a project: resources, time, money, and most importantly, scope.

All these elements are interrelated. Each must be managed effectively. All must be managed together if the project is to be a success. The resource that can be leveraged to the greatest extent in all projects is the people involved.

THE PROJECT MANAGER

A person with a diverse set of skills – management, leadership, technical, conflict management, and customer relationship – who is responsible for:

- initiating,
- planning,
- executing,
- controlling,
- monitoring,
- and closing down a project.

Project Managers are essentially jugglers. They must make sure that everything keeps to task, that potential issues are quickly eliminated and the project is delivered on time, all the while making sure everyone knows what is happening and the project quality and budget are acceptable. Specifically they:

- direct all activities required to successfully meet the project objectives
- manage risk - scanning ahead for potential issues and resolving them before they become a problem
- solve problems - recommending alternative approaches to problems that arise and providing guidance to the Project Sponsor
- track and report project progress
- communicate to all stakeholders in the project

- Ultimately responsible for the Project's Success

- Plan and Act
- Focus on the project's end
- Be a manager & leader

PROJECT INITIATION

The Initiation phase of the project is the most important phase. The success of the entire project depends on how clearly and completely the Terms of References are established.

- Project Sponsor
- Lines of Authority
- Participants
- Objectives

PROJECT INITIATION con't

- Constraints
- Costs/Budget

- Resources
- Deliverables
- Phases & Time Scales
- Strategy
- Risks
- Roles & Responsibilities

CHARACTERISTICS OF PROJECTS

- A project contains a well defined objective. The project objective is defined in terms of scope (or requirements), schedule, and cost.
- A project is carried out via a set of interdependent tasks.
- A project uses various resources to carry out these tasks.
- A project has a definite start date and an expected completion date. The actual completion date may not always be the same as the expected date.
- A project is a one time or unique endeavor.
- A project has a customer.
- So why do projects fail?

WHY DO PROJECTS FAIL?

1. Poor project and program management discipline
2. Lack of executive-level support
3. Wrong team members
4. Poor communication
5. No measures for evaluating the success of the project
6. No risk management
7. Inability to manage change

A project has a degree of UNCERTAINTY. In project planning many assumptions are made regarding:

- access to resources.
- resource capability.
- impact of environmental factors.

- These assumptions are not always accurate.
- Requires project managers to re-assess and trade-offs between requirements, costs, and time. Above all, be PRO-ACTIVE.

MEASURING PROJECT SUCCESS

We measure the success of a project using 4 major project constraints, specifically:

- Scope.
- Cost.
- Schedule (Time).
- Customer satisfaction (quality and performance).

PROJECT CONSTRAINTS

- Project scope - Have all the project requirements (i.e., deliverables) been completed?
- Project cost - Is the cost of the project close to the amount the customer has agreed to pay?
- Schedule - Was the project completed on time?
- Customer satisfaction - Is the customer happy with the quality of the project?

PROJECT SUCCESS - 12 Golden Rules

☐ Rule #1

Thou shalt gain consensus on project outcome.

☐ Rule #2

Thou shalt build the best team possible.

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Thou shalt develop a comprehensive, viable plan and keep it up-to-date.

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Thou won't try to do more than can

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Critical Path Analysis 14.1

14

Critical Path Analysis

LEARNING OBJECTIVES :

After studying this unit you will be able to :

I Explain the generic term network analysis

I Draw diagram & chart representing inter-relationships between various elements of the project and finally establishing critical path

I Identify Dummy Activities

I Identify earliest and latest event times of each event. Identify different types of Float
Help in Decision Making about project times and resource allocation Calculating probability of completing network in a given time

14.1 INTRODUCTION

Planning, Scheduling and Controlling are three important functions of management. Planning involves the formulation of objectives and goals that are subsequently translated into specific plans and projects. Scheduling is concerned about the implementation of activities necessary to achieve the laid down plans. The function of control is to institute a mechanism that can trigger a warning signal if actual performance is deviating (in terms of time, cost and some other measures of effectiveness) from the plan. If such a deviation is unacceptable to the concerned manager, he will be required to take corrective action to bring performance in conformity with the plans. The PERT and CPM models are extremely useful for the purpose of planning, scheduling and controlling the progress and completion of large and complex projects or for carrying out the analysis of these three managerial functions. Before we describe the basic concepts used in the construction and analysis of these models, let us first understand the meaning of a project.

What is a project?

A project can be defined as a set of large number of activities or jobs that are performed in a certain sequence determined logically or technologically and it has to be completed within (i) a specified time, (ii) a specified cost and (iii) meeting the performance standards. Examples of a project from fairly diverse fields are given below:

1. Introducing a new product in the market.
2. Construction of a new bridge over a river or construction of a 25 storied building.

14.2 Advanced Management Accounting

3. Executing a large and complex order on jobbing production.
4. Sending a spacecraft to the mars.

14.2 GENERAL FRAMEWORK OF PERT/CPM

A network is a graphical representation of a project, depicting the flow as well as the sequence of well-defined activities and events. Developed during the 1950s, both CPM (Critical Path Method) and PERT (Programme Evaluation and Review Technique) are network techniques/models. The network approach helps project managers in planning, scheduling and controlling. As a planning tool it helps the manager to estimate the requirements of resources viz., materials, equipment, manpower, cost and time for each activity or tasks of the project. This approach cannot make decisions by its own. It only provides additional information to executives to facilitate decision making process. Also it does not provide solution to every management problem. It certainly helps in identification of those activities, jobs or events which control the completion of the project.

The working methodology of critical path analysis (CPA) which includes both CPM and

PERT, consists of following five steps:

1. Analyse and break down the project in terms of specific activities and/ or events.
2. Determine the interdependence and sequence of specific activities and prepare a network.
3. Assign estimates of time, cost or both to all the activities of the network.
4. Identify the longest or critical path through the network.
5. Monitor, evaluate and control the progress of the project by replanning, rescheduling and reassignment of resources.

The central task in the control aspect of these models is to identify the longest path through the network. The longest path is the critical path because it equals the minimum time required to complete the project. All other paths other than the critical path (i.e. non-critical or slack paths) offer flexibility in scheduling and transferring resources, because they take less time to complete than the critical path.

14.3 ADVANTAGES OF CRITICAL PATH ANALYSIS

There are a number of advantages in using critical path analysis.

1. It allows for a comprehensive view of the entire project. Because of the sequential and concurrent relationships, time scheduling becomes very effective. Identifying the critical activities keeps the executive alert and in a state of preparedness, with alternative plans ready in case these are needed. Breaking down the project into smaller components permits better and closer control.
2. Critical path analysis offers economical and effective system of control based on the principle of management by exception i.e. need for corrective action arises only in exceptional situations and in most of other cases, performance is in conformity with the plans.
3. It is a dynamic tool of management which calls for constant review, a reformulation of

Critical Path Analysis 14.3

the network, and finding the current path of relevance and optimum resources allocation.

14.4 FUNDAMENTALS OF A CPA NETWORK

◆ Activity

An activity is any portion of a project which consumes time or resources and has a definable beginning and ending. For example, "laying of pipe" is an activity requiring the use of resource mainly effort. Activity may involve labour, paper work, contractual negotiations, machinery operations, etc. Commonly used terms synonymous with "activity" are "task" and "job".

Tail Head

7

1 3

Figures 1 and 2

10

3 4

14

14

11 24

3 5

7

1 4 6

2

Figures 3 to 6

14.4 Advanced Management Accounting

Activities are graphically represented by arrows, usually with description and time estimates written along the arrows. The tail of the arrow portraying an activity represents the starting point of the activity and its head represents its completion. The arrow may be straight slanting, or bent but not broken (see figure-1). The arrow is not a vector and need not be drawn to scale.

◆ Events

The beginning and ending points of an activity or a group of activities are called events. Synonyms of an event are "node" and "connectors" An event is often represented graphically by a numbered circle (see figure-2), although any geometric figure such as square, oval, rectangle etc. will serve the purpose. We shall, however, stick to the most commonly used convention for representing an event viz, the circle. A few examples of events are as follows: (i) Material procured, (ii) Design completed, (iii) Project started, (iv) Bricks laid, etc. All activities in a network must commence from some event. Such events are called the tail events because they are connected to the tail of an activity. These are shown in figure 3. Similarly, all activities in a network must have terminal points called the head event, because it is at the head of an activity. These are shown in figure-4. Figure-5 depicts the tail and head events connected by arrows representing activities i.e. it depicts the dual role of an event. Event 14 is the head event for one activity and tail event for another.

In a network, symbol "i" is used for the tail event (also called preceding event) and "j" for the head event (or succeeding event) of an activity. The activity, then being i-j.

If an event represents the joint completion of more than one activity, it is called a merge event. If an event represents the joint initiation of more than one activity, it is called a burst event.

A network is, then, a graphical representation of a project plan, showing the inter-relationship of the various activities. Networks are also called arrow diagrams (see figure-6). When the results of time estimates and computations have been added to a network, it may be used as a project schedule.

14.4.1 Conventions adopted in drawing networks: There are two conventions normally adopted while drawing networks. In the early stages of network drawing, it is suggested that the conventions should be respected until sufficient experience has been gained to justify

dropping them. These conventions are:

- (a) Time flows from left to right.
- (b) Head events always have a number higher than that of the tail events.

The above stated conventions allow activities to be referred uniquely by their tail and head event numbers, so that "activity 3-4" means only "the activity which starts from event 3 and proceeds to event 4"; it cannot mean "the activity which starts from event 4 and finishes at event 3".

Critical Path Analysis 14.5

14.4.2 Graphical representation of events and activities: Events are represented by numbers within circles. Activities are represented by arrows, the arrow-heads represent the completion of the activities. The length and orientation of the arrow are of no significance whatsoever (chosen only for the convenience of drawing). The activity of leaving place A and walking to place B can equally well be represented by figure-7.

14.4.3 Fundamental properties governing the representation of events and activities: The representation of events and activities is governed by one simple dependency rule which requires that an activity which depends upon another activity is shown to emerge from the head event of the activity upon which it depends and that only dependent activities are drawn in this way. Thus, if activity B depends upon activity A, then the two activities are drawn in figure-8.

The dependency rule gives rise to two fundamental properties of events and activities:

1 2 1

1 2

2

2

1 2

1

Figure 7

A B

1 2 3

Figure 8

14.6 Advanced Management Accounting

1. An event cannot occur until all activities leading to it are complete.
2. No activity can start until its tail event is reached.

The above two properties can be combined into a single one, namely that "no activity may start until all previous activities in the same chain are completed.

14.4.4 Logical sequencing and connection of activities: A project entails several activities. The arrows are arranged to show the plan of logical sequence in which the activities of the project are to be accomplished. The sequence is ascertained for each activity by answering the following three queries viz:

- (i) Which activity or activities must be completed before the start of a particular activity?
- (ii) Which activity or activities should follow this?
- (iii) Which activities can be accomplished simultaneously?

The activity or activities which immediately come before another activity without any intervening activities are called predecessor activities to that activity. The activities which follow another activity without any intervening activities are called successor activities to that activity.

In a project of laying a pipe line, the three activities involved may be trenching, laying pipe and welding pipe. To decide the logical connection between these three activities, it is necessary that they be carried out in series, the reasoning being that the pipe cannot be laid until trenching has been done and welding cannot be undertaken until the pipe has been laid. This way we decide the logical sequencing between different activities.

Errors in logical sequencing: Two types of errors in logic may arise while drawing a network, particularly when it is a complicated one. These are known as looping and dangling.

(1) Looping: Normally in a network, the arrow points from left to right. This convention is to be strictly adhered, as this would avoid illogical looping, as shown wrongly below:

1 2 3 4

Figure 9

Critical Path Analysis 14.7

(2) Dangling: The situation represented by the following diagram is also at fault, since the activity represented by the dangling arrow 9-11 is undertaken with no result.

A

8 9 10

11

Figure 10

To overcome the problem arising due to dangling arrows, following rules may be adopted.

- (i) All events, except the first and the last, must have at least one activity entering and one activity leaving them.
- (ii) All activities must start and finish with an event.
- (3) Duplicate activities: Consider the following figure 11:

A

x Y 8

6 7

B

Figure 11

In the above figure, activities A and B may be called duplicate activities because they have same head event (i.e 6) and the same tail event (i.e.7). One remedy for such a situation is the introduction of a dummy activity.

(4) Dummy activity: It is a hypothetical activity which consumes no resource and time. It is represented by dotted lines and is inserted in the network to clarify activity pattern under the following situations:

- (i) It is created to make activities with common starting and finishing events distinguishable.
- (ii) to identify and maintain the proper precedence relationship between activities that are not connected by events.

14.8 Advanced Management Accounting

- (iii) to bring all "loose ends" to a single initial and a single terminal event in each network using dummies, if necessary.

For example, problem of duplicate activities in the figure-11 above may be circumvented as shown in figure-12.

A
x Y
6 8
B
7

Figure 12

Figure-13 shows three cases for the following set of dependency relationships:

Activity C is dependent upon both A and B.

Activity D is dependent upon A alone.

1 4 2 B 3 C 4
A C
3 A
B D A
1 D 5
3
2 5
B C
2 3 5
A D
1 4 6

Figure 13

The first portrayal (on top left of figure-13) is clearly wrong since it shows D as dependent upon not only A but also B which is not desired. The other portrayal (ii) is also wrong since A is being shown twice and thus contravenes the fundamental axiom of network that there must be one arrow for each activity. The way out to this dilemma is the representation by means of the dummy activity. In the third portrayal of figure -13, C is dependent upon both A and B (via dummy) whereas D is dependent upon just A.

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14.4.5 Numbering the events: The event numbers in a network should in some respect reflect their logical sequences. When a complicated network has been drawn then the problem of assigning numbers to the events involved in the network arises. A rule devised by D.R. Fulkerson, involving the following steps may be followed to resolve the problem of numbering the events.

- (i) An "initial" event is one which has arrow/arrows coming out of it and none of the arrow entering it. In a network there will be only one such event. Call it "1".
- (ii) Delete all arrows coming out from the event 1. This will give us at least one more "initial event".
- (iii) Number these events as "2, 3...."
- (iv) Delete all emerging arrows from these numbered events which will create new initial events. Then follow step (iii).
- (v) Continue the above steps till last event is obtained which has no arrows coming out of it. Consider the numbering of events in the following figure.

y 4

1 2 5 6 7 8

3

x

Figure 14

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A F A F

B B G

C G H

C

A F

A F

B G

B G

C H

C H

A F

A F

B G

B G

C H

C H

Figure 15

Critical Path Analysis 14.11

Here we proceed from left to right. The event with least x-co-ordinate is assigned the smallest integer, say 1. Other events are assigned progressively higher integers with regard to x-co-ordinate. If two or more events (4 and 5 above) have the same x-co-ordinate, the one towards arrow should have higher number.

Furthermore, it is not necessary, and in fact also not desirable to number the events consecutively. It would be a better scheme to number the events as 10, 20, 30, 40, 50, 60, 70, in the above diagram instead of 1,2,3,4,5,6,7. This affords insertion of more activities and events omitted by oversight or having become necessary in view of certain logic revisions. It was mentioned earlier that it is desirable that all the activity arrows point from left to right. If the arrow is vertical it may point downwards or upwards.

For the sake of presentability it is to be recommended that activities emanating from one event or converging to another may make as great angles between themselves as possible.

A few more conventions are given below:

(i) Keep the arrow to the extreme right. (ii) As far as possible avoid drawing arrows that cross each other. Usually by suitable 'stretching' the network diagram it is possible to avoid this. (iii) Where, however, crossing is unavoidable, bridging may be done. This applies to dummies as well. Draw boldly a big network. Smaller ones are confusing. Use of pencil and rubber is recommended.

Exercise : Depict the following dependency relationships by means of network diagrams.

The Alphabets stand for activities.

1. A and B control F; B and C control G.
2. A and B control F; B controls G while C controls G and H.
3. A controls F and G; B controls G; while C controls G and H.
4. A controls F and G; B and C control G with H depending upon C.
5. F and G are controlled by A, G and H are controlled by B with H controlled by B and C.
6. A controls F, G and H; B controls G and H with H controlled by C.

Answer. The required networks are given in figure-15

B E

H

A C F

G

D

Figure 16

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J K L

M

Figure 17

F G

A B C D E

H J L

M

I K

Figure 18

Exercise : Find out the superfluous (unnecessary) dummy activities in the networks below.

14.4.6 Basic steps involved in drawing a CPM/PERT network : Network is defined as a diagram representing the activities and events of a project, their sequence and inter-relationships.

The basic steps involved in drawing a network are:

- (i) Breaking up of the entire project into smaller systems known as tasks.
- (ii) For each task ascertain the activities and events to be performed.
- (iii) For each activity determine the preceding and succeeding activities.
- (iv) For each activity determine or estimate the time and other resources needed.
- (v) Draw a network depicting the assembly of tasks into a project.

14.4.7 Network Construction

Illustration

The activities involved in the computer installation process are detailed below. You are required to draw the network.

Activity Predecessor Activities

- A. Physical Preparation None
- B. Organisational Planning None
- C. Personnel Selection B
- D. Equipment Installation A
- E. Personnel Training C

Critical Path Analysis 14.13

- F. Detailed Systems Design C
- G. File Conversion F
- H. Establish Standards and Controls F
- I. Programme Preparation H
- J. Programme Testing I
- K. Parallel Operations D, E, G, J.
- L. Finalise Systems Documentation I
- M. Follow up. K, L

A A B < C (ii)

NONE < A

(i)

NONE < B B

B

C

D

D
 A A C < E (iv)
 A<D C<F
 B (iii) B
 C C
 E
 D J F
 F<G
 F<H
 A (v)
 E G I
 H<I
 B C F H I <J
 K D, E, G and
 J < K (vi)
 D J
 E G
 A I
 B C F H

Figure 19

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Solution

The network is shown in figure 19. The construction has been carried out step by step which is explained below. The symbol '<' is to be read as "precedes" in these figures.

(i) Since A and B are preceded by no activity, they are shown emanating from the start event. We do not put arrows and the head events for A and B because they have been drawn as such only tentatively in the following step (s) their position may have to be changed.

(ii) Here C is shown as dependent upon B. Again we neither show arrow and nor the head event of C because its position is tentative.

(iii) D is shown as dependent upon A.

(iv) We show here C preceding E and F.

(v) F is shown preceding G and H, H precedes I and I precedes J.

(vi) K is shown as dependent upon D,E,G and J, the latter four therefore, are made to converge on the same head event.

(vii) L is shown as dependent upon I and M as dependent on K and L. Note carefully that K and L are the terminal activities and, therefore, converge on the last event.

About this point the student is cautioned since the beginners do tend to show the last activities dangling and do not converge them to the common last event.

Of course, the network would be constructed in just one diagram (as shown in Figure-19).

We drew so many for explanation only. However, for a stylish presentation it is desirable that this one diagram is suitably sketched i.e. faired up in another diagram. The student may number the events as an exercise.

M

K I < L

L < M

L

D J

E G

A I

B C F H

Figure 19 contd.

Critical Path Analysis 14.15

Example : Construct the network for the following activity data:

Activity Preceded by

A None

B None

C A

D B,C

E D

F E

G B,C

H F

I F,G

J H,I

K B

L F,G,K

NONE < A C

A

NONE < B

B

A A < C

B

D

E F

C

D

A B < D C

C < D D < F

B A E < F

B

H

E F

D

B, C < G

C

F < H

G

A

B Figure 20

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H I

E F < H, I

F

G < I

D

C G

A

B

H

I J

I, H < J

E F

D

C G

A

B

J

0 1

B < K

F, G, K < L

- .

L

+ /

* K

Figure 20 (Contd.)

(The students may remember the events as an exercise)

Critical Path Analysis 14.17

Solution: The network is shown in figure-20 step by step with dependency relationships

that are incorporated at each step which the student should, by now, be able to follow.

Please carefully review the steps where dummies are incorporated.

14.4.8 Concurrent activities: Activities may not always be discrete i.e. they may be done in part allowing the subsequent activities to commence before the preceding activity is fully completed. Activities of this kind are to be frequently encountered in batch production. If, for example, a batch of 50 spindles is to be processed on two machines obviously it is not necessary to process all the items of the batch on the first machine and then transfer these to the next machine. A few items processed on the first machine may be transferred to the second machine before completion of the entire batch on the first machine. Since this is a matter of great practical importance we shall dwell upon it at a greater length. Such simultaneous or concurrent activities are to be encountered in sewage work e.g.,, trenching, laying pipe, welding pipe and back filling, all going on simultaneously with suitable lags on construction work.

Illustration

A batch of 4 axles is to be proceeded on the following three machines in this sequence:

Lathe (L), Milling (M) and Grinding (G)

Instead of first working on these 4 axles on lathe and then on milling and finally on grinding in sequence, it is desired to process the first axle on the lathe and as and when it is processed, it is taken up on milling and the 2nd axle on the lathe, and so on. In other words, each of the three activities L,M and G have been quartered for the sake of concurrent operations. You are required to draw the network.

Solution

The dependency relationships are sorted out hereunder:

Quartered Activity Preceded by

L1 None

L2 L1

L3 L2

L4 L3

M1 L1

M2 L 2, M 1

M3 L 3, M 2

M4 L 4, M 3

G1 M1

G2 M 2, G 1

G3 M 3, G 2

G4 M 4,G 3

The network is now constructed below. (The student is urged to draw his own network. Mere understanding of our solution is not enough.)

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1 2 4 8

L1 L2 L3 L4

3 5 6 9 10 12

M1 M2 M3 M4

7 11 13 14

G1 G2 G3 G4

Figure 21

(N.B.: The concurrent activities so drawn are known: as 'ladders' in the network jargon.)

Illustration

Draw a network diagram for the following data.

Task Immediate Predecessor

A —

B —

C B

D B

E B

F E

G A, D, C.

Solution

The required network is given below.

4

A G

C

B 2 D

1 3 6

E

F

5

Figure 22

Critical Path Analysis 14.19

Illustration

Draw a network diagram for the following data.

Activity Preceding activities

A —

B A

C A

D B

E A

F B, E

G C

H D, F

I G

J H, I

Solution: The required network is given below:

5

E F

1 A B D H J

2 4 6 8 9

C I

3 G

7

Figure 23

14.5 CRITICAL PATH ANALYSIS

You have already been familiarised with the logic of the critical path analysis by way of introduction of this chapter.

The purpose of the analysis is two-fold: (i) to find the critical path, i.e. the sequence of activities with the longest duration. Once it is found it is marked in bold sequence of arrows on the network. For a simple network as of figure -24 the various sequences can be enumerated and the durations of activities encompassed by them simply added, to find the critical sequence. As stated earlier, one could indeed end up with more than one critical sequence; and (ii) to find the float associated with each non-critical activity.

♦ Systematic analysis: The enumeration method would be too cumbersome computationally for any real life project, Even the computer would be hard pressed to proceed this way. There is a systematic way that cuts short the analysis time to manageable proportions.

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It is accomplished by performing the following steps:

1. Calculate the time schedule for each activity. This represents the time by which an activity must begin and the time before which it must be completed. The time schedule data for each activity includes, the earliest start, earliest finish, the latest start, latest finish, and finally the float, which is the spare time associated with an activity.
2. Calculate the time schedule for the completion of entire project. This represents the scheduled date for the completion of the entire project and the probability of completing the project on or before the deadline.
3. Identify the critical activities. These activities are the ones which must be started and completed on schedule, or else the project is likely to be delayed.
4. Determine the critical path for the network. This path represents the critical activities which must be closely followed in order to complete the project on time.

14.5.1 Scheduling computations: After preparing the network diagram, we wish to know how long it will take to complete the project and also to identify the activities in the network that are to be placed under strict control. The basic scheduling computations involve a forward and then a backward pass through the network. The process of tracing the network from START to END is called forward pass, and from END to START is called backward

pass. Based on a specified occurrence time for the initial network event, the forward pass computations give the earliest start and finish times for each activity. By the specification of the latest allowable occurrence time for the terminal network event, the backward pass computation will give the latest allowable start and finish times for each activity.

14.5.2 Forward pass computations: As stated above, the purpose of the forward pass is to compute the earliest start (EST) and finish time (EFT) for each activity. The EST time indicates the earliest time that a given activity can be scheduled. Earliest finish time for an activity indicates the time by which the activity can be completed, at the earliest. To compute these time estimates, we will first of all compute the earliest allowable occurrence time for various events of the network.

It is a convention to keep the earliest allowable occurrence time of the START event as zero. To understand, how this time estimate for other events is computed, let us consider the following network diagram.

$E_3 = 14$
 3
 8 6
 6 16
 1 2 0 5 6 $E_6 = 52$
 $E_1 = 0$ 10 20 $E_5 = 36$
 $E_2 = 6$
 4
 $E_4 = 16$

Figure 24

Critical Path Analysis 14.21

♦ Earliest allowable occurrence time of an event or Earliest Event Time :

In the network shown above, event 1 stands for the beginning of the activity 1 - 2 and we can say that it occurs at the time zero i.e. $E_1 = 0$. Event 2 stands for the finish of the activity 1-2 thus event 2 can occur at the earliest time E_2 which is computed as

$$E_2 = 0 + D_{12} = 0 + 6 = 6$$

where D_{12} stands for the duration of activity 1 - 2

Event 3 stands for finish of the activity 2-3 and its earliest time is

$$E_3 = E_2 + D_{23} = 6 + 8 = 14$$

The event 4 can occur either at the end of the activity 3 - 4 or at the finish of activity 2 - 4. In this case, there will be two time estimates as follows:

$$E_4 = E_3 + D_{34} = 14 + 0 = 14$$

$$E_4 = E_2 + D_{24} = 6 + 10 = 16$$

In case two or more time estimates exist for a particular event, then the time estimate with maximum value is retained as the earliest event time and other values are discarded. This maximum value represents the completion of all the activities ending at the event under consideration. In the above example, the earliest event time for event 4 will be 16.

A general rule can also be given here for determining the earliest event time as below:

$$E_j = \text{Max} (E_i + D_{ij})$$

where E_j is the earliest time for event j , E_i is the earliest time for event i , and D_{ij} is the duration of the activity $i-j$.

◆ Earliest start and earliest finish times of an activity

After computing the earliest event time of various events, one can easily compute the earliest start and finish times of all the activities on the network. The earliest start time of an activity is given by the earliest allowable occurrence time of the tail event of that activity. Thus, in our example, the earliest start time of the activity 1-2 will be given by the earliest time of the event 1 i.e. it will be 0. The earliest start time for the activities 2 - 3 and 2 - 4 will be given by the earliest time of event 2 which is equal to 6. The earliest time for the activities 3-4 and 3-5 will be 14 which is the earliest time for the event 3 and so on.

The earliest finish time of an activity will be simply equal to the earliest start time of the activity plus the duration of that activity. Hence, in our example, earliest finish time of activity 1-2 will be $0+6 = 6$, for activity 2-3, it will be $6+8 = 14$ and for activity 2-4, it will be $6+10 = 16$ and so on.

The complete computations for all the activities are shown in columns (3) and (4) of table -1 given on the next page.

(*Note: In various networks we have omitted subscripts for E_i 's and L_j 's)

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14.5.3 Backward pass computations: The purpose of the backward pass is to compute the latest start and finish times for each activity. These computations are precisely a "mirror image" of the forward pass computations. The term "latest allowable occurrence time of an event" (denoted by L_i) is used in the sense that the project terminal event must occur on or before some arbitrary scheduled time. Thus, the backward pass computations are started rolling back by arbitrarily specifying the latest allowable occurrence time for the project terminal event. If no schedule date for the completion of the project is specified, then the convention of setting the latest allowable time for the terminal event equal to its earliest time, determined in the forward pass, is usually followed, i.e. $L = E^*$ for the terminal event of the project. This convention is called the zero slack convention. Following this, one can also interpret the latest allowable activity finish time (LFT) as the time to which the completion of an activity can be delayed without directly causing any increase in the total time to complete the project.

To explain the computation, let us again consider the network diagram in figure-24. The terminal event is 6 so we set $L_6 = E_6 = 52$ and we start rolling back. The latest allowable occurrence time for the events 5 and 4 are $L_5 = 52 - 16 = 36$ and $L_4 = 36 - 20 = 16$ respectively. It may be noted here that we can roll back to event 3 via activity 3-5 as well as activity 4-3. So there are two latest allowable occurrence times for the event 3 as given below;

$$L_3 = L_4 - D_{34} = 16 - 0 = 16$$

$$L_3 = L_5 - D_{35} = 36 - 6 = 30$$

We retain the minimum value as the latest occurrence time for the event 3 and ignore other values. Therefore, the latest allowable occurrence time for the event 3 is 16. Similarly

$$L_2 = L_3 - D_{23} = 16 - 8 = 8$$

$$\text{and } L_2 = L_4 - D_{24} = 16 - 10 = 6$$

The latest occurrence time for the event 2 is thus 6 and the latest occurrence time for the event 1 is equal to its earliest time i.e. zero.

In general, the latest allowable occurrence time of an event can be calculated by selecting an appropriate formula among the following two:

$$L_i = L_j - D_{ij}$$

$$\text{or } L_i = \text{minimum } (L_j - D_{ij})$$

The second formula is used for the event having two or more latest allowable occurrence time estimates.

◆ Latest start and latest finish times of an activity

After computing latest allowable occurrence time for various events, one can compute the latest start and latest finish times of an activity. The latest finish time of an activity is equal to the latest allowable occurrence time of the head event of that activity.

$$\text{i.e. } LFT(i - j) = L_j$$

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The latest start time of an activity is equal to its latest finish time minus its duration.

$$\text{i.e. } LST(i - j) = LFT(i - j) - D_{ij}$$

These computations are shown in column (4) and column (6) of table-1 given below:

Table 1

START FINISH

Activity Duration Earliest time Latest time Earliest time Latest Time

(1) (2) (3) (4) (5) (6)

1-2 6 0 0 6 6

2-3 8 6 8 14 16

2-4 10 6 6 16 16

3-4 0 14 16 14 16

3-5 6 14 30 20 36

4-5 20 16 16 36 36

5-6 16 36 36 52 52

14.5.4 The critical path determination: After having computed various time estimates, we are now interested in finding the critical path of the network. A network will consist of a number of paths. A path is a continuous series of activities through the network that leads from the initial event (or node) of the network to its terminal event. For finding the critical path, we list out all possible paths through a network along with their duration. In the network under consideration, various paths have been listed below.

Path Length (in days)

1-2-3-5-6 36

1-2-4-5-6 52

1-2-3-4-5-6 50

♦ Critical path: A path in a project network is called critical if it is the longest path. The activities lying on the critical path are called the critical activities.

In the above example, the path 1-2-4-5-6 with the longest duration of 52 days is the critical path and the activities 1-2, 2-4, 4-5 and 5-6 are the critical activities.

14.5.5 Calculation of Floats: It may be observed that for every critical activity in a network, the earliest start and latest start time are the same. This is so since the critical activities cannot be scheduled later than their earliest schedule time without delaying the total project duration, they do not have any flexibility in scheduling. However, non-critical activities do have some flexibility i.e. these activities can be delayed for some time without affecting the project duration. This flexibility is termed as slack in case of an event and as float in case of an activity.

(*Note: In various networks we have omitted subscripts for E_i and L_j)

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Some people do not make any distinction between a slack and a float.

♦ Slack time for an event

The slack time or slack of an event in a network is the difference between the latest event time and the earliest event time. Mathematically it may be calculated using the formula $L_i - E_i$, where L_i is the latest allowable occurrence time and E_i is the earliest allowable occurrence time of an event i .

♦ Total float of an activity: The total activity float is equal to the difference between the earliest and latest allowable start or finish times for the activity in question. Thus, for an activity $(i-j)$, the total float is given by

$$TF_{ij} = LST - EST \text{ or } TF_{ij} = LFT - EFT$$

In other words, it is the difference between the maximum time available for the activity and the actual time it takes to complete. Thus, total float indicates the amount of time by which the actual completion of an activity can exceed its earliest expected completion time without causing any delay in the project duration.

♦ Free Float: It is defined as that portion of the total float within which an activity can be manipulated without affecting the float of the succeeding activities. It can be determined by subtracting the head event slack from the total float of an activity.

$$\text{i.e. } FF_{ij} = TF_{ij} - (\text{slack of event } j)$$

The free float indicates the value by which an activity in question can be delayed beyond the earliest starting point without affecting the earliest start, and therefore, the total float of the activities following it.

♦ Independent float : It is defined as that portion of the total float within which an activity can be delayed for start without affecting float of the preceding activities. It is computed by subtracting the tail event slack from the free float of an activity.

$$IF_{ij} = FF_{ij} - (\text{slack of event } i)$$

The independent float is always either equal to or less than the free float of an activity. If a

negative value is obtained, the independent float is taken to be zero .

♦ Interfering float: Utilization of the float of an activity can affect the float of subsequent activities in the network. Thus, interfering float can be defined as that part of the total float which causes a reduction in the float of the successor activities. In other words, it can be defined as the difference between the latest finish time of the activity under consideration and the earliest start time of the following activity, or zero, whichever is larger. Thus, interfering float refers to that portion of the activity float which cannot be consumed without affecting adversely the float of the subsequent activity or activities.

Illustration

Activity Duration

1-2 4 days

1-3 12 days

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1-4 10 days

2-4 8 days

2-5 6 days

3-6 8 days

4-6 10 days

5-7 10 days

6-7 0 days

6-8 8 days

7-8 10 days

8-9 6 days

With the help of the activities given above draw a network. Determine its critical path, earliest start time, earliest finish time, latest start time, latest finish time, total float, free float and independent float.

Solution

The network based on the activities given in the example is as follows:

E=4

L=4 E = 10

6 L = 12

2 5

4 10

8 77 E = 22

L = 22

10 10 0 10

E=0 1 6

4 16 8 9

L=0 8 E = 38

E = 12 E = 22 E = 32

L = 32 L = 38

$$12 L = 12 \quad 8 L = 22$$

$$3 E = 12$$

$$L = 14$$

Figure - 25

The time estimates and floats are calculated as below:

Job or Duration EST EFT LST LFT Slack of

activity of an (= E_i) ($E_i + D_{ij}$) ($L_j - D_{ij}$) (= L_j) tail head Total float Free float Independent

activity event event (LST- (Total Float (Free float-

D_{ij} ($L_i - E_i$) ($L_j - E_j$) EST) -Slack of slack of

head event) tail event)

$$1-2 \quad 4 \quad 0 \quad 4 \quad 0 \quad 4 \quad 0 \quad 0 \quad 0 \quad 0$$

$$1-3 \quad 12 \quad 0 \quad 12 \quad 2 \quad 14 \quad 0 \quad 2 \quad 2 \quad 0$$

$$1-4 \quad 10 \quad 0 \quad 10 \quad 2 \quad 12 \quad 0 \quad 0 \quad 2 \quad 2$$

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$$2-4 \quad 8 \quad 4 \quad 12 \quad 4 \quad 12 \quad 0 \quad 0 \quad 0 \quad 0$$

$$2-5 \quad 6 \quad 4 \quad 10 \quad 6 \quad 12 \quad 0 \quad 2 \quad 2 \quad 0$$

$$3-6 \quad 8 \quad 12 \quad 20 \quad 14 \quad 22 \quad 2 \quad 0 \quad 2 \quad 0$$

$$4-6 \quad 10 \quad 12 \quad 22 \quad 12 \quad 22 \quad 0 \quad 0 \quad 0 \quad 0$$

$$5-7 \quad 10 \quad 10 \quad 20 \quad 12 \quad 22 \quad 2 \quad 0 \quad 2 \quad 0$$

$$6-7 \quad 0 \quad 22 \quad 22 \quad 22 \quad 22 \quad 0 \quad 0 \quad 0 \quad 0$$

$$6-8 \quad 8 \quad 22 \quad 30 \quad 24 \quad 32 \quad 0 \quad 0 \quad 2 \quad 2$$

$$7-8 \quad 10 \quad 22 \quad 32 \quad 22 \quad 32 \quad 0 \quad 0 \quad 0 \quad 0$$

$$8-9 \quad 6 \quad 32 \quad 38 \quad 32 \quad 38 \quad 0 \quad 0 \quad 0 \quad 0$$

Critical path is represented by 1-2-4-6-7-8-9.

(Note: In this table we have given detailed calculations merely for explanation.)

Critical Path Analysis 14.27

Illustration

Analyse the network below for the critical path and for different floats

$$5$$

$$2 \quad 6$$

$$E=3 \quad E = 19$$

$$L=3 \quad L = 19$$

$$3 \quad 10 \quad 6 \quad 1$$

$$E=0 \quad 1 \quad 3 \quad 15 \quad E = 17$$

$$L=0 \quad 4 \quad 4 \quad L = 18$$

$$E = 13$$

$$L = 13$$

$$14 \quad 1$$

$$4 \quad E = 14$$

$L = 17$

Figure 26

Network Analysis Table

Activity Duration Start Finish Float

E L E L Total FreeInd. Interfering

1-2 3 0 0 3 3 0 0 0 0

1-3 4 0 9 4 13 9 9 9 0

1-4 14 0 3 14 17 3 0 0 3

2-3 10 3 3 13 13 0 0 0 0

2-6 5 3 14 8 19 11 11 11 0

3-5 4 13 14 17 18 1 0 0 1

3-6 6 13 13 19 19 0 0 0 0

4-5 1 14 17 15 18 3 2 0 1

5-6 1 17 18 18 19 1 1 0 0

(N.B.: Activities must always be arranged in the i-j sequence.)

Explanation: Consider activity 1 - 3. The slack of head event 3 is $13 - 13 = 0$. Therefore, free float = $9 - 0 = 9$. Likewise slack of tail event 1 is $0 - 0 = 0$. Therefore independent float = 9.

Illustration

Analyse the network below for the critical path and calculate total float, free float and independent float for each activity.

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$E = 43$ $E = 63$ $E = 83$

$E = 36$ $L = 43$ $L = 63$ $L = 83$

$L = 41$

4 8 9 10

A/36 D/2 K/20 M/20

0 J/9

1 2 3 E/1 L/8

B/4 C/2

$E=0$ $E=4$ F/15 7

G/9

$L=0$ $E=6$ $E = 34$

$L=4$ $L=6$ $L = 34$

5

I/9

$E = 21$ H/4 Critical Path: 2-3-5-6-7-8-9-10

$L = 21$

6 $E = 25$

$L = 25$

Figure - 27

Solution

Network Analysis Table

i-j D Start Finish Floats

E L E L T F I

1- 2 4 0 0 4 4 0 0 0

1-4 36 0 5 36 41 5 0 0

2-3 2 4 4 6 6 0 0 0

3-5 15 6 6 21 21 0 0 0

3-8 10 6 33 16 43 27 27 27

4-8 2 36 41 38 43 5 5 0

5-6 4 21 21 25 25 0 0 0

5-8 9 21 34 30 43 13 13 13

6-7 9 25 25 34 34 0 0 0

7-8 9 34 34 43 43 0 0 0

7-9 8 34 55 42 63 21 21 21

8-9 20 43 43 63 63 0 0 0

9-10 20 63 63 83 83 0 0 0

Critical Path : BCHIJKM = 83 weeks.

Illustration

A small assembly plant assembles PCs through 9 interlinked stages according to following precedence / process:

Critical Path Analysis 14.29

Stage Duration

From to (Hours)

1-2 4

1-3 12

1-4 10

2-4 8

2-5 6

3-6 8

4-6 10

5-7 10

6-7 0

6-8 8

7-8 10

8-9 6

(i) Draw an arrow diagram (network) representing the above assembly work.

(ii) Tabulate earliest start, earliest finish, latest start and latest finish time for all the stages.

(iii) Find the critical path and the assembly duration.

(iv) Tabulate total float, free float and independent float.

Solution

(i) The network diagram representing the given assembly work is as follows:

E=4 6 E = 10

> 5

L=4 2 L = 12

10 >

4 7

8

E=0 0 10

L=0

10 10 8 6

1 > 4 > 6 > 8 > 9

E = 12 E = 22 E = 32 E = 38

L = 12 L = 22 L = 32

12 L = 38

8

3 E = 12

L = 14

Figure 28

14.30 Advanced Management Accounting

(ii) The earliest start, earliest finish, latest start and latest finish times for all the stages are calculated in the table given below:

Stage or Duration	Earliest Start (EST)	Earliest Finish (EFT)	Latest Start (LST)	Latest Finish (LFT)	Total Float (LFT-EST)	Free Float (LFT-EFT)	Independent Slack of Head event	Slack of tail event
Activity (Hours)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1-2	4	0	4	0	4	0	0	0
1-3	12	0	12	2	14	2	0	0
1-4	10	0	10	2	12	2	2	2
2-4	8	4	12	4	12	0	0	0
2-5	6	4	10	6	12	2	0	0
3-6	8	12	20	14	22	2	2	0
4-6	10	22	12	22	0	0	0	0
5-7	10	12	20	12	22	2	2	0
6-7	0	10	22	22	22	0	0	0
6-8	8	22	30	24	32	2	2	2
7-8	10	22	32	22	32	0	0	0
8-9	6	22	38	32	38	0	0	0

(iii) The critical path for the above network is given by 1 - 2 - 4 - 6 - 7 - 8 - 9. The assembly duration is 38 hours.

(iv) The total float, free float and independent float are given by columns (7), (8) and (9) respectively of the above table.

14.6 Distinction between PERT & CPM

PERT CPM

PERT is used for non-repetitive Jobs like planning the CPM is used for non-Repetitive Job assembly of the space platform like building a house

It is a probabilistic Model It is a deterministic Model

It is event oriented It is activity oriented

It incorporates Statistical Analysis It does not incorporates Statistical Analysis

It serve as useful control device It is difficult to use CPM as a control device

Critical Path Analysis 14.31

SUMMARY :

- Two most commonly used network Models
 - ü Critical Path Method(CPM)
 - ü Programme Evaluation Review Technique(PERT)
- Construction of Network diagrams
- Calculation of Critical Path and Project Completion time together with Floats, earliest start and finishing times
- Application of technique to crash cost problems
- Critical path should be longest route through the network
- Resource schedule problems and use of bar charts
- Identifying the project duration
- Cheapest critical activities to crash and not reduce project completion time by reducing non-critical activities

SELF-EXAMINATION QUESTIONS

1. Define a project and briefly explain the four common implications which characterise a project and state the five steps of the working methodology of critical path analysis.
2. Distinguish between PERT and CPM.
3. What kinds of decision making situation may be analysed by using CPM technique.
4. Draw a network corresponding to the following information. Obtain the early and late start and completion times, and determine the critical activities.

Activity Duration

1-2 4 days

1-3 6 days

2-6 8 days

3-4 7 days

3-5 4 days

4-6 6 days

5-6 5 days

5-7 19 days

6-7 10 days

14.32 Advanced Management Accounting

5. A work project consists of twelve activities labelled A through L. Upon being asked to specify the order in which the jobs had to be done, the managers answered as follows :

A, B and C are the first activities of the project and can start simultaneously and immediately;

A and B precede D while B precedes E, F and H. Activities F and C precede G while E and

H precede I and J. C, D, F and J precede K which in turn, precedes L. Further I, G and L are the terminal activities of the project.

The completion times of the various activities are listed here :

Activity Time (days)

A 6

B 4

C 10

D 1

E 6

F 3

G 14

H 1

I 9

J 2

K 7

L 5

(a) Draw a network diagram corresponding to this project.

(b) Obtain the lengths of all the paths and determine critical path.

ANSWER TO SELF-EXAMINATION QUESTIONS

5. The network diagram is shown in Figure below :

E 4

H

2 6

I

B F

C G

1 5 9

A J L

D K

3 7 8

Figure 29

Critical Path Analysis 14.33

The various paths and their lengths are as follows :

Path Length

1-3-7-8-9 19

1-2-3-7-8-9 17

1-2-4-5-7-8-9 19

1-2-5-7-8-9 24* Critical

1-2-5-9 19

1-2-4-5-9 14

1-2-6-7-8-9 19

1-2-6-9 21

1-6-7-8-9 22

1-6-9 24* Critical

ESD.36 System Project Management

Lecture 1

- Class Introduction

Instructor(s)

Prof. Olivier de Weck

Dr. James Lyneis, Prof. Dan Braha

Today's Agenda

- Welcome and Introductions
- Definitions – Initial Discussion
- Course Objectives
- Schedule
- Term Project, Homeworks
- Questions?

2

Introductions

- Olivier de Weck
- Dipl. Ing. Industrial Engineering – ETH Zurich '93
- 1993-1997 Engineering Program Manager Swiss F/A-18 Project, McDonnell Douglas, St. Louis
- S.M. '99 Ph.D. '01 Aerospace Systems – MIT
- Associate Professor – dual appointment AA and ESD, Executive Director Production in Innovation Economy (PIE) Study
- Research:

□ Systems Engineering for Changeability
and Commonality

□ <http://strategic.mit.edu>

□ Space Logistics

□ <http://spacelogistics.mit.edu>

3

Introductions

□ James Lyneis

□ S.B.s MIT – EE and System Dynamics ('71)

□ PhD – Univ. of Michigan (Mgt. Science; '74)

□ Sloan Faculty '74-'78; Senior Lecturer '98-Present

□ Professor of the Practice, WPI '02-Present

□ Consultant with Pugh-Roberts Associates, '78-'02,
working on many project models

4

Introductions

□ Dan Braha

□ Professor at University of Massachusetts

□ Sabbatical at MIT ESD in AY 2012/13

□ Affiliation with New England Complex
Systems Institute (NECSI)

□ Specializes in Complexity research, with
application to complex projects and human
organizations

The Structure and Diffusion Dynamics
of Large Scale Organizational

Networks Social Network Analysis of
ProductDesign and Development
Organizational Networks

5

Introductions

□ Let's go around the room & remote
sites

□ Name

□ Company and Job (present or past)

□ One observation on project management

6

Project Definition

- A Project is a set of tasks that
- Are related to each other
- Have a specific objective to be completed within certain specifications
- Have defined start and end dates
- Have funding limits
- Consume resources

7

The “Iron Triangle”

Cost Scope

Project

- Why “iron” triangle?

Schedule □ Risk if all three are constrained !

8

System Definition

- A System is a set of physical or virtual objects whose interrelationships enable desired function(s).
- more than the sum of its parts
- Undesired (emergent) functions often exist
- System complexity scales with the number of objects as well as the type and number of interconnections between them
- Instantaneously available functions, versus “lifecycle” properties (scalability, flexibility, robustness ...)
- A Product is a “System” sold for profit

9

Example System: F/A-18 Aircraft

- Clean sheet design ~ 1978 F/A-18 A/B
- Re-designs: C/D (1987), E/F (1999)
- Hardware, Software, Humans ...
- What is inside the system boundary?

10

Project Management

- Project Management comprises a body of methods and tools that facilitate the achievement of

project objectives

- ☐ Within time
- ☐ Within cost
- ☐ Within scope
- ☐ At the desired performance/specification level
- ☐ While effectively and efficiently utilizing resources
- ☐ While carefully managing risks and opportunities

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Research and Development

R D

Research, Technology Product/System

Development Development

- ☐ Unstructured methods ☐ Structured methods
- ☐ Difficult to plan ☐ Generally planned
- ☐ Predictable
- ☐ Unpredictable

Our focus is on

downstream

development.

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+ Discussion Point 1:

- Why is complex (S)PM hard?

- ☐ Main obstacles to easy project success:
- ☐ Poorly defined project objectives or shifting system requirements
- ☐ What do you think? ...

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Concept Question 1 (DEMO)

- ☐ A project is
- ☐ A - ongoing management of facilities
- ☐ B - a finite undertaking with a specific goal
- ☐ C - task-based design
- ☐ D - doomed to failure from the outset
- ☐ E - all of the above
- ☐ F - none of the above

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Relationship w/other SDM core classes

- ☐ System Architecture (ESD.34) is about the “DNA” of the

ARTIFACTS themselves – atomic unit: object

□ Concept, form, function, decomposition ...

□ Systems Engineering (ESD.33) is about the PROCESSES to understand and design systems – atomic unit: process

□ QFD, DOE, Requirements Analysis and Verification, ...

□ Integrating the Lean Enterprise (ESD.61J) is about the PEOPLE and ORGANIZATIONS – atomic unit: person

□ Principles of lean manufacturing, organizational models

□ System Project Management (ESD.36) is about how to best utilize resources to implement a set of objectives – atomic unit: task

□ CPM, DSM, System Dynamics

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Task as an Object-Process-Diagram

System

Inputs n r Elements

o Task m

Resources Executing Deliverables

p q

Time

Personnel Facilities

Money (\$)

Consumables Using OPM Nomenclature

16

Ref: OPM, D. Dori, 2002

Project = set of related tasks

m

o A n

r

m

q

p

B n

r

m

q D

n

r

q

F m

Deliverable from A
is an input to C B and D use the p q
same people

r

m

C n r F requires deliverables

From D and E

p q E m

p

C and E need access

to the same facility

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+ System Project Management

ESD.36 Framework

Project

Preparation Project Project

Planning Monitoring

Project

Control

Enterprise has

chosen what product Project

or system to develop Adaptation Project

Completion

Next

Project Project

Learning

Start

18

- Course Objectives

□ Introduce advanced methods and tools of Project
Management in a development context

□ CPM/PERT, Critical Chain, Design Structure Matrix

□ System Dynamics

□ Earned Value Management

□ Understand how methods work (strengths, limitations)

□ Industry Examples

□ Gain appreciation for organizational and human aspects

□ Case Studies

□ Managing International Projects, Portfolios of Projects ...

□ Learn from each other

- Class Discussions
- Project Assignments
- □ Improve development projects in your career/firm

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+ Class

Schedule

(next 7 weeks, refer to syllabus)

Business Trip 20

Readings

- Required Readings
- NO Paper Class Reader Packet
- Read ahead of lecture ~ 1-2 papers/chapters per session
- Check reading assignments in the syllabus
- Next:
 - MIT Press Book Chapter: Introduction to PM
 - ABCs of the Critical Path Method (1963)
- Optional Readings
- Textbooks
 - Available at MIT Library (Dewey)
 - Purchase only if you think useful beyond class
 - (e.g. MIT COOP, amazon.com etc)

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Draft Textbook

- “Successfully Designing and Managing Complex Projects”
- de Weck, Lyneis
- MIT Press, draft in development
- textbook to support SDM core class ESD.36
- current draft ~ 300 pages

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Case Studies

- Case 1: Civil Engineering Project
 - BAE Systems: DIA Baggage Handling System
 - HBS-9-396-311
 - or “live case” BP Wind Farm Development
- Case 2: Aerospace Project
 - Raise and Fall of Iridium (or Mission to Mars HBS-9-603-083)
 - HBS-9-601-040
 - Presented by Prof. Joel Schindall
- Case 3: Software Project

- Microsoft .Net (and/or Microsoft Office 2000)
- HBS-9-602-086
- Presented by Prof. Paulo Gomes
- Case 4: Voted by Students

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Project Assignment

- Apply Design Structure Matrix (DSM) method, generally at your sponsor company site
- System Dynamics Project (incl. simulation)
- Survey of Methods & Tools in company
- Analyze Success or Failure of a significant Past Product/System Development Project

Product/System Development Project

- Work in teams of 4 (nominally)
- 1-page project proposals due on 9/18
- Get approval by 9/25
- Project Update due on: 10/25
- Final presentation in class on December 4 or 6

24

Previous Project Examples

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
36 37 38 39 40 41

Marketing & Competition Research x 1 1

Cost Risk Analysis to Bid 1 x 1

Customer Requirement Negotiation 1 1 x

DSM

Technical Screening for Qualification 1 1 1 x

Project Authorization & Kick off 1 1 1 x

Failure Effect Mode Analysis x

Control of Engineering Changes

Project

Design for Catalyst Volume & Emissions 1 1 1 1 1

CAE for Catalyst Volume & Emission 1 1 1 x 1

Design for Tuning Volume, Pressure & Flow 1 1 1 x 1 1

CAE for Tuning Volume, Pressure & Flow 1 1 1 x 1

Design for Legal Noise and Sound Quality 1 1 1 x 1 1

CAE for Legal Noise and Sound Quality 1 1 1 x 1

Design for Durability Target 1 1 1 x 1 1 1 1

CAE for Durability Target 1 1 1 x 1 1 1

Cost Analysis for Manufacturing 1 1 1 1 x 1

Exhaust

Prototype for DV Test & Pre-prod Build 1 1 1 1 1 1 1 1 1 x

Design Verification Test 1 1 x

Testing for Emission 1 1 x

Testing for Tuning Volume, Pressure & Flow 1 1 x

System

Testing for Legal Noise and Sound Quality 1 1 x

Sub-tier Supplier DV Bench Testing 1 1 1 1 1 x

Vehicular Proving Ground Testing 1 1 1 1 1 1 1 1 1 1 1 x

Data Analysis and Editing 1 x

Design Sub-System Durability Testing

Design Release

Release of Accurate 3D Geometry

1 1

1

1

1

1

1

1

1

1 1 x

x

x

1

Issues Sales Authority to Release (SAR) 1 1 x

Creation and Release of all 2D drawings 1 1 1 x

Mfg Facility Lineup & Tooling Process 1 1 x 1 1

Mfg Cell Setup and Fine Tuning 1 1 1 x 1 1 1

PSO & "Run-at-Rate" Demonstration 1 1 1 1 x

Quality Control for Manufacturing 1 1 1 1 1 x

Mfg Process Evaluation 1 1 1 1 1 1 x

Product Verification Testing 1 1 1 1 1 1 1 x

Sub-tier Supplier PV Testing 1 1 1 1 1 1 1 1 x

Production Part Approval Process 1 1 1 1 x

Production 1 1 1 x

Warranty Cost Analysis 1 1 1 1 1 1 1 1 1 1 x

Customer Satisfaction Evaluation 1 1 1 1 1 1 1 1 1 x

- Understand Iterations

- Reduce Expected Project Duration

25

Understanding the sources of change --
 External factors; management responses; side effects
 Out-of-Sequence
 Work, Worksite
 Congestion,
 Coordination
 Fatigue, Problems,
 Burnout Morale Problems
 Average P eo p le P ro d u ctivity Q u ality
 Employee Skill
 and Quality P ro g ress
 W o rk W o rk
 Overtime To Be Done R e a lly D o n e
 Scope Customer
 Growth Changes Schedule
 Know n U n d is c o v e r e d
 Acceleration
 R e w o rk R e w o rk
 R e w o rk
 D isco very
 Hiring
 Apparent
 Progress
 26
 + Understand the dynamics of a cancelled
 project at NASA
 Baseline Plan Staffing
 FY2000 Replan
 700
 Actuals Through FY02 & Projections FY02-
 FY06 (FY2000 Re-plan)
 600 Model for Baseline
 Project
 Cancelled
 500
 September
 2002
 400
 Staff - EP
 300

200
FY00 Re-plan
First Launch
100 Baseline First
Hypergolic Launch
Maintenance
0 Facility Set
Operational
M1-3
M4-6
M7-9
M10-12
M13-15
M16-18
M19-21
M22-24
M25-27
M28-30
M31-33
M34-36
M37-39
M40-42
M43-45
M46-48
M49-51
M52-54
M55-57
M58-60
M61-63
M64-66
M67-69
M70-72
M73-75
M76-78
M79-81
M82-84
M85-87
M88-90
M91-93
M94-96
M97-99

M100-102

M103-105

M106-108

M109-111

M112-114

M115-117

M118-120

M121-123

M124-126

M127-129

M130-132

FY1996 FY1997 FY1998 FY1999 FY2000 FY2001 FY2002 FY2003 FY2004 FY2005 FY2006

Sep 4, 2008

Time 27

Discussion Point 2: Failure?

- ☐ What can lead to projects failing ?
- ☐ What is success/failure?
- ☐ Project Manager is unqualified and overwhelmed.
- ☐ What else...

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Homework Assignments

- ☐ 6 Individual assignments,
but can cooperate
(acknowledge !)

- ☐ Don't spend more than
~10-15 hours per HW !

Electrical CityCar

Design Project

HW1: Critical Path and Network of Tasks

HW2: Design Structure Matrix and Iterations

HW3: System Dynamics – Initial Model

HW4: Budgeting and Earned Value

HW5: System Dynamics – Brook's Law

HW6: Project Organizational Design

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Grading

- ☐ Homeworks 60%
- ☐ Project Assignment 25%
- ☐ Active Participation 15%

- Total 100%
- Each HW counts 10%
- All project team members receive same project grade, work together
- People in this class do get A,B,C ...even F ... you want a good grade? ... you have to earn it !

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The Course Site

- Getting Started

- SDM students should add themselves to the class list (if not already there)
- Non-SDM students get permission from instructors
- Contact TA via email to be added
- Course number is ESD.36
- Make sure you go to the Fall 2012 version

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The Course Site

- Main features

- Syllabus (under > Materials > General)
- Calendar
- Schedule
- Handouts
- Go to “Materials” (store all shared files)
- will be posted before each lecture
- Homework
- Submit assignments through the course site
- Receive comments and grades
- FAQ / Forums
- Forum will be used for various topics

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- Remember

- Read the syllabus!
- Answers many of your questions
- Take a look at what's already uploaded
- Reading for next class
- Sign up on the course site if needed
- Non-SDM students need permission of instructor to enroll.

- Use your name cards every class.
 - PLEASE ... okay, pay attention
 - Laptops needed to answer concept questions
 - IM TA's during lecture from remote sites if problems surface
- 33

MIT OpenCourseWare

<http://ocw.mit.edu>

ESD.66 6\VWHP3URMHFW0DQDJHPHQW

Fall 2012

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ESD.36 System & Project Management

Lecture 2

- Critical Path Method (CPM)

Instructor(s)

Prof. Olivier de Weck

Today's Agenda

- Overview of PM methods and tools
- CPM 101
- Critical Paths, Slack
- Task "Crashing" and Cost
- Conclusions and Class Discussions
- Introduce HW3
- ESD.36 SPM 2

History of Project Management

- Big Projects since antiquity
- Pyramids (Egypt), Great Wall (China)
- Enormous workforce, but little documented evidence of formal project management
- Formal Project Management
- Henry Gantt (1861-1919) □ bar chart 1910
- 1957 Sputnik Crisis □ revival of "scientific management"
- Polaris (1958) □ Project Evaluation and Review Technique (PERT)
- DuPont Company (1960) □ Critical Path Method (CPM)
- 1960's NASA projects: Mercury, Gemini, Apollo
- Work Breakdown Structures (WBS)

□ Cost and Schedule Tracking, Configuration Management

- ESD.36 SPM 3

Comments about early PM

□ Project decomposition necessary due to complexity

□ Resource allocation and workload smoothing

□ Schedule urgency .."before the decade is out"

JFK

□ Circumstances

□ Complex Relations between Government and Contractors

□ "Shielded" from Society, Competition, Regulations

□ Cold War Pressures for Nuclear Power, Space Race ..

□ Other Innovations

□ Project Manager as a central figure

□ Beginnings of Matrix Organization

□ "Earned Value" – adopted by USAF (1963)

□ Professionalization since 1969

□ Diffusion into other industries: computers, automotive ...

□ Project Management Institute (PMI) founded – PMBOK

□ ISO 10006:2003 Quality in Project Management

□ Recent criticism about PM standards as "bureaucratic"

- ESD.36 SPM 4

Fundamental Approaches

□ How to represent task relationships?

□ Network-based (graph theory) methods

□ CPM, PERT,

a b

□ Task is a node or an arc d e

c

□ Matrix-based methods

□ DSM - Tasks are columns and rows x x

□ Interrelationships are off-diagonal entries

xx

x x

x x

□ System Dynamics x x

□ Feedback loops, causal relationships

□ Stocks and flows simulation

□ Tasks that are done or waiting to be done

PEOPLE PRODUCTIVITY

are stocks – "amount of work" WORK WORK

WORK

□ Doing project work causes a “flow”

TO BE

BEING DONE DONE

DONE

- ESD.36 SPM 5

Gantt Charts

□ Attributed to Henry Gantt – most popular PM tool (80%)

□ Used to plan big shipbuilding projects (cargo ships WWI)

□ Graphical way of showing task durations, project schedule

□ Does not explicitly show relationships between tasks

□ Limited use for project tracking

□ Easy to understand

today calendar

completion

Gantt Chart Builder System (Excel) 1.6

08 Sep'03 22 Sep'03 06 Oct'03

Bus Unit Project % Start Finish 08/09 15/09 22/09 29/09 06/10 13/10

Project "XYZ" 30 9-Sep-03 6-Oct-03

Mkt Customer Clinic 100 09-Sep-2003 12-Sep-2003

milestone

Sys Requirements Definition 100 11-Sep-2003 15-Sep-2003

Eng Parts Design 50 15-Sep-2003 22-Sep-2003

Sys Design Review 0 23-Sep-2003 23-Sep-2003

Mfg Manufacturing 0 24-Sep-2003 05-Oct-2003

Sys Product Release 0 06-Oct-2003 06-Oct-2003

tasks actual planned

- ESD.36 SPM D 6

CPM 101

□ Represent a project (set of task) as a network using graph theory

□ Capture task durations

□ Capture task logic (dependencies)

(Parallel)

Task ID (Series) Expected

Duration B,8

A,5 B,8 D,3

A,5

C,2

“B can only start after A is completed” “B and C do not depend on each other”

- ESD.36 SPM 7

CPM Assumptions

- Project consists of a collection of well defined tasks (jobs)
- Project ends when all jobs completed
- Jobs may be started and stopped independently of each other within a given sequence (no “continuous-flow” processes)
- Jobs are ordered □ “technological sequence”

- ESD.36 SPM 8

Task Representations

- Tasks as Nodes of a Graph
- Circles

ES EF

- Boxes

A,5 ID:A Dur:5

LS LF

- Tasks as Arcs of a Graph
- Tasks are uni-directional arrows
- Nodes now represent “states” of a project
- Kelley-Walker form

A, 5

broken fixed

- ESD.36 SPM 9

Concept Question 1

What in project management is named after Henry Gantt (1861-1919)?

- A new method for managing budgets in shipyards
- A matrix representation of the task relationships known as the Gantt-Matrix
- A horizontal bar chart known as the Gantt-Chart
- Gantt-teams, a new form of organizational structure
- Gantt was a mechanical engineer and did not care about projects

- ESD.36 SPM 10

Work Breakdown Structure

- Used to create the task (job) list
- Tree-decomposition of project tasks
- WBS identifies “terminal elements”
- The key starting point for project planning
- Required by US Govt as part of SOW
- Can be activity-oriented or deliverable- oriented
- Use “sticky-notes” method early on
- Carl L. Pritchard. Nuts and Bolts Series 1: How to Build a Work Breakdown Structure. ISBN 1890367125

Job A Job B Job X Job G

- ESD.36 SPM 11

WBS – Painting a Room

- 1 Prepare materials
 - 1.1 Buy paint
 - 1.2 Buy brushes/rollers
 - 1.3 Buy wallpaper remover
- 2. Prepare room
 - 2.1 Remove old wallpaper
 - 2.2 Remove detachable decorations
 - 2.3 Cover floor with old newspapers
 - 2.4 Cover electrical outlets/switches with tape
 - 2.5 Cover furniture with sheets
- 3. Paint the room
- 4. Clean up the room
 - 4.1 Dispose or store left over paint
 - 4.2 Clean brushes/rollers
 - 4.3 Dispose of old newspapers
 - 4.4 Remove covers Source: <http://www.wikipedia.org>

- ESD.36 SPM 12

+ WBS of MIT/NASA Space

- Logistics Project

IP S C M & L A

1 . 0 P r o j e c t 2 . 0 T e r r e s t r i a l 3 . 0 S p a c e L o g i s t i c s 4 . 0 E x p l o r a t i o n 5 . 0 I n t e r p l a n e t a r y 6 . 0 P r o j e c t

M a n a g e m e n t A n a l o g i e s N e t w o r k M o d e l S u p p l y / D e m a n d S C A r c h T r a d e I n t e g r a t i o n a n d

(T a s k 1) (T a s k 2) M o d e l i n g S t u d i e s D i s s e m i n a t i o n

(Task 3) (Task 4)

1.1 MIT Project 2.1 Low Density Equipment 3.1 Model Formulation 4.1 Supply / Demand 5.1 Trade Studies 6.1 NASA Workshop I

Management SCMLessons Learned Data Modeling Planning and Setup

2.2 Long-Range Military 6.2 NASA Workshop II

1.1.1 MIT Project SCMLessons Learned 3.1.1 MIT Segment 4.1.1 MIT Segment 5.1.1 MIT Segment

Manager / PI

2.3 Remote Environments 6.3 Internal Integration

1.1.2 MIT Travel SCMLessons Learned 3.1.2 JPL Segment 4.1.2 JPL Segment 5.1.2 JPL Segment

2.4 ISS SCM

1.1.3 MIT Resource Lessons Learned 3.2 Model 4.1.3 USA Segment 5.2 Trade Studies 6.3.1 MIT Segment

Administration Development Execution

2.5 Shuttle / ISS

1.1.4 Reserves Lessons Learned 4.2 Data 5.3 Trade Studies 6.3.2 JPL Segment

3.2.1 MIT Segment Collection Reporting

1.2 JPL Project 6.3.3 USA Segment

Management 3.2.2 JPL Segment 4.2.1 JPL Segment

6.4 Public Education

1.2.1 JPL Project 3.3 Model 4.2.2 USA Segment and Outreach

Manager / Co-PI Validation

1.2.2 JPL Travel 4.3 Data 6.4.1 MIT Segment

3.3.1 MIT Segment Validation

1.2.3 JPL Resource 6.4.2 JPL Segment

Administration 3.3.2 JPL Segment 4.3.1 JPL Segment

1.2.4 Reserves 6.4.3 USA Segment

3.3.3 USA Segment 4.3.2 USA Segment

1.3 USA Project 6.5 Space SCM

Management Executive Short Course

1.3.1 USA Project

MIT (incl. PSI): cyan

Manager / Co-PI

JPL: gray

1.3.2 USA Travel USA: green

Color code WBS by

Organization in charge

1.3.3 USA Resource

Administration

- ESD.36 SPM 13

1.3.4 Reserves

Discussion Point 1

□ Why is it difficult to come up with a good WBS (task list, task structure) in a complex project?

□ Not all tasks known ahead of time if completely new product/system

□ Others...?

- ESD.36 SPM 14

WBS Guidelines

□ No more than 100-200 terminal elements, if more □ use subprojects

□ Can be up to 3-4 Levels deep

□ Not more than 5-9 jobs at one level

□ Human cognitive “bandwidth” only 3 bits=2³=8

□ Short term memory for most people 5-9 items

□ Poorer planning if “too-fine grained” – dilution of attention

□ The more tasks there are, the more intricate dependencies there will be to keep track of

□ Jobs should be of similar size/complexity

□ Manageable chunks □ sense of progress

□ Level of graininess □ very difficult to answer

- ESD.36 SPM 15

Task List

□ List all tasks in a table with

□ Identifying symbol (tag, ID number)

□ Task description

□ Immediate prerequisite jobs

□ Expected task duration

□ Arrange jobs in “technological order”

□ No job appears in the list until all its predecessors have been listed

□ Iterations are NOT allowed □ “cycle error”

□ Job a precedes b precedes c precedes a

□ We will discuss iterations a lot in this class !!!

- ESD.36 SPM 16

Simple Example: Job List

□ Two Parts X and Y: Manufacture and Assembly

Job # Description Immediate Time

Predecessors [min]

A Start 0

B Get materials for X A 10

C Get materials for Y A 20

D Turn X on lathe B,C 30

E Turn Y on lathe B,C 20

F Polish Y E 40

G Assemble X and Y D,F 20

H Finish G 0

- ESD.36 SPM 17

Project Graph

□ Each job is drawn on a graph as a circle*

□ Connect each job with immediate predecessor(s) – unidirectional arrows “→”

□ Jobs with no predecessor connect to “Start”

□ Jobs with no successors connect to “Finish”

□ “Start” and “Finish” are pseudo-jobs of length 0

□ A finite number of “arrow paths” from “Start” to “Finish” will be the result

□ Total time of each path is the sum of job times

□ Path with the longest total time □ “critical path”

□ There can be multiple critical paths □ minimum time to complete project

* or other symbol, see before

- ESD.36 SPM 18

Project Graph

C,20 E,20 F,40

critical path

A,0 G,20 H,0

Start Finish

B,10 D,30

4 unique paths: A,C,E,F,G,H; A,C,D,G,H; A,B,D,G,H; A,B,E,F,G,H

100 70 60 90

- ESD.36 SPM 19

Critical Path

- CP is the “bottleneck route”
- Shortening or lengthening tasks on the critical path directly affects project finish
- Duration of “non-critical” tasks is irrelevant
- “Crashing” all jobs is ineffective, focus on the few % of jobs that are on the CP
- “Crashing” tasks can shift the CP to a different task
- Shortening tasks – technical and economical challenge
- How can it be done?
- Previously non-critical tasks can become critical
- Lengthening of non-critical tasks can also shift the critical path (see HW1).

- ESD.36 SPM 20

Discussion Point 2

- What is the usefulness of knowing the CP in a project?
- Tells which task to shorten to finish project earlier.
- Others ...?

- ESD.36 SPM 21

Critical Path Algorithm

- For large projects there are many paths
- Need a algorithm to identify the CP efficiently
- Develop information about each task in context of the overall project
- Times
- Start time (S)
- For each job: Earliest Start (ES)
- Earliest start time of a job if all its predecessors start at ES
- Job duration: t
- Earliest Finish (EF) = $(ES) + t$
- Finish time (F) – earliest finish time of the overall project
- Show algorithm using project graph

- ESD.36 SPM 22

CP Algorithm

1. Mark the value of S to left and right of Start
2. Consider any new unmarked job, all of whose predecessors have been marked. Mark to the left of the new job the largest number to the

right of its immediate predecessors: (ES)

3. Add to ES the job time t and mark result to the right (EF)

4. Stop when Finish has been reached

- ESD.36 SPM 23

CP Algorithm - Graphical

critical path

E,20 F,40

0 C,20 20 20 40 40 80

Finish

S=0 A,0 0 G,20 H,0

80 100 100 F=100

Start

0 B,10 10 20 D,30 50

- ESD.36 SPM 24

Concept Question 2

Possible

A project starts with (A,5). Task (B,10)

can start after A is completed. This is Answers

also true for task (E,5). Task (C,8) □20

depends only on (B,10), while task (F,10)

□22

depends on both (B,10) and (E,5). Task

(D,5) is the last task in the project and it □25

can start once (C,8) and (F,10) have □27

been finished.

□30

The Earliest Finish (EF) time for the □35

whole project is:

□40

- ESD.36 SPM 25

Latest Start and Finish Times

□ Set target finish time for project: $T \geq F$

□ Usually target is a specific calendar date, e.g. October 1, 2007

□ When is the latest date the project can be started?

□ Late Finish (LF) - latest time a job can be finished, without delaying the project

beyond its target time (T)

□ Late Start: $LS = LF - t$

- ESD.36 SPM 26

Determine LF and LS

□ Work from the end of the project: T

1. Mark value of T to left and right of Finish

2. Consider any new unmarked job, all of whose successors have been marked - mark to the right the smallest LS time marked to the left of any of its immediate successors

3. Subtract from this number, LF, the job time t and mark result to the left of the job: LS

4. Continue upstream until Start has been reached, then stop

- ESD.36 SPM 27

LS and LF : Project Graph

0 20 20 40 40 80

0 20 20 40 40 80

C,20 E,20 F,40

Finish

S=0 A,0 G,20 H,0

Start 80 100 100 100

80 100 100 100

0 0

0 0 0 B,10 10 D,30

Legend

0 10 20 50 early ES EF

10 20 50 80 late LS LF

- ESD.36 SPM 28

Slack

□ Some tasks have $ES=LS$ --> no slack

□ Total Slack of a task $TS=LS-ES$

□ Maximum amount of time a task may be delayed beyond its early start without delaying project completion

□ Slack time is precious ... managerial freedom, don't squander it unnecessarily

□ e.g. resource, work load smoothing

□ When $T=F$ then all critical tasks have $TS=0$

□ At least one path from Start->Finish with critical jobs only

□ When $T > F$, then all critical jobs have $TS = T - F$

- ESD.36 SPM 29

Project Graph - Slack

0 20 20 40 40 80

0 20 20 40 40 80

C,20 E,20 F,40

Finish

$TS=0$ $TS=0$ $TS=0$

$TS=0$ $TS=0$

$S=0$ A,0 G,20 H,0

Start 80 100 100 100

$TS=10$ $TS=30$ 80 100 100 100

0 0

0 0 0 B,10 10 D,30

Legend

0 10 20 50 early ES EF

10 20 50 80 late LS LF

- ESD.36 SPM 30

Task Times Detail - Task i

$ES(i)$ $LS(i)$ $EF(i)$ $LF(i)$

Duration $t(i)$

Total Slack Duration $t(i)$

j is the immediate

$TS(i)$ successor of i with

the smallest ES

$FS(i)$ $ES(j)$ $j > i$

Free Slack

□ Free Slack (FS) is the amount a job can be delayed without delaying the Early Start (ES) of any other job.

$FS \leq TS$ always

- ESD.36 SPM 31

Main CPM Errors

□ Estimated job times are wrong

□ Predecessor relationships may contain cycles □ "cycle error"

□ List of prerequisites contains more than the immediate

predecessors, e.g. $a \rightarrow b$, $b \rightarrow c$ and $a, b \rightarrow c$

- Overlooked some predecessor relationships
- Some predecessor relationships may be listed that are spurious
- and Some tasks/jobs may be missing !!!

- ESD.36 SPM 32

Gradual Refinement of CPM

- Job Times
- Given rough time estimates construct CPM chart
- Re-estimate times for CP and those with very small TS
- Iterate until the critical path is stable
- Focus attention on a subset of tasks
- Predecessor Relationships
- Check algorithmically for cycle errors and pre-predecessor errors
- Cancel all except immediate predecessor relationships
- Wrong or Missing Facts
- Cannot be detected by computers!

- ESD.36 SPM 33

Sample CPM (2007)

A&D High Tech Case, Online Store Project, 60 tasks, HW2, 2007

- ESD.36 SPM 34

Crashing Tasks

- What is we want to speed up project completion?
- Options
- Work overtime
- Put more resources on the critical tasks
- Parallelize tasks that are really serial (later in class)
- Cost of speedup?
- Is there a net savings resulting from reduction in overall project time?

- ESD.36 SPM 35

Cost Calculations

- Can compute project costs if cost of each job is included in the task data
- (Potentially) shorten crew jobs by adding personnel or working overtime with existing personnel
- Speedup carries price tag: "normal time", "crash time"

- Assign some critical jobs to their “crash time”
- Direct costs will increase as we “crash” critical tasks
- Indirect (fixed, overhead) costs will decrease as the overall project duration decreases – “standing army phenomenon”
- Minimize the sum of fixed and direct costs

- ESD.36 SPM 36

Typical Cost Pattern

Project Total Costs

Costs [\$]

“natural pace”

savings

Cum Fixed Costs

Direct Costs

crash Total Project

A B

Time [days]

- ESD.36 SPM 37

CPM Judgment

- +
- Focuses attention on a subset of critical tasks
- Determine effect of shortening/lengthening tasks
- Evaluate costs of a “crash” program
- -
- Doesn’t capture task iterations, in fact ...
- Prohibits iterations = “cycle error”
- Treats task durations as deterministic

- ESD.36 SPM 38

Summary

- CPM is useful, despite criticism, to identify the critical path - focus on a subset of the project
- Slack (TS and FS) is precious
- apply flexibility to smooth resource/schedules
- PERT treats task times as probabilistic
- Next lecture
- Selective “crashing” of critical tasks can reduce (or increase) total project cost
- CPM does not allow for task iterations

- ESD.36 SPM 39

Class Frustrations

- Poor examples set by project managers
- Perception of PMs as bureaucratic “box-checkers” and “ankle-biters”

Why?

- Traditional Project Management ...
- Doesn’t acknowledge the existence of iterations
- Is inflexible, “changing the plan” considered a failure
- Does not think of projects in a probabilistic sense
- “Hostage” to existing project management software
- In a reactive mode – no “early warning” systems

- ESD.36 SPM 40

HW1 Introduction

- You are Project Manager for the new CityCar development project
- Plan the project
- Task list
- Create project graph
- Critical path
- Slack times
- Re-planning after change
- “managerial”-type questions

- ESD.36 SPM 41

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Module B

Transportation and
Assignment Solution
Methods

B-1

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B-2 Module B Transportation and Assignment Solution Methods

Solution of the Transportation Model

The following example was used in Chapter 6 of the text to demonstrate the formulation of the transportation model. Wheat is harvested in the Midwest and stored in grain elevators in three different cities—Kansas City, Omaha, and Des Moines. These grain elevators supply three flour mills, located in Chicago, St. Louis, and Cincinnati. Grain is shipped to the mills in railroad cars, each of which is capable of holding one ton of wheat. Each grain elevator is able to supply the following number of tons (i.e., railroad cars) of wheat to the mills on a monthly basis:

Grain Elevator Supply

1. Kansas City 150
2. Omaha 175
3. Des Moines 275

Total 600 tons

Each mill demands the following number of tons of wheat per month.

Mill Demand

- A. Chicago 200
- B. St. Louis 100
- C. Cincinnati 300

Total 600 tons

The cost of transporting one ton of wheat from each grain elevator (source) to each mill (destination) differs according to the distance and rail system. These costs are shown in the following table. For example, the cost of shipping one ton of wheat from the grain elevator at Omaha to the mill at Chicago is \$7.

Mill

Grain Elevator A. Chicago B. St. Louis C. Cincinnati

1. Kansas City \$6 \$ 8 \$10
2. Omaha 7 11 11
3. Des Moines 4 5 12

The problem is to determine how many tons of wheat to transport from each grain elevator to each mill on a monthly basis in order to minimize the total cost of transportation.

The linear programming model for this problem is formulated in the equations that follow:

minimize $Z = \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C}$

subject to

$$x_{1A} + x_{1B} + x_{1C} = 150$$

$$x_{2A} + x_{2B} + x_{2C} = 175$$

$$x_{3A} + x_{3B} + x_{3C} = 275$$

$$x_{1A} + x_{2A} + x_{3A} = 200$$

$$x_{1B} + x_{2B} + x_{3B} = 100$$

$$x_{1C} + x_{2C} + x_{3C} = 300$$

$$x_{ij} \geq 0$$

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Solution of the Transportation Model B-3

In this model the decision variables, x_{ij} , represent the number of tons of wheat transported from each grain elevator, i (where $i = 1, 2, 3$), to each mill, j (where $j = A, B, C$). The objective function represents the total transportation cost for each route. Each term in the objective function reflects the cost of the tonnage transported for one route. For example, if 20 tons are transported from elevator 1 to mill A, the cost of \$6 is multiplied by $x_{1A}(=20)$, which equals \$120.

The first three constraints in the linear programming model represent the supply at each elevator; the last three constraints represent the demand at each mill. As an example, consider the first supply constraint, $x_{1A} + x_{1B} + x_{1C} = 150$. This constraint represents the tons of wheat transported from Kansas City to all three mills: Chicago (x_{1A}), St. Louis (x_{1B}), and Cincinnati (x_{1C}). The amount transported from Kansas City is limited to the 150 tons available. Note that this constraint (as well as all others) is an equation ($=$) rather than a ... inequality because all the tons of wheat available will be needed to meet the total demand of 600 tons. In other words, the three mills demand 600 total tons, which is the exact amount that can be supplied by the three grain elevators. Thus, all that can be supplied will be, in order to meet demand. This type of model, in which supply exactly equals demand, is referred to as a balanced transportation model. The balanced model will be used to demonstrate the solution of a transportation problem.

Transportation problems are Transportation models are solved manually within the context of a tableau, as

solved manually within a tableau in the simplex method. The tableau for our wheat transportation model is shown in

format. Table B-1.

Table B-1

To

The Transportation Tableau

From A B C Supply

6 8 10

1 150

7 11 11

2 175

4 5 12

3 275

Demand 200 100 300 600

Each cell in a transportation tableau Each cell in the tableau represents the amount transported from one source to one des-

is analogous to a decision variable tination. Thus, the amount placed in each cell is the value of a decision variable for that cell.

that indicates the amount allocated. For example, the cell at the intersection of row 1 and column A represents the decision variable x_{1A} . The smaller box within each cell contains the unit transportation cost for that route. For example, in cell 1A the value, \$6, is the cost of transporting one ton of wheat along the outside rim of a tableau from Kansas City to Chicago. Along the outer rim of the tableau are the supply and demand values, which are referred to as rim requirements. The two methods for solving a transportation model are the stepping-stone method and the modified distribution method (also known as MODI). In applying the simplex method, an initial solution had to be established in the initial simplex tableau. This same condition at the origin where all decision must be met in solving a transportation model. In a transportation model, an initial feasible solution can be found by several alternative methods, including the northwest corner method, the minimum cell cost method, and Vogel's approximation model.

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B-4 Module B Transportation and Assignment Solution Methods

The Northwest Corner Method

In the northwest corner method With the northwest corner method, an initial allocation is made to the cell in the upper

the largest possible allocation is left-hand corner of the tableau (i.e., the "northwest corner"). The amount allocated is the

made to the cell in the upper left- most possible, subject to the supply and demand constraints for that cell. In our example,

hand corner of the tableau,

we first allocate as much as possible to cell 1A (the northwest corner). This amount is 150

followed by allocations to adjacent

feasible cells.

tons, since that is the maximum that can be supplied by grain elevator 1 at Kansas City,

even though 200 tons are demanded by mill A at Chicago. This initial allocation is shown in

Table B-2.

We next allocate to a cell adjacent to cell 1A, in this case either cell 2A or cell 1B.

However, cell 1B no longer represents a feasible allocation, because the total tonnage of wheat available at source 1 (i.e., 150 tons) has already been allocated. Thus, cell 2A represents the only feasible alternative, and as much as possible is allocated to this cell. The

amount allocated at 2A can be either 175 tons, the supply available from source 2 (Omaha),

or 50 tons, the amount now demanded at destination A. (Recall that 150 of the 200 tons

demanded at A have already been supplied.) Because 50 tons is the most constrained

amount, it is allocated to cell 2A, as shown in Table B-2.

Table B-2

To

The Initial NW Corner Solution

From A B C Supply

6 8 10

1 150 150

7 11 11

2 50 100 25 175

4 5 12

3 275 275

Demand 200 100 300 600

The third allocation is made in the same way as the second allocation. The only feasible cell adjacent to cell 2A is cell 2B. The most that can be allocated is either 100 tons (the amount demanded at mill B) or 125 tons (175 tons minus the 50 tons allocated to cell 2A).

The smaller (most constrained) amount, 100 tons, is allocated to cell 2B, as shown in Table B-2.

The initial solution is complete The fourth allocation is 25 tons to cell 2C, and the fifth allocation is 275 tons to cell 3C,

when all rim requirements are both of which are shown in Table B-2. Notice that all of the row and column allocations

satisfied. add up to the appropriate rim requirements.

The transportation cost of this solution is computed by substituting the cell allocations (i.e., the amounts transported),

$$x_{1A} = 150$$

$$x_{2A} = 50$$

$$x_{2B} = 100$$

$$x_{2C} = 25$$

$$x_{3C} = 275$$

into the objective function:

$$\begin{aligned} Z &= \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C} \\ &= 6(150) + 8(0) + 10(0) + 7(50) + 11(100) + 11(25) + 4(0) + 5(0) + 12(275) \\ &= \$5,925 \end{aligned}$$

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Solution of the Transportation Model B-5

The steps of the northwest corner method are summarized here:

1. Allocate as much as possible to the cell in the upper left-hand corner, subject to the supply and demand constraints.
2. Allocate as much as possible to the next adjacent feasible cell.
3. Repeat step 2 until all rim requirements have been met.

The Minimum Cell Cost Method

In the minimum cell cost method With the minimum cell cost method, the basic logic is to allocate to the cells with the low-

as much as possible is allocated to est costs. The initial allocation is made to the cell in the tableau having the lowest cost. In

the cell with the minimum cost. the transportation tableau for our example problem, cell 3A has the minimum cost of \$4.

As much as possible is allocated to this cell; the choice is either 200 tons or 275 tons. Even though 275 tons could be supplied to cell 3A, the most we can allocate is 200 tons, since only 200 tons are demanded. This allocation is shown in Table B-3.

Table B-3

To

The Initial Minimum Cell From A B C Supply

Cost Allocation

6 8 10

1 150

7 11 11

2 175

4 5 12

3 200 275

Demand 200 100 300 600

Notice that all the remaining cells in column A have now been eliminated, because all the wheat demanded at destination A, Chicago, has now been supplied by source 3, Des Moines.

The next allocation is made to the cell that has the minimum cost and also is feasible.

This is cell 3B, which has a cost of \$5. The most that can be allocated is 75 tons (275 tons minus the 200 tons already supplied). This allocation is shown in Table B-4.

Table B-4

To

The Second Minimum Cell From A B C Supply

Cost Allocation

6 8 10

1 150

7 11 11

2 175

4 5 12

3 200 75 275

Demand 200 100 300 600

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B-6 Module B Transportation and Assignment Solution Methods

The third allocation is made to cell 1B, which has the minimum cost of \$8. (Notice that cells with lower costs, such as 1A and 2A, are not considered because they were previously ruled out as infeasible.) The amount allocated is 25 tons. The fourth allocation of 125 tons is made to cell 1C, and the last allocation of 175 tons is made to cell 2C. These allocations, which complete the initial minimum cell cost solution, are shown in Table B-5.

Table B-5

To

The Initial Solution From A B C Supply

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The minimum cell cost method The total cost of this initial solution is \$4,550, as compared with a total cost of \$5,925 for

will provide a solution with a the initial northwest corner solution. It is not a coincidence that a lower total cost is derived

lower cost than the northwest using the minimum cell cost method; it is a logical occurrence. The northwest corner

corner solution because it

considers cost in the

method does not consider cost at all in making allocations—the minimum cell cost

allocation process. method does. It is therefore quite natural that a lower initial cost will be attained using the

latter method. Thus, the initial solution achieved by using the minimum cell cost method is usually better in that, because it has a lower cost, it is closer to the optimal solution; fewer subsequent iterations will be required to achieve the optimal solution.

The specific steps of the minimum cell cost method are summarized next:

1. Allocate as much as possible to the feasible cell with the minimum transportation cost, and adjust the rim requirements.
2. Repeat step 1 until all rim requirements have been met.

Vogel's Approximation Model

The third method for determining an initial solution, Vogel's approximation model (also

A penalty cost is the difference called VAM), is based on the concept of penalty cost or regret. If a decision maker incor-

between the largest and rectly chooses from several alternative courses of action, a penalty may be suffered (and the

next largest cell cost in a row decision maker may regret the decision that was made). In a transportation problem, the (or column).

courses of action are the alternative routes, and a wrong decision is allocating to a cell that does not contain the lowest cost.

VAM allocates as much as possible In the VAM method, the first step is to develop a penalty cost for each source and desti-

to the minimum cost cell in the nation. For example, consider column A in Table B-6. Destination A, Chicago, can be sup-

row or column with the largest plied by Kansas City, Omaha, and Des Moines. The best decision would be to supply

penalty cost.

Chicago from source 3 because cell 3A has the minimum cost of \$4. If a wrong decision was made and the next higher cost of \$6 was selected at cell 1A, a “penalty” of \$2 per ton would result (i.e., $6 - 4 = \$2$). This demonstrates how the penalty cost is determined for each row and column of the tableau. The general rule for computing a penalty cost is to subtract the minimum cell cost from the next higher cell cost in each row and column. The penalty costs for our example are shown at the right and at the bottom of Table B-6.

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Solution of the Transportation Model B-7

Table B-6

To

The VAM Penalty Costs From A B C Supply

6 8 10

2

1 150

7 11 11

4

2 175

4 5 12

1

3 275

Demand 200 100 300 600

2 3 1

The initial allocation in the VAM method is made in the row or column that has the highest penalty cost. In Table B-6, row 2 has the highest penalty cost of \$4. We allocate as much as possible to the feasible cell in this row with the minimum cost. In row 2, cell 2A has the lowest cost of \$7, and the most that can be allocated to cell 2A is 175 tons. With this allocation the greatest penalty cost of \$4 has been avoided because the best course of action has been selected. The allocation is shown in Table B-7.

Table B-7

To

The Initial VAM Allocation From A B C Supply

6 8 10

2

1 150

7 11 11

2 175 175

4 5 12

1

3 275

Demand 200 100 300 600

2 3 2

After each VAM cell allocation, After the initial allocation is made, all the penalty costs must be recomputed. In some

all row and column penalty costs cases the penalty costs will change; in other cases they will not change. For example, the

are recomputed. penalty cost for column C in Table B-7 changed from \$1 to \$2 (because cell 2C is no longer

considered in computing penalty cost), and the penalty cost in row 2 was eliminated altogether (because no more allocations are possible for that row).

Next, we repeat the previous step and allocate to the row or column with the highest penalty cost, which is now column B with a penalty cost of \$3 (see Table B-7). The cell in column B with the lowest cost is 3B, and we allocate as much as possible to this cell, 100 tons. This allocation is shown in Table B-8.

Note that all penalty costs have been recomputed in Table B-8. Since the highest penalty cost is now \$8 for row 3 and since cell 3A has the minimum cost of \$4, we allocate 25 tons to this cell, as shown in Table B-9.

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B-8 Module B Transportation and Assignment Solution Methods

Table B-8

To

The Second VAM Allocation From A B C Supply

6 8 10

4

1 150

7 11 11

2 175 175

4 5 12

8

3 100 275

Demand 200 100 300 600

2 2

Table B-9

To

The Third VAM Allocation From A B C Supply

6 8 10

1 150

7 11 11

2 175 175

4 5 12

3 25 100 275

Demand 200 100 300 600

2

Table B-9 also shows the recomputed penalty costs after the third allocation. Notice that by now only column C has a penalty cost. Rows 1 and 3 have only one feasible cell, so a penalty does not exist for these rows. Thus, the last two allocations are made to column C. First, 150 tons are allocated to cell 1C because it has the lowest cell cost. This leaves only cell 3C as a feasible possibility, so 150 tons are allocated to this cell. Both of these allocations are shown in Table B-10.

Table B-10

To

The Initial VAM Solution From A B C Supply

6 8 10

1 150 150

7 11 11

2 175 175

4 5 12

3 25 100 150 275

Demand 200 100 300 600

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Solution of the Transportation Model B-9

VAM and minimum cell cost both The total cost of this initial Vogel's approximation model solution is \$5,125, which is

provide better initial solutions not as high as the northwest corner initial solution of \$5,925. It is also not as low as the

than the northwest corner method. minimum cell cost solution of \$4,550. Like the minimum cell cost method, VAM typi-

cally results in a lower cost for the initial solution than does the northwest corner

method.

The steps of Vogel's approximation model can be summarized in the following list:

1. Determine the penalty cost for each row and column by subtracting the lowest cell cost in the row or column from the next lowest cell cost in the same row or column.
2. Select the row or column with the highest penalty cost (breaking ties arbitrarily or choosing the lowest-cost cell).
3. Allocate as much as possible to the feasible cell with the lowest transportation cost in the row or column with the highest penalty cost.
4. Repeat steps 1, 2, and 3 until all rim requirements have been met.

The Stepping-Stone Solution Method

Once an initial basic feasible solution has been determined by any of the previous three methods, the next step is to solve the model for the optimal (i.e., minimum total cost) solu-

Once an initial solution is derived, tion. There are two basic solution methods: the stepping-stone solution method and the

the problem must be solved using modified distribution method (MODI). The stepping-stone solution method will be

either the stepping-stone method demonstrated first. Because the initial solution obtained by the minimum cell cost method

or the modified distribution

had the lowest total cost of the three initial solutions, we will use it as the starting solution.

method (MODI).

Table B-11 repeats the initial solution that was developed from the minimum cell cost method.

Table B-11

To

The Minimum Cell From A B C Supply

Cost Solution

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The stepping-stone method The basic solution principle in a transportation problem is to determine whether a

determines whether there is a cell transportation route not at present being used (i.e., an empty cell) would result in a

with no allocation that would lower total cost if it were used. For example, Table B-11 shows four empty cells (1A, 2A,

reduce cost if used.

2B, and 3C) representing unused routes. Our first step in the stepping-stone method is to evaluate these empty cells to see whether the use of any of them would reduce total cost. If we find such a route, then we will allocate as much as possible to it.

First, let us consider allocating one ton of wheat to cell 1A. If one ton is allocated to cell 1A, cost will be increased by \$6—the transportation cost for cell 1A. However, by allocating one ton to cell 1A, we increase the supply in row 1 to 151 tons, as shown in Table B-12.

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B-10 Module B Transportation and Assignment Solution Methods

Table B-12

To

The Allocation of One Ton to From A B C Supply

Cell 1A

+1 6 8 10

1 25 125 150 151

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The constraints of the problem cannot be violated, and feasibility must be maintained.

If we add one ton to cell 1A, we must subtract one ton from another allocation along that row. Cell 1B is a logical candidate because it contains 25 tons. By subtracting one ton from cell 1B, we now have 150 tons in row 1, and we have satisfied the supply constraint again. At the same time, subtracting one ton from cell 1B has reduced total cost by \$8.

However, by subtracting one ton from cell 1B, we now have only 99 tons allocated to column B, where 100 tons are demanded, as shown in Table B-13. To compensate for this constraint violation, one ton must be added to a cell that already has an allocation. Since cell 3B has 75 tons, we will add one ton to this cell, which again satisfies the demand constraint of 100 tons.

Table B-13

To

The Subtraction of One Ton From A B C Supply

from Cell 1B

+1 6 -1 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

99

A requirement of this solution method is that units can be added to and subtracted from only those cells that already have allocations. That is why one ton was added to cell 3B and not to cell 2B. It is from this requirement that the method derives its name. The process of adding and subtracting units from allocated cells is analogous to crossing a pond by stepping on stones (i.e., only allocated-to cells).

By allocating one extra ton to cell 3B we have increased cost by \$5, the transportation cost for that cell. However, we have also increased the supply in row 3 to 276 tons, a violation of the supply constraint for this source. As before, this violation can be remedied by subtracting one ton from cell 3A, which contains an allocation of 200 tons. This satisfies the supply constraint again for row 3, and it also reduces the total cost by \$4, the transportation cost for cell 3A. These allocations and deletions are shown in Table B-14.

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Solution of the Transportation Model B-11

Table B-14

To

The Addition of One Ton to Cell From A B C Supply

3B and the Subtraction of One

Ton from Cell 3A +1 6 -1 8 10

1 25 125 150

7 11 11

2 175 175

-1 4 +1 5 12

3 200 75 275

Demand 200 100 300 600

Notice in Table B-14 that by subtracting one ton from cell 3A, we did not violate the demand constraint for column A, since we previously added one ton to cell 1A.

Now let us review the increases and reductions in costs resulting from this process. We initially increased cost by \$6 at cell 1A, then reduced cost by \$8 at cell 1B, then increased cost by \$5 at cell 3B, and, finally, reduced cost by \$4 at cell 3A:

1A : 1B : 3B : 3A

+ \$6 - 8 + 5 - 4 = - \$1

An empty cell that will reduce cost In other words, for each ton allocated to cell 1A (a route not at present used), total cost

is a potential entering variable. will be reduced by \$1. This indicates that the initial solution is not optimal because a lower

cost can be achieved by allocating additional tons of wheat to cell 1A (i.e., cell 1A is analogous to a pivot column in the simplex method). Our goal is to determine the cell or enter-

ing “variable” that will reduce cost the most. Another variable (empty cell) may result in an even greater decrease in cost than cell 1A. If such a cell exists, it will be selected as the entering variable; if not, cell 1A will be selected. To identify the appropriate entering variable, the remaining empty cells must be tested as cell 1A was.

To evaluate the cost reduction Before testing the remaining empty cells, let us identify a few of the general characteris-

potential of an empty cell, a closed tics of the stepping-stone process. First, we always start with an empty cell and form a closed

path connecting used cells to the path of cells that now have allocations. In developing the path, it is possible to skip over

empty cell is identified.

both unused and used cells. In any row or column there can be only one addition and one subtraction. (For example, in row 1, wheat is added at cell 1A and is subtracted at cell 1B.)

Let us test cell 2A to see if it results in a cost reduction. The stepping-stone closed path for cell 2A is shown in Table B-15. Notice that the path for cell 2A is slightly more complex

Table B-15

To

The Stepping-Stone Path for From A B C Supply

Cell 2A

6 - 8 + 10

1 25 125 150

+ 7 11 - 11

2 175 175

- 4 + 5 12

3 200 75 275

Demand 200 100 300 600

2A : 2C : 1C : 1B : 3B : 3A

+ \$7 - 11 + 10 - 8 + 5 - 4 = - \$1

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B-12 Module B Transportation and Assignment Solution Methods

than the path for cell 1A. Notice also that the path crosses itself at one point, which is perfectly acceptable. An allocation to cell 2A will reduce cost by \$1, as shown in the computation in Table B-15. Thus, we have located another possible entering variable, although it is no better than cell 1A.

The remaining stepping-stone paths and the resulting computations for cells 2B and 3C are shown in Tables B-16 and B-17, respectively.

Table B-16

To

The Stepping-Stone Path for From A B C Supply

Cell 2B

6 + 8 - 10

1 25 125 150

7 - 11 + 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

2B : 2C : 1C : 1B

+ \$11 - 11 + 10 - 8 = + \$2

Table B-17

To

The Stepping-Stone Path for From A B C Supply

Cell 3C

6 + 8 - 10

1 25 125 150

7 11 11

2 175 175

4 - 5 + 12

3 200 75 275

Demand 200 100 300 600

3C : 1C : 1B : 3B

+ \$12 - 10 + 8 - 5 = + \$5

After all empty cells are evaluated, Notice that after all four unused routes are evaluated, there is a tie for the entering vari-

the one with the greatest able between cells 1A and 2A. Both show a reduction in cost of \$1 per ton allocated to that

cost reduction potential is the route. The tie can be broken arbitrarily. We will select cell 1A (i.e., x1A) to enter the solution.

entering variable.

Because the total cost of the model will be reduced by \$1 for each ton we can reallocate to cell 1A, we naturally want to reallocate as much as possible. To determine how much to allocate, we need to look at the path for cell 1A again, as shown in Table B-18.

When reallocating units to the The stepping-stone path in Table B-18 shows that tons of wheat must be subtracted at

entering variable (cell), the cells 1B and 3A to meet the rim requirements and thus satisfy the model constraints.

amount is the minimum Because we cannot subtract more than is available in a cell, we are limited by the 25 tons in

amount subtracted on the

cell 1B. In other words, if we allocate more than 25 tons to cell 1A, then we must subtract stepping-stone path.

more than 25 tons from 1B, which is impossible because only 25 tons are available. Therefore, 25 tons is the amount we reallocate to cell 1A according to our path. That is, 25 tons are added to 1A, subtracted from 1B, added to 3B, and subtracted from 3A. This reallocation is shown in Table B-19.

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Solution of the Transportation Model B-13

Table B-18

To

The Stepping-Stone Path for From A B C Supply

Cell 1A

+ 6 - 8 10

1 25 125 150

7 11 11

2 175 175

- 4 + 5 12

3 200 75 275

Demand 200 100 300 600

Table B-19

To

The Second Iteration of the From A B C Supply

Stepping-Stone Method

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 175 100 275

Demand 200 100 300 600

The process culminating in Table B-19 represents one iteration of the stepping-stone method. We selected x_{1A} as the entering variable, and it turned out that x_{1B} was the leaving variable (because it now has a value of zero in Table B-19). Thus, at each iteration one variable enters and one leaves (just as in the simplex method).

Now we must check to see whether the solution shown in Table B-19 is, in fact, optimal.

We do this by plotting the paths for the unused routes (i.e., empty cells 2A, 1B, 2B, and 3C) that are shown in Table B-19. These paths are shown in Tables B-20 through B-23.

Table B-20

To

The Stepping-Stone Path for From A B C Supply

Cell 2A

- 6 8 + 10

1 25 125 150

+ 7 11 - 11

2 175 175

4 5 12

3 175 100 275

Demand 200 100 300 600

2A : 2C : 1C : 1A

+ \$7 - 11 + 10 - 6 = \$0

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B-14 Module B Transportation and Assignment Solution Methods

Table B-21

To

The Stepping-Stone Path for From A B C Supply

Cell 1B

- 6 + 8 10

1 25 125 150

7 11 11

2 175 175

+ 4 - 5 12

3 175 100 275

Demand 200 100 300 600

1B : 3B : 3A : 1A

+ \$8 - 5 + 4 - 6 = + \$1

Table B-22

To

The Stepping-Stone Path for From A B C Supply

Cell 2B

- 6 8 + 10

1 25 125 150

7 + 11 - 11

2 175 175

+ 4 - 5 12

3 175 100 275

Demand 200 100 300 600

2B : 3B : 3A : 1A : 1C : 2C

+ \$11 - 5 + 4 - 6 + 10 - 11 = + \$3

Table B-23

To

The Stepping-Stone Path for

From A B C Supply

Cell 3C

+ 6 8 - 10

1 25 125 150

7 11 11

2 175 175

- 4 5 + 12

3 175 100 275

Demand 200 100 300 600

3C : 3A : 1A : 1C

+ \$12 - 4 + 6 - 10 = + \$4

The stepping-stone process is Our evaluation of the four paths indicates no cost reductions; therefore, the solution

repeated until none of the shown in Table B-19 is optimal. The solution and total minimum cost are empty cells will reduce cost

(i.e., an optimal solution). $x_{1A} = 25$ tons $x_{2C} = 175$ tons $x_{3A} = 175$ tons

$x_{1C} = 125$ tons $x_{3B} = 100$ tons

$Z = \$6(25) + 8(0) + 10(125) + 7(0) + 11(0) + 11(175) + 4(175) + 5(100) + 12(0)$

$= \$4,525$

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Solution of the Transportation Model B-15

Multiple optimal solutions However, notice in Table B-20 that the path for cell 2A resulted in a cost change of \$0. In

occur when an empty cell has a other words, allocating to this cell would neither increase nor decrease total cost. This situ-

cost change of zero and all other ation indicates that the problem has multiple optimal solutions.

Thus, x_{2A} could be

empty cells are positive.

entered into the solution and there would not be a change in the total minimum cost of

An alternative optimal solution is

\$4,525. To identify the alternative solution, we would allocate as much as possible to cell 2A,

determined by allocating to the

empty cell a zero cost change. which in this case is 25 tons of wheat. The alternative solution is shown in Table B-24.

Table B-24 To

The Alternative Optimal From A B C Supply

Solution

6 8 10

1 150 150

7 11 11

2 25 150 175

4 5 12

3 175 100 275

Demand 200 100 300 600

The solution in Table B-24 also results in a total minimum cost of \$4,525. The steps of the stepping-stone method are summarized here:

1. Determine the stepping-stone paths and cost changes for each empty cell in the tableau.
2. Allocate as much as possible to the empty cell with the greatest net decrease in cost.
3. Repeat steps 1 and 2 until all empty cells have positive cost changes that indicate an optimal solution.

The Modified Distribution Method

The modified distribution method (MODI) is basically a modified version of the stepping-stone method. However, in the MODI method the individual cell cost changes are determined mathematically, without identifying all the stepping-stone paths for the empty cells.

MODI is a modified version of

the stepping-stone method in To demonstrate MODI, we will again use the initial solution obtained by the minimum

which math equations replace the cell cost method. The tableau for the initial solution with the modifications required by

stepping-stone paths. MODI is shown in Table B-25.

Table B-25 v_j $v_A = v_B = v_C =$

The Minimum Cell Cost

To

Initial Solution

u_i From A B C Supply

6 8 10

$u_1 = 1$ 25 125 150

7 11 11

$u_2 = 2$ 175 175

4 5 12

$u_3 = 3$ 200 75 275

Demand 200 100 300 600

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B-16 Module B Transportation and Assignment Solution Methods

The extra left-hand column with the u_i symbols and the extra top row with the v_j symbols represent column and row values that must be computed in MODI. These values are computed for all cells with allocations by using the following formula:

$$u_i + v_j = c_{ij}$$

The value c_{ij} is the unit transportation cost for cell ij . For example, the formula for cell 1B is

$$u_1 + v_B = c_{1B}$$

and, since $c_{1B} = 8$,

$$u_1 + v_B = 8$$

The formulas for the remaining cells that presently contain allocations are

$$x_{1C}: u_1 + v_C = 10$$

$$x_{2C}: u_2 + v_C = 11$$

$$x_{3A}: u_3 + v_A = 4$$

$$x_{3B}: u_3 + v_B = 5$$

Now there are five equations with six unknowns. To solve these equations, it is necessary to assign only one of the unknowns a value of zero. Thus, if we let $u_1 = 0$, we can solve for all remaining u_i and v_j values.

$$x_{1B}: u_1 + v_B = 8$$

$$0 + v_B = 8$$

$$v_B = 8$$

$$x_{1C}: u_1 + v_C = 10$$

$$0 + v_C = 10$$

$$v_C = 10$$

$$x_{2C}: u_2 + v_C = 11$$

$$u_2 + 10 = 11$$

$$u_2 = 1$$

$$x_{3B}: u_3 + v_B = 5$$

$$u_3 + 8 = 5$$

$$u_3 = -3$$

$$x_{3A}: u_3 + v_A = 4$$

$$-3 + v_A = 4$$

$$v_A = 7$$

Notice that the equation for cell 3B had to be solved before the cell 3A equation could be solved. Now all the u_i and v_j values can be substituted into the tableau, as shown in Table B-26.

Table B-26 $v_A = 7$ $v_B = 8$ $v_C = 10$

The Initial Solution with All u_i

To

and v_j Values

u_i From A B C Supply

6 8 10

$u_1 = 0$ 1 25 125 150

7 11 11

$u_2 = 1$ 2 175 175

4 5 12

$u_3 = -3$ 3 200 75 275

Demand 200 100 300 600

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Solution of the Transportation Model B-17

Next, we use the following formula to evaluate all empty cells:

$$c_{ij} - u_i - v_j = k_{ij}$$

where k_{ij} equals the cost increase or decrease that would occur by allocating to a cell.

For the empty cells in Table B-26, the formula yields the following values:

$$x1A: k1A = c1A - u1 - vA = 6 - 0 - 7 = -1$$

$$x2A: k2A = c2A - u2 - vA = 7 - 1 - 7 = -1$$

$$x2B: k2B = c2B - u2 - vB = 11 - 1 - 8 = +2$$

$$x3C: k3C = c3C - u3 - vC = 12 - (-3) - 10 = +5$$

Each MODI allocation replicates These calculations indicate that either cell 1A or cell 2A will decrease cost by \$1 per allo-

the stepping-stone allocation. cated ton. Notice that those are exactly the same cost changes for all four empty cells as

were computed in the stepping-stone method. That is, the same information is obtained by evaluating the paths in the stepping-stone method and by using the mathematical formulas of the MODI.

After each allocation to an empty We can select either cell 1A or 2A to allocate to because they are tied at -1. If cell 1A is

cell, the u_i and v_j values must be selected as the entering nonbasic variable, then the stepping-stone path for that cell must

recomputed. be determined so that we know how much to reallocate. This is the same path previously

identified in Table B-18. Reallocating along this path results in the tableau shown in Table B-27 (and previously shown in Table B-19).

Table B-27 v_j $vA = vB = vC =$

The Second Iteration of the

To

MODI Solution Method

u_i From A B C Supply

6 8 10

$u1 = 1$ 25 125 150

7 11 11

$u2 = 2$ 175 175

4 5 12

$u3 = 3$ 175 100 275

Demand 200 100 300 600

The u_i and v_j values for Table B-27 must now be recomputed using our formula for the allocated-to cells:

$$x1A: u1 + vA = 6$$

$$0 + v_A = 6$$

$$v_A = 6$$

$$x_{1C}: u_1 + v_C = 10$$

$$0 + v_C = 10$$

$$v_C = 10$$

$$x_{2C}: u_2 + v_C = 11$$

$$u_2 + 10 = 11$$

$$u_2 = 1$$

$$x_{3A}: u_3 + v_A = 4$$

$$u_3 + 6 = 4$$

$$u_3 = -2$$

$$x_{3B}: u_3 + v_B = 5$$

$$-2 + v_B = 5$$

$$v_B = 7$$

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B-18 Module B Transportation and Assignment Solution Methods

These new u_i and v_j values are shown in Table B-28.

Table B-28 v_j $v_A = 6$ $v_B = 7$ $v_C = 10$

The New u_i and v_j Values for

To

the Second Iteration

u_i From A B C Supply

6 8 10

$u_1 = 0$ 1 25 125 150

7 11 11

$u_2 = 1$ 2 175 175

4 5 12

$u_3 = -2$ 3 175 100 275

Demand 200 100 300 600

The cost changes for the empty cells are now computed using the formula

$c_{ij} - u_i - v_j = k_{ij}$:

$$x_{1B}: k_{1B} = c_{1B} - u_1 - v_B = 8 - 0 - 7 = +1$$

$$x_{2A}: k_{2A} = c_{2A} - u_2 - v_A = 7 - 1 - 6 = 0$$

$$x_{2B}: k_{2B} = c_{2B} - u_2 - v_B = 11 - 1 - 7 = +3$$

$$x_{3C}: k_{3C} = c_{3C} - u_3 - v_C = 12 - (-2) - 10 = +4$$

Because none of these values is negative, the solution shown in Table B-28 is optimal.

However, as in the stepping-stone method, cell 2A with a zero cost change indicates multiple optimal solutions.

The steps of the modified distribution method can be summarized as follows:

1. Develop an initial solution using one of the three methods available.

2. Compute u_i and v_j values for each row and column by applying the formula $u_i + v_j = c_{ij}$ to each cell that has an allocation.
3. Compute the cost change, k_{ij} , for each empty cell using $c_{ij} - u_i - v_j = k_{ij}$.
4. Allocate as much as possible to the empty cell that will result in the greatest net decrease in cost (most negative k_{ij}). Allocate according to the stepping-stone path for the selected cell.
5. Repeat steps 2 through 4 until all k_{ij} values are positive or zero.

The Unbalanced Transportation Model

Thus far, the methods for determining an initial solution and an optimal solution have been demonstrated within the context of a balanced transportation model. Realistically, however, an unbalanced problem is a more likely occurrence. Consider our example of transporting wheat. By changing the demand at Cincinnati to 350 tons, we create a situation in which total demand is 650 tons and total supply is 600 tons.

When demand exceeds supply, To compensate for this difference in the transportation tableau, a “dummy” row is added

a dummy row is added to the to the tableau, as shown in Table B-29. The dummy row is assigned a supply of 50 tons

tableau. to balance the model. The additional 50 tons demanded, which cannot be supplied, will be allocated to a cell in the dummy row. The transportation costs for the cells in the dummy row are zero because the tons allocated to these cells are not amounts really transported but the amounts by which demand was not met. These dummy cells are, in effect, slack variables.

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Solution of the Transportation Model B-19

Table B-29

To

An Unbalanced Model From A B C Supply

(Demand 7 Supply)

6 8 10

1 150

7 11 11

2 175

4 5 12

3 275

0 0 0

Dummy 50

Demand 200 100 350 650

When a supply exceeds demand, Now consider our example with the supply at Des Moines increased to 375 tons. This

a dummy column is added to the increases total supply to 700 tons, while total demand remains at 600 tons. To compensate for tableau. this imbalance, we add a dummy column instead of a dummy row, as shown in Table B-30.

Table B-30

To

An Unbalanced Model From A B C Dummy Supply

(Supply 7 Demand)

6 8 10 0

1 150

7 11 11 0

2 175

4 5 12 0

3 375

Demand 200 100 300 100 700

The addition of a dummy row or a dummy column has no effect on the initial solution methods or on the methods for determining an optimal solution. The dummy row or column cells are treated the same as any other tableau cell. For example, in the minimum cell cost method, three cells would be tied for the minimum cost cell, each with a cost of zero. In this case (or any time there is a tie between cells) the tie would be broken arbitrarily.

Degeneracy

In all the tableaus showing a solution to the wheat transportation problem, the following condition was met:

$m \text{ rows} + n \text{ columns} - 1 = \text{the number of cells with allocations}$

For example, in any of the balanced tableaus for wheat transportation, the number of rows was three (i.e., $m = 3$) and the number of columns was three (i.e., $n = 3$); thus,

In a transportation tableau

$3 + 3 - 1 = 5$ cells with allocations.

with m rows and n columns, there These tableaus always had five cells with allocations; thus, our condition for normal

must be $m + n - 1$ cells with solution was met. When this condition is not met and fewer than $m + n - 1$ cells have

allocations; if not, it is degenerate. allocations, the tableau is said to be degenerate.

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B-20 Module B Transportation and Assignment Solution Methods

Consider the wheat transportation example with the supply values changed to the amounts shown in Table B-31. The initial solution shown in this tableau was developed using the minimum cell cost method.

Table B-31

To

The Minimum Cell Cost Initial

From A B C Supply

Solution

6 8 10

1 100 50 150

7 11 11

2 250 250

4 5 12

3 200 200

Demand 200 100 300 600

The tableau shown in Table B-31 does not meet the condition

$m + n - 1 = \text{the number of cells with allocations}$

$3 + 3 - 1 = 5 \text{ cells}$

In a degenerate tableau, not all of because there are only four cells with allocations. The difficulty resulting from a degenerate

the stepping-stone paths or MODI solution is that neither the stepping-stone method nor MODI will work unless the preced-

equations can be developed. ing condition is met (there is an appropriate number of cells with allocations). When the

tableau is degenerate, a closed path cannot be completed for all cells in the stepping-stone method, and not all the $u_i + v_j = c_{ij}$ computations can be completed in MODI. For example, a closed path cannot be determined for cell 1A in Table B-31.

To rectify a degenerate tableau, an To create a closed path, one of the empty cells must be artificially designated as a cell

empty cell must artificially be with an allocation. Cell 1A in Table B-32 is designated arbitrarily as a cell with artificial

treated as an occupied cell. allocation of zero. (However, any symbol, such as $\square$, could be used to signify the artificial

allocation.) This indicates that this cell will be treated as a cell with an allocation in determining stepping-stone paths or MODI formulas, although there is no real allocation in this cell. Notice that the location of 0 was arbitrary because there is no general rule for allocating the artificial cell. Allocating zero to a cell does not guarantee that all the stepping-stone paths can be determined.

Table B-32

To

The Initial Solution From A B C Supply

6 8 10

1 0 100 50 150

7 11 11

2 250 250

4 5 12

3 200 200

Demand 200 100 300 600

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Solution of the Transportation Model B-21

For example, if zero had been allocated to cell 2B instead of to cell 1A, none of the stepping-stone paths could have been determined, even though technically the tableau would no longer be degenerate. In such a case, the zero must be reallocated to another cell and all paths determined again. This process must be repeated until an artificial allocation has been made that will enable the determination of all paths. In most cases, however, there is more than one possible cell to which such an allocation can be made.

The stepping-stone paths and cost changes for this tableau follow:

2A 2C 1C 1A

x2A: $7 - 11 + 10 - 6 = 0$

2B 2C 1C 1B

x2B: $11 - 11 + 10 - 8 = +2$

3B 1B 1A 3A

x3B: $5 - 8 + 6 - 4 = -1$

3C 1C 1A 3A

x3C: $12 - 10 + 6 - 4 = +4$

Because cell 3B shows a \$1 decrease in cost for every ton of wheat allocated to it, we will allocate 100 tons to cell 3B. This results in the tableau shown in Table B-33.

Table B-33

To

The Second Stepping-Stone From A B C Supply

Iteration

6 8 10

1 100 50 150

7 11 11

2 250 250

4 5 12

3 100 100 200

Demand 200 100 300 600

A normal problem can become Notice that the solution in Table B-33 now meets the condition $m + n - 1 = 5$. Thus,

degenerate at any iteration and in applying the stepping-stone method (or MODI) to this tableau, it is not necessary to

vice versa. make an artificial allocation to an empty cell. It is quite possible to begin the solution process

with a normal tableau and have it become degenerate or begin with a degenerate tableau and have it become normal. If it had been indicated that the cell with the zero should have units subtracted from it, no actual units could have been subtracted. In that case the zero would

have been moved to the cell that represents the entering variable. (The solution shown in Table B-33 is optimal; however, multiple optimal solutions exist at cell 2A.)

Prohibited Routes

Sometimes one or more of the routes in the transportation model are prohibited. That is, units cannot be transported from a particular source to a particular destination. When this situation occurs, we must make sure that no units in the optimal solution are allocated to the cell representing this route. In our study of the simplex tableau, we learned that assigning a large relative cost or a coefficient of M to a variable would keep it out of the final solution. A prohibited route is assigned this same principle can be used in a transportation model for a prohibited route. A

a large cost such as M so that its value of M is assigned as the transportation cost for a cell that represents a prohibited

route. Thus, when the prohibited cell is evaluated, it will always contain a large positive cost

change of M , which will keep it from being selected as an entering variable.

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B-22 Module B Transportation and Assignment Solution Methods

Solution of the Assignment Model

An assignment model is a special form of a linear programming model that is similar to

the transportation model. There are differences, however. In the assignment model, the

supply at each source and the demand at each destination are limited to one unit each.

demand values equal one.

The following example from the text will be used to demonstrate the assignment model and its special solution method. The Atlantic Coast Conference has four basketball games on a particular night. The conference office wants to assign four teams of officials to the four games in a way that will minimize the total distance traveled by the officials. The distances in miles for each team of officials to each game location are shown in Table B-34.

Table B-34

Game Sites

The Travel Distances to Each

Game for Each Team of Officials RALEIGH ATLANTA DURHAM CLEMSON

Officials

A 210 90 180 160

B 100 70 130 200

C 175 105 140 170

D 80 65 105 120

The supply is always one team of officials, and the demand is for only one team of officials at each game. Table B-34 is already in the proper form for the assignment.

An opportunity cost table is The first step in the assignment method of solution is to develop an opportunity cost developed by first subtracting table. We accomplish this by first subtracting the minimum value in each row from every the minimum value in each value in the row. These computations are referred to as row reductions. We applied a similar row from all other row values and principle in the VAM method when we determined penalty costs. In other words, the best then repeating this process for each column. course of action is determined for each row, and the penalty or “lost opportunity” is devel- oped for all other row values. The row reductions for this example are shown in Table B-35.

Table B-35

Game Sites

The Assignment Tableau with

Row Reductions Officials RALEIGH ATLANTA DURHAM CLEMSON

A 120 0 90 70

B 30 0 60 130

C 70 0 35 65

D 15 0 40 55

Next, the minimum value in each column is subtracted from all column values. These computations are called column reductions and are shown in Table B-36, which represents the completed opportunity cost table for our example. Assignments can be made in this table wherever a zero is present. For example, team A can be assigned to Atlanta. An optimal solution results when each of the four teams can be uniquely assigned to a different game.

Table B-36

Game Sites

The Tableau with

Column Reductions Officials RALEIGH ATLANTA DURHAM CLEMSON

A 105 0 55 15

B 15 0 25 75

C 55 0 0 10

D 0 0 5 0

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Solution of the Assignment Model B-23

Assignments are made to locations Notice in Table B-36 that the assignment of team A to Atlanta means that no other team with zeros in the opportunity cost can be assigned to that game. Once this assignment is made, the zero in row B is infeasible, table. which indicates that there is not a unique optimal assignment for team B. Therefore, Table B-36 does not contain an optimal solution.

An optimal solution occurs when A test to determine whether four unique assignments exist in Table B-36 is to draw the minimum number of independent unique minimum number of horizontal or vertical lines necessary to cross out all zeros through assignments equals the number of the rows and columns of the table. For example, Table B-37 shows that three lines are required to cross out all zeros.

Table B-37

Game Sites

The Opportunity Cost Table

with the Line Test Officials RALEIGH ATLANTA DURHAM CLEMSON

A 105 0 55 15

B 15 0 25 75

C 55 0 0 10

D 0 0 5 0

If the number of unique The three lines indicate that there are only three unique assignments, whereas four are assignments is less than the required for an optimal solution. (Note that even if the three lines could have been drawn number of rows (or columns), differently, the subsequent solution method would not be affected.) Next, subtract the minimum value that is not crossed out from all values not crossed out. Then, add this minimum value to those cells where two lines intersect. The minimum value not crossed out in Table B-37 is 15. The second iteration for this model with the appropriate changes is shown in Table B-38.

Table B-38

Game Sites

The Second Iteration

Officials RALEIGH ATLANTA DURHAM CLEMSON

A 90 0 40 0

B 0 0 10 60

C 55 15 0 10

D 0 15 5 0

In a line test, all zeros are crossed No matter how the lines are drawn in Table B-38, at least four are required to cross out all the zeros. This indicates that four unique assignments can be made and that an optimal solution has been reached. Now let us make the assignments from Table B-38. value is subtracted from all

First, team A can be assigned to either the Atlanta game or the Clemson game. We will uncrossed values and added to cell values where two lines cross.

assign team A to Atlanta first. This means that team A cannot be assigned to any other game, and no other team can be assigned to Atlanta. Therefore, row A and the Atlanta column can be eliminated. Next, team B is assigned to Raleigh. (Team B cannot be assigned to Atlanta, which has already been eliminated.) The third assignment is of team C to the Durham game. This leaves team D for the Clemson game. These assignments and their respective distances (from Table B-34) are summarized as follows:

Assignment Distance

Team A : Atlanta 90

Team B : Raleigh 100

Team C : Durham 140

Team D : Clemson _120

450 miles

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B-24 Module B Transportation and Assignment Solution Methods

Now let us go back and make the initial assignment of team A to Clemson (the alternative assignment we did not initially make). This will result in the following set of assignments:

Assignment Distance

Team A : Clemson 160

Team B : Atlanta 70

Team C : Durham 140

Team D : Raleigh __80

450 miles

These two assignments represent multiple optimal solutions for our example problem.

Both assignments will result in the officials traveling a minimum total distance of 450 miles.

When supply exceeds demands, a Like a transportation problem, an assignment model can be unbalanced when supply

dummy column is added to the exceeds demand or demand exceeds supply. For example, assume that, instead of four

assignment tableau. teams of officials, there are five teams to be assigned to the four games. In this case

a dummy column is added to the assignment tableau to balance the model, as shown in Table B-39.

Table B-39 Game Sites

An Unbalanced Assignment

Tableau with a Dummy Column Officials RALEIGH ATLANTA DURHAM CLEMSON DUMMY

A 210 90 180 160 0

B 100 70 130 200 0
 C 175 105 140 170 0
 D 80 65 105 120 0
 E 95 115 120 100 0

When demand exceeds supply, a In solving this model, one team of officials would be assigned to the dummy column. If

dummy row is added to the there were five games and only four teams of officials, a dummy row would be added

assignment tableau. instead of a dummy column. The addition of a dummy row or column does not affect the

solution method.

A prohibited assignment is given Prohibited assignments are also possible in an assignment problem, just as prohibited

a large relative cost of M so that it routes can occur in a transportation model. In the transportation model, an M value was

will never be selected. assigned as a large cost for the cell representing the prohibited route. This same method is

used for a prohibited assignment. A value of M is placed in the cell that represents the prohibited assignment.

The steps of the assignment solution method are summarized here:

1. Perform row reductions by subtracting the minimum value in each row from all row values.
2. Perform column reductions by subtracting the minimum value in each column from all column values.
3. In the completed opportunity cost table, cross out all zeros, using the minimum number of horizontal or vertical lines.
4. If fewer than m lines are required (where m = the number of rows or columns), subtract the minimum uncrossed value from all uncrossed values, and add this same minimum value to all cells where two lines intersect. Leave all other values unchanged, and repeat step 3.
5. If m lines are required, the tableau contains the optimal solution and m unique assignments can be made. If fewer than m lines are required, repeat step 4.

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Problems B-25

Problems

1. Green Valley Mills produces carpet at plants in St. Louis and Richmond. The carpet is then shipped

to two outlets located in Chicago and Atlanta. The cost per ton of shipping carpet from each of the two plants to the two warehouses is as follows:

To

From Chicago Atlanta

St. Louis \$40 \$65

Richmond 70 30

The plant at St. Louis can supply 250 tons of carpet per week; the plant at Richmond can supply 400

tons per week. The Chicago outlet has a demand of 300 tons per week, and the outlet at Atlanta demands 350 tons per week. The company wants to know the number of tons of carpet to ship from each plant to each outlet in order to minimize the total shipping cost. Solve this transportation problem.

2. A transportation problem involves the following costs, supply, and demand:

To

From 1 2 3 4 Supply

1 \$500 \$750 \$300 \$450 12

2 650 800 400 600 17

3 400 700 500 550 11

Demand 10 10 10 10

- Find the initial solution using the northwest corner method, the minimum cell cost method, and Vogel's approximation model. Compute total cost for each.
- Using the VAM initial solution, find the optimal solution using the modified distribution method (MODI).

3. Consider the following transportation tableau and solution:

To

From A B C Supply

12 10 6

1 600 600

4 15 3

2 400 400

9 7 M

3 300 300

11 8 6

4 500 300 800

0 0 0

Dummy 200 200

Demand 900 500 900 2,300

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- Is this a balanced or an unbalanced transportation problem? Explain.
- Is this solution degenerate? Explain. If it is degenerate, show how it would be put into proper form.
- Is there a prohibited route in this problem?
- Compute the total cost of this solution.

e. What is the value of x_{2B} in this solution?

4. Solve the following transportation problem:

To

From 1 2 3 Supply

1 \$ 40 \$ 10 \$ 20 800

2 15 20 10 500

3 20 25 30 600

Demand 1,050 500 650

5. Given a transportation problem with the following costs, supply, and demand, find the initial solution using the minimum cell cost method and Vogel's approximation model. Is the VAM solution optimal?

To

From 1 2 3 Supply

A \$ 6 \$ 7 \$ 4 100

B 5 3 6 180

C 8 5 7 200

Demand 135 175 170

6. Consider the following transportation problem:

To

From 1 2 3 Supply

A \$ 6 9 M 130

B 12 3 5 70

C 4 8 11 100

Demand 80 110 60

a. Find the initial solution by using VAM and then solve it using the stepping-stone method.

b. Formulate this problem as a general linear programming model.

7. Solve the following linear programming problem:

$$\text{minimize } Z = 3x_{11} + 12x_{12} + 8x_{13} + 10x_{21} + 5x_{22} + 6x_{23} + 6x_{31} + 7x_{32} + 10x_{33}$$

subject to

$$x_{11} + x_{12} + x_{13} = 90 \quad x_{11} + x_{21} + x_{31} \leq 70$$

$$x_{21} + x_{22} + x_{23} = 30 \quad x_{12} + x_{22} + x_{32} \leq 110$$

$$x_{31} + x_{32} + x_{33} = 100 \quad x_{13} + x_{23} + x_{33} \leq 80$$

$$x_{ij} \geq 0$$

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Problems B-27

8. Consider the following transportation problem:

To

From 1 2 3 Supply

A \$ 6 \$ 9 \$ 7 130

B 12 3 5 70

C 4 11 11 100

Demand 80 110 60

- Find the initial solution using the minimum cell cost method.
- Solve using the stepping-stone method.

9. Steel mills in three cities produce the following amounts of steel:

Location Weekly Production (tons)

A. Bethlehem 150

B. Birmingham 210

C. Gary 320

680

These mills supply steel to four cities where manufacturing plants have the following demand:

Location Weekly Demand (tons)

1. Detroit 130

2. St. Louis 70

3. Chicago 180

4. Norfolk 240

620

Shipping costs per ton of steel are as follows:

To

From 1 2 3 4

A \$14 9 16 18

B 11 8 7 16

C 16 12 10 22

Because of a truckers' strike, shipments are at present prohibited from Birmingham to Chicago.

- Set up a transportation tableau for this problem and determine the initial solution. Identify the method used to find the initial solution.
- Solve this problem using MODI.
- Are there multiple optimal solutions? Explain. If so, identify them.
- Formulate this problem as a general linear programming model.

10. In Problem 9, what would be the effect on the optimal solution of a reduction in production capacity at the Gary mill from 320 tons to 290 tons per week?

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11. Coal is mined and processed at the following four mines in Kentucky, West Virginia, and Virginia:

Location Capacity (tons)

- A. Cabin Creek 90
 - B. Surry 50
 - C. Old Fort 80
 - D. McCoy __60
- 280

These mines supply the following amount of coal to utility power plants in three cities:

Plant Demand (tons)

- 1. Richmond 120
- 2. Winston-Salem 100
- 3. Durham 110

330

The railroad shipping costs (\$1,000s) per ton of coal are shown in the following table. Because of railroad construction, shipments are now prohibited from Cabin Creek to Richmond:

To

From 1 2 3

A \$ 7 \$10 \$ 5

B 12 9 4

C 7 3 11

D 9 5 7

- a. Set up the transportation tableau for this problem, determine the initial solution using VAM, and compute total cost.
- b. Solve using MODI.
- c. Are there multiple optimal solutions? Explain. If there are alternative solutions, identify them.
- d. Formulate this problem as a linear programming model.

12. Oranges are grown, picked, and then stored in warehouses in Tampa, Miami, and Fresno. These warehouses supply oranges to markets in New York, Philadelphia, Chicago, and Boston. The following table shows the shipping costs per truckload (\$100s), supply, and demand. Because of an agreement between distributors, shipments are prohibited from Miami to Chicago.

To

From New York Philadelphia Chicago Boston Supply

Tampa \$ 9 \$ 14 \$ 12 \$ 17 200

Miami 11 10 6 10 200

Fresno 12 8 15 7 200

Demand 130 170 100 150

- a. Set up the transportation tableau for this problem and determine the initial solution using the minimum cell cost method.
- b. Solve using MODI.
- c. Are there multiple optimal solutions? Explain. If so, identify them.
- d. Formulate this problem as a linear programming model.

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Problems B-29

13. A manufacturing firm produces diesel engines in four cities—Phoenix, Seattle, St. Louis, and Detroit. The company is able to produce the following numbers of engines per month:

Plant Production

1. Phoenix 5
2. Seattle 25
3. St. Louis 20
4. Detroit 25

Three trucking firms purchase the following numbers of engines for their plants in three cities:

Firm Demand

- A. Greensboro 10
- B. Charlotte 20
- C. Louisville 15

The transportation costs per engine (\$100s) from sources to destinations are shown in the following table. However, the Charlotte firm will not accept engines made in Seattle, and the Louisville firm will not accept engines from Detroit; therefore, these routes are prohibited.

To

From A B C

1 \$ 7 \$ 8 \$ 5

2 6 10 6

3 10 4 5

4 3 9 11

- a. Set up the transportation tableau for this problem. Find the initial solution using VAM.
- b. Solve for the optimal solution using the stepping-stone method. Compute the total minimum cost.
- c. Formulate this problem as a linear programming model.

14. The Interstate Truck Rental firm has accumulated extra trucks at three of its truck leasing outlets,

as shown in the following table:

Extra

Leasing Outlet Trucks

1. Atlanta 70
2. St. Louis 115
3. Greensboro 60

Total 245

The firm also has four outlets with shortages of rental trucks, as follows:

Truck

Leasing Outlet Shortage

- A. New Orleans 80
- B. Cincinnati 50
- C. Louisville 90
- D. Pittsburgh 25

Total 245

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The firm wants to transfer trucks from those outlets with extras to those with shortages at the minimum total cost. The following costs of transporting these trucks from city to city have been determined:

To

From A B C D

1 \$ 70 80 45 90

2 120 40 30 75

3 110 60 70 80

- a. Find the initial solution using the minimum cell cost method.
- b. Solve using the stepping-stone method.

15. The Sholtz Beer Company has breweries in two cities; the breweries can supply the following numbers of barrels of draft beer to the company's distributors each month:

Brewery Monthly Supply (bbl)

A. Tampa 3,500

B. St. Louis 5,000

Total 8,500

The distributors, which are spread throughout six states, have the following total monthly demand:

Distributor Monthly Demand (bbl)

1. Tennessee 1,600

2. Georgia 1,800

3. North Carolina 1,500

4. South Carolina 950

5. Kentucky 1,250

6. Virginia 1,400

Total 8,500

The company must pay the following shipping costs per barrel:

To

From 1 2 3 4 5 6

A \$0.50 0.35 0.60 0.45 0.80 0.75

B 0.25 0.65 0.40 0.55 0.20 0.65

- a. Find the initial solution using VAM.
- b. Solve using the stepping-stone method.

16. In Problem 15, the Sholtz Beer Company management has negotiated a new shipping contract with a trucking firm between its Tampa brewery and its distributor in Kentucky that reduces the shipping cost per barrel from \$0.80 per barrel to \$0.65 per barrel. How will this cost change affect the optimal solution?

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Problems B-31

17. Computers Unlimited sells microcomputers to universities and colleges on the East Coast and ships

them from three distribution warehouses. The firm is able to supply the following numbers of microcomputers to the universities by the beginning of the academic year:

Distribution Supply

Warehouse (microcomputers)

1. Richmond 420
2. Atlanta 610
3. Washington, D.C. 340

Total 1,370

Four universities have ordered microcomputers that must be delivered and installed by the beginning of the academic year:

Demand

University (microcomputers)

- A. Tech 520
- B. A and M 250
- C. State 400
- D. Central 380

Total 1,550

The shipping and installation costs per microcomputer from each distributor to each university are as follows:

To

From A B C D

1 \$22 17 30 18

2 15 35 20 25

3 28 21 16 14

- a. Find the initial solution using VAM.
- b. Solve using MODI.

18. In Problem 17, Computers Unlimited wants to better meet demand at the four universities it sup-

plies. It is considering two alternatives: (1) expand its warehouse at Richmond to a capacity of 600

at a cost equivalent to an additional \$6 in handling and shipping per unit or (2) purchase a new warehouse in Charlotte that can supply 300 units with shipping costs of \$19 to Tech, \$26 to A and M, \$22 to State, and \$16 to Central. Which alternative should management select based solely on transportation costs (i.e., no capital costs)?

19. Computers Unlimited in Problem 17 has determined that when it is unable to meet the demand for

microcomputers at the universities it supplies, the universities tend to purchase microcomputers elsewhere in the future. Thus, the firm has estimated a shortage cost for each microcomputer demanded but not supplied that reflects the loss of future sales and goodwill. These costs for each university are as follows:

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University Cost/Microcomputer

- A. Tech \$40
- B. A and M 65
- C. State 25
- D. Central 50

Solve Problem 17 with these shortage costs included. Compute the total transportation cost and the total shortage cost.

20. A severe winter ice storm has swept across North Carolina and Virginia, followed by over a foot of

snow and frigid, single-digit temperatures. These weather conditions have resulted in numerous downed power lines and power outages in the region causing dangerous conditions for much of the population. Local utility companies have been overwhelmed and have requested assistance from unaffected utility companies across the Southeast. The following table shows the number of utility trucks with crews available from five different companies in Georgia, South Carolina, and Florida; the demand for crews in seven different areas that local companies cannot get to; and the weekly cost (\$1,000s) of a crew going to a specific area (based on the visiting company's normal charges, the distance the crew has to come, and living expenses in an area):

Area (Cost = \$1,000s)

Crews

Crew NC-E NC-SW NC-P NC-W VA-SW VA-C VA-T Available

GA-1 15.2 14.3 13.9 13.5 14.7 16.5 18.7 12

GA-2 12.8 11.3 10.6 12.0 12.7 13.2 15.6 10

SC-1 12.4 10.8 9.4 11.3 13.1 12.8 14.5 14

FL-1 18.2 19.4 18.2 17.9 20.5 20.7 22.7 15

FL-2 19.3 20.2 19.5 20.2 21.2 21.3 23.5 12

Crews

Needed 9 7 6 8 10 9 7

Determine the number of crews that should be sent from each utility to each affected area that

will minimize total costs.

21. A large manufacturing company is closing three of its existing plants and intends to transfer some of its more skilled employees to three plants that will remain open. The number of employees available for transfer from each closing plant is as follows:

Closing Plant Transferable Employees

1 60

2 105

3 70

Total 235

The following number of employees can be accommodated at the three plants remaining open:

Open Plants Employees Demanded

A 45

B 90

C 35

Total 170

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Problems B-33

Each transferred employee will increase product output per day at each plant as shown in the following table. The company wants to transfer employees so as to ensure the maximum increase in product output.

To

From A B C

1 5 8 6

2 10 9 12

3 7 6 8

a. Find the initial solution using VAM.

b. Solve using MODI.

22. The Sav-Us Rental Car Agency has six lots in Nashville, and it wants to have a certain number of cars available at each lot at the beginning of each day for local rental. The agency would like a model it could quickly solve at the end of each day that would tell it how to redistribute the cars among the six lots in the minimum total time. The times required to travel between the six lots are as follows:

To (minutes)

From 1 2 3 4 5 6

1 — 12 17 18 10 20

2 14 — 10 19 16 15

3 14 10 — 12 8 9

4 8 16 14 — 12 15

5 11 21 16 18 — 10

6 24 12 9 17 15 —

The agency would like the following numbers of cars at each lot at the end of the day. Also shown are the numbers of available cars at each lot at the end of a particular day. Determine the optimal reallocation of rental cars using any initial solution approach and any solution method.

Lot

Cars 1 2 3 4 5 6

Available 37 20 14 26 40 28

Desired 30 25 20 40 30 20

23. Bayville has built a new elementary school so that the town now has a total of four schools—Addison, Beeks, Canfield, and Daley. Each has a capacity of 400 students. The school wants to assign children to schools so that their travel time by bus is as short as possible. The school has partitioned the town into five districts conforming to population density—north, south, east, west, and central. The average bus travel time from each district to each school is shown as follows:

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Travel Time (min.)

Student

District Addison Beeks Canfield Daley Population

North 12 23 35 17 250

South 26 15 21 27 340

East 18 20 22 31 310

West 29 24 35 10 210

Central 15 10 23 16 290

Determine the number of children that should be assigned from each district to each school to minimize total student travel time.

24. In Problem 23, the school board has determined that it does not want any one school to be more

crowded than any other school. It would like to assign students from each district to each school so that enrollments are evenly balanced among the four schools. However, the school board is concerned that this might significantly increase travel time. Determine the number of students to be assigned from each district to each school so that school enrollments are evenly balanced. Does this

new solution appear to result in a significant increase in travel time per student?

25. The Easy Time Grocery chain operates in major metropolitan areas on the eastern seaboard. The

stores have a “no-frills” approach, with low overhead and high volume. They generally buy their stock in volume at low prices. However, in some cases they actually buy stock at stores in other areas

and ship it in. They can do this because of high prices in the cities they operate in compared with

costs in other locations. One example is baby food. Easy Time purchases baby food at stores in Albany, Binghamton, Claremont, Dover, and Edison and then trucks it to six stores in and around New York City. The stores in the outlying areas know what Easy Time is up to, so they limit the number of cases of baby food Easy Time can purchase. The following table shows the profit Easy Time makes per case of baby food based on where the chain purchases it and at which store it's sold, plus the available baby food per week at purchase locations and the shelf space available at each Easy Time store per week:

Easy Time Store

Purchase

Location 1 2 3 4 5 6 Supply

Albany 9 8 11 12 7 8 26

Binghamton 10 10 8 6 9 7 40

Claremont 8 6 6 5 7 4 20

Dover 4 6 9 5 8 10 40

Edison 12 10 8 9 6 7 45

Demand 25 15 30 18 27 35

Determine where Easy Time should purchase baby food and how the food should be distributed in order to maximize profit. Use any initial solution approach and any solution method.

26. Suppose that in Problem 25 Easy Time can purchase all the baby food it needs from a New York City distributor at a price that will result in a profit of \$9 per case at stores 1, 3, and 4, \$8 per case at

stores 2 and 6, and \$7 per case at store 5. Should Easy Time purchase all, none, or some of its baby food from the distributor rather than purchasing it at other stores and trucking it in?

27. In Problem 25, if Easy Time could arrange to purchase more baby food from one of the outlying locations, which should it be, how many additional cases could be purchased, and how much would this increase profit?

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Problems B-35

28. The Roadnet Transport Company has expanded its shipping capacity by purchasing 90 trailer trucks from a competitor that went bankrupt. The company subsequently located 30 of the purchased trucks at each of its shipping warehouses in Charlotte, Memphis, and Louisville. The company makes shipments from each of these warehouses to terminals in St. Louis, Atlanta, and New York. Each truck is capable of making one shipment per week. The terminal managers have indicated their capacity of extra shipments. The manager at St. Louis can accommodate 40 additional trucks per week, the manager at Atlanta can accommodate 60 additional trucks, and the manager at

New York can accommodate 50 additional trucks. The company makes the following profit per truckload shipment from each warehouse to each terminal. The profits differ as a result of differences in products shipped, shipping costs, and transport rates:

Terminal

Warehouse St. Louis Atlanta New York

Charlotte \$1,800 \$2,100 \$1,600

Memphis 1,000 700 900

Louisville 1,400 800 2,200

Determine how many trucks to assign to each route (i.e., warehouse to terminal) in order to maximize profit.

29. During the Gulf War, Operation Desert Storm required large amounts of military matériel and supplies to be shipped daily from supply depots in the United States to bases in the Middle East. The critical factor in the movement of these supplies was speed. The following table shows the number of planeloads of supplies available each day from each of six supply depots and the number of daily loads demanded at each of five bases. (Each planeload is approximately equal in tonnage.) Also included are the transport hours per plane, including loading and fueling, actual flight time, and unloading and refueling:

Military Base	Supply	Depot A	B	C	D	E	Supply
1	36	40	32	43	29	7	
2	28	27	29	40	38	10	
3	34	35	41	29	31	8	
4	41	42	35	27	36	8	
5	25	28	40	34	38	9	
6	31	30	43	38	40	6	
Demand	9	6	12	8	10		

Determine the optimal daily flight schedule that will minimize total transport time.

30. PM Computer Services produces personal computers from component parts it buys on the open market. The company can produce a maximum of 300 personal computers per month. PM wants to determine its production schedule for the first six months of the new year. The cost to produce a personal computer in January will be \$1,200. However, PM knows the cost of component parts will decline each month such that the overall cost to produce a PC will be 5% less each month. The cost of holding a computer in inventory is \$15 per unit per month. Following is the demand for the company's computers each month:

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Month Demand Month Demand

January 180 April 210

February 260 May 400

March 340 June 320

Determine a production schedule for PM that will minimize total cost.

31. In Problem 30, suppose the demand for personal computers increased each month as follows:

Month Demand

January 410
 February 320
 March 500
 April 620
 May 430
 June 380

In addition to the regular production capacity of 300 units per month, PM Computer Services can also produce an additional 200 computers per month using overtime. Overtime production adds 20% to the cost of a personal computer.

Determine a production schedule for PM that will minimize total cost.

32. National Foods Company has five plants where it processes and packages fruits and vegetables. It has suppliers in six cities in California, Texas, Alabama, and Florida. The company has owned and operated its own trucking system in the past for transporting fruits and vegetables from its suppliers to its plants. However, it is now considering outsourcing all its shipping to outside trucking firms and getting rid of its own trucks. It currently spends \$245,000 per month to operate its own trucking system. It has determined monthly shipping costs (in \$1,000s per ton) using outside shippers from each of its suppliers to each of its plants as shown in the following table:

Processing Plants (\$1,000s per ton)

Suppliers Denver St. Paul Louisville Akron Topeka Supply (tons)

Sacramento 3.7 4.6 4.9 5.5 4.3 18

Bakersfield 3.4 5.1 4.4 5.9 5.2 15

San Antonio 3.3 4.1 3.7 2.9 2.6 10

Montgomery 1.9 4.2 2.7 5.4 3.9 12

Jacksonville 6.1 5.1 3.8 2.5 4.1 20

Ocala 6.6 4.8 3.5 3.6 4.5 15

Demand (tons) 20 15 15 15 20 90

Should National Foods continue to operate its own shipping network or sell its trucks and out-source its shipping to independent trucking firms?

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33. In Problem 32, National Foods would like to know what the effect would be on the optimal solution and the company's decision regarding its shipping if it negotiates with its suppliers in Sacramento, Jacksonville, and Ocala to increase their capacity to 25 tons per month? What would be the effect of negotiating instead with its suppliers at San Antonio and Montgomery to increase their capacity to 25 tons each?

34. Orient Express is a global distribution company that transports its clients' products to customers in

Hong Kong, Singapore, and Taipei. All the products Orient Express ships are stored at three distribution centers, one in Los Angeles, one in Savannah, and one in Galveston. For the coming month

the company has 450 containers of computer components available at the Los Angeles center, 600 containers available at Savannah, and 350 containers available in Galveston. The company has orders for 600 containers from Hong Kong, 500 containers from Singapore, and 500 containers from Taipei. The shipping costs per container from each U.S. port to each of the overseas ports are shown in the following table:

Overseas Port			
U.S. Distribution Center	Hong Kong	Singapore	Taipei
Los Angeles	\$300	\$210	\$340
Savannah	490	520	610
Galveston	360	320	500

The Orient Express, as the overseas broker for its U.S. customers, is responsible for unfulfilled orders, and it incurs stiff penalty costs from overseas customers if it does not meet an order. The Hong Kong customers charge a penalty cost of \$800 per container for unfulfilled demand, Singapore customers charge a penalty cost of \$920 per container, and Taipei customers charge \$1,100 per container. Formulate and solve a transportation model to determine the shipments from each U.S. distribution center to each overseas port that will minimize shipping costs. Indicate what portion of the total cost is a result of penalties.

35. Binford Tools manufactures garden tools. It uses inventory, overtime, and subcontracting to absorb demand fluctuations. Expected demand, regular and overtime production capacity, and subcontracting capacity are provided in the following table for the next four quarters for its basic line of steel garden tools:

	Regular	Overtime	Subcontracting
Quarter	Demand	Capacity	Capacity
1	9,000	9,000	1,000
2	12,000	10,000	1,500
3	16,000	12,000	2,000
4	19,000	12,000	2,000

The regular production cost per unit is \$20, the overtime cost per unit is \$25, the cost to subcontract a unit is \$27, and the inventory carrying cost is \$2 per unit. The company has 300 units in inventory at the beginning of the year.

Determine the optimal production schedule for the four quarters that will minimize total costs.

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36. Solve the following linear programming problem:

$$\begin{aligned}
 \text{minimize } Z = & 18x_{11} + 30x_{12} + 20x_{13} + 18x_{14} + 25x_{21} + 27x_{22} + 22x_{23} \\
 & + 16x_{24} + 30x_{31} + 26x_{32} + 19x_{33} + 32x_{34} + 40x_{41} + 36x_{42} \\
 & + 27x_{43} + 29x_{44} + 30x_{51} + 26x_{52} + 18x_{53} + 24x_{54}
 \end{aligned}$$

subject to

$$x_{11} + x_{12} + x_{13} + x_{14} \dots 1$$

$$x_{21} + x_{22} + x_{23} + x_{24} \dots 1$$

$$x_{31} + x_{32} + x_{33} + x_{34} \dots 1$$

$$x_{41} + x_{42} + x_{43} + x_{44} \dots 1$$

$$x_{51} + x_{52} + x_{53} + x_{54} \dots 1$$

$$x_{11} + x_{21} + x_{31} + x_{41} + x_{51} = 1$$

$$x_{12} + x_{22} + x_{32} + x_{42} + x_{52} = 1$$

$$x_{13} + x_{23} + x_{33} + x_{43} + x_{53} = 1$$

$$x_{14} + x_{24} + x_{34} + x_{44} + x_{54} = 1$$

$$x_{ij} \geq 0$$

37. A plant has four operators to be assigned to four machines. The time (minutes) required by each

worker to produce a product on each machine is shown in the following table. Determine the optimal assignment and compute total minimum time.

Machine

Operator A B C D

1 10 12 9 11

2 5 10 7 8

3 12 14 13 11

4 8 15 11 9

38. A shop has four machinists to be assigned to four machines. The hourly cost of having each machine operated by each machinist is as follows:

Machine

Machinist A B C D

1 \$12 11 8 14

2 10 9 10 8

3 14 8 7 11

4 6 8 10 9

However, because he does not have enough experience, machinist 3 cannot operate machine B.

a. Determine the optimal assignment and compute total minimum cost.

b. Formulate this problem as a general linear programming model.

39. The Omega pharmaceutical firm has five salespersons, whom the firm wants to assign to five sales regions. Given their various previous contacts, the salespersons are able to cover the

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regions in different amounts of time. The amount of time (days) required by each salesperson to cover each city is shown in the following table. Which salesperson should be assigned to each region to minimize total time? Identify the optimal assignments and compute total minimum

time.

Region

Salesperson A B C D E

1 17 10 15 16 20

2 12 9 16 9 14

3 11 16 14 15 12

4 14 10 10 18 17

5 13 12 9 15 11

40. The Bunker Manufacturing firm has five employees and six machines and wants to assign the employees to the machines to minimize cost. A cost table showing the cost incurred by each employee on each machine follows. Because of union rules regarding departmental transfers, employee 3 cannot be assigned to machine E and employee 4 cannot be assigned to machine B. Solve this problem, indicate the optimal assignment, and compute total minimum cost.

Machine

Employee A B C D E F

1 \$12 \$ 7 \$20 \$14 \$ 8 \$10

2 10 14 13 20 9 11

3 5 3 6 9 7 10

4 9 11 7 16 9 10

5 10 6 14 8 10 12

41. Given the following cost table for an assignment problem, determine the optimal assignment and compute total minimum cost. Identify all alternative solutions if there are multiple optimal solutions.

Machine

Operator A B C D

1 \$10 \$2 \$ 8 \$ 6

2 9 5 11 9

3 12 7 14 14

4 3 1 4 2

42. An electronics firm produces electronic components, which it supplies to various electrical manufacturers. Quality control records indicate that different employees produce different numbers of defective items. The average number of defects produced by each employee for each of six components

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is given in the following table. Determine the optimal assignment that will minimize the total average number of defects produced by the firm per month.

Component

Employee A B C D E F

1 30 24 16 26 30 22

2 22 28 14 30 20 13

3 18 16 25 14 12 22

4 14 22 18 23 21 30

5 25 18 14 16 16 28

6 32 14 10 14 18 20

43. A dispatcher for the Citywide Taxi Company has six taxicabs at different locations and five customers who have called for service. The mileage from each taxi's present location to each customer is shown in the following table. Determine the optimal assignment(s) that will minimize the total mileage traveled.

Customer

Cab 1 2 3 4 5

A 7 2 4 10 7

B 5 1 5 6 6

C 8 7 6 5 5

D 2 5 2 4 5

E 3 3 5 8 4

F 6 2 4 3 4

44. The Southeastern Conference has nine basketball officials who must be assigned to three conference games, three to each game. The conference office wants to assign the officials so that the total distance they travel will be minimized. The distance (in miles) each official would travel to each game is given in the following table:

Game

Official Athens Columbia Nashville

1 165 90 130

2 75 210 320

3 180 170 140

4 220 80 60

5 410 140 80

6 150 170 190

7 170 110 150

8 105 125 160

9 240 200 155

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a. Should this problem be solved by the transportation method or by the assignment method?

Explain.

b. Determine the optimal assignment(s) that will minimize the total distance traveled by the officials.

45. In Problem 44, officials 2 and 8 had a recent confrontation with one of the coaches in the game in

Athens. They were forced to eject the coach after several technical fouls. The conference office decided that it would not be a good idea to have these two officials work the Athens game so soon after this confrontation, so they decided that officials 2 and 8 will not be assigned to the Athens game. How will this affect the optimal solution to this problem?

46. State University has planned six special catered events for the November Saturday of its homecom-

ing football game. The events include an alumni brunch, a parents' brunch, a booster club luncheon, a postgame party for season ticket holders, a lettermen's dinner, and a fund-raising dinner for major contributors. The university wants to use local catering firms as well as the university catering service to cater these events and it has asked the caterers to bid on each event. The bids (in

\$1,000s) based on menu guidelines for the events prepared by the university are shown in the following table:

Event

Alumni Parents' Booster Postgame Lettermen's Contributor's

Caterer Brunch Brunch Club Lunch Party Dinner Dinner

Al's \$12.6 \$10.3 \$14.0 \$19.5 \$25.0 \$30.0

Bon Apétit 14.5 13.0 16.5 17.0 22.5 32.0

Custom 13.0 14.0 17.6 21.5 23.0 35.0

Divine 11.5 12.6 13.0 18.7 26.2 33.5

Epicurean 10.8 11.9 12.9 17.5 21.9 28.5

Fouchés 13.5 13.5 15.5 22.3 24.5 36.0

University 12.5 14.3 16.0 22.0 26.7 34.0

The Bon Apétit, Custom, and University caterers can handle two events, whereas the other four caterers can handle only one. The university is confident all the caterers will do a high-quality job, so it wants to select the caterers for the events that will result in the lowest total cost.

Determine the optimal selection of caterers that will minimize total cost.

47. A university department head has five instructors to be assigned to four different courses. All the

instructors have taught the courses in the past and have been evaluated by the students. The rating

for each instructor for each course is given in the following table (a perfect score is 100). The department head wants to know the optimal assignment of instructors to courses that will maximize the overall average evaluation. The instructor who is not assigned to teach a course will be assigned to grade exams. Solve this problem using the assignment method.

Course

Instructor A B C D

1 80 75 90 85

2 95 90 90 97

3 85 95 88 91

4 93 91 80 84

5 91 92 93 88

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48. The coach of the women's swim team at State University is preparing for the conference swim meet

and must choose the four swimmers she will assign to the 800-meter medley relay team. The medley relay consists of four strokes—the backstroke, breaststroke, butterfly, and freestyle. The coach has computed the average times (in minutes) each of her top six swimmers has achieved in each of the four strokes for 200 meters in previous swim meets during the season as follows:

Stroke (min)

Swimmer Backstroke Breaststroke Butterfly Freestyle

Annie 2.56 3.07 2.90 2.26

Beth 2.63 3.01 3.12 2.35

Carla 2.71 2.95 2.96 2.29

Debbie 2.60 2.87 3.08 2.41

Erin 2.68 2.97 3.16 2.25

Fay 2.75 3.10 2.93 2.38

Determine the medley relay team and its total expected relay time for the coach.

49. Biggio's Department Store has six employees available to assign to four departments in the store—

home furnishings, china, appliances, and jewelry. Most of the six employees have worked in each of the four departments on several occasions in the past and have demonstrated that they perform better in some departments than in others. The average daily sales for each of the six employees in each of the four departments is shown in the following table.

Department Sales (\$)

Home

Employee Furnishings China Appliances Jewelry

1 340 160 610 290

2 560 370 520 450

3 270 — 350 420

4 360 220 630 150

5 450 190 570 310

6 280 320 490 360

Employee 3 has not worked in the china department before, so the manager does not want to assign

this employee to china.

Determine which employee to assign to each department and indicate the total expected daily sales.

50. The Vanguard Publishing Company has eight college students it hires as salespeople to sell encyclo-

pedias during the summer. The company desires to allocate them to three sales territories.

Territory

1 requires three salespeople, and territories 2 and 3 require two salespeople each. It is estimated that

each salesperson will be able to generate the amounts of dollar sales per day in each of the three ter-

ritories as given in the following table. The company desires to allocate the salespeople to the three territories so that sales will be maximized. Solve this problem using any method to determine the initial solution and any solution method. Compute the maximum total sales per day.

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Problems B-43

Territory

Salesperson 1 2 3

A \$110 \$150 \$130

B 90 120 80

C 205 160 175

D 125 100 115

E 140 105 150

F 100 140 120

G 180 210 160

H 110 120 90

51. Carolina Airlines, a small commuter airline in North Carolina, has six flight attendants whom it wants to assign to six monthly flight schedules in a way that will minimize the number of nights they will be away from their homes. The numbers of nights each attendant must be away from home with each schedule are given in the following table. Identify the optimal assignments that will minimize the total number of nights the attendants will be away from home.

Schedule

Attendant A B C D E F

1 7 4 5 10 5 7

2 4 5 4 12 7 5

3 9 9 10 7 10 7

4 11 6 7 5 9 9

5 5 8 5 10 7 5

6 10 12 10 9 9 9

52. The football coaching staff at Tech focuses its recruiting on several key states, including Georgia,

Florida, Virginia, Pennsylvania, New York, and New Jersey. The staff includes seven assistant

coaches, two of whom are responsible for Florida, a high school talent-rich state, whereas one coach is assigned to each of the other five states. The staff has been together for a long time and at one time or another all the coaches have recruited all of the states. The head coach has accumulated

some data on the past success rate (i.e., percentage of targeted recruits signed) for each coach in each state as shown in the following table:

State

Coach GA FL VA PA NY NJ

Allen 62 56 65 71 55 63

Bush 65 70 63 81 75 72

Crumb 46 53 62 55 64 50

Doyle 58 66 70 67 71 49

Evans 77 73 69 80 80 74

Fouch 68 73 72 80 78 57

Goins 72 60 74 72 62 61

Determine the optimal assignment of coaches to recruiting regions that will maximize the overall success rate and indicate the average percentage success rate for the staff with this assignment.

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Transportation, Assignment, and

Transshipment Problems

In this chapter, we discuss three special types of linear programming problems: transportation, assignment, and transshipment. Each of these can be solved by the simplex algorithm, but specialized algorithms for each type of problem are much more efficient.

7.1 Formulating Transportation Problems

We begin our discussion of transportation problems by formulating a linear programming model of the following situation.

EXAMPLE 1 Powerco Formulation

Powerco has three electric power plants that supply the needs of four cities.† Each power plant can supply the following numbers of kilowatt-hours (kwh) of electricity: plant 1—35 million; plant 2—50 million; plant 3—40 million (see Table 1). The peak power demands in these cities, which occur at the same time (2 P.M.), are as follows (in kwh): city 1—45 million; city 2—20 million; city 3—30 million; city 4—30 million. The costs of sending 1 million kwh of electricity from plant to city depend on the distance the electricity must travel. Formulate an LP to minimize the cost of meeting each city's peak power demand.

Solution To formulate Powerco's problem as an LP, we begin by defining a variable for each decision that Powerco must make. Because Powerco must determine how much power is sent from each plant to each city, we define (for $i \in 1, 2, 3$ and $j \in 1, 2, 3, 4$)

x_{ij} = number of (million) kwh produced at plant i and sent to city j

In terms of these variables, the total cost of supplying the peak power demands to cities 1–4 may be written as

$8x_{11} + 6x_{12} + 10x_{13} + 9x_{14}$ (Cost of shipping power from plant 1)

$9x_{21} + 12x_{22} + 13x_{23} + 7x_{24}$ (Cost of shipping power from plant 2)

$14x_{31} + 9x_{32} + 16x_{33} + 5x_{34}$ (Cost of shipping power from plant 3)

Powerco faces two types of constraints. First, the total power supplied by each plant cannot exceed the plant's capacity. For example, the total amount of power sent from plant

This example is based on Aarvik and Randolph (1975).

TABLE 1

Shipping Costs, Supply, and Demand for Powerco

To

Supply

From City 1 City 2 City 3 City 4 (million kwh)

Plant 1 \$8 \$6 \$10 \$9 35

Plant 2 \$9 \$12 \$13 \$7 50

Plant 3 \$14 \$9 \$16 \$5 40

Demand 45 20 30 30

(million kwh)

1 to the four cities cannot exceed 35 million kwh. Each variable with first subscript 1 represents a shipment of power from plant 1, so we may express this restriction by the LP constraint

$x_{11} + x_{12} + x_{13} + x_{14} \leq 35$

In a similar fashion, we can find constraints that reflect plant 2's and plant 3's capacities. Because power is supplied by the power plants, each is a supply point. Analogously, a constraint that ensures that the total quantity shipped from a plant does not exceed plant capacity is a supply constraint. The LP formulation of Powerco's problem contains the following three supply constraints:

$x_{11} + x_{12} + x_{13} + x_{14} \leq 35$ (Plant 1 supply constraint)

$x_{21} + x_{22} + x_{23} + x_{24} \leq 50$ (Plant 2 supply constraint)

$x_{31} + x_{32} + x_{33} + x_{34} \leq 40$ (Plant 3 supply constraint)

Second, we need constraints that ensure that each city will receive sufficient power to meet its peak demand. Each city demands power, so each is a demand point. For example, city 1 must receive at least 45 million kwh. Each variable with second subscript 1 represents a shipment of power to city 1, so we obtain the following constraint:

$x_{11} + x_{21} + x_{31} \geq 45$

Similarly, we obtain a constraint for each of cities 2, 3, and 4. A constraint that ensures that a location receives its demand is a demand constraint. Powerco must satisfy the following four demand constraints:

$x_{11} + x_{21} + x_{31} \geq 45$ (City 1 demand constraint)

$x_{12} + x_{22} + x_{32} \geq 20$ (City 2 demand constraint)

$x_{13} + x_{23} + x_{33} = 30$ (City 3 demand constraint)

$x_{14} + x_{24} + x_{34} = 30$ (City 4 demand constraint)

Because all the x_{ij} 's must be nonnegative, we add the sign restrictions $x_{ij} \geq 0$ ($i = 1, 2, 3; j = 1, 2, 3, 4$).

Combining the objective function, supply constraints, demand constraints, and sign restrictions yields the following LP formulation of Powerco's problem:

min $z = 8x_{11} + 6x_{12} + 10x_{13} + 9x_{14} + 9x_{21} + 12x_{22} + 13x_{23} + 7x_{24}$

$+ 14x_{31} + 9x_{32} + 16x_{33} + 5x_{34}$

s.t. $x_{11} + x_{12} + x_{13} + x_{14} = 35$ (Supply constraints)

s.t. $x_{21} + x_{22} + x_{23} + x_{24} = 50$

s.t. $x_{31} + x_{32} + x_{33} + x_{34} = 40$

7. 1 Formulating Transportation Problems 361

Supply points Demand points

City 1 $d_1 = 45$

$x_{11} = 0$

$s_1 = 35$ Plant 1 $x_{12} = 10$

$x_{13} = 25$

$x_{14} = 0$

FIGURE 1 City 2 $d_2 = 20$

Graphical $x_{21} = 45$

$x_{22} = 0$

Representation of $s_1 = 50$ Plant 2 $x_{23} = 5$

Powerco Problem and $x_{24} = 0$

Its Optimal Solution City 3 $d_3 = 30$

$x_{31} = 0$

$x_{32} = 10$

$s_1 = 40$ Plant 3 $x_{33} = 0$

$x_{34} = 30$

City 4 $d_4 = 30$

s.t. $x_{11} + x_{21} + x_{31} + x_{34} = 45$ (Demand constraints)

s.t. $x_{12} + x_{22} + x_{32} + x_{34} = 20$

s.t. $x_{13} + x_{23} + x_{33} + x_{34} = 30$

s.t. $x_{14} + x_{24} + x_{34} + x_{34} = 30$

$x_{ij} \geq 0$ ($i = 1, 2, 3; j = 1, 2, 3, 4$)

In Section 7.3, we will find that the optimal solution to this LP is $z = 1020$, $x_{12} = 10$, $x_{13} = 25$, $x_{21} = 45$, $x_{23} = 5$, $x_{32} = 10$, $x_{34} = 30$. Figure 1 is a graphical representation of the Powerco problem and its optimal solution. The variable x_{ij} is represented by a line, or arc, joining the i th supply point (plant i) and the j th demand point (city j).

General Description of a Transportation Problem

In general, a transportation problem is specified by the following information:

- 1 A set of m supply points from which a good is shipped. Supply point i can supply at most s_i units. In the Powerco example, $m = 3$, $s_1 = 35$, $s_2 = 50$, and $s_3 = 40$.
- 2 A set of n demand points to which the good is shipped. Demand point j must receive at least d_j units of the shipped good. In the Powerco example, $n = 4$, $d_1 = 45$, $d_2 = 20$, $d_3 = 30$, and $d_4 = 30$.
- 3 Each unit produced at supply point i and shipped to demand point j incurs a variable cost of c_{ij} . In the Powerco example, $c_{12} = 6$.

Let

x_{ij} = number of units shipped from supply point i to demand point j

then the general formulation of a transportation problem is

$$i=1 \dots m, j=1 \dots n$$

$$\min \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$i=1 \dots m, j=1 \dots n$$

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$$j=1 \dots n$$

s.t. $\sum_{j=1}^n x_{ij} \leq s_i$ ($i = 1, 2, \dots, m$) (Supply constraints)

$$j=1 \dots n$$

$$i=1 \dots m$$

s.t. $\sum_{i=1}^m x_{ij} \geq d_j$ ($j = 1, 2, \dots, n$)

$$i=1 \dots m$$

(Demand constraints)

$$x_{ij} \geq 0 \text{ (} i = 1, 2, \dots, m; j = 1, 2, \dots, n \text{)}$$

If a problem has the constraints given in (1) and is a maximization problem, then it is still a transportation problem (see Problem 7 at the end of this section). If

$$i=1 \dots m, j=1 \dots n$$

$$\sum_{i=1}^m s_i = \sum_{j=1}^n d_j$$

$$i=1 \dots m, j=1 \dots n$$

then total supply equals total demand, and the problem is said to be a balanced transportation problem.

For the Powerco problem, total supply and total demand both equal 125, so this is a balanced transportation problem. In a balanced transportation problem, all the constraints must be binding. For example, in the Powerco problem, if any supply constraint were non-binding, then the remaining available power would not be sufficient to meet the needs of all four cities. For a balanced transportation problem, (1) may be written as

$$i=1 \dots m, j=1 \dots n$$

$$\min \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$i=1 \dots m, j=1 \dots n$$

$$j=1 \dots n$$

s.t. $\sum_{j=1}^n x_{ij} = s_i$ ($i = 1, 2, \dots, m$) (Supply constraints)

$$j \in \{1, 2, \dots, n\}$$

$$i \in \{1, 2, \dots, m\}$$

$$\text{s.t. } x_{ij} \leq d_j \quad (j = 1, 2, \dots, n)$$

$$i \in \{1, 2, \dots, m\}$$

(Demand constraints)

$$x_{ij} \geq 0 \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, n)$$

Later in this chapter, we will see that it is relatively simple to find a basic feasible solution for a balanced transportation problem. Also, simplex pivots for these problems do not involve multiplication and reduce to additions and subtractions. For these reasons, it is desirable to formulate a transportation problem as a balanced transportation problem.

Balancing a Transportation Problem

If Total Supply Exceeds Total Demand

If total supply exceeds total demand, we can balance a transportation problem by creating a dummy demand point that has a demand equal to the amount of excess supply. Because shipments to the dummy demand point are not real shipments, they are assigned a cost of zero. Shipments to the dummy demand point indicate unused supply capacity. To understand the use of a dummy demand point, suppose that in the Powerco problem, the demand for city 1 were reduced to 40 million kwh. To balance the Powerco problem, we would add a dummy demand point (point 5) with a demand of $125 - 120 = 5$ million kwh. From each plant, the cost of shipping 1 million kwh to the dummy is 0. The optimal solution to this balanced transportation problem is $z = 975$, $x_{13} = 20$, $x_{12} = 15$, $x_{21} = 40$, $x_{23} = 10$, $x_{32} = 5$, $x_{34} = 30$, and $x_{35} = 5$. Because $x_{35} = 5$, 5 million kwh of plant 3 capacity will be unused (see Figure 2).

A transportation problem is specified by the supply, the demand, and the shipping costs, so the relevant data can be summarized in a transportation tableau (see Table 2). The square, or cell, in row i and column j of a transportation tableau corresponds to the

7.1 Formulating Transportation Problems 363

Supply points Demand points

 $x_{11} = 0$

City 1 $d_1 = 40$

 $x_{21} = 40$

$s_1 = 35$ Plant 1 $x_{31} = 0$

 $x_{12} = 15$

FIGURE 2

 $x_{22} = 0$ City 2 $d_2 = 20$

Graphical $x_{32} = 5$

Representation of

Unbalanced Powerco $s_2 = 50$ Plant 2 $x_{13} = 20$

Problem and Its $x_{23} = 10$

Optimal Solution (with $x_{33} = 0$ City 3 $d_3 = 30$)

Dummy Demand Point)

$x_{14} = 0$

$s_3 = 40$ Plant 3 $x_{24} = 0$

$x_{34} = 30$

City 4 $d_4 = 30$

$x_{25} = 0$ $x_{15} = 0$

$x_{35} = 5$

Dummy

$d_5 = 5$

City 5

TABLE 2 c_{11} c_{12} c_{1n} Supply

A Transportation Tableau s_1

c_{21} c_{22} c_{2n}

s_2

c_{m1} c_{m2} c_{mn}

s_m

d_1 d_2 d_n

Demand

TABLE 3 City 1 City 2 City 3 City 4 Supply

Transportation Tableau 8 6 10 9

for Powerco

Plant 1 10 25 35

9 12 13 7

Plant 2 45 5 50

14 9 16 5

Plant 3 10 30 40

Demand 45 20 30 30

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variable x_{ij} . If x_{ij} is a basic variable, its value is placed in the lower left-hand corner of the ij th cell of the tableau. For example, the balanced Powerco problem and its optimal solution could be displayed as shown in Table 3. The tableau format implicitly expresses the supply and demand constraints through the fact that the sum of the variables in row i must equal s_i and the sum of the variables in column j must equal d_j .

Balancing a Transportation Problem

If Total Supply Is Less Than Total Demand

If a transportation problem has a total supply that is strictly less than total demand, then the problem has no feasible solution. For example, if plant 1 had only 30 million kwh of capacity, then a total of only 120 million kwh would be available. This amount of power would be insufficient to meet the total demand of 125 million kwh, and the Powerco prob-

lem would no longer have a feasible solution.

When total supply is less than total demand, it is sometimes desirable to allow the possibility of leaving some demand unmet. In such a situation, a penalty is often associated with unmet demand. Example 2 illustrates how such a situation can yield a balanced transportation problem.

EXAMPLE 2 Handling Shortages

Two reservoirs are available to supply the water needs of three cities. Each reservoir can supply up to 50 million gallons of water per day. Each city would like to receive 40 million gallons per day. For each million gallons per day of unmet demand, there is a penalty. At city 1, the penalty is \$20; at city 2, the penalty is \$22; and at city 3, the penalty is \$23. The cost of transporting 1 million gallons of water from each reservoir to each city is shown in Table 4. Formulate a balanced transportation problem that can be used to minimize the sum of shortage and transport costs.

Solution In this problem,

Daily supply $\square$ 50 $\square$ 50 $\square$ 100 million gallons per day

Daily demand $\square$ 40 $\square$ 40 $\square$ 40 $\square$ 120 million gallons per day

To balance the problem, we add a dummy (or shortage) supply point having a supply of $120 \square 100 \square 20$ million gallons per day. The cost of shipping 1 million gallons from the dummy supply point to a city is just the shortage cost per million gallons for that city. Table 5 shows the balanced transportation problem and its optimal solution. Reservoir 1 should send 20 million gallons per day to city 1 and 30 million gallons per day to city 2, whereas reservoir 2 should send 10 million gallons per day to city 2 and 40 million gallons per day to city 3. Twenty million gallons per day of city 1's demand will be unsatisfied.

TABLE 4

Shipping Costs for Reservoir

To

From City 1 City 2 City 3

Reservoir 1 \$7 \$8 \$10

Reservoir 2 \$9 \$7 \$8

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TABLE 5 City 1 City 2 City 3 Supply

Transportation Tableau 7 8 10

for Reservoir

Reservoir 1 20 30 50

9 7 8

Reservoir 2 10 40 50

20 22 23

Dummy (shortage) 20 20

Demand 40 40 40

Modeling Inventory Problems as Transportation Problems

Many inventory planning problems can be modeled as balanced transportation problems. To illustrate, we formulate a balanced transportation model of the Sailco problem of Section 3.10.

EXAMPLE 3 Setting Up an Inventory Problem as a Transportation Problem

Sailco Corporation must determine how many sailboats should be produced during each of the next four quarters (one quarter is three months). Demand is as follows: first quarter, 40 sailboats; second quarter, 60 sailboats; third quarter, 75 sailboats; fourth quarter, 25 sailboats. Sailco must meet demand on time. At the beginning of the first quarter, Sailco has an inventory of 10 sailboats. At the beginning of each quarter, Sailco must decide how many sailboats should be produced during the current quarter. For simplicity, we assume that sailboats manufactured during a quarter can be used to meet demand for the current quarter. During each quarter, Sailco can produce up to 40 sailboats at a cost of \$400 per sailboat. By having employees work overtime during a quarter, Sailco can produce additional sailboats at a cost of \$450 per sailboat. At the end of each quarter (after production has occurred and the current quarter's demand has been satisfied), a carrying or holding cost of \$20 per sailboat is incurred. Formulate a balanced transportation problem to minimize the sum of production and inventory costs during the next four quarters.

Solution We define supply and demand points as follows:

Supply Points Point 1 □ initial inventory (s_1 □ 10)

Supply Points Point 2 □ quarter 1 regular-time (RT) production (s_2 □ 40)

Supply Points Point 3 □ quarter 1 overtime (OT) production (s_3 □ 150)

Supply Points Point 4 □ quarter 2 RT production (s_4 □ 40)

Supply Points Point 5 □ quarter 2 OT production (s_5 □ 150)

Supply Points Point 6 □ quarter 3 RT production (s_6 □ 40)

Supply Points Point 7 □ quarter 3 OT production (s_7 □ 150)

Supply Points Point 8 □ quarter 4 RT production (s_8 □ 40)

Supply Points Point 9 □ quarter 4 OT production (s_9 □ 150)

There is a supply point corresponding to each source from which demand for sailboats can be met:

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Demand Points Point 1 □ quarter 1 demand (d_1 □ 40)

Demand Points Point 2 □ quarter 2 demand (d_2 □ 60)

Demand Points Point 3 □ quarter 3 demand (d_3 □ 75)

Demand Points Point 4 □ quarter 4 demand (d_4 □ 25)

Demand Points Point 5 □ dummy demand point (d_5 □ $770 - 200$ □ 570)

A shipment from, say, quarter 1 RT to quarter 3 demand means producing 1 unit on regular time during quarter 1 that is used to meet 1 unit of quarter 3's demand. To determine, say, c_{13} , observe that producing 1 unit during quarter 1 RT and using that unit to meet quarter 3 demand incurs a cost equal to the cost of producing 1 unit on quarter 1 RT plus the

cost of holding a unit in inventory for 3 $\times$ 1 $\times$ 2 quarters. Thus, $c_{13} = 400 + 2(20) = 440$. Because there is no limit on the overtime production during any quarter, it is not clear what value should be chosen for the supply at each overtime production point. Total demand = 200, so at most $200 - 10 = 190$ (-10 is for initial inventory) units will be produced during any quarter. Because 40 units must be produced on regular time before any units are produced on overtime, overtime production during any quarter will never exceed $190 - 40 = 150$ units. Any unused overtime capacity will be “shipped” to the dummy demand point. To ensure that no sailboats are used to meet demand during a quarter prior to their production, a cost of M (M is a large positive number) is assigned to any cell that corresponds to using production to meet demand for an earlier quarter.

TABLE 6 1 2 3 4 Dummy Supply

Transportation Tableau

0 20 40 60 0

for Sailco

Initial 10 10

400 420 440 460 0

Qtr 1 RT 30 10 40

450 470 490 510 0

Qtr 1 OT 150 150

M 400 420 440 0

Qtr 2 RT 40 40

M 450 470 490 0

Qtr 2 OT 10 140 150

M M 400 420 0

Qtr 3 RT 40 40

M M 450 470 0

Qtr 3 OT 35 115 150

M M M 400 0

Qtr 4 RT 25 15 40

M M M 450 0

Qtr 4 OT 150 150

Demand 40 60 75 25 570

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Total supply = 770 and total demand = 200, so we must add a dummy demand point with a demand of $770 - 200 = 570$ to balance the problem. The cost of shipping a unit from any supply point to the dummy demand point is 0.

Combining these observations yields the balanced transportation problem and its optimal solution shown in Table 6. Thus, Sailco should meet quarter 1 demand with 10 units of initial inventory and 30 units of quarter 1 RT production; quarter 2 demand with 10

units of quarter 1 RT, 40 units of quarter 2 RT, and 10 units of quarter 2 OT production; quarter 3 demand with 40 units of quarter 3 RT and 35 units of quarter 3 OT production; and finally, quarter 4 demand with 25 units of quarter 4 RT production.

In Problem 12 at the end of this section, we show how this formulation can be modified to incorporate other aspects of inventory problems (backlogged demand, perishable inventory, and so on).

Solving Transportation Problems on the Computer

To solve a transportation problem with LINDO, type in the objective function, supply constraints, and demand constraints. Other menu-driven programs are available that accept the shipping costs, supply values, and demand values. From these values, the program can generate the objective function and constraints.

LINGO can be used to easily solve any transportation problem. The following LINGO

Trans.lng

model can be used to solve the Powerco example (file Trans.lng).

MODEL:

1]SETS:

2]PLANTS/P1,P2,P3/:CAP;

3]CITIES/C1,C2,C3,C4/:DEM;

4]LINKS(PLANTS,CITIES):COST,SHIP;

5]ENDSETS

6]MIN=@SUM(LINKS:COST*SHIP);

7]@FOR(CITIES(J):

8]@SUM(PLANTS(I):SHIP(I,J))>DEM(J));

9]@FOR(PLANTS(I):

10]@SUM(CITIES(J):SHIP(I,J))<CAP(I));

11]DATA:

12]CAP=35,50,40;

13]DEM=45,20,30,30;

14]COST=8,6,10,9,

15]9,12,13,7,

16]14,9,16,5;

17]ENDDATA

END

Lines 1–5 define the SETS needed to generate the objective function and constraints.

In line 2, we create the three power plants (the supply points) and specify that each has a capacity (given in the DATA section). In line 3, we create the four cities (the demand points) and specify that each has a demand (given in the DATA section). The LINK statement in line 4 creates a LINK(I,J) as I runs over all PLANTS and J runs over all CITIES.

Thus, objects LINK(1,1), LINK (1,2), LINK(1,3), LINK(1,4), LINK(2,1), LINK (2,2), LINK(2,3), LINK(2,4), LINK(3,1), LINK (3,2), LINK(3,3), LINK(3,4) are created and stored in this order. Attributes with multiple subscripts are stored so that the rightmost

subscripts advance most rapidly. Each LINK has two attributes: a per-unit shipping cost [(COST), given in the DATA section] and the amount shipped (SHIP), for which LINGO will solve.

Line 6 creates the objective function. We sum over all links the product of the unit shipping cost and the amount shipped. Using the @FOR and @SUM operators, lines 7–8

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generate all demand constraints. They ensure that for each city, the sum of the amount shipped into the city will be at least as large as the city's demand. Note that the extra parenthesis after SHIP(I,J) in line 8 is to close the @SUM operator, and the extra parenthesis after DEM(J) is to close the @FOR operator. Using the @FOR and @SUM operators, lines 9–10 generate all supply constraints. They ensure that for each plant, the total shipped out of the plant will not exceed the plant's capacity.

Lines 11–17 contain the data needed for the problem. Line 12 defines each plant's capacity, and line 13 defines each city's demand. Lines 14–16 contain the unit shipping cost from each plant to each city. These costs correspond to the ordering of the links described previously. ENDDATA ends the data section, and END ends the program. Typing GO will solve the problem.

This program can be used to solve any transportation problem. If, for example, we wanted to solve a problem with 15 supply points and 10 demand points, we would change line 2 to create 15 supply points and line 3 to create 10 demand points. Moving to line 12, we would type in the 15 plant capacities. In line 13, we would type in the demands for the 10 demand points. Then in line 14, we would type in the 150 shipping costs. Observe that the part of the program (lines 6–10) that generates the objective function and constraints remains unchanged! Notice also that our LINGO formulation does not require that the transportation problem be balanced.

Obtaining LINGO Data from an Excel Spreadsheet

Often it is easier to obtain data for a LINGO model from a spreadsheet. For example, shipping costs for a transportation problem may be the end result of many computations. As an example, suppose we have created the capacities, demands, and shipping costs for Powerco.xls the Powerco model in the file Powerco.xls (see Figure 3). We have created capacities in the cell range F9:F11 and named the range Cap. As you probably know, you can name a range of cells in Excel by selecting the range and clicking in the name box in the upper left-hand corner of your spreadsheet. Then type the range name and hit the Enter key. In a similar fashion, name the city demands (in cells B12:E12) with the name Demand and the unit shipping costs (in cells B4:E6) with the name Costs.

FIGURE 3

A B C D E F G H

1 OPTIMAL SOLUTION FOR POWERCO COSTS

2 COSTS CITY 1020

3 PLANT 1 2 3 4

```

4 1 8 6 10 9
5 2 9 12 13 7
6 3 14 9 16 5
7 SHIPMENTS CITY SHIPPED SUPPLIES
8 PLANT 1 2 3 4
9 1 0 10 25 0 35 <= 35
10 2 45 0 5 0 50 <= 50
11 3 0 10 0 30 40 <= 40
12 RECEIVED 45 20 30 30
13 >= >= >= >=
14 DEMANDS 45 20 30 30

```

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Using an @OLE statement, LINGO can read from a spreadsheet the values of data that are defined in the Sets portion of a program. The LINGO program (see file Transp-Transpspread.lng spread.lng) needed to read our input data from the Powerco.xls file is shown below.

MODEL:

SETS:

PLANTS/P1,P2,P3/:CAP;

CITIES/C1,C2,C3,C4/:DEM;

LINKS(PLANTS,CITIES):COST,SHIP;

ENDSETS

MIN=@SUM(LINKS:COST*SHIP);

@FOR(CITIES(J):

@SUM(PLANTS(I):SHIP(I,J))>DEM(J));

@FOR(PLANTS(I);

@SUM(CITIES(J):SHIP(I,J))<CAP(I));

DATA:

CAP, DEM, COST=@OLE('C:\MPROG\POWERCO.XLS','Cap','Demand','Costs');

ENDDATA

END

The key statement is

CAP, DEM, COST=@OLE('C:\MPROG\POWERCO.XLS','Cap','Demand','Costs');

This statement reads the defined data sets CAP, DEM, and COSTS from the Powerco.xls spreadsheet. Note that the full path location of our Excel file (enclosed in single quotes) must be given first followed by the spreadsheet range names that contain the needed data. The range names are paired with the data sets in the order listed. Therefore, CAP values are found in range Cap and so on. The @OLE statement is very powerful, because a spreadsheet will usually greatly simplify the creation of data for a LINGO program.

Spreadsheet Solution of Transportation Problems

Powerco.xls In the file Powerco.xls, we show how easy it is to use the Excel Solver to find the optimal solution to a transportation problem. After entering the plant capacities, city demands, and unit shipping costs as shown, we enter trial values of the units shipped from each plant to each city in the range B9:E11. Then we proceed as follows:

Step 1 Compute the total amount shipped out of each city by copying from F9 to F10:F11 the formula

`=SUM(B9:E9)`

Step 2 Compute the total received by each city by copying from B12 to C12:E12 the formula

`=SUM(B9:B11)`

Step 3 Compute the total shipping cost in cell F2 with the formula

`=SUMPRODUCT(B9:E11,Costs)`

Note that the `SUMPRODUCT` function works on rectangles as well as rows or columns of numbers. Also, we have named the range of unit shipping costs (B4:E6) as COSTS.

Step 4 We now fill in the Solver window shown in Figure 4. We minimize total shipping costs (F2) by changing units shipped from each plant to each city (B9:E11). We constrain amount received by each city (B12:E12) to be at least each city's demand (range name Demand). We constrain the amount shipped out of each plant (F9:F11) to be at most each plant's capacity (range name Cap). After checking the Assume Nonnegative option and Assume Linear Model option, we obtain the optimal solution shown in Figure 3. Note, of course, that the objective function of the optimal solution found by Excel equals the ob-

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FIGURE 4

jective function value found by LINGO and our hand solution. If the problem had multiple optimal solutions, then it is possible that the values of the shipments found by LINGO, Excel, and our hand solution might be different.

PROBLEMS

Group A

1 A company supplies goods to three customers, who each overtime production is limited to 100 pairs. It costs \$1 per

require 30 units. The company has two warehouses. month to hold a pair of shoes in inventory.

Formulate a

Warehouse 1 has 40 units available, and warehouse 2 has 30 balanced transportation problem to minimize the total cost

units available. The costs of shipping 1 unit from warehouse of meeting the next six months of demand on time.

to customer are shown in Table 7. There is a penalty for each

4 Steelco manufactures three types of steel at different

unmet customer unit of demand: With customer 1, a penalty

plants. The time required to manufacture 1 ton of steel

cost of \$90 is incurred; with customer 2, \$80; and with

(regardless of type) and the costs at each plant are shown in customer 3, \$110. Formulate a balanced transportation

Table 8. Each week, 100 tons of each type of steel (1, 2, and problem to minimize the sum of shortage and shipping costs.

3) must be produced. Each plant is open 40 hours per week.

2 Referring to Problem 1, suppose that extra units could a Formulate a balanced transportation problem to min-

be purchased and shipped to either warehouse for a total imize the cost of meeting Steelco's weekly requirements.

cost of \$100 per unit and that all customer demand must be b Suppose the time required to produce 1 ton of steel

met. Formulate a balanced transportation problem to depends on the type of steel as well as on the plant at

minimize the sum of purchasing and shipping costs. which it is produced (see Table 9, page 372). Could a

3 A shoe company forecasts the following demands during transportation problem still be formulated?

the next six months: month 1—200; month 2—260; month 5 A hospital needs to purchase 3 gallons of a perishable

3—240; month 4—340; month 5—190; month 6—150. It medicine for use during the current month and 4 gallons for

costs \$7 to produce a pair of shoes with regular-time labor use during the next month. Because the medicine is

(RT) and \$11 with overtime labor (OT). During each month, regular production is limited to 200 pairs of shoes, and

TA B L E 8

TA B L E 7 Cost (\$)

Time

To Plant Steel 1 Steel 2 Steel 3 (minutes)

From Customer 1 Customer 2 Customer 3

1 60 40 28 20

Warehouse 1 \$15 \$35 \$25 2 50 30 30 16

Warehouse 2 \$10 \$50 \$40 3 43 20 20 15

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TA B L E 9 TA B L E 12

Time (minutes) To (\$)

Plant Steel 1 Steel 2 Steel 3 From (\$)

England Japan

1 15 12 15 Field 1 1 2

2 15 15 20 Field 2 2 1

3 10 10 15

TA B L E 13

perishable, it can only be used during the month of purchase. Project (\$)

Two companies (Daisy and Laroach) sell the medicine. The Auditor 1 2 3 medicine is in short supply. Thus, during the next two months, the hospital is limited to buying at most 5 gallons 1 120 150 190 from each company. The companies charge the prices shown 2 140 130 120 in Table 10. Formulate a balanced transportation model to 3 160 140 150 minimize the cost of purchasing the needed medicine.

6 A bank has two sites at which checks are processed. Site 1 can process 10,000 checks per day, and site 2 can process

8 The Ayatola Oil Company controls two oil fields. Field 6,000 checks per day. The bank processes three types of 1 can produce up to 40 million barrels of oil per day, and checks: vendor, salary, and personal. The processing cost field 2 can produce up to 50 million barrels of oil per day. per check depends on the site (see Table 11). Each day, At field 1, it costs \$3 to extract and refine a barrel of oil; at 5,000 checks of each type must be processed. Formulate a field 2, the cost is \$2. Ayatola sells oil to two countries: balanced transportation problem to minimize the daily cost England and Japan. The shipping cost per barrel is shown of processing checks.

in Table 12. Each day, England is willing to buy up to 40

7† The U.S. government is auctioning off oil leases at two million barrels (at \$6 per barrel), and Japan is willing to

sites: 1 and 2. At each site, 100,000 acres of land are to be buy up to 30 million barrels (at \$6.50 per barrel). Formulate

auctioned. Cliff Ewing, Blake Barnes, and Alexis Pickens are a balanced transportation problem to maximize Ayatola's

bidding for the oil. Government rules state that no bidder can profits.

receive more than 40% of the land being auctioned. Cliff has

9 For the examples and problems of this section, discuss

bid \$1,000/acre for site 1 land and \$2,000/acre for site 2 land.

whether it is reasonable to assume that the proportionality

Blake has bid \$900/acre for site 1 land and \$2,200/acre for site

assumption holds for the objective function.

2 land. Alexis has bid \$1,100/acre for site 1 land and

\$1,900/acre for site 2 land. Formulate a balanced transportation 10 Touche Young has three auditors. Each can work as

model to maximize the government's revenue. many as 160 hours during the next month, during which

time three projects must be completed. Project 1 will take

130 hours; project 2, 140 hours; and project 3, 160 hours.

The amount per hour that can be billed for assigning each

TABLE 10 auditor to each project is given in Table 13. Formulate a balanced transportation problem to maximize total billings Current Month's Next Month's during the next month.

Price per Price per

Company Gallon (\$) Gallon (\$)

Group B

Daisy 800 720

Laroach 710 750 11† Paperco recycles newsprint, uncoated paper, and coated paper into recycled newsprint, recycled uncoated paper, and recycled coated paper. Recycled newsprint can be produced by processing newsprint or uncoated paper.

Recycled coated paper can be produced by recycling any

TABLE 11 type of paper. Recycled uncoated paper can be produced by

Site (¢)

processing uncoated paper or coated paper. The process

used to produce recycled newsprint removes 20% of the

Checks 1 2 input's pulp, leaving 80% of the input's pulp for recycled

Vendor 5 3 paper. The process used to produce recycled coated paper

removes 10% of the input's pulp. The process used to

Salary 4 4 produce recycled uncoated paper removes 15% of the input's

Personal 2 5 pulp. The purchasing costs, processing costs, and availability of each type of paper are shown in Table 14. To meet demand,

† ‡

This problem is based on Jackson (1980). This problem is based on Glassey and Gupta (1974).

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TABLE 14 Paperco must produce at least 250 tons of recycled newsprint pulp, at least 300 tons of recycled uncoated paper pulp, and

Purchase Processing

Cost per Ton Cost per Ton at least 150 tons of recycled coated paper pulp. Formulate of Pulp (\$) of Input (\$) Availability a balanced transportation problem that can be used to minimize the cost of meeting Paperco's demands.

Newsprint 10 500

12 Explain how each of the following would modify the

Coated paper 19 300

formulation of the Sailco problem as a balanced trans-

Uncoated paper 18 200 portation problem:

NP used for RNP 3 a Suppose demand could be backlogged at a cost of

NP used for RCP 4 \$30/sailboat/month. (Hint: Now it is permissible to ship

UCP used for RNP 4 from, say, month 2 production to month 1 demand.)

UCP used for RUP 1 b If demand for a sailboat is not met on time, the sale

UCP used for RCP 6 is lost and an opportunity cost of \$450 is incurred.

CP used for RUP 5 c Sailboats can be held in inventory for a maximum of two months.

CP used for RCP 3

d At a cost of \$440/sailboat, Sailco can purchase up to 10 sailboats/month from a subcontractor.

7.2 Finding Basic Feasible Solutions for Transportation Problems

Consider a balanced transportation problem with m supply points and n demand points. From (2), we see that such a problem contains $m \times n$ equality constraints. From our experience with the Big M method and the two-phase simplex method, we know it is difficult to find a bfs if all of an LP's constraints are equalities. Fortunately, the special structure of a balanced transportation problem makes it easy for us to find a bfs.

Before describing three methods commonly used to find a bfs to a balanced transportation problem, we need to make the following important observation. If a set of values for the x_{ij} 's satisfies all but one of the constraints of a balanced transportation problem, then the values for the x_{ij} 's will automatically satisfy the other constraint. For example, in the Powerco problem, suppose a set of values for the x_{ij} 's is known to satisfy all the constraints with the exception of the first supply constraint. Then this set of x_{ij} 's must supply $d_1 \times d_2 \times d_3 \times d_4 = 125$ million kwh to cities 1–4 and supply $s_2 \times s_3 \times 125 \times s_1 = 90$ million kwh from plants 2 and 3. Thus, plant 1 must supply $125 \times (125 \times s_1) = 35$ million kwh, so the x_{ij} 's must also satisfy the first supply constraint. The preceding discussion shows that when we solve a balanced transportation problem, we may omit from consideration any one of the problem's constraints and solve an LP having $m \times n - 1$ constraints. We (arbitrarily) assume that the first supply constraint is omitted from consideration.

In trying to find a bfs to the remaining $m \times n - 1$ constraints, you might think that any collection of $m \times n - 1$ variables would yield a basic solution. Unfortunately, this is not the case. For example, consider (3), a balanced transportation problem. (We omit the costs because they are not needed to find a bfs.)

4

(3)

5

3 2 4

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In matrix form, the constraints for this balanced transportation problem may be written as

$\begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} \end{bmatrix}$

x_{11}

1 1 1 0 0 4

x_{12}

0 0 0 1 1 5

x_{13}

1 0 0 1 0 0 □ 3 (3□)

x21

0 1 0 0 1 0 2

x22

0 0 1 0 0 1 4

x23

After dropping the first supply constraint, we obtain the following linear system:

□ □ □ □ □

x11

0 0 0 1 1 1 x12 5

1 0 0 1 0 0 x13 3

□ (3□)

0 1 0 0 1 0 x21 2

0 0 1 0 0 1 x22 4

x23

A basic solution to (3□) must have four basic variables. Suppose we try $BV = \{x11, x12, x21, x22\}$. Then

0 0 1 1

1 0 1 0

B□

0 1 0 1

0 0 0 0

For $\{x11, x12, x21, x22\}$ to yield a basic solution, it must be possible to use EROs to transform B to I4. Because $\text{rank } B = 3$ and EROs do not change the rank of a matrix, there is no way that EROs can be used to transform B into I4. Thus, $BV = \{x11, x12, x21, x22\}$ cannot yield a basic solution to (3□). Fortunately, the simple concept of a loop may be used to determine whether an arbitrary set of $m - n + 1$ variables yields a basic solution to a balanced transportation problem.

DEFINITION ■ An ordered sequence of at least four different cells is called a loop if

- 1 Any two consecutive cells lie in either the same row or same column
- 2 No three consecutive cells lie in the same row or column
- 3 The last cell in the sequence has a row or column in common with the first cell in the sequence ■

In the definition of a loop, the first cell is considered to follow the last cell, so the loop may be thought of as a closed path. Here are some examples of the preceding definition: Figure 5 represents the loop (2, 1)–(2, 4)–(4, 4)–(4, 1). Figure 6 represents the loop (1, 1)–(1, 2)–(2, 2)–(2, 3)–(4, 3)–(4, 5)–(3, 5)–(3, 1). In Figure 7, the path (1, 1)–(1, 2)–(2, 3)–(2, 1) does not represent a loop, because (1, 2) and (2, 3) do not lie in the same row or column. In Figure 8, the path (1, 2)–(1, 3)–(1, 4)–(2, 4)–(2, 2) does not represent a loop, because (1, 2), (1, 3), and (1, 4) all lie in the same row.

Theorem 1 (which we state without proof) shows why the concept of a loop is important.

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FIGURE 5

FIGURE 6

FIGURE 7

FIGURE 8

THEOREM 1

In a balanced transportation problem with m supply points and n demand points, the cells corresponding to a set of $m + n - 1$ variables contain no loop if and only if the $m + n - 1$ variables yield a basic solution.

Theorem 1 follows from the fact that a set of $m + n - 1$ cells contains no loop if and only if the $m + n - 1$ columns corresponding to these cells are linearly independent. Because $(1, 1)-(1, 2)-(2, 2)-(2, 1)$ is a loop, Theorem 1 tells us that $\{x_{11}, x_{12}, x_{22}, x_{21}\}$ cannot yield a basic solution for (3) . On the other hand, no loop can be formed with the cells $(1, 1)-(1, 2)-(1, 3)-(2, 1)$, so $\{x_{11}, x_{12}, x_{13}, x_{21}\}$ will yield a basic solution to (3) .

We are now ready to discuss three methods that can be used to find a basic feasible solution for a balanced transportation problem:

- 1 northwest corner method
- 2 minimum-cost method
- 3 Vogel's method

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Northwest Corner Method for Finding a Basic Feasible Solution

To find a bfs by the northwest corner method, we begin in the upper left (or northwest) corner of the transportation tableau and set x_{11} as large as possible. Clearly, x_{11} can be no larger than the smaller of s_1 and d_1 . If $x_{11} = s_1$, cross out the first row of the transportation tableau; this indicates that no more basic variables will come from row 1. Also change d_1 to $d_1 - s_1$. If $x_{11} = d_1$, cross out the first column of the transportation tableau; this indicates that no more basic variables will come from column 1. Also change s_1 to $s_1 - d_1$. If $x_{11} = s_1 = d_1$, cross out either row 1 or column 1 (but not both). If you cross out row 1, change d_1 to 0; if you cross out column 1, change s_1 to 0.

Continue applying this procedure to the most northwest cell in the tableau that does not lie in a crossed-out row or column. Eventually, you will come to a point where there is only one cell that can be assigned a value. Assign this cell a value equal to its row or column demand, and cross out both the cell's row and column. A basic feasible solution has now been obtained.

We illustrate the use of the northwest corner method by finding a bfs for the balanced transportation problem in Table 15. (We do not list the costs because they are not needed to apply the algorithm.) We indicate the crossing out of a row or column by placing an $\square$

by the row's supply or column's demand.

To begin, we set $x_{11} = \min\{5, 2\} = 2$. Then we cross out column 1 and change s_1 to $5 - 2 = 3$. This yields Table 16. The most northwest remaining variable is x_{12} . We set $x_{12} = \min\{3, 4\} = 3$. Then we cross out row 1 and change d_2 to $4 - 3 = 1$. This yields Table 17. The most northwest available variable is now x_{22} . We set $x_{22} = \min\{1, 1\} = 1$. Because both the supply and demand corresponding to the cell are equal, we may cross out either row 2 or column 2 (but not both). For no particular reason, we choose to cross out row 2. Then d_2 must be changed to $1 - 1 = 0$. The resulting tableau is Table 18. At the next step, this will lead to a degenerate bfs.

TABLE 15

5
1
3
2 4 2 1

TABLE 16

2 3
1
3
 $\square$ 4 2 1

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TABLE 17

2 3 $\square$
1
3
 $\square$ 1 2 1

TABLE 18

2 3 $\square$
1 $\square$
3
 $\square$ 0 2 1

The most northwest available cell is now x_{32} , so we set $x_{32} = \min\{3, 0\} = 0$. Then we cross out column 2 and change s_3 to $3 - 0 = 3$. The resulting tableau is Table 19. We now set $x_{33} = \min\{3, 2\} = 2$. Then we cross out column 3 and reduce s_3 to $3 - 2 = 1$. The resulting tableau is Table 20. The only available cell is x_{34} . We set $x_{34} = \min\{1, 1\} = 1$. Then we cross out row 3 and column 4. No cells are available, so we are finished. We have obtained the bfs $x_{11} = 2$, $x_{12} = 3$, $x_{22} = 1$, $x_{32} = 0$, $x_{33} = 2$, $x_{34} = 1$.

Why does the northwest corner method yield a bfs? The method ensures that no basic variable will be assigned a negative value (because no right-hand side ever becomes nega-

TABLE 19

2 3

1

0 3

2 1

TABLE 20

2 3

1

0 2 1

1

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tive) and also that each supply and demand constraint is satisfied (because every row and column is eventually crossed out). Thus, the northwest corner method yields a feasible solution.

To complete the northwest corner method, $m - n$ rows and columns must be crossed out. The last variable assigned a value results in a row and column being crossed out, so the northwest corner method will assign values to $m - n + 1$ variables. The variables chosen by the northwest corner method cannot form a loop, so Theorem 1 implies that the northwest corner method must yield a bfs.

Minimum-Cost Method for Finding a Basic Feasible Solution

The northwest corner method does not utilize shipping costs, so it can yield an initial bfs that has a very high shipping cost. Then determining an optimal solution may require several pivots. The minimum-cost method uses the shipping costs in an effort to produce a bfs that has a lower total cost. Hopefully, fewer pivots will then be required to find the problem's optimal solution.

To begin the minimum-cost method, find the variable with the smallest shipping cost (call it x_{ij}). Then assign x_{ij} its largest possible value, $\min\{s_i, d_j\}$. As in the northwest corner method, cross out row i or column j and reduce the supply or demand of the noncrossed-out row or column by the value of x_{ij} . Then choose from the cells that do not lie in a crossed-out row or column the cell with the minimum shipping cost and repeat the procedure. Continue until there is only one cell that can be chosen. In this case, cross out both the cell's row and column. Remember that (with the exception of the last variable) if a variable satisfies both a supply and demand constraint, only cross out a row or column, not both. To illustrate the minimum cost method, we find a bfs for the balanced transportation problem in Table 21. The variable with the minimum shipping cost is x_{22} . We set $x_{22} = \min\{10, 8\} = 8$. Then we cross out column 2 and reduce s_2 to $10 - 8 = 2$ (Table 22). We could now choose either x_{11} or x_{21} (both having shipping costs of 2). We arbitrarily choose x_{21} and set $x_{21} = \min\{2, 12\} = 2$. Then we cross out row 2 and change d_1 to $12 - 2 = 10$ (Table 23). Now we set $x_{11} = \min\{5, 10\} = 5$, cross out row 1, and change d_1 to $10 - 5 = 5$ (Table 24). The minimum cost that does not lie in a crossed-out row or column is x_{31} . We set $x_{31} =$

$\min\{15, 5\} = 5$, cross out column 1, and reduce s_3 to $15 - 5 = 10$ (Table 25). Now we set $x_{33} = \min\{10, 4\} = 4$, cross out column 3, and reduce s_3 to $10 - 4 = 6$ (Table 26). The only cell that we can choose is x_{34} . We set $x_{34} = \min\{6, 6\}$ and cross out both row 3 and column 4. We have now obtained the bfs: $x_{11} = 5$, $x_{21} = 2$, $x_{22} = 8$, $x_{31} = 5$, $x_{33} = 4$, and $x_{34} = 6$. Because the minimum-cost method chooses variables with small shipping costs to be basic variables, you might think that this method would always yield a bfs with a relatively low total shipping cost. The following problem shows how the minimum-cost method can be fooled into choosing a relatively high-cost bfs.

TABLE 21 2 3 5 6

5

2 1 3 5

10

3 8 4 6

15

12 8 4 6

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TABLE 22 2 3 5 6

5

2 1 3 5

8 2

3 8 4 6

15

12 4 6

TABLE 23

2 3 5 6

5

2 1 3 5

2 8 4

3 8 4 6

15

10 4 6

TABLE 24 2 3 5 6

5

2 1 3 5

2 8 4

3 8 4 6

15

5 4 6

TABLE 25

2 3 5 6

5

2 1 3 5

2 8 □

3 8 4 6

5 10

□ □ 4 6

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TABLE 26 2 3 5 6

5

2 1 3 5

2 8 □

3 8 4 6

5 4 6

□ □ □ 6

TABLE 27 6 7 8

10

15 80 78

15

15 5 5

If we apply the minimum-cost method to Table 27, we set $x_{11} = 10$ and cross out row 1. This forces us to make x_{22} and x_{23} basic variables, thereby incurring their high shipping costs. Thus, the minimum-cost method will yield a costly bfs. Vogel's method for finding a bfs usually avoids extremely high shipping costs.

Vogel's Method for Finding a Basic Feasible Solution

Begin by computing for each row (and column) a "penalty" equal to the difference between the two smallest costs in the row (column). Next find the row or column with the largest penalty. Choose as the first basic variable the variable in this row or column that has the smallest shipping cost. As described in the northwest corner and minimum-cost methods, make this variable as large as possible, cross out a row or column, and change the supply or demand associated with the basic variable. Now recompute new penalties (using only cells that do not lie in a crossed-out row or column), and repeat the procedure until only one uncrossed cell remains. Set this variable equal to the supply or demand associated with the variable, and cross out the variable's row and column. A bfs has now been obtained.

We illustrate Vogel's method by finding a bfs to Table 28. Column 2 has the largest penalty, so we set $x_{12} = \min\{10, 5\} = 5$. Then we cross out column 2 and reduce s_1 to $10 - 5 = 5$. After recomputing the new penalties (observe that after a column is crossed

out, the column penalties will remain unchanged), we obtain Table 29. The largest penalty now occurs in column 3, so we set $x_{13} \leq \min\{5, 5\}$. We may cross out either row 1 or column 3. We arbitrarily choose to cross out column 3, and we reduce s_1 to $5 - 5 = 0$. Because each row has only one cell that is not crossed out, there are no row penalties. The resulting tableau is Table 30. Column 1 has the only (and, of course, the largest) penalty. We set $x_{11} \leq \min\{0, 15\} = 0$, cross out row 1, and change d_1 to $15 - 0 = 15$. The result is Table 31. No penalties can be computed, and the only cell that is not in a crossed-out row or column is x_{21} . Therefore, we set $x_{21} = 15$ and cross out both column 1 and row 2. Our application of Vogel's method is complete, and we have obtained the bfs: $x_{11} = 0$, $x_{12} = 5$, $x_{13} = 5$, and $x_{21} = 15$ (see Table 32).

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TABLE 28 Supply Row Penalty

6 7 8

10 7-6=1

15 80 78

15 78 - 15 = 63

Demand 15 5 5

Column 15 - 6 = 9 80 - 7 = 73 78 - 8 = 70

Penalty

TABLE 29 Supply Row Penalty

6 7 8

5 5 8-6=2

15 80 78

15 78 - 15 = 63

Demand 15 5

Column

Penalty 9 --- 70

TABLE 30 Supply Row Penalty

6 7 8

5 5 0 ---

15 80 78

15 ---

Demand 15 5

Column 9 --- ---

Penalty

TABLE 31 Supply Row Penalty

6 7 8

0 5 5 5 ---

15 80 78

15 ---

Demand 15 □ □

Column

Penalty --- --- ---

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TABLE 32.6.7.8

0 5 5 10

15 80 78

15 15

15 5 5

Observe that Vogel's method avoids the costly shipments associated with x_{22} and x_{23} . This is because the high shipping costs resulted in large penalties that caused Vogel's method to choose other variables to satisfy the second and third demand constraints. Of the three methods we have discussed for finding a bfs, the northwest corner method requires the least effort, and Vogel's method requires the most effort. Extensive research [Glover et al. (1974)] has shown, however, that when Vogel's method is used to find an initial bfs, it usually takes substantially fewer pivots than if the other two methods had been used. For this reason, the northwest corner and minimum-cost methods are rarely used to find a basic feasible solution to a large transportation problem.

PROBLEMS

Group A

1 Use the northwest corner method to find a bfs for 3 Use Vogel's method to find a bfs for Problems 5 and 6

Problems 1, 2, and 3 of Section 7.1. of Section 7.1.

2 Use the minimum-cost method to find a bfs for Problems 4 How should Vogel's method be modified to solve a

4, 7, and 8 of Section 7.1. (Hint: For a maximization maximization problem?

problem, call the minimum-cost method the maximum-profit method or the maximum-revenue method.)

7.3 The Transportation Simplex Method

In this section, we show how the simplex algorithm simplifies when a transportation problem is solved. We begin by discussing the pivoting procedure for a transportation problem. Recall that when the pivot row was used to eliminate the entering basic variable from other constraints and row 0, many multiplications were usually required. In solving a transportation problem, however, pivots require only additions and subtractions.

How to Pivot in a Transportation Problem

By using the following procedure, the pivots for a transportation problem may be performed within the confines of the transportation tableau:

Step 1 Determine (by a criterion to be developed shortly) the variable that should enter

the basis.

Step 2 Find the loop (it can be shown that there is only one loop) involving the entering variable and some of the basic variables.

Step 3 Counting only cells in the loop, label those found in step 2 that are an even num-

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ber (0, 2, 4, and so on) of cells away from the entering variable as even cells. Also label those that are an odd number of cells away from the entering variable as odd cells.

Step 4 Find the odd cell whose variable assumes the smallest value. Call this value .

The variable corresponding to this odd cell will leave the basis. To perform the pivot, decrease the value of each odd cell by and increase the value of each even cell by . The values of variables not in the loop remain unchanged. The pivot is now complete. If $\theta = 0$, then the entering variable will equal 0, and an odd variable that has a current value of 0 will leave the basis. In this case, a degenerate bfs existed before and will result after the pivot. If more than one odd cell in the loop equals θ , you may arbitrarily choose one of these odd cells to leave the basis; again, a degenerate bfs will result.

We illustrate the pivoting procedure on the Powerco example. When the northwest corner method is applied to the Powerco example, the bfs in Table 33 is found. For this bfs, the basic variables are $x_{11} = 35$, $x_{21} = 10$, $x_{22} = 20$, $x_{23} = 20$, $x_{33} = 10$, and $x_{34} = 30$.

Suppose we want to find the bfs that would result if x_{14} were entered into the basis.

The loop involving x_{14} and some of the basic variables is

E O E O E O

(1, 4)–(3, 4)–(3, 3)–(2, 3)–(2, 1)–(1, 1)

In this loop, (1, 4), (3, 3), and (2, 1) are the even cells, and (1, 1), (3, 4), and (2, 3) are the odd cells. The odd cell with the smallest value is $x_{23} = 20$. Thus, after the pivot, x_{23} will have left the basis. We now add 20 to each of the even cells and subtract 20 from each of the odd cells. The bfs in Table 34 results. Because each row and column has as many ≤ 20 s as ≥ 20 s, the new solution will satisfy each supply and demand constraint.

By choosing the smallest odd variable (x_{23}) to leave the basis, we have ensured that all variables will remain nonnegative. Thus, the new solution is feasible. There is no loop involving the cells (1, 1), (1, 4), (2, 1), (2, 2), (3, 3), and (3, 4), so the new solution is a bfs. After the pivot, the new bfs is $x_{11} = 15$, $x_{14} = 20$, $x_{21} = 30$, $x_{22} = 20$, $x_{33} = 30$, and $x_{34} = 10$, and all other variables equal 0.

TABLE 33

Northwest Corner Basic

Feasible Solution for Powerco

10 20 20 50

10 30 40

45 20 30 30

TABLE 34

New Basic Feasible Solution $35 - 20 = 15$ $0 + 20 = 20$

After x14 Is Pivoted into Basis

10 + 20 20 20 - 20 50

(nonbasic)

10 + 20 30 - 20 40

45 20 30 30

7. 3 The Transportation Simplex Method 383

The preceding illustration of the pivoting procedure makes it clear that each pivot in a transportation problem involves only additions and subtractions. Using this fact, we can show that if all the supplies and demands for a transportation problem are integers, then the transportation problem will have an optimal solution in which all the variables are integers. Begin by observing that, by the northwest corner method, we can find a bfs in which each variable is an integer. Each pivot involves only additions and subtractions, so each bfs obtained by performing the simplex algorithm (including the optimal solution) will assign all variables integer values. The fact that a transportation problem with integer supplies and demands has an optimal integer solution is useful, because it ensures that we need not worry about whether the Divisibility Assumption is justified.

Pricing Out Nonbasic Variables (Based on Chapter 6)

To complete our discussion of the transportation simplex, we now show how to compute row 0 for any bfs. From Section 6.2, we know that for a bfs in which the set of basic variables is BV , the coefficient of the variable x_{ij} (call it c_{ij}) in the tableau's row 0 is given by $c_{ij} - c_{BV}B^{-1}a_{ij} = \bar{c}_{ij}$

where c_{ij} is the objective function coefficient for x_{ij} and a_{ij} is the column for x_{ij} in the original LP (we are assuming that the first supply constraint has been dropped).

Because we are solving a minimization problem, the current bfs will be optimal if all the $\bar{c}_{ij}$'s are nonpositive; otherwise, we enter into the basis the variable with the most positive $\bar{c}_{ij}$.

After determining $c_{BV}B^{-1}$, we can easily determine $\bar{c}_{ij}$. Because the first constraint has been dropped, $c_{BV}B^{-1}$ will have $m - n + 1$ elements. We write

$$c_{BV}B^{-1} = [u_2 \ u_3 \ \dots \ u_m \ v_1 \ v_2 \ v_n]$$

where $u_2, u_3, \dots, u_m$ are the elements of $c_{BV}B^{-1}$ corresponding to the $m - 1$ supply constraints, and $v_1, v_2, \dots, v_n$ are the elements of $c_{BV}B^{-1}$ corresponding to the n demand constraints.

To determine $c_{BV}B^{-1}$, we use the fact that in any tableau, each basic variable x_{ij} must have $\bar{c}_{ij} = 0$. Thus, for each of the $m - n + 1$ variables in BV ,

$$c_{BV}B^{-1}a_{ij} = c_{ij} = 0 \quad (4)$$

For a transportation problem, the equations in (4) are very easy to solve. To illustrate the solution of (4), we find $c_{BV}B^{-1}$ for (5), by applying the northwest corner method bfs to the Powerco problem.

8 6 10 9

35

35

9 12 13 7

10 20 20 50 (5)

14 9 16 5

10 30 40

45 20 30 30

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For this bfs, $BV = \{x_{11}, x_{21}, x_{22}, x_{23}, x_{33}, x_{34}\}$. Applying (4) we obtain

0

0

1

 $c_{11} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 8 \quad v_1 = 8 \quad 0$

0

0

0

1

0

1

 $c_{21} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 9 \quad u_2 = v_1 = 9 \quad 0$

0

0

0

1

0

0

 $c_{22} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 12 \quad u_2 = v_2 = 12 \quad 0$

1

0

0

1

0

0

 $c_{23} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 13 \quad u_2 = v_3 = 13 \quad 0$

0

1

0

0

1

0

$$c_{33} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 16 \ u_3 \ v_3 = 16 \ 0$$

0

1

0

0

1

0

$$c_{34} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 5 \ u_3 \ v_4 = 5 \ 0$$

0

0

1

For each basic variable x_{ij} (except those having $i = 1$), we see that (4) reduces to $u_i - v_j = c_{ij}$. If we define $u_1 = 0$, we see that (4) reduces to $u_i - v_j = c_{ij}$ for all basic variables. Thus, to solve for c_{BVB}^{-1} , we must solve the following system of $m - n$ equations: $u_1 = 0$, $u_i - v_j = c_{ij}$ for all basic variables.

For (5), we find c_{BVB}^{-1} by solving

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$$u_1 - u_1 = 0 \quad (6)$$

$$u_1 - v_1 = 8 \quad (7)$$

$$u_2 - v_1 = 9 \quad (8)$$

$$u_2 - v_2 = 12 \quad (9)$$

$$u_2 - v_3 = 13 \quad (10)$$

$$u_3 - v_3 = 16 \quad (11)$$

$$u_3 - v_4 = 5 \quad (12)$$

From (7), $v_1 = 8$. From (8), $u_2 = 1$. Then (9) yields $v_2 = 11$, and (10) yields $v_3 = 12$.

From (11), $u_3 = 4$. Finally, (12) yields $v_4 = 1$. For each nonbasic variable, we now compute $c_{ij} - u_i - v_j = c_{ij}$. We obtain

$$c_{12} - 0 - 11 = 6 \quad 5 \quad c_{13} - 0 - 12 = 10 \quad 2$$

$$c_{14} - 0 - 1 = 9 \quad 8 \quad c_{24} - 1 - 1 = 7 \quad 5$$

$$c_{31} - 4 - 8 = 14 \quad 2 \quad c_{32} - 4 - 11 = 9 \quad 6$$

Because c_{32} is the most positive c_{ij} , we would next enter x_{32} into the basis. Each unit of x_{32} that is entered into the basis will decrease Powerco's cost by \$6.

How to Determine the Entering Nonbasic Variable

(Based on Chapter 5)

For readers who have not covered Chapter 6, we now discuss how to determine whether a bfs is optimal, and, if it is not, how to determine which nonbasic variable should enter the basis. Let u_i ($i = 1, 2, \dots, m$) be the shadow price of the i th supply constraint, and let v_j ($j = 1, 2, \dots, n$) be the shadow price of the j th demand constraint. We assume that the first supply constraint has been dropped, so we may set $u_1 = 0$. From the definition of shadow

price, if we were to increase the right-hand side of the i th supply and j th demand constraint by 1, the optimal z -value would decrease by $-u_i - v_j$. Equivalently, if we were to decrease the right-hand side of the i th supply and j th demand constraint by 1, the optimal z -value would increase by $-u_i - v_j$. Now suppose x_{ij} is a nonbasic variable. Should we enter x_{ij} into the basis? Observe that if we increase x_{ij} by 1, costs directly increase by c_{ij} . Also, increasing x_{ij} by 1 means that one less unit will be shipped from supply point i and one less unit will be shipped to demand point j . This is equivalent to reducing the right-hand sides of the i th supply constraint and j th demand constraint by 1. This will increase z by $-u_i - v_j$. Thus, increasing x_{ij} by 1 will increase z by a total of $c_{ij} - u_i - v_j$. So if $c_{ij} - u_i - v_j \geq 0$ (or $u_i - v_j \leq c_{ij} - 0$) for all nonbasic variables, the current bfs will be optimal. If, however, a nonbasic variable x_{ij} has $c_{ij} - u_i - v_j < 0$ (or $u_i - v_j > c_{ij} - 0$), then z can be decreased by $u_i - v_j - c_{ij}$ per unit of x_{ij} by entering x_{ij} into the basis. Thus, we may conclude that if $u_i - v_j \leq c_{ij} - 0$ for all nonbasic variables, then the current bfs is optimal. Otherwise, the nonbasic variable with the most positive value of $u_i - v_j - c_{ij}$ should enter the basis. How do we find the u_i 's and v_j 's? The coefficient of a nonbasic variable x_{ij} in row 0 of any tableau is the amount by which a unit increase in x_{ij} will decrease z , so we can conclude that the coefficient of any nonbasic variable (and, it turns out, any basic variable) in row 0 is $u_i - v_j - c_{ij}$. So we may solve for the u_i 's and v_j 's by solving the following system of equations: $u_1 = 0$ and $u_i - v_j - c_{ij} = 0$ for all basic variables.

To illustrate the previous discussion, consider the bfs for the Powerco problem shown in (5).

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8 6 10 9

35

35

9 12 13 7 (5)

10 20 20 50

14 9 16 5

10 30 40

45 20 30 30

We

find the u_i 's and v_j 's by solving

$$u_1 - u_1 = 0 \quad (6)$$

$$u_1 - v_1 = 8 \quad (7)$$

$$u_2 - v_1 = 9 \quad (8)$$

$$u_2 - v_2 = 12 \quad (9)$$

$$u_2 - v_3 = 13 \quad (10)$$

$$u_3 - v_3 = 16 \quad (11)$$

$$u_3 - v_4 = 5 \quad (12)$$

From (7), $v_1 = 8$. From (8), $u_2 = 1$. Then (9) yields $v_2 = 11$, and (10) yields $v_3 = 12$.

From (11), $u_3 = 4$. Finally, (12) yields $v_4 = 1$. For each nonbasic variable, we now compute $c_{ij} - u_i - v_j = c_{ij}$. We obtain

$$c_{12} - 0 - 11 = 6 \quad c_{13} - 0 - 12 = 10 \quad 2$$

$$c_{14} - 0 - 1 = 9 \quad c_{24} - 1 - 1 = 7 \quad 5$$

$$c_{31} - 4 - 8 = 14 \quad c_{32} - 4 - 11 = 9 \quad 6$$

Because c_{32} is the most positive c_{ij} , we would next enter x_{32} into the basis. Each unit of x_{32} that is entered into the basis will decrease Powerco's cost by \$6.

We can now summarize the procedure for using the transportation simplex to solve a transportation (min) problem.

Summary and Illustration

of the Transportation Simplex Method

Step 1 If the problem is unbalanced, balance it.

Step 2 Use one of the methods described in Section 7.2 to find a bfs.

Step 3 Use the fact that $u_1 = 0$ and $u_i - v_j = c_{ij}$ for all basic variables to find the $[u_1 \ u_2 \ \dots \ u_m \ v_1 \ v_2 \ \dots \ v_n]$ for the current bfs.

Step 4 If $u_i - v_j = c_{ij} \leq 0$ for all nonbasic variables, then the current bfs is optimal. If this is not the case, then we enter the variable with the most positive $u_i - v_j = c_{ij}$ into the basis using the pivoting procedure. This yields a new bfs.

Step 5 Using the new bfs, return to steps 3 and 4.

For a maximization problem, proceed as stated, but replace step 4 by step 4'.

Step 4' If $u_i - v_j = c_{ij} \geq 0$ for all nonbasic variables, then the current bfs is optimal.

Otherwise, enter the variable with the most negative $u_i - v_j = c_{ij}$ into the basis using the pivoting procedure described earlier.

7.3 The Transportation Simplex Method 387

We illustrate the procedure for solving a transportation problem by solving the Powerco problem. We begin with the bfs (5). We have already determined that x_{32} should enter the basis. As shown in Table 35, the loop involving x_{32} and some of the basic variables is (3, 2)–(3, 3)–(2, 3)–(2, 2). The odd cells in this loop are (3, 3) and (2, 2). Because $x_{33} = 10$ and $x_{22} = 20$, the pivot will decrease the value of x_{33} and x_{22} by 10 and increase the value of x_{32} and x_{23} by 10. The resulting bfs is shown in Table 36. The u_i 's and v_j 's for the new bfs were obtained by solving

$$u_2 - u_1 = 0 \quad u_2 - v_3 = 13$$

$$u_2 - v_2 = 12 \quad u_2 - v_1 = 9$$

$$u_3 - v_4 = 5 \quad u_3 - v_2 = 9$$

$$u_1 - v_1 = 82 \quad u_2 - v_1 = 9$$

In computing $c_{ij} - u_i - v_j = c_{ij}$ for each nonbasic variable, we find that $c_{12} = 5$, $c_{24} = 1$, and $c_{13} = 2$ are the only positive c_{ij} 's. Thus, we next enter x_{12} into the basis. The loop involving x_{12} and some of the basic variables is (1, 2)–(2, 2)–(2, 1)–(1, 1). The odd cells are (2, 2) and (1, 1). Because $x_{22} = 10$ is the smallest entry in an odd cell, we decrease x_{22} and x_{11} by 10 and increase x_{12} and x_{21} by 10. The resulting bfs is shown in Table 37. For this bfs, the u_i 's and v_j 's were determined by solving

$$u_1 - u_1 = 0 \quad u_1 - v_2 = 6$$

$$u_2 - v_1 = 9 \quad u_3 - v_2 = 9$$

$$u_1 - v_1 = 8 \quad u_3 - v_4 = 5$$

$$u_2 - v_3 = 13 \quad u_3 - v_4 = 5$$

In computing c_{ij} for each nonbasic variable, we find that the only positive c_{ij} is $c_{13} = 2$. Thus, x_{13} enters the basis. The loop involving x_{13} and some of the basic variables is

TABLE 35 8 6 10 9

Loop Involving 35

Entering Variable x_{32} 35

$$9 \ 12 \ 13 \ 7$$

$$10 \ 20 \ 20 \ 50$$

$$14 \ 9 \ 16 \ 5$$

$$10 \ 30 \ 40$$

$$45 \ 20 \ 30 \ 30$$

TABLE 36 $v_j - j$

$$= 8 \ 11$$

$$11 \ 12$$

$$12 \ 77$$

x_{32} Has Entered the Basis,

$$8 \ 6 \ 10 \ 9$$

and x_{12} Enters Next

$$u_{ii} = 00 \ 35$$

$$35$$

$$9 \ 12 \ 13 \ 7$$

$$11 \ 10 \ 10 \ 30 \ 50$$

$$14 \ 9 \ 16 \ 5$$

$$-2$$

$$2 \ 10 \ 30 \ 40$$

$$45 \ 20 \ 30 \ 30$$

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TABLE 37 $v_j - j$

$$= 88 \ 66 \ 12$$

$$12 \ 22$$

x_{12} Has Entered the Basis,

$$8 \ 6 \ 10 \ 9$$

and x_{13} Enters Next

$$u_{ii} = 00 \ 25 \ 10$$

$$35$$

$$9 \ 12 \ 13 \ 7$$

11 20 30 50

14 9 16 5

33 10 30 40

45 20 30 30

TABLE 38

$u_i = 66$ 66 10

10 22

Optimal Tableau for Powerco

8 6 10 9

$u_i = 00$ 10 25

35

9 12 13 7

33 45 5 50

14 9 16 5

33 10 30 40

45 20 30 30

(1, 3)-(2, 3)-(2, 1)-(1, 1). The odd cells are x_{23} and x_{11} . Because $x_{11} = 25$ is the smallest entry in an odd cell, we decrease x_{23} and x_{11} by 25 and increase x_{13} and x_{21} by 25. The resulting bfs is shown in Table 38. For this bfs, the u_i 's and v_j 's were obtained by solving

$$u_2 - u_1 = 0 \quad u_2 - v_3 = 13$$

$$u_2 - v_1 = 9 \quad u_1 - v_3 = 10$$

$$u_3 - v_4 = 5 \quad u_3 - v_2 = 9$$

$$u_1 - v_2 = 6 \quad u_3 - v_2 = 9$$

The reader should check that for this bfs, all $c_{ij} \geq 0$, so an optimal solution has been obtained. Thus, the optimal solution to the Powerco problem is $x_{12} = 10$, $x_{13} = 25$, $x_{21} = 45$, $x_{23} = 5$, $x_{32} = 10$, $x_{34} = 30$, and

$$z = 6(10) + 10(25) + 9(45) + 13(5) + 9(10) + 5(30) = \$1,020$$

PROBLEMS

Group A

Use the transportation simplex to solve Problems 1-8 in Section 7.1. Begin with the bfs found in Section 7.2.

7.3 The Transportation Simplex Method 389

7.4 Sensitivity Analysis for Transportation Problems†

We have already seen that for a transportation problem, the determination of a bfs and of row 0 for a given set of basic variables, as well as the pivoting procedure, all simplify. It should therefore be no surprise that certain aspects of the sensitivity analysis discussed in Section 6.3 can be simplified. In this section, we discuss the following three aspects of sensitivity analysis for the transportation problem:

Change 1 Changing the objective function coefficient of a nonbasic variable.

Change 2 Changing the objective function coefficient of a basic variable.

Change 3 Increasing a single supply by Δ and a single demand by Δ .

We illustrate three changes using the Powerco problem. Recall from Section 7.3 that the optimal solution for the Powerco problem was $z = \$1,020$; the optimal tableau is Table 39.

Changing the Objective Function Coefficient of a Nonbasic Variable

As in Section 6.3, changing the objective function coefficient of a nonbasic variable x_{ij} will leave the right-hand side of the optimal tableau unchanged. Thus, the current basis will still be feasible. We are not changing c_{BV}^1 , so the u_i 's and v_j 's remain unchanged. In row 0, only the coefficient of x_{ij} will change. Thus, as long as the coefficient of x_{ij} in the optimal row 0 is nonpositive, the current basis remains optimal.

To illustrate the method, we answer the following question: For what range of values of the cost of shipping 1 million kwh of electricity from plant 1 to city 1 will the current basis remain optimal? Suppose we change c_{11} from 8 to $8 + \Delta$. For what values of Δ will the current basis remain optimal? Now $\Delta c_{11} = u_1 - v_1 - c_{11} = 0 - 6 - (8 + \Delta) = -2 - \Delta$. Thus, the current basis remains optimal for $-2 - \Delta \leq 0$, or $\Delta \geq -2$, and $c_{11} = 8 - 2 \leq 6$.

Changing the Objective Function Coefficient of a Basic Variable

Because we are changing c_{BV}^1 , the coefficient of each nonbasic variable in row 0 may change, and to determine whether the current basis remains optimal, we must find the new u_i 's and v_j 's and use these values to price out all nonbasic variables. The current basis remains optimal as long as all nonbasic variables price out nonpositive. To illustrate the idea, we determine for the Powerco problem the range of values of the cost of shipping 1 million kwh from plant 1 to city 3 for which the current basis remains optimal.

Suppose we change c_{13} from 10 to $10 + \Delta$. Then the equation $\Delta c_{13} = 0$ changes from $u_1 - v_3 = 10$ to $u_1 - v_3 = 10 + \Delta$. Thus, to find the u_i 's and v_j 's, we must solve the following equations:

$$u_2 - u_1 = 0 \quad u_3 - v_2 = 90 \quad \Delta$$

$$u_2 - v_1 = 9 \quad u_1 - v_3 = 10 + \Delta$$

$$u_1 - v_2 = 6 \quad u_3 - v_4 = 50 \quad \Delta$$

$$u_2 - v_3 = 13 \quad u_1 - v_3 = 10 + \Delta$$

This section covers topics that may be omitted with no loss of continuity.

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TABLE 39 City 1 City 2 City 3 City 4 Supply

Optimal Tableau for Powerco

$$v_j = 6 \quad 6 \quad 10 \quad 2$$

$$8 \quad 6 \quad 10 \quad 9$$

$$\text{Plant 1 } u_i = 0 \quad 10 \quad 25$$

$$35$$

$$9 \quad 12 \quad 13 \quad 7$$

$$\text{Plant 2 } 3 \quad 45 \quad 5 \quad 50$$

14 9 16 5

Plant 3 3 10 30 40

Demand 45 20 30 30

Solving these equations, we obtain $u_1 = 0$, $v_2 = 6$, $v_3 = 10$, $v_1 = 6$, $u_2 = 3$, $u_3 = 3$, and $v_4 = 2$.

We now price out each nonbasic variable. The current basis will remain optimal as long as each nonbasic variable has a nonpositive coefficient in row 0.

$c_{11} - u_1 - v_1 = 8 - 0 - 6 = 2 > 0$ for x_{11}

$c_{14} - u_1 - v_4 = 9 - 0 - 2 = 7$

$c_{22} - u_2 - v_2 = 12 - 3 - 6 = 3 > 0$ for x_{22}

$c_{24} - u_2 - v_4 = 7 - 3 - 2 = 2 > 0$ for x_{24}

$c_{31} - u_3 - v_1 = 14 - 3 - 6 = 5 > 0$ for x_{31}

$c_{33} - u_3 - v_3 = 16 - 3 - 10 = 3 > 0$ for x_{33}

Thus, the current basis remains optimal for $\Delta_2 \leq 2$, or $8 - 10 \leq 2 \leq c_{13} - 10 \leq 2 \leq 12$.

Increasing Both Supply s_i and Demand d_j by Δ

Observe that this change maintains a balanced transportation problem. Because the u_i 's and v_j 's may be thought of as the negative of each constraint's shadow prices, we know from (37) of Chapter 6 that if the current basis remains optimal,

New z-value = old z-value + $\Delta u_i + \Delta v_j$

For example, if we increase plant 1's supply and city 2's demand by 1 unit, then (new cost) = $1,020 + 1(0) + 1(6) = \$1,026$.

We may also find the new values of the decision variables as follows:

1 If x_{ij} is a basic variable in the optimal solution, then increase x_{ij} by Δ .

2 If x_{ij} is a nonbasic variable in the optimal solution, then find the loop involving x_{ij} and some of the basic variables. Find an odd cell in the loop that is in row i . Increase the value of this odd cell by Δ and go around the loop, alternately increasing and then decreasing current basic variables in the loop by Δ .

To illustrate the first situation, suppose we increase s_1 and d_2 by 2. Because x_{12} is a basic variable in the optimal solution, the new optimal solution will be the one shown in Table 40. The new optimal z-value is $1,020 + 2u_1 + 2v_2 = \$1,032$. To illustrate the second situation, suppose we increase both s_1 and d_1 by 1. Because x_{11} is a nonbasic variable in the current optimal solution, we must find the loop involving x_{11} and some of the

7. 4 Sensitivity Analysis for Transportation Problems 391

TA B L E 40 City 1 City 2 City 3 City 4 Supply

Optimal Tableau for Powerco If

$v_j = 6 \ 6 \ 10 \ 2$

$s_1 = 35 - 2 = 37$ and

$d_2 = 20 + 2 = 22 \ 8 \ 6 \ 10 \ 9$

Plant 1 = 00

uui = 12 25

37

9 12 13 7

Plant 2 33 45 5 50

14 9 16 5

Plant 3 3 10 30 40

Demand 45 22 30 30

TA B L E 41 City 1 City 2 City 3 City 4 Supply

Optimal Tableau for Powerco If

$v_j \leq 6 \ 6 \ 10 \ 2$

$s_1 \leq 35 \ 1 \leq 36$ and

$d_1 \leq 45 \ 1 \leq 46 \ 8 \ 6 \ 10 \ 9$

Plant 1 uuii = 00 10 26

36

9 12 13 7

Plant 2 33 46 4 50

14 9 16 5

Plant 3 33 10 30 40

Demand 46 20 30 30

basic variables. The loop is (1, 1)–(1, 3)–(2, 3)–(2, 1). The odd cell in the loop and row 1 is x_{13} . Thus, the new optimal solution will be obtained by increasing both x_{13} and x_{21} by 1 and decreasing x_{23} by 1. This yields the optimal solution shown in Table 41. The new optimal z-value is found from (new z-value) $\leq 1,020 \leq v_1 \leq v_1 \leq \$1,026$. Observe that if both s_1 and d_1 were increased by 6, the current basis would be infeasible. (Why?)

PROBLEMS

Group A

The following problems refer to the Powerco example. a The company's goal is to minimize the cost of meet-

ing customers' demands. Find two optimal bfs for this

1 Determine the range of values of c_{14} for which the transportation problem.

current basis remains optimal.

b Suppose that customer 2's demand increased by one

2 Determine the range of values of c_{34} for which the unit. By how much would costs increase? current basis remains optimal.

3 If s_2 and d_3 are both increased by 3, what is the new TA B L E 42 optimal solution?

To

4 If s_3 and d_3 are both decreased by 2, what is the new optimal solution? From Customer 1 Customer 2 Customer 3 Supply

5 Two plants supply three customers with medical Plant 1 \$55 \$65 \$80 35 supplies. The unit costs of shipping from the plants to the Plant 2 \$10 \$15 \$25 50 customers, along with the supplies and demands, are given Demand 10 10 10 in Table 42.

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7.5 Assignment Problems

Although the transportation simplex appears to be very efficient, there is a certain class of transportation problems, called assignment problems, for which the transportation simplex is often very inefficient. In this section, we define assignment problems and discuss an efficient method that can be used to solve them.

EXAMPLE 4 Machine Assignment Problem

Machineco has four machines and four jobs to be completed. Each machine must be assigned to complete one job. The time required to set up each machine for completing each job is shown in Table 43. Machineco wants to minimize the total setup time needed to complete the four jobs. Use linear programming to solve this problem.

Solution Machineco must determine which machine should be assigned to each job. We define (for $i, j \in 1, 2, 3, 4$)

$x_{ij} \in 1$ if machine i is assigned to meet the demands of job j

$x_{ij} \in 0$ if machine i is not assigned to meet the demands of job j

Then Machineco's problem may be formulated as

$$\min z = 14x_{11} + 5x_{12} + 8x_{13} + 7x_{14} + 2x_{21} + 12x_{22} + 6x_{23} + 5x_{24}$$

$$\min z = 7x_{31} + 8x_{32} + 3x_{33} + 9x_{34} + 2x_{41} + 4x_{42} + 6x_{43} + 10x_{44}$$

$$\text{s.t. } x_{11} + x_{12} + x_{13} + x_{14} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{21} + x_{22} + x_{23} + x_{24} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{31} + x_{32} + x_{33} + x_{34} = 1 \text{ (Machine constraints)}$$

(13)

$$\text{s.t. } x_{41} + x_{42} + x_{43} + x_{44} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{11} + x_{21} + x_{31} + x_{41} = 1 \text{ (Job constraints)}$$

$$\text{s.t. } x_{12} + x_{22} + x_{32} + x_{42} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{13} + x_{23} + x_{33} + x_{43} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{14} + x_{24} + x_{34} + x_{44} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{ij} \in 0 \text{ or } x_{ij} \in 1 \text{ (Machine constraints)}$$

The first four constraints in (13) ensure that each machine is assigned to a job, and the last four ensure that each job is completed. If $x_{ij} \in 1$, then the objective function will pick up the time required to set up machine i for job j ; if $x_{ij} \in 0$, then the objective function will not pick up the time required.

Ignoring for the moment the $x_{ij} \in 0$ or $x_{ij} \in 1$ restrictions, we see that Machineco faces a balanced transportation problem in which each supply point has a supply of 1 and each

TABLE 43

Setup Times for Machineco

Time (Hours)

Machine Job 1 Job 2 Job 3 Job 4

1 14 5 8 7

2 2 12 6 5

3 7 8 3 9

4 2 4 6 10

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demand point has a demand of 1. In general, an assignment problem is a balanced transportation problem in which all supplies and demands are equal to 1. Thus, an assignment problem is characterized by knowledge of the cost of assigning each supply point to each demand point. The assignment problem's matrix of costs is its cost matrix.

All the supplies and demands for the Machineco problem (and for any assignment problem) are integers, so our discussion in Section 7.3 implies that all variables in Machineco's optimal solution must be integers. Because the right-hand side of each constraint is equal to 1, each x_{ij} must be a nonnegative integer that is no larger than 1, so each x_{ij} must equal 0 or 1. This means that we can ignore the restrictions that $x_{ij} \geq 0$ or 1 and solve (13) as a balanced transportation problem. By the minimum cost method, we obtain the bfs in Table 44. The current bfs is highly degenerate. (In any bfs to an $m \times m$ assignment problem, there will always be m basic variables that equal 1 and $m - 1$ basic variables that equal 0.)

We find that $c_{43} = 1$ is the only positive c_{ij} . We therefore enter x_{43} into the basis. The loop involving x_{43} and some of the basic variables is $(4, 3)-(1, 3)-(1, 2)-(4, 2)$. The odd variables in the loop are x_{13} and x_{42} . Because $x_{13} = x_{42} = 0$, either x_{13} or x_{42} will leave

TABLE 44 Job 1 Job 2 Job 3 Job 4

Basic Feasible Solution

$v_j = 3 \ 4 \ 8 \ 7$

for Machineco

14 5 8 7

Machine 1 $u_i = 0 \ 1 \ 0 \ 0$

1

2 12 6 5

Machine 2 $= 2 \ 1 \ 1$

7 8 3 9

Machine 3 $= 5 \ 1 \ 1$

2 4 6 10

Machine 4 $= 1 \ 1 \ 0 \ 1$

1 1 1 1

TABLE 45 Job 1 Job 2 Job 3 Job 4

x_{43} Has Entered the Basis

$v_j = 3 \ 5 \ 7 \ 7$

14 5 8 7

Machine 1 ui $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$

1

2 12 6 5

Machine 2 $\begin{bmatrix} 2 & 1 & 1 \end{bmatrix}$

7 8 3 9

Machine 3 $\begin{bmatrix} 4 & 1 & 1 \end{bmatrix}$

2 4 6 10

Machine 4 $\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \end{bmatrix}$

1 1 1 1

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the basis. We arbitrarily choose x_{13} to leave the basis. After performing the pivot, we obtain the bfs in Table 45. All $\bar{c}_{ij}$'s are now nonpositive, so we have obtained an optimal assignment: $x_{12} = 1$, $x_{24} = 1$, $x_{33} = 1$, and $x_{41} = 1$. Thus, machine 1 is assigned to job 2, machine 2 is assigned to job 4, machine 3 is assigned to job 3, and machine 4 is assigned to job 1. A total setup time of $5 + 5 + 3 + 2 = 15$ hours is required.

The Hungarian Method

Looking back at our initial bfs, we see that it was an optimal solution. We did not know that it was optimal, however, until performing one iteration of the transportation simplex. This suggests that the high degree of degeneracy in an assignment problem may cause the transportation simplex to be an inefficient way of solving assignment problems. For this reason (and the fact that the algorithm is even simpler than the transportation simplex), the Hungarian method is usually used to solve assignment (min) problems:

Step 1 Find the minimum element in each row of the $m \times m$ cost matrix. Construct a new matrix by subtracting from each cost the minimum cost in its row. For this new matrix, find the minimum cost in each column. Construct a new matrix (called the reduced cost matrix) by subtracting from each cost the minimum cost in its column.

Step 2 Draw the minimum number of lines (horizontal, vertical, or both) that are needed to cover all the zeros in the reduced cost matrix. If m lines are required, then an optimal solution is available among the covered zeros in the matrix. If fewer than m lines are needed, then proceed to step 3.

Step 3 Find the smallest nonzero element (call its value k) in the reduced cost matrix that is uncovered by the lines drawn in step 2. Now subtract k from each uncovered element of the reduced cost matrix and add k to each element that is covered by two lines. Return to step 2.

REMARKS 1 To solve an assignment problem in which the goal is to maximize the objective function, mul-

tiply the profits matrix through by -1 and solve the problem as a minimization problem.

2 If the number of rows and columns in the cost matrix are unequal, then the assignment problem is unbalanced. The Hungarian method may yield an incorrect solution if the problem is unbalanced.

Thus, any assignment problem should be balanced (by the addition of one or more dummy points) before it is solved by the Hungarian method.

3 In a large problem, it may not be easy to find the minimum number of lines needed to cover all zeros in the current cost matrix. For a discussion of how to find the minimum number of lines needed, see Gillett (1976). It can be shown that if j lines are required, then only j “jobs” can be assigned to zero costs in the current matrix. This explains why the algorithm terminates when m lines are required.

Solution of Machineco Example by the Hungarian Method

We illustrate the Hungarian method by solving the Machineco problem (see Table 46).

Step 1 For each row, we subtract the row minimum from each element in the row, obtaining Table 47. We now subtract 2 from each cost in column 4, obtaining Table 48.

Step 2 As shown, lines through row 1, row 3, and column 1 cover all the zeros in the reduced cost matrix. From remark 3, it follows that only three jobs can be assigned to zero costs in the current cost matrix. Fewer than four lines are required to cover all the zeros, so we proceed to step 3.

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TA B L E 46 Row Minimum

Cost Matrix for Machineco 14 5 8 7 5

2 12 6 5 2

7 8 3 9 3

2 4 6 10 2

TA B L E 47

Cost Matrix After Row 9 0 3 2

Minimums Are Subtracted

0 10 4 3

4 5 0 6

0 2 4 8

0

Column Minimum 0 0 2

TA B L E 48

Cost Matrix After Column 9 0 3 0

Minimums Are Subtracted

0 10 4 1

4 5 0 4

0 2 4 6

Step 3 The smallest uncovered element equals 1, so we now subtract 1 from each uncovered element in the reduced cost matrix and add 1 to each twice-covered element. The resulting matrix is Table 49. Four lines are now required to cover all the zeros. Thus, an op-

timal solution is available. To find an optimal assignment, observe that the only covered 0 in column 3 is x_{33} , so we must have $x_{33} = 1$. Also, the only available covered zero in column 2 is x_{12} , so we set $x_{12} = 1$ and observe that neither row 1 nor column 2 can be used again. Now the only available covered zero in column 4 is x_{24} . Thus, we choose $x_{24} = 1$ (which now excludes both row 2 and column 4 from further use). Finally, we choose $x_{41} = 1$.

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TABLE 49

Four Lines Required; Optimal 10 0 3 0

Solution Is Available

0 9 3 0

5 5 0 4

0 1 3 5

Thus, we have found the optimal assignment $x_{12} = 1$, $x_{24} = 1$, $x_{33} = 1$, and $x_{41} = 1$.

Of course, this agrees with the result obtained by the transportation simplex.

Intuitive Justification of the Hungarian Method

To give an intuitive explanation of why the Hungarian algorithm works, we need to discuss the following result: If a constant is added to each cost in a row (or column) of a balanced transportation problem, then the optimal solution to the problem is unchanged.

To show why the result is true, suppose we add k to each cost in the first row of the Machineco problem. Then

New objective function $\square$ old objective function $\square k(x_{11} \square x_{12} \square x_{13} \square x_{14})$

Because any feasible solution to the Machineco problem must have $x_{11} \square x_{12} \square x_{13} \square x_{14} = 1$,

New objective function $\square$ old objective function $\square k$

Thus, the optimal solution to the Machineco problem remains unchanged if a constant k is added to each cost in the first row. A similar argument applies to any other row or column.

Step 1 of the Hungarian method consists (for each row and column) of subtracting a constant from each element in the row or column. Thus, step 1 creates a new cost matrix having the same optimal solution as the original problem. Step 3 of the Hungarian method is equivalent (see Problem 7 at the end of this section) to adding k to each cost that lies in a covered row and subtracting k from each cost that lies in an uncovered column (or vice versa). Thus, step 3 creates a new cost matrix with the same optimal solution as the initial assignment problem. Each time step 3 is performed, at least one new zero is created in the cost matrix.

Steps 1 and 3 also ensure that all costs remain nonnegative. Thus, the net effect of steps 1 and 3 of the Hungarian method is to create a sequence of assignment problems (with nonnegative costs) that all have the same optimal solution as the original assignment problem. Now consider an assignment problem in which all costs are nonnegative. Any feasible assignment in which all the x_{ij} 's that equal 1 have zero costs must be optimal for such an assignment problem. Thus, when step 2 indicates that m lines are required to cover all

the zeros in the cost matrix, an optimal solution to the original problem has been found.

Computer Solution of Assignment Problems

To solve assignment problems in LINDO, type in the objective function and constraints. Also, many menu-driven programs require the user to input only a list of supply and de-

7. 5 Assignment Problems 397

mand points (such as jobs and machines, respectively) and a cost matrix. LINGO can also be used to easily solve assignment problems, including the following model to solve the Assign.Ing Machineco example (file Assign.Ing).

MODEL:

1]SETS:

2]MACHINES/1..4/;

3]JOBS/1..4/;

4]LINKS(MACHINES,JOBS):COST,ASSIGN;

5]ENDSETS

6]MIN=@SUM(LINKS:COST*ASSIGN);

7]@FOR(MACHINES(I):

8]@SUM(JOBS(J):ASSIGN(I,J))<1);

9]@FOR(JOBS(J):

10]@SUM(MACHINES(I):ASSIGN(I,J))>1);

11]DATA:

12]COST = 14,5,8,7,

13]2,12,6,5,

14]7,8,3,9,

15]2,4,6,10;

16]ENDDATA

END

Line 2 defines the four supply points (machines), and line 3 defines the four demand points (jobs). In line 4, we define each possible combination of jobs and machines (16 in all) and associate with each combination an assignment cost [for example COST(1, 2) = 5] and a variable ASSIGN(I,J). ASSIGN(I,J) equals 1 if machine i is used to perform job j; it equals 0 otherwise. Line 5 ends the definition of sets.

Line 6 expresses the objective function by summing over all possible (I,J) combinations the product of the assignment cost and ASSIGN(I,J). Lines 7-8 limit each MACHINE to performing at most one job by forcing (for each machine) the sum of ASSIGN(I,J) over all JOBS to be at most 1. Lines 9-10 require that each JOB be completed by forcing (for each job) the sum of ASSIGN(I,J) over all MACHINES to be at least 1.

Lines 12-16 input the cost matrix.

Observe that this LINGO program can (with simple editing) be used to solve any assignment problem (even if it is not balanced!). For example, if you had 10 machines available to perform 8 jobs, you would edit line 2 to indicate that there are 10 machines (re-

place 1..4 with 1..10). Then edit line 3 to indicate that there are 8 jobs. Finally, in line 12, you would type the 80 entries of your cost matrix, following “COST” and you would be ready to roll!

REMARK 1 From our discussion of the Machineco example, it is unnecessary to force the ASSIGN(I,J) to equal 0 or 1; this will happen automatically!

PROBLEMS

Group A

1 Five employees are available to perform four jobs. The TABLE 50 time it takes each person to perform each job is given in Table 50. Determine the assignment of employees to jobs Time (hours) that minimizes the total time required to perform the four Person Job 1 Job 2 Job 3 Job 4 jobs.

1 22 18 30 18

2† Doc Councillman is putting together a relay team for 2 18 — 27 22 the 400-meter relay. Each swimmer must swim 100 meters

3 26 20 28 28

of breaststroke, backstroke, butterfly, or freestyle. Doc believes that each swimmer will attain the times given in 4 16 22 — 14 5 21 — 25 28

This problem is based on Machol (1970). Note: Dashes indicate person cannot do that particular job.

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Table 51. To minimize the team’s time for the race, which did not bid on the given job). Person 1 can do only one job,

swimmer should swim which stroke? but persons 2 and 3 can each do as many as two jobs.

3 Tom Cruise, Freddy Prinze Jr., Harrison Ford, and Matt Determine the minimum cost assignment of persons to jobs.

LeBlanc are marooned on a desert island with Jennifer 5 Greydog Bus Company operates buses between Boston

Aniston, Courteney Cox, Gwyneth Paltrow, and Julia and Washington, D.C. A bus trip between these two cities

Roberts. The “compatibility measures” in Table 52 indicate takes 6 hours. Federal law requires that a driver rest for four

how much happiness each couple would experience if they or more hours between trips. A driver’s workday consists of

spent all their time together. The happiness earned by a two trips: one from Boston to Washington and one from

couple is proportional to the fraction of time they spend Washington to Boston. Table 54 gives the departure times

together. For example, if Freddie and Gwyneth spend half for the buses. Greydog’s goal is to minimize the total

their time together, they earn happiness of $\frac{1}{12}(9) = 4.5$. downtime for all drivers. How should Greydog assign crews

a Let x_{ij} be the fraction of time that the i th man spends to trips? Note: It is permissible for a driver's "day" to overlap

with the j th woman. The goal of the eight people is to midnight. For example, a Washington-based driver can be

maximize the total happiness of the people on the is- assigned to the Washington-Boston 3 P.M. trip and the

land. Formulate an LP whose optimal solution will yield Boston-Washington 6 A.M. trip.

the optimal values of the x_{ij} 's. 6 Five male characters (Billie, John, Fish, Glen, and Larry)

b Explain why the optimal solution in part (a) will and five female characters (Ally, Georgia, Jane, Rene, and

have four $x_{ij} = 1$ and twelve $x_{ij} = 0$. The optimal solu- Nell) from Ally McBeal are marooned on a desert island.

tion requires that each person spend all his or her time The problem is to determine what percentage of time each

with one person of the opposite sex, so this result is of- woman on the island should spend with each man. For

ten referred to as the Marriage Theorem. example, Ally could spend 100% of her time with John or

c Determine the marriage partner for each person. she could "play the field" by spending 20% of her time with

d Do you think the Proportionality Assumption of lin- each man. Table 55 shows a "happiness index" for each

ear programming is valid in this situation? potential pairing of a man and woman. For example, if Larry

and Rene spend all their time together, they earn 8 units of

4 A company is taking bids on four construction jobs. Three happiness for the island.

people have placed bids on the jobs. Their bids (in thousands

a Play matchmaker and determine an allocation of

of dollars) are given in Table 53 (a * indicates that the person

each man and woman's time that earns the maximum to-

tal happiness for the island. Assume that happiness

earned by a couple is proportional to the amount of time

TA B L E 51 they spend together.

Time (seconds) b Explain why the optimal solution to this problem

Swimmer Free Breast Fly Back will, for any matrix of "happiness indices," always involve each woman spending all her time with one man.

Gary Hall 54 54 51 53

Mark Spitz 51 57 52 52

Jim Montgomery 50 53 54 56

Chet Jastremski 56 54 55 53

TA B L E 54

Departure Departure

Trip Time Trip Time

TA B L E 52

Boston 1 6 A.M. Washington 1 5:30 A.M.

JA CC GP JR

Boston 2 7:30 A.M. Washington 2 9 A.M.

TC 7 5 8 2 Boston 3 11.30 A.M. Washington 3 3 P.M.

FP 7 8 9 4 Boston 4 7 P.M. Washington 4 6:30 P.M.

HF 3 5 7 9 Boston 5 12:30 A.M. Washington 5 12 midnight

ML 5 5 6 7

TA B L E 55

TA B L E 53

Ally Georgia Jane Rene Nell

Job

Billie 8 6 4 7 5

Person 1 2 3 4

John 5 7 6 4 9

1 50 46 42 40 Fish 10 6 5 2 10

2 51 48 44 * Glen 1 0 0 0 0

3 * 47 45 45 Larry 5 7 9 8 6

7. 5 Assignment Problems 399

c What assumption made in the problem is needed for TA B L E 57
the Marriage Theorem to hold?

Bids

Route Route Route Route

Group B Company 1 2 3 4

7 Any transportation problem can be formulated as an

1 \$4,000 \$5,000 — —

assignment problem. To illustrate the idea, determine an

assignment problem that could be used to find the optimal 2 — \$4,000 — \$4,000

solution to the transportation problem in Table 56. (Hint: 3 \$3,000 — \$2,000 —

You will need five supply and five demand points). 4 — — \$4,000 \$5,000

8 The Chicago board of education is taking bids on the

city's four school bus routes. Four companies have made the

bids in Table 57. b Suppose that each company can be assigned two

a Suppose each bidder can be assigned only one route. routes. Use the assignment method to
minimize

Use the assignment method to minimize Chicago's cost Chicago's cost of running the four bus
routes. (Hint:

of running the four bus routes. Two supply points will be needed for each company.)

9 Show that step 3 of the Hungarian method is equivalent

to performing the following operations: (1) Add k to each

TABLE 56 cost that lies in a covered row. (2) Subtract k from each cost

3 1 that lies in an uncovered column.

2 10 Suppose c_{ij} is the smallest cost in row i and column j of an assignment problem. Must $x_{ij} = 1$ in any optimal

2 3 assignment?

3

1 4

7.6 Transshipment Problems

A transportation problem allows only shipments that go directly from a supply point to a demand point. In many situations, shipments are allowed between supply points or between demand points. Sometimes there may also be points (called transshipment points) through which goods can be transshipped on their journey from a supply point to a demand point. Shipping problems with any or all of these characteristics are transshipment problems. Fortunately, the optimal solution to a transshipment problem can be found by solving a transportation problem.

In what follows, we define a supply point to be a point that can send goods to another point but cannot receive goods from any other point. Similarly, a demand point is a point that can receive goods from other points but cannot send goods to any other point. A transshipment point is a point that can both receive goods from other points and send goods to other points. The following example illustrates these definitions (“—” indicates that a shipment is impossible).

EXAMPLE 5 Transshipment

Widgetco manufactures widgets at two factories, one in Memphis and one in Denver. The Memphis factory can produce as many as 150 widgets per day, and the Denver factory can produce as many as 200 widgets per day. Widgets are shipped by air to customers in Los Angeles and Boston. The customers in each city require 130 widgets per day. Because of the deregulation of airfares, Widgetco believes that it may be cheaper to first fly some widgets to New York or Chicago and then fly them to their final destinations. The costs of flying a widget are shown in Table 58. Widgetco wants to minimize the total cost of shipping the required widgets to its customers.

400 CHAPTER 7 Transportation, Assignment, and Transshipment Problems

TABLE 58

Shipping Costs for Transshipments

To (\$)

From Memphis Denver N.Y. Chicago L.A. Boston

Memphis 0 — 8 13 25 28

Denver — 0 15 12 26 25

N.Y. — — 0 6 16 17

Chicago — — 6 0 14 16

L.A. — — — — 0 —

Boston — — — — 0

In this problem, Memphis and Denver are supply points, with supplies of 150 and 200 widgets per day, respectively. New York and Chicago are transshipment points. Los Angeles and Boston are demand points, each with a demand of 130 widgets per day. A graphical representation of possible shipments is given in Figure 9.

We now describe how the optimal solution to a transshipment problem can be found by solving a transportation problem. Given a transshipment problem, we create a balanced transportation problem by the following procedure (assume that total supply exceeds total demand):

Step 1 If necessary, add a dummy demand point (with a supply of 0 and a demand equal to the problem’s excess supply) to balance the problem. Shipments to the dummy and from a point to itself will, of course, have a zero shipping cost. Let s = total available supply.

Step 2 Construct a transportation tableau as follows: A row in the tableau will be needed for each supply point and transshipment point, and a column will be needed for each demand point and transshipment point. Each supply point will have a supply equal to its original supply, and each demand point will have a demand equal to its original demand. Let s = total available supply. Then each transshipment point will have a supply equal to (point’s original supply) $+ s$ and a demand equal to (point’s original demand) $+ s$. This ensures that any transshipment point that is a net supplier will have a net outflow equal to the point’s original supply, and, similarly, a net demander will have a net inflow equal to the point’s original demand. Although we don’t know how much will be shipped through each transshipment point, we can be sure that the total amount will not exceed s . This explains why we add s to the supply and demand at each transshipment point. By adding the same amounts to the supply and demand, we ensure that the net outflow at each transshipment point will be correct, and we also maintain a balanced transportation tableau.

Los
Memphis New York
Angeles
FIGURE 9
A Transshipment
Problem
Denver Chicago Boston

7. 6 Transshipment Problems 401

For the Widgetco example, this procedure yields the transportation tableau and its optimal solution given in Table 59. Because s = (total supply) $= 150 + 200 = 350$ and (total demand) $= 130 + 130 = 260$, the dummy demand point has a demand of $350 - 260 = 90$. The other supplies and demands in the transportation tableau are obtained by adding $s = 350$ to each transshipment point’s supply and demand.

In interpreting the solution to the transportation problem created from a transshipment problem, we simply ignore the shipments to the dummy and from a point to itself. From

Table 59, we find that Widgetco should produce 130 widgets at Memphis, ship them to New York, and transship them from New York to Los Angeles. The 130 widgets produced at Denver should be shipped directly to Boston. The net outflow from each city is

Memphis: $220 - 130 - 20 = 220 - 150$

Denver: $220 - 130 - 70 = 220 - 200$

N.Y.: $220 - 130 - 130 = 220 - 0$

Chicago: $350 - 350 - 130 = 220 - 0$

L.A.: $350 - 350 - 130 = 220 - 0$

Boston: $350 - 350 - 130 = 220 - 0$

Dummy: $-20 - 70 - 130 = 220 - 90$

A negative net outflow represents an inflow. Observe that each transshipment point (New York and Chicago) has a net outflow of 0; whatever flows into the transshipment point must leave the transshipment point. A graphical representation of the optimal solution to the Widgetco example is given in Figure 10.

Suppose that we modify the Widgetco example and allow shipments between Memphis and Denver. This would make Memphis and Denver transshipment points and would add columns for Memphis and Denver to the Table 59 tableau. The Memphis row in the tableau would now have a supply of $150 + 350 = 500$, and the Denver row would have

TABLE 59 N.Y. Chicago L.A. Boston Dummy Supply

Representation of 8 13 25 28 0

Transshipment Problem Memphis 130 20 150

as Balanced

Transportation Problem 15 12 26 25 0

Denver

130 70 200

0 6 16 17 0

220 130 350

N.Y.

6 0 14 16 0

350 350

Chicago

Demand

350 350 130 130 90

130 130 Los

Memphis New York

Angeles

FIGURE 10

Optimal Solution

to Widgetco Denver Chicago Boston

130

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a supply of 200 □ 350 □ 550. The new Memphis column would have a demand of 0 □ 350 □ 350, and the new Denver column would have a demand of 0 □ 350 □ 350. Finally, suppose that shipments between demand points L.A. and Boston were allowed. This would make L.A. and Boston transshipment points and add rows for L.A. and Boston. The supply for both the L.A. and Boston rows would be 0 □ 350 □ 350. The demand for both the L.A. and Boston columns would now be 130 □ 350 □ 480.

PROBLEMS

Group A

1 General Ford produces cars at L.A. and Detroit and has TA B L E 61

a warehouse in Atlanta; the company supplies cars to

customers in Houston and Tampa. The cost of shipping a car To (\$)

between points is given in Table 60 (“—” means that a From Well 1 Well 2 Mobile Galveston N.Y. L.A.

shipment is not allowed). L.A. can produce as many as

Well 1 0 — 10 13 25 28

1,100 cars, and Detroit can produce as many as 2,900 cars.

Houston must receive 2,400 cars, and Tampa must receive Well 2 — 0 15 12 26 25

1,500 cars. Mobile — — 0 6 16 17

a Formulate a balanced transportation problem that Galveston — — 6 0 14 16

can be used to minimize the shipping costs incurred in N.Y. — — — — 0 15

meeting demands at Houston and Tampa. L.A. — — — — 15 0

b Modify the answer to part (a) if shipments between

L.A. and Detroit are not allowed. Note: Dashes indicate shipments that are not allowed.

c Modify the answer to part (a) if shipments between

Houston and Tampa are allowed at a cost of \$5.

2 Sunco Oil produces oil at two wells. Well 1 can produce formulate a transshipment and balanced transportation model

as many as 150,000 barrels per day, and well 2 can produce to minimize the daily cost of transporting and refining the oil

as many as 200,000 barrels per day. It is possible to ship oil requirements of Los Angeles and New York.

directly from the wells to Sunco’s customers in Los Angeles

and New York. Alternatively, Sunco could transport oil to 4 Rework Problem 3 under the assumption that Galveston

the ports of Mobile and Galveston and then ship it by tanker has a refinery capacity of 150,000 barrels per day and Mobile

to New York or Los Angeles. Los Angeles requires 160,000 has one of 180,000 barrels per day. (Hint: Modify the method

barrels per day, and New York requires 140,000 barrels per used to determine the supply and demand at each

day. The costs of shipping 1,000 barrels between two points transshipment point to incorporate the refinery capacity

are shown in Table 61. Formulate a transshipment model restrictions, but make sure to keep the problem balanced.)

(and equivalent transportation model) that could be used to 5 General Ford has two plants, two warehouses, and three

minimize the transport costs in meeting the oil demands of customers. The locations of these are as follows:

Los Angeles and New York.

Plants: Detroit and Atlanta

3 In Problem 2, assume that before being shipped to Los Warehouses: Denver and New York Angeles or New York, all oil produced at the wells must be Customers: Los Angeles, Chicago, and Philadelphia

refined at either Galveston or Mobile. To refine 1,000 barrels

of oil costs \$12 at Mobile and \$10 at Galveston. Assuming Cars are produced at plants, then shipped to warehouses,

that both Mobile and Galveston have infinite refinery capacity, and finally shipped to customers. Detroit can produce 150

cars per week, and Atlanta can produce 100 cars per week.

Los Angeles requires 80 cars per week; Chicago, 70; and

Philadelphia, 60. It costs \$10,000 to produce a car at each

TA B L E 60 plant, and the cost of shipping a car between two cities is

given in Table 62. Determine how to meet General Ford's

To (\$)

weekly demands at minimum cost.

From L.A. Detroit Atlanta Houston Tampa

L.A. 0 140 100 90 225 Group B

Detroit 145 0 111 110 119 6† A company must meet the following demands for cash

Atlanta 105 115 0 113 78 at the beginning of each of the next six months: month 1,

Houston 89 109 121 0 —

Tampa 210 117 82 — 0

Based on Srinivasan (1974).

7. 6 Transshipment Problems 403

TA B L E 62 TA B L E 63

To (\$) Month of Sale

From Denver New York Bond 1 2 3 4 5 6

Detroit 1,253 637 1 \$0.21 \$0.19 \$0.17 \$0.13 \$0.09 \$0.05

Atlanta 1,398 841 2 \$0.50 \$0.50 \$0.50 \$0.33 \$0.00 \$0.00

To (\$) 3 \$1.00 \$1.00 \$1.00 \$1.00 \$1.00 \$0.00

From Los Angeles Chicago Philadelphia

Denver 1,059 996 1,691

New York 2,786 802 100

used to minimize the cost of meeting the cash demands for the next six months.

b Assume that payment of bills can be made after they are due, but a penalty of 5¢ per month is assessed for each dollar of cash demands that is postponed for one month. Assuming all bills must be paid by the end of month 6, develop a transshipment model that can be used to minimize the cost of paying the next six months' worth of bond 2, and \$400 worth of bond 3. The company bills. (Hint: Transshipment points are needed, in the form C_t cash available at beginning of month t after will be charged for any bonds sold before the end of month t bonds for month t have been sold, but before month $t + 1$. The penalties for selling \$1 worth of each bond are as demand is met. Shipments into C_t occur from bond sales shown in Table 63.

and $C_t \rightarrow 1$. Shipments out of C_t occur to $C_t \rightarrow 1$ and

a Assuming that all bills must be paid on time, formulate a balanced transportation problem that can be

SUMMARY Notation

m = number of supply points

n = number of demand points

x_{ij} = number of units shipped from supply point i to demand point j

c_{ij} = cost of shipping 1 unit from supply point i to demand point j

s_i = supply at supply point i

d_j = demand at demand point j

c_{ij} = coefficient of x_{ij} in row 0 of a given tableau

a_{ij} = column for x_{ij} in transportation constraints

A transportation problem is balanced if total supply equals total demand. To use the methods of this chapter to solve a transportation problem, the problem must first be balanced by use of a dummy supply or a dummy demand point. A balanced transportation problem may be written as

$$i=1 \dots m \quad j=1 \dots n$$

$$\min \sum_{i=1}^m \sum_{j=1}^n c_{ij}x_{ij}$$

$$i=1 \dots m$$

$$j=1 \dots n$$

$$\text{s.t. } \sum_{j=1}^n x_{ij} = s_i \quad (i = 1, 2, \dots, m)$$

$$j=1 \dots n$$

(Supply constraints)

$i \leq m$
 $s.t. \sum_{j=1}^n x_{ij} = d_j \quad (j = 1, 2, \dots, n)$
 $i \leq 1$

(Demand constraints)

 $x_{ij} \geq 0 \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, n)$

404 CHAPTER 7 Transportation, Assignment, and Transshipment Problems

Finding Basic Feasible Solutions

for Balanced Transportation Problems

We can find a bfs for a balanced transportation problem by the northwest corner method, the minimum-cost method, or Vogel's method. To find a bfs by the northwest corner method, begin in the upper left-hand (or northwest) corner of the transportation tableau and set x_{11} as large as possible. Clearly, x_{11} can be no larger than the smaller of s_1 and d_1 . If $x_{11} = s_1$, then cross out the first row of the transportation tableau; this indicates that no more basic variables will come from row 1 of the tableau. Also change d_1 to $d_1 - s_1$. If $x_{11} = d_1$, then cross out the first column of the transportation tableau and change s_1 to $s_1 - d_1$. If $x_{11} = s_1 = d_1$, cross out either row 1 or column 1 (but not both) of the transportation tableau. If you cross out row 1, change d_1 to 0; if you cross out column 1, change s_1 to 0. Continue applying this procedure to the most northwest cell in the tableau that does not lie in a crossed-out row or column. Eventually, you will come to a point where there is only one cell that can be assigned a value. Assign this cell a value equal to its row or column demand, and cross out both the cell's row and its column. A basic feasible solution has now been obtained.

Finding the Optimal Solution

for a Transportation Problem

Step 1 If the problem is unbalanced, balance it.

Step 2 Use one of the methods described in Section 7.2 to find a bfs.

Step 3 Use the fact that $u_i \geq 0$ and $u_i + v_j \leq c_{ij}$ for all basic variables to find the $[u_1 \ u_2 \ \dots \ u_m \ v_1 \ v_2 \ \dots \ v_n]$ for the current bfs.

Step 4 If $u_i + v_j \leq c_{ij}$ for all nonbasic variables, then the current bfs is optimal. If this is not the case, then we enter the variable with the most positive $u_i + v_j - c_{ij}$ into the basis. To do this, find the loop. Then, counting only cells in the loop, label the even cells. Also label the odd cells. Now find the odd cell whose variable assumes the smallest value, θ . The variable corresponding to this odd cell will leave the basis. To perform the pivot, decrease the value of each odd cell by θ and increase the value of each even cell by θ . The values of variables not in the loop remain unchanged. The pivot is now complete. If $\theta = 0$, then the entering variable will equal 0, and an odd variable that has a current value of 0 will leave the basis. In this case, a degenerate bfs will result. If more than one odd cell in the loop equals θ , you may arbitrarily choose one of these odd cells to leave the basis; again, a degenerate bfs will result. The pivoting yields a new bfs.

Step 5 Using the new bfs, return to steps 3 and 4.

For a maximization problem, proceed as stated, but replace step 4 by step 4 .

Step 4 If $u_i - v_j - c_{ij} \leq 0$ for all nonbasic variables, the current bfs is optimal. Otherwise, enter the variable with the most negative $u_i - v_j - c_{ij}$ into the basis using the pivoting procedure.

Assignment Problems

An assignment problem is a balanced transportation problem in which all supplies and demands equal 1. An $m \times m$ assignment problem may be efficiently solved by the Hungarian method:

7.1 Summary 405

Step 1 Find the minimum element in each row of the cost matrix. Construct a new matrix by subtracting from each cost the minimum cost in its row. For this new matrix, find the minimum cost in each column. Construct a new matrix (reduced cost matrix) by subtracting from each cost the minimum cost in its column.

Step 2 Cover all the zeros in the reduced cost matrix using the minimum number of lines needed. If m lines are required, then an optimal solution is available among the covered zeros in the matrix. If fewer than m lines are needed, then proceed to step 3.

Step 3 Find the smallest nonzero element (k) in the reduced cost matrix that is uncovered by the lines drawn in step 2. Now subtract k from each uncovered element and add k to each element that is covered by two lines. Return to step 2.

REMARKS 1 To solve an assignment problem in which the goal is to maximize the objective function, mul-

tiply the profits matrix through by -1 and solve it as a minimization problem.

2 If the number of rows and columns in the cost matrix are unequal, then the problem is unbalanced. The Hungarian method may yield an incorrect solution if the problem is unbalanced. Thus, any assignment problem should be balanced (by the addition of one or more dummy points) before it is solved by the Hungarian method.

Transshipment Problems

A transshipment problem allows shipment between supply points and between demand points, and it may also contain transshipment points through which goods may be shipped on their way from a supply point to a demand point. Using the following method, a transshipment problem may be transformed into a balanced transportation problem.

Step 1 If necessary, add a dummy demand point (with a supply of 0 and a demand equal to the problem's excess supply) to balance the problem. Shipments to the dummy and from a point to itself will, of course, have a zero shipping cost. Let s = total available supply.

Step 2 Construct a transportation tableau creating a row for each supply point and transshipment point, and a column for each demand point and transshipment point. Each supply point will have a supply equal to its original supply, and each demand point will have a demand equal to its original demand. Let s = total available supply. Then each transshipment point will have a supply equal to (point's original supply) $- s$ and a demand equal to (point's original demand) $+ s$.

Sensitivity Analysis for Transportation Problems

Following the discussion of sensitivity analysis in Chapter 6, we can analyze how a change in a transportation problem affects the problem's optimal solution.

Change 1 Changing the objective function coefficient of a nonbasic variable. As long as the coefficient of x_{ij} in the optimal row 0 is nonpositive, the current basis remains optimal.

Change 2 Changing the objective function coefficient of a basic variable. To see whether the current basis remains optimal, find the new u_i 's and v_j 's and use these values to price out all nonbasic variables. The current basis remains optimal as long as all nonbasic variables have a nonpositive coefficient in row 0.

Change 3 Increasing both supply s_i and demand d_j by Δ .

New z-value Δ old z-value $\Delta \sum u_i \Delta \sum v_j$

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We may find the new values of the decision variables as follows:

1 If x_{ij} is a basic variable in the optimal solution, then increase x_{ij} by Δ .

2 If x_{ij} is a nonbasic variable in the optimal solution, find the loop involving x_{ij} and some of the basic variables. Find an odd cell in the loop that is in row i . Increase the value of this odd cell by Δ and go around the loop, alternately increasing and then decreasing current basic variables in the loop by Δ .

REVIEW PROBLEMS

Group A

1 Televco produces TV picture tubes at three plants. Plant 3 A company must meet the following demands for

1 can produce 50 tubes per week; plant 2, 100 tubes per a product: January, 30 units; February, 30 units; March,

week; and plant 3, 50 tubes per week. Tubes are shipped to 20 units. Demand may be backlogged at a cost of

three customers. The profit earned per tube depends on the \$5/unit/month. All demand must be met by the end of March.

site where the tube was produced and on the customer who Thus, if 1 unit of January demand is met during March, a

purchases the tube (see Table 64). Customer 1 is willing to backlogging cost of 5(2) = \$10 is incurred. Monthly

purchase as many as 80 tubes per week; customer 2, as production capacity and unit production cost during each

many as 90; and customer 3, as many as 100. Televco wants month are given in Table 66. A holding cost of \$20/unit is

to find a shipping and production plan that will maximize assessed on the inventory at the end of each month.

profits. a Formulate a balanced transportation problem that

a Formulate a balanced transportation problem that could be used to determine how to minimize the total

can be used to maximize Televco's profits. cost (including backlogging, holding, and production

b Use the northwest corner method to find a bfs to the costs) of meeting demand.

problem. b Use Vogel's method to find a basic feasible solution.

c Use the transportation simplex to find an optimal so- c Use the transportation simplex to determine how to

lution to the problem. meet each month's demand. Make sure to give an inter-

pretation of your optimal solution (for example, 20 units

2 Five workers are available to perform four jobs. The

of month 2 demand is met from month 1 production).

time it takes each worker to perform each job is given in

Table 65. The goal is to assign workers to jobs so as to 4 Appletree Cleaning has five maids. To complete cleaning

minimize the total time required to perform the four jobs. my house, they must vacuum, clean the kitchen, clean the

Use the Hungarian method to solve the problem. bathroom, and do general straightening up. The time it takes

each maid to do each job is shown in Table 67. Each maid

TA B L E 64

TA B L E 66

To (\$)

Production Unit Production

From Customer 1 Customer 2 Customer 3 Month Capacity Cost

Plant 1 75 60 69 January 35 \$400

Plant 2 79 73 68 February 30 \$420

Plant 3 85 76 70 March 35 \$410

TA B L E 67

TA B L E 65

Time (Hours)

Time (Hours)

Clean Clean Straighten

Worker Job 1 Job 2 Job 3 Job 4 Maid Vacuum Kitchen Bathroom Up

1 10 15 10 15 1 6 5 2 1

2 12 8 20 16 2 9 8 7 3

3 12 9 12 18 3 8 5 9 4

4 6 12 15 18 4 7 7 8 3

5 16 12 8 12 5 5 5 6 4

7. 1 Review Problems 407

is assigned one job. Use the Hungarian method to determine TA B L E 70

assignments that minimize the total number of maid-hours

needed to clean my house. No. of Students Distance to (Miles)

District Whites Blacks District 2 District 3

5† Currently, State University can store 200 files on hard

disk, 100 files in computer memory, and 300 files on tape. 1 210 120 3 5

Users want to store 300 word-processing files, 100 2 210 30 — 4

packaged-program files, and 100 data files. Each month a

typical word-processing file is accessed eight times; a typical 3 180 150 — —

packaged-program file, four times; and a typical data file,

two times. When a file is accessed, the time it takes for the

file to be retrieved depends on the type of file and on the

storage medium (see Table 68). Formulate a balanced transportation problem that can be

used to determine the minimum total distance that students

a If the goal is to minimize the total time per month must be bused while still satisfying the Supreme Court's

that users spend accessing their files, formulate a bal- requirements. Assume that a student who remains in his or

anced transportation problem that can be used to deter- her own district will not be bused.

mine where files should be stored.

b Use the minimum cost method to find a bfs. 8 Using the northwest corner method to find a bfs, find

(via the transportation simplex) an optimal solution to the

c Use the transportation simplex to find an optimal

transportation (minimization) problem shown in Table 71.

solution.

9 Solve the following LP:

6 The Gotham City police have just received three calls for

police. Five cars are available. The distance (in city blocks) $\min z = 2x_1 + 3x_2 + 4x_3 + 3x_4$

of each car from each call is given in Table 69. Gotham City s.t. $x_1 + x_2 + x_3 + x_4 = 4$

wants to minimize the total distance cars must travel to s.t. $x_1 + x_2 + x_3 + x_4 = 5$

respond to the three police calls. Use the Hungarian method s.t. $x_1 + x_2 + x_3 + x_4 = 3$

to determine which car should respond to which call.

s.t. $x_1 + x_2 + x_3 + x_4 = 6$

7 There are three school districts in the town of Busville. $\min x_j = 0 \ (j = 1, 2, 3, 4)$

The number of black and white students in each district are

shown in Table 70. The Supreme Court requires the schools 10 Find the optimal solution to the balanced transportation

in Busville to be racially balanced. Thus, each school must problem in Table 72 (minimization).

have exactly 300 students, and each school must have the 11 In Problem 10, suppose we increase s_i to 16 and d_3 to

same number of black students. The distances between 11. The problem is still balanced, and because 31 units

districts are shown in Table 70. (instead of 30 units) must be shipped, one would think that

the total shipping costs would be increased. Show that the

total shipping cost has actually decreased by \$2, however.

This is called the "more for less" paradox. Explain why

TABLE 68

increasing both the supply and the demand has decreased

Time (Minutes) cost. Using the theory of shadow prices, explain how one

Storage Word Packaged could have predicted that increasing s_1 and d_3 by 1 would

Medium Processing Program Data decrease total cost by \$2.

Hard disk 5 4 4 12 Use the northwest corner method, the minimum-cost

method, and Vogel's method to find basic feasible solutions

Memory 2 1 1

to the transportation problem in Table 73.

Tape 10 8 6

13 Find the optimal solution to Problem 12.

TABLE 69 TABLE 71

Distance (Blocks) 12 14 16

Car Call 1 Call 2 Call 3

60

1 10 11 18

14 13 19

2 6 7 7

3 7 8 5 50

4 5 6 4

17 15 18

5 9 4 7

40

This problem is based on Evans (1984). 40 70 10

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TABLE 72 TABLE 74

4 2 4 To (\$)

15 From Dallas Houston N.Y. Chicago

L.A. 300 110 — —

12 8 4

San Diego 420 100 — —

15 Dallas — — 450 550

Houston — — 470 530

10 10 10

TABLE 73 TABLE 75

20 11 3 6 Plant Cars Available

5 Atlanta 5,000

Boston 6,000

5 9 10 2 Chicago 4,000

10 L.A. 3,000

18 7 4 1

15

TA B L E 76

3 3 12 12 Warehouse Cars Required

Memphis 6,000

Milwaukee 4,000

14 Oilco has oil fields in San Diego and Los Angeles. The

San Diego field can produce 500,000 barrels per day, and N.Y. 4,000

the Los Angeles field can produce 400,000 barrels per day. Denver 2,000

Oil is sent from the fields to a refinery, either in Dallas or San Francisco 2,000

in Houston (assume that each refinery has unlimited

capacity). It costs \$700 to refine 100,000 barrels of oil at

Dallas and \$900 at Houston. Refined oil is shipped to

customers in Chicago and New York. Chicago customers TA B L E 77

require 400,000 barrels per day of refined oil; New York

Memphis Milwaukee N.Y. Denver S.F.

customers require 300,000. The costs of shipping 100,000

barrels of oil (refined or unrefined) between cities are given Atlanta 371 761 841 1,398 2,496

in Table 74. Formulate a balanced transportation model of Boston 1,296 1,050 206 1,949 3,095

this situation.

Chicago 530 87 802 996 2,142

15 For the Powerco problem, find the range of values of L.A. 1,817 2,012 2,786 1,059 379

c24 for which the current basis remains optimal.

16 For the Powerco problem, find the range of values of

c23 for which the current basis remains optimal.

18 During the next three quarters, Airco faces the following

17 A company produces cars in Atlanta, Boston, Chicago, demands for air conditioner compressors: quarter 1—200;

and Los Angeles. The cars are then shipped to warehouses quarter 2—300; quarter 3—100. As many as 240 air

in Memphis, Milwaukee, New York City, Denver, and San compressors can be produced during each quarter. Production

Francisco. The number of cars available at each plant is costs/compressor during each quarter are given in Table 78.

given in Table 75. The cost of holding an air compressor in inventory is

Each warehouse needs to have available the number of \$100/quarter. Demand may be backlogged (as long as it is met

cars given in Table 76. by the end of quarter 3) at a cost of \$60/compressor/quarter.

The distance (in miles) between the cities is given in Formulate the tableau for a balanced transportation problem

Table 77. whose solution tells Airco how to minimize the total cost of

a Assuming that the cost (in dollars) of shipping a car meeting the demands for quarters 1–3.

equals the distance between two cities, determine an op-

19 A company is considering hiring people for four types
timal shipping schedule.

of jobs. It would like to hire the number of people in Table

b Assuming that the cost (in dollars) of shipping a car 79 for each type of job.

equals the square root of the distance between two cities, Four types of people can be hired by the
company. Each

determine an optimal shipping schedule. type is qualified to perform two types of jobs according to

7. 1 Review Problems 409

TA B L E 78 TA B L E 81

Quarter 1 Quarter 2 Quarter 3 Leave Arrive Leave Arrive

Flight Chicago New York Flight New York Chicago

\$200 \$180 \$240

1 6 A.M. 10 A.M. 1 7 A.M. 9 A.M.

2 9 A.M. 1 P.M. 2 8 A.M. 10 A.M.

3 12 noon 4 P.M. 3 10 A.M. 12 noon

TA B L E 79 4 3 P.M. 7 P.M. 4 12 noon 2 P.M.

Job 5 5 P.M. 9 P.M. 5 2 P.M. 4 P.M.

1 2 3 4 6 7 P.M. 11 P.M. 6 4 P.M. 6 P.M.

7 8 P.M. 12 midnight 7 6 P.M. 8 P.M.

Number of people 30 30 40 20

TA B L E 80 York flight with at least 1 hour of downtime between flights.

Braneast wants to schedule the crews to minimize the total

Type of Person downtime. Set up an assignment problem that can be used

1 2 3 4 to accomplish this goal. (Hint: Let $x_{ij} = 1$ if the crew that

flies flight i also flies flight j , and $x_{ij} = 0$ otherwise. If $x_{ij} =$

Jobs qualified for 1 and 3 2 and 3 3 and 4 1 and 4 1, then a cost c_{ij} is incurred, corresponding to the
downtime

associated with a crew flying flight i and flight j .) Of course,

some assignments are not possible. Find the flight

assignments that minimize the total downtime. How many

Table 80. A total of 20 Type 1, 30 Type 2, 40 Type 3, and crews should be based in each city?

Assume that at the end

20 Type 4 people have applied for jobs. Formulate a balanced of the day, each crew must be in its
home city.

transportation problem whose solution will tell the company

how to maximize the number of employees assigned to 23 A firm producing a single product has
three plants and

suitable jobs. (Note: Each person can be assigned to at most four customers. The three plants will
produce 3,000, 5,000,

one job.) and 5,000 units, respectively, during the next time period.

The firm has made a commitment to sell 4,000 units to

20 During each of the next two months you can produce customer 1, 3,000 units to customer 2, and at least 3,000

as many as 50 units/month of a product at a cost of \$12/unit units to customer 3. Both customers 3 and 4 also want to

during month 1 and \$15/unit during month 2. The customer buy as many of the remaining units as possible. The profit

is willing to buy as many as 60 units/month during each of associated with shipping a unit from plant i to customer j is

the next two months. The customer will pay \$20/unit during given in Table 82. Formulate a balanced transportation

month 1, and \$16/unit during month 2. It costs \$1/unit to problem that can be used to maximize the company's profit.

hold a unit in inventory for a month. Formulate a balanced

transportation problem whose solution will tell you how to 24 A company can produce as many as 35 units/month.

maximize profit. The demands of its primary customers must be met on time

each month; if it wishes, the company may also sell units to

secondary customers each month. A \$1/unit holding cost is

Group B assessed against each month's ending inventory. The relevant

21† The Carter Caterer Company must have the following data are shown in Table 83. Formulate a balanced

number of clean napkins available at the beginning of each transportation problem that can be used to maximize profits

of the next four days: day 1—15; day 2—12; day 3—18; earned during the next three months.

day 4—6. After being used, a napkin can be cleaned by one 25 My home has four valuable paintings that are up for

of two methods: fast service or slow service. Fast service sale. Four customers are bidding for the paintings. Customer

costs 10¢ per napkin, and a napkin cleaned via fast service 1 is willing to buy two paintings, but each other customer

is available for use the day after it is last used. Slow service is willing to purchase at most one painting. The prices that

costs 6¢ per napkin, and these napkins can be reused two each customer is willing to pay are given in Table 84. Use

days after they are last used. New napkins can be purchased

for a cost of 20¢ per napkin. Formulate a balanced

transportation problem to minimize the cost of meeting the

demand for napkins during the next four days. TA B L E 82

22 Braneast Airlines must staff the daily flights between To Customer (\$) From 1 2 3 4

New York and Chicago shown in Table 81. Each of Braneast's

crews lives in either New York or Chicago. Each day a crew

must fly one New York–Chicago and one Chicago–New Plant 1 65 63 62 64

Plant 2 68 67 65 62

Plant 3 63 60 59 60

This problem is based on Jacobs (1954).

410 CHAPTER 7 Transportation, Assignment, and Transshipment Problems

TABLE 83 TABLE 86

Available for To (\$)

Production Primary Secondary Sales

Month Cost/Unit (\$) Demand Demand Price/Unit (\$) From NE NW MW SE SW

1 13 20 15 15 Atlanta 8.41 21.52 6.74 3.00 7.89

2 12 15 20 14 Houston 15.20 13.87 10.67 7.89 3.00

3 13 25 15 16

27† During the month of July, Pittsburgh resident B. Fly must make four round-trip flights between Pittsburgh and

TABLE 84

Chicago. The dates of the trips are as shown in Table 87.

Bid for (\$) B. Fly must purchase four round-trip tickets. Without a

Customer Painting 1 Painting 2 Painting 3 Painting 4 discounted fare, a round-trip ticket between Pittsburgh and

Chicago costs \$500. If Fly's stay in a city includes a weekend,

1 8 11 — — then he gets a 20% discount on the round-trip fare. If his

2 9 13 12 7 stay in a city is at least 21 days, then he receives a 35%

3 9 — 11 — discount; and if his stay is more than 10 days, then he receives a 30% discount. Of course, only one discount can

4 — — 12 9

be applied toward the purchase of any ticket. Formulate and

solve an assignment problem that minimizes the total cost

of purchasing the four round-trip tickets. (Hint: Let $x_{ij} \in \{0, 1\}$

the Hungarian method to determine how to maximize the if a round-trip ticket is purchased for use on the i th flight

total revenue received from the sale of the paintings. out of Pittsburgh and the j th flight out of Chicago. Also

think about where Fly should buy a ticket if, for example,

26 Powerhouse produces capacitors at three locations: Los Angeles, Chicago, and New York. Capacitors are shipped

from these locations to public utilities in five regions of the

28 Three professors must be assigned to teach six sections

country: northeast (NE), northwest (NW), midwest (MW), of finance. Each professor must teach two sections of

southeast (SE), and southwest (SW). The cost of producing finance, and each has ranked the six time periods during

and shipping a capacitor from each plant to each region of which finance is taught, as shown in Table 88. A ranking of

the country is given in Table 85. Each plant has an annual 10 means that the professor wants to teach that time, and a

production capacity of 100,000 capacitors. Each year, each ranking of 1 means that he or she does not want to teach at

region of the country must receive the following number of that time. Determine an assignment of professors to sections

capacitors: NE, 55,000; NW, 50,000; MW, 60,000; SE, that will maximize the total satisfaction of the professors.

60,000; SW, 45,000. Powerhouse feels shipping costs are 29¢ Three fires have just broken out in New York. Fires 1

too high, and the company is therefore considering building and 2 each require two fire engines, and fire 3 requires three

one or two more production plants. Possible sites are Atlanta fire engines. The “cost” of responding to each fire depends

and Houston. The costs of producing a capacitor and on the time at which the fire engines arrive. Let t_{ij} be the

shipping it to each region of the country are given in Table time (in minutes) when the j th engine arrives at fire i . Then

86. It costs \$3 million (in current dollars) to build a new the cost of responding to each fire is as follows:

plant, and operating each plant incurs a fixed cost (in

addition to variable shipping and production costs) of Fire 1: $6t_{11} + 4t_{12} + 5t_{13}$

\$50,000 per year. A plant at Atlanta or Houston will have Fire 2: $7t_{21} + 3t_{22} + 5t_{23}$

the capacity to produce 100,000 capacitors per year. Fire 3: $9t_{31} + 8t_{32} + 5t_{33}$

Assume that future demand patterns and production costs

Three fire companies can respond to the three fires.

will remain unchanged. If costs are discounted at a rate of

1 Company 1 has three engines available, and companies 2

$11\frac{9}{10}\%$ per year, how can Powerhouse minimize the present

value of all costs associated with meeting current and future demands?

TA B L E 87

Leave Pittsburgh Leave Chicago

TA B L E 85

Monday, July 1 Friday, July 5

To (\$) Tuesday, July 9 Thursday, July 11

From NE NW MW SE SW Monday, July 15 Friday, July 19

L.A. 27.86 4.00 20.54 21.52 13.87 Wednesday, July 24 Thursday, July 25

Chicago 8.02 20.54 2.00 6.74 10.67

N.Y. 2.00 27.86 8.02 8.41 15.20 †

Based on Hansen and Wendell (1982).

Based on Denardo, Rothblum, and Swersey (1988).

7. 1 Review Problems 411

TA B L E 88 TA B L E 89

Professor 9 A.M. 10 A.M. 11 A.M. 1 P.M. 2 P.M. 3 P.M. Company Fire 1 Fire 2 Fire 3

1 8 7 6 5 7 6 1 6 7 9
 2 9 9 8 8 4 4 2 5 8 11
 3 7 6 9 6 9 9 3 6 9 10

and 3 each have two engines available. The time (in minutes) signing the fire engines. (Hint: Seven demand points

it takes an engine to travel from each company to each fire will be needed.)

is shown in Table 89. b Would the formulation in part (a) still be valid if the

a Formulate and solve a transportation problem that cost of fire 1 were $4t_{11} \square 6t_{12}$?

can be used to minimize the cost associated with as-

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Paper 4 of 5 — CSC 425 Project Management 2025

Level 400 • Second Semester • Handout

INTRODUCTION TO PROJECT MANAGEMENT

OVERVIEW

The purpose of presentation is to provide leaders and team members of projects, committees or task forces with advanced techniques and practical skills for initiating, planning, tracking, controlling and evaluating any kind or size of project.

Introduction to Project Management

- ☐ On time
- ☐ On budget
- ☐ Meeting the goals that have been agreed upon

WHAT IS A PROJECT?

- A project is an activity that :
 - is temporary having a start and end date
 - is unique
 - brings about change
 - has unknown elements, which therefore create risk
- Generally projects are formed to solve a problem or take advantage of an opportunity.
- Business as usual activities can often be mistaken for projects.
- Generally it is the uniqueness of the activity that is the deciding factor – do we do this every year? If so, then it is not truly a project – although you can use project methods to get it done.

QUIZ – ARE THESE PROJECTS?

- ☐ Building a deck Yes No
- ☐ Implementing a new system Yes No
- ☐ Mowing the lawn Yes No

- ☐ Planning a wedding Yes No
- ☐ Planning a fundraiser Yes No
- ☐ Planning a student graduation Yes No

COMMON PROJECT TERMS

- Deliverables: Tangible ‘things’ that the project produces
- Milestones: Dates by which major activities are performed.
- Tasks: Also called Actions. Activities undertaken during the project
- Risks: Potential problems that may arise
- Issues: Risks that have happened
- Gantt Chart: A specific type of chart showing time and tasks. Usually created by a Project Management program like MS Project.
- Stakeholder: Any person or group of people who may be affected by your project

EXAMPLE: Building a deck

- Deliverables: A plan, a consent form, the deck
- Milestones: Plan drafted 1 Jun

Plan approved 15 Jun

Plan submitted 16 Jun

Plan approved 19 Jun

Materials purchased 16 Jul

Resources booked 16 Jul

Equipment identified 16 Jul

Deck constructed 20 Jul

Deck tested 24 Jul

Deck quality approved 24 Jul

“Deck warming” completed 28 Jul

tasks Subtasks

- Tasks:
- Plan drafted Requirement gathered
- Best practice researched
- Draft 1 prepared
- Distributed to stakeholders
- Plan approved Feedback gathered
- Amendments made
- Final plan prepared
- Distributed to stakeholders
- Sign-off obtained

WORK-BREAKDOWN STRUCTURE

- WBS
- Hierarchy of tasks required to complete project
- Each task is broken into smaller tasks that can be managed and estimated
- Define task dependencies
 - Some tasks must begin at the same time, some must end at the same time and some cannot start until the other tasks have finished.
- Estimate task durations and cost
- May be inputted into project management software
 - Final WBS plan is called baseline WBS
 - Risks:
 - Plan is not approved after first round of feedback
 - Resources are not available at the required time
 - Plan is not given consent
 - For each of the above, you should have a contingency plan, or do some activity that may prevent it happening in the first place.
 - Issues:
 - If any of the above actually happens, then it becomes an issue to solve.
 - Gantt Chart:
 - Stakeholder:
 - House owner, Builder, Council, ???

A successful Project Manager must simultaneously manage the four basic elements of a project: resources, time, money, and most importantly, scope.

All these elements are interrelated. Each must be managed effectively. All must be managed together if the project is to be a success. The resource that can be leveraged to the greatest extent in all projects is the people involved.

THE PROJECT MANAGER

A person with a diverse set of skills – management, leadership, technical, conflict management, and customer relationship – who is responsible for:

- initiating,

- planning,
- executing,
- controlling,
- monitoring,
- and closing down a project.

Project Managers are essentially jugglers. They must make sure that everything keeps to task, that potential issues are quickly eliminated and the project is delivered on time, all the while making sure everyone knows what is happening and the project quality and budget are acceptable. Specifically they:

- direct all activities required to successfully meet the project objectives
- manage risk - scanning ahead for potential issues and resolving them before they become a problem
- solve problems - recommending alternative approaches to problems that arise and providing guidance to the Project Sponsor
- track and report project progress
- communicate to all stakeholders in the project
- Ultimately responsible for the Project's Success
- Plan and Act
- Focus on the project's end
- Be a manager & leader

PROJECT INITIATION

The Initiation phase of the project is the most important phase. The success of the entire project depends on how clearly and completely the Terms of References are established.

- Project Sponsor
- Lines of Authority
- Participants
- Objectives

PROJECT INITIATION con't

- Constraints
- Costs/Budget
- Resources

- Deliverables
- Phases & Time Scales
- Strategy
- Risks
- Roles & Responsibilities

CHARACTERISTICS OF PROJECTS

- A project contains a well defined objective. The project objective is defined in terms of scope (or requirements), schedule, and cost.
- A project is carried out via a set of interdependent tasks.
- A project uses various resources to carry out these tasks.
- A project has a definite start date and an expected completion date. The actual completion date may not always be the same as the expected date.
- A project is a one time or unique endeavor.
- A project has a customer.
- So why do projects fail?

WHY DO PROJECTS FAIL?

1. Poor project and program management discipline
2. Lack of executive-level support
3. Wrong team members
4. Poor communication
5. No measures for evaluating the success of the project
6. No risk management
7. Inability to manage change

A project has a degree of UNCERTAINTY. In project planning many assumptions are made regarding:

- access to resources.
- resource capability.
- impact of environmental factors.
- These assumptions are not always accurate.

- Requires project managers to re-assess and trade-offs between requirements, costs, and time. Above all, be PRO-ACTIVE.

MEASURING PROJECT SUCCESS

We measure the success of a project using 4 major project constraints, specifically:

- Scope.
- Cost.
- Schedule (Time).
- Customer satisfaction (quality and performance).

PROJECT CONSTRAINTS

- Project scope – Have all the project requirements (i.e., deliverables) been completed?
- Project cost – Is the cost of the project close to the amount the customer has agreed to pay?
- Schedule – Was the project completed on time?
- Customer satisfaction – Is the customer happy with the quality of the project?

PROJECT SUCCESS – 12 Golden Rules

☐ Rule #1

Thou shalt gain consensus on project outcome.

☐ Rule #2

Thou shalt build the best team possible.

☐ Rule #3

Thou shalt develop a comprehensive, viable plan and keep it up-to-date.

☐ Rule #4

Thou shalt determine how much stuff you really need to get things done.

☐ Rule #5

Thou shalt have a realistic schedule.

☐ Rule #6

Thou won't try to do more than can be done.

□ Rule #7

Thou will remember that people count.

□ Rule #8

Thou will gain the formal and ongoing support of management and stakeholders.

□ Rule #9

Thou must be willing to change.

□ Rule #10

Thou must keep others informed of what you're up to.

□ Rule #11

Thou must be willing to try new things.

□ Rule #12

Thou must become a leader.

□ □ □ □ □ □ □ □ □ □

Transportation, Assignment, and Transshipment Problems

In this chapter, we discuss three special types of linear programming problems: transportation, assignment, and transshipment. Each of these can be solved by the simplex algorithm, but specialized algorithms for each type of problem are much more efficient.

7.1 Formulating Transportation Problems

We begin our discussion of transportation problems by formulating a linear programming model of the following situation.

EXAMPLE 1 Powerco Formulation

Powerco has three electric power plants that supply the needs of four cities.† Each power plant can supply the following numbers of kilowatt-hours (kwh) of electricity: plant 1—35 million; plant 2—50 million; plant 3—40 million (see Table 1). The peak power demands in these cities, which occur at the same time (2 P.M.), are as follows (in kwh): city 1—45 million; city 2—20 million; city 3—30 million; city 4—30 million. The costs of sending 1 million kwh of electricity from plant to city depend on the distance the electricity must travel. Formulate an LP to minimize the cost of meeting each city's peak power demand.

Solution To formulate Powerco's problem as an LP, we begin by defining a variable for each decision that Powerco must make. Because Powerco must determine how much power is sent from each plant to each city, we define (for $i \in \{1, 2, 3\}$ and $j \in \{1, 2, 3, 4\}$)

x_{ij} = number of (million) kwh produced at plant i and sent to city j

In terms of these variables, the total cost of supplying the peak power demands to cities

1–4 may be written as

$x_{11} \ x_{12} \ x_{13} \ x_{14}$ (Cost of shipping power from plant 1)

$x_{21} \ x_{22} \ x_{23} \ x_{24}$ (Cost of shipping power from plant 2)

$x_{31} \ x_{32} \ x_{33} \ x_{34}$ (Cost of shipping power from plant 3)

Powerco faces two types of constraints. First, the total power supplied by each plant cannot exceed the plant's capacity. For example, the total amount of power sent from plant

This example is based on Aarvik and Randolph (1975).

TABLE 1

Shipping Costs, Supply, and Demand for Powerco

To

Supply

From City 1 City 2 City 3 City 4 (million kwh)

Plant 1 \$8 \$6 \$10 \$9 35

Plant 2 \$9 \$12 \$13 \$7 50

Plant 3 \$14 \$9 \$16 \$5 40

Demand 45 20 30 30

(million kwh)

1 to the four cities cannot exceed 35 million kwh. Each variable with first subscript 1 represents a shipment of power from plant 1, so we may express this restriction by the LP constraint

$$x_{11} + x_{12} + x_{13} + x_{14} \leq 35$$

In a similar fashion, we can find constraints that reflect plant 2's and plant 3's capacities. Because power is supplied by the power plants, each is a supply point. Analogously, a constraint that ensures that the total quantity shipped from a plant does not exceed plant capacity is a supply constraint. The LP formulation of Powerco's problem contains the following three supply constraints:

$$x_{11} + x_{12} + x_{13} + x_{14} \leq 35 \text{ (Plant 1 supply constraint)}$$

$$x_{21} + x_{22} + x_{23} + x_{24} \leq 50 \text{ (Plant 2 supply constraint)}$$

$$x_{31} + x_{32} + x_{33} + x_{34} \leq 40 \text{ (Plant 3 supply constraint)}$$

Second, we need constraints that ensure that each city will receive sufficient power to meet its peak demand. Each city demands power, so each is a demand point. For example, city 1 must receive at least 45 million kwh. Each variable with second subscript 1 represents a shipment of power to city 1, so we obtain the following constraint:

$$x_{11} + x_{21} + x_{31} \geq 45$$

Similarly, we obtain a constraint for each of cities 2, 3, and 4. A constraint that ensures that a location receives its demand is a demand constraint. Powerco must satisfy the following four demand constraints:

$$x_{11} + x_{21} + x_{31} \geq 45 \text{ (City 1 demand constraint)}$$

$$x_{12} + x_{22} + x_{32} \geq 20 \text{ (City 2 demand constraint)}$$

$$x_{13} + x_{23} + x_{33} \geq 30 \text{ (City 3 demand constraint)}$$

$$x_{14} + x_{24} + x_{34} \geq 30 \text{ (City 4 demand constraint)}$$

Because all the x_{ij} 's must be nonnegative, we add the sign restrictions $x_{ij} \geq 0$ ($i = 1, 2, 3; j = 1, 2, 3, 4$).

Combining the objective function, supply constraints, demand constraints, and sign restrictions yields the following LP formulation of Powerco's problem:

$$\begin{aligned} \min z &= 8x_{11} + 6x_{12} + 10x_{13} + 9x_{14} + 9x_{21} + 12x_{22} + 13x_{23} + 7x_{24} \\ &+ 14x_{31} + 9x_{32} + 16x_{33} + 5x_{34} \\ \text{s.t. } &x_{11} + x_{12} + x_{13} + x_{14} = 35 \text{ (Supply constraints)} \\ &x_{21} + x_{22} + x_{23} + x_{24} = 50 \\ &x_{31} + x_{32} + x_{33} + x_{34} = 40 \end{aligned}$$

7.1 Formulating Transportation Problems 361

Supply points Demand points

City 1 $d_1 = 45$

$x_{11} = 0$

$s_1 = 35$ Plant 1 $x_{12} = 10$

$x_{13} = 25$

$x_{14} = 0$

FIGURE 1 City 2 $d_2 = 20$

Graphical $x_{21} = 45$

$x_{22} = 0$

Representation of $s_1 = 50$ Plant 2 $x_{23} = 5$

Powerco Problem and $x_{24} = 0$

Its Optimal Solution City 3 $d_3 = 30$

$x_{31} = 0$

$x_{32} = 10$

$s_1 = 40$ Plant 3 $x_{33} = 0$

$x_{34} = 30$

City 4 $d_4 = 30$

s.t. $x_{11} + x_{21} + x_{31} + x_{34} = 45$ (Demand constraints)

s.t. $x_{12} + x_{22} + x_{32} + x_{34} = 20$

s.t. $x_{13} + x_{23} + x_{33} + x_{34} = 30$

s.t. $x_{14} + x_{24} + x_{34} + x_{34} = 30$

$x_{ij} \geq 0$ ($i = 1, 2, 3; j = 1, 2, 3, 4$)

In Section 7.3, we will find that the optimal solution to this LP is $z = 1020$, $x_{12} = 10$, $x_{13} = 25$, $x_{21} = 45$, $x_{23} = 5$, $x_{32} = 10$, $x_{34} = 30$. Figure 1 is a graphical representation of the Powerco problem and its optimal solution. The variable x_{ij} is represented by a line, or arc, joining the i th supply point (plant i) and the j th demand point (city j).

General Description of a Transportation Problem

In general, a transportation problem is specified by the following information:

1 A set of m supply points from which a good is shipped. Supply point i can supply at

most s_i units. In the Powerco example, $m = 3$, $s_1 = 35$, $s_2 = 50$, and $s_3 = 40$.

2 A set of n demand points to which the good is shipped. Demand point j must receive at least d_j units of the shipped good. In the Powerco example, $n = 4$, $d_1 = 45$, $d_2 = 20$, $d_3 = 30$, and $d_4 = 30$.

3 Each unit produced at supply point i and shipped to demand point j incurs a variable cost of c_{ij} . In the Powerco example, $c_{12} = 6$.

Let

x_{ij} = number of units shipped from supply point i to demand point j

then the general formulation of a transportation problem is

$$i = 1, \dots, m \quad j = 1, \dots, n$$

$$\min \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$i = 1, \dots, m \quad j = 1, \dots, n$$

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$$j = 1, \dots, n$$

s.t. $\sum_{j=1}^n x_{ij} \leq s_i \quad (i = 1, 2, \dots, m)$ (Supply constraints)

$$j = 1, \dots, n$$

$$i = 1, \dots, m$$

s.t. $\sum_{i=1}^m x_{ij} \geq d_j \quad (j = 1, 2, \dots, n)$

$$i = 1, \dots, m$$

(Demand constraints)

$$x_{ij} \geq 0 \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, n)$$

If a problem has the constraints given in (1) and is a maximization problem, then it is still a transportation problem (see Problem 7 at the end of this section). If

$$i = 1, \dots, m \quad j = 1, \dots, n$$

$$\sum_{i=1}^m s_i = \sum_{j=1}^n d_j$$

$$i = 1, \dots, m \quad j = 1, \dots, n$$

then total supply equals total demand, and the problem is said to be a balanced transportation problem.

For the Powerco problem, total supply and total demand both equal 125, so this is a balanced transportation problem. In a balanced transportation problem, all the constraints must be binding. For example, in the Powerco problem, if any supply constraint were non-binding, then the remaining available power would not be sufficient to meet the needs of all four cities. For a balanced transportation problem, (1) may be written as

$$i = 1, \dots, m \quad j = 1, \dots, n$$

$$\min \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$i = 1, \dots, m \quad j = 1, \dots, n$$

$$j = 1, \dots, n$$

s.t. $\sum_{j=1}^n x_{ij} = s_i \quad (i = 1, 2, \dots, m)$ (Supply constraints)

$$j = 1, \dots, n$$

$$i = 1, \dots, m$$

s.t. $\sum_{i=1}^m x_{ij} \leq d_j \quad (j = 1, 2, \dots, n)$

$i \geq 1$

(Demand constraints)

$x_{ij} \geq 0 \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, n)$

Later in this chapter, we will see that it is relatively simple to find a basic feasible solution for a balanced transportation problem. Also, simplex pivots for these problems do not involve multiplication and reduce to additions and subtractions. For these reasons, it is desirable to formulate a transportation problem as a balanced transportation problem.

Balancing a Transportation Problem

If Total Supply Exceeds Total Demand

If total supply exceeds total demand, we can balance a transportation problem by creating a dummy demand point that has a demand equal to the amount of excess supply. Because shipments to the dummy demand point are not real shipments, they are assigned a cost of zero. Shipments to the dummy demand point indicate unused supply capacity. To understand the use of a dummy demand point, suppose that in the Powerco problem, the demand for city 1 were reduced to 40 million kwh. To balance the Powerco problem, we would add a dummy demand point (point 5) with a demand of $125 - 120 = 5$ million kwh. From each plant, the cost of shipping 1 million kwh to the dummy is 0. The optimal solution to this balanced transportation problem is $z = 975$, $x_{13} = 20$, $x_{12} = 15$, $x_{21} = 40$, $x_{23} = 10$, $x_{32} = 5$, $x_{34} = 30$, and $x_{35} = 5$. Because $x_{35} = 5$, 5 million kwh of plant 3 capacity will be unused (see Figure 2).

A transportation problem is specified by the supply, the demand, and the shipping costs, so the relevant data can be summarized in a transportation tableau (see Table 2). The square, or cell, in row i and column j of a transportation tableau corresponds to the

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Supply points Demand points

$x_{11} = 0$

City 1 $d_1 = 40$

$x_{21} = 40$

$s_1 = 35$ Plant 1 $x_{31} = 0$

$x_{12} = 15$

FIGURE 2

$x_{22} = 0$ City 2 $d_2 = 20$

Graphical $x_{32} = 5$

Representation of

Unbalanced Powerco $s_2 = 50$ Plant 2 $x_{13} = 20$

Problem and Its $x_{23} = 10$

Optimal Solution (with $x_{33} = 0$ City 3 $d_3 = 30$

Dummy Demand Point)

$x_{14} = 0$

$s_3 = 40$ Plant 3 $x_{24} = 0$

$x_{34} = 30$

City 4 $d_4 = 30$

$x_{25} = 0$ $x_{15} = 0$

$x_{35} = 5$

Dummy

$d_5 = 5$

City 5

TABLE 2 c_{11} c_{12} c_{1n} Supply

A Transportation Tableau s_1

c_{21} c_{22} c_{2n}

s_2

c_{m1} c_{m2} c_{mn}

s_m

d_1 d_2 d_n

Demand

TABLE 3 City 1 City 2 City 3 City 4 Supply

Transportation Tableau 8 6 10 9

for Powerco

Plant 1 10 25 35

9 12 13 7

Plant 2 45 5 50

14 9 16 5

Plant 3 10 30 40

Demand 45 20 30 30

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variable x_{ij} . If x_{ij} is a basic variable, its value is placed in the lower left-hand corner of the ij th cell of the tableau. For example, the balanced Powerco problem and its optimal solution could be displayed as shown in Table 3. The tableau format implicitly expresses the supply and demand constraints through the fact that the sum of the variables in row i must equal s_i and the sum of the variables in column j must equal d_j .

Balancing a Transportation Problem

If Total Supply Is Less Than Total Demand

If a transportation problem has a total supply that is strictly less than total demand, then the problem has no feasible solution. For example, if plant 1 had only 30 million kwh of capacity, then a total of only 120 million kwh would be available. This amount of power would be insufficient to meet the total demand of 125 million kwh, and the Powerco problem would no longer have a feasible solution.

When total supply is less than total demand, it is sometimes desirable to allow the pos-

sibility of leaving some demand unmet. In such a situation, a penalty is often associated with unmet demand. Example 2 illustrates how such a situation can yield a balanced transportation problem.

EXAMPLE 2 Handling Shortages

Two reservoirs are available to supply the water needs of three cities. Each reservoir can supply up to 50 million gallons of water per day. Each city would like to receive 40 million gallons per day. For each million gallons per day of unmet demand, there is a penalty. At city 1, the penalty is \$20; at city 2, the penalty is \$22; and at city 3, the penalty is \$23. The cost of transporting 1 million gallons of water from each reservoir to each city is shown in Table 4. Formulate a balanced transportation problem that can be used to minimize the sum of shortage and transport costs.

Solution In this problem,

Daily supply 50 50 100 million gallons per day

Daily demand 40 40 40 120 million gallons per day

To balance the problem, we add a dummy (or shortage) supply point having a supply of 120 100 20 million gallons per day. The cost of shipping 1 million gallons from the dummy supply point to a city is just the shortage cost per million gallons for that city. Table 5 shows the balanced transportation problem and its optimal solution. Reservoir 1 should send 20 million gallons per day to city 1 and 30 million gallons per day to city 2, whereas reservoir 2 should send 10 million gallons per day to city 2 and 40 million gallons per day to city 3. Twenty million gallons per day of city 1’s demand will be unsatisfied.

TABLE 4

Shipping Costs for Reservoir

To

From City 1 City 2 City 3

Reservoir 1 \$7 \$8 \$10

Reservoir 2 \$9 \$7 \$8

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TABLE 5 City 1 City 2 City 3 Supply

Transportation Tableau 7 8 10

for Reservoir

Reservoir 1 20 30 50

9 7 8

Reservoir 2 10 40 50

20 22 23

Dummy (shortage) 20 20

Demand 40 40 40

Modeling Inventory Problems as Transportation Problems

Many inventory planning problems can be modeled as balanced transportation problems.

To illustrate, we formulate a balanced transportation model of the Sailco problem of Section 3.10.

EXAMPLE 3 Setting Up an Inventory Problem as a Transportation Problem

Sailco Corporation must determine how many sailboats should be produced during each of the next four quarters (one quarter is three months). Demand is as follows: first quarter, 40 sailboats; second quarter, 60 sailboats; third quarter, 75 sailboats; fourth quarter, 25 sailboats. Sailco must meet demand on time. At the beginning of the first quarter, Sailco has an inventory of 10 sailboats. At the beginning of each quarter, Sailco must decide how many sailboats should be produced during the current quarter. For simplicity, we assume that sailboats manufactured during a quarter can be used to meet demand for the current quarter. During each quarter, Sailco can produce up to 40 sailboats at a cost of \$400 per sailboat. By having employees work overtime during a quarter, Sailco can produce additional sailboats at a cost of \$450 per sailboat. At the end of each quarter (after production has occurred and the current quarter's demand has been satisfied), a carrying or holding cost of \$20 per sailboat is incurred. Formulate a balanced transportation problem to minimize the sum of production and inventory costs during the next four quarters.

Solution We define supply and demand points as follows:

Supply Points Point 1 □ initial inventory ($s_1 = 10$)

Supply Points Point 2 □ quarter 1 regular-time (RT) production ($s_2 = 40$)

Supply Points Point 3 □ quarter 1 overtime (OT) production ($s_3 = 150$)

Supply Points Point 4 □ quarter 2 RT production ($s_4 = 40$)

Supply Points Point 5 □ quarter 2 OT production ($s_5 = 150$)

Supply Points Point 6 □ quarter 3 RT production ($s_6 = 40$)

Supply Points Point 7 □ quarter 3 OT production ($s_7 = 150$)

Supply Points Point 8 □ quarter 4 RT production ($s_8 = 40$)

Supply Points Point 9 □ quarter 4 OT production ($s_9 = 150$)

There is a supply point corresponding to each source from which demand for sailboats can be met:

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Demand Points Point 1 □ quarter 1 demand ($d_1 = 40$)

Demand Points Point 2 □ quarter 2 demand ($d_2 = 60$)

Demand Points Point 3 □ quarter 3 demand ($d_3 = 75$)

Demand Points Point 4 □ quarter 4 demand ($d_4 = 25$)

Demand Points Point 5 □ dummy demand point ($d_5 = 770 - 200 = 570$)

A shipment from, say, quarter 1 RT to quarter 3 demand means producing 1 unit on regular time during quarter 1 that is used to meet 1 unit of quarter 3's demand. To determine, say, c_{13} , observe that producing 1 unit during quarter 1 RT and using that unit to meet quarter 3 demand incurs a cost equal to the cost of producing 1 unit on quarter 1 RT plus the cost of holding a unit in inventory for $3 - 1 = 2$ quarters. Thus, $c_{13} = 400 + 2(20) = 440$. Because there is no limit on the overtime production during any quarter, it is not clear what value should be chosen for the supply at each overtime production point. Total de-

mand ≤ 200 , so at most $200 - 10 = 190$ (≤ 10 is for initial inventory) units will be produced during any quarter. Because 40 units must be produced on regular time before any units are produced on overtime, overtime production during any quarter will never exceed $190 - 40 = 150$ units. Any unused overtime capacity will be “shipped” to the dummy demand point. To ensure that no sailboats are used to meet demand during a quarter prior to their production, a cost of M (M is a large positive number) is assigned to any cell that corresponds to using production to meet demand for an earlier quarter.

TABLE 6 1 2 3 4 Dummy Supply

Transportation Tableau

0 20 40 60 0

for Sailco

Initial 10 10

400 420 440 460 0

Qtr 1 RT 30 10 40

450 470 490 510 0

Qtr 1 OT 150 150

M 400 420 440 0

Qtr 2 RT 40 40

M 450 470 490 0

Qtr 2 OT 10 140 150

M M 400 420 0

Qtr 3 RT 40 40

M M 450 470 0

Qtr 3 OT 35 115 150

M M M 400 0

Qtr 4 RT 25 15 40

M M M 450 0

Qtr 4 OT 150 150

Demand 40 60 75 25 570

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Total supply ≤ 770 and total demand ≤ 200 , so we must add a dummy demand point with a demand of $770 - 200 = 570$ to balance the problem. The cost of shipping a unit from any supply point to the dummy demand point is 0.

Combining these observations yields the balanced transportation problem and its optimal solution shown in Table 6. Thus, Sailco should meet quarter 1 demand with 10 units of initial inventory and 30 units of quarter 1 RT production; quarter 2 demand with 10 units of quarter 1 RT, 40 units of quarter 2 RT, and 10 units of quarter 2 OT production; quarter 3 demand with 40 units of quarter 3 RT and 35 units of quarter 3 OT production; and finally, quarter 4 demand with 25 units of quarter 4 RT production.

In Problem 12 at the end of this section, we show how this formulation can be modified to incorporate other aspects of inventory problems (backlogged demand, perishable inventory, and so on).

Solving Transportation Problems on the Computer

To solve a transportation problem with LINDO, type in the objective function, supply constraints, and demand constraints. Other menu-driven programs are available that accept the shipping costs, supply values, and demand values. From these values, the program can generate the objective function and constraints.

LINGO can be used to easily solve any transportation problem. The following LINGO

Trans.lng

model can be used to solve the Powerco example (file Trans.lng).

MODEL:

1]SETS:

2]PLANTS/P1,P2,P3/:CAP;

3]CITIES/C1,C2,C3,C4/:DEM;

4]LINKS(PLANTS,CITIES):COST,SHIP;

5]ENDSETS

6]MIN=@SUM(LINKS:COST*SHIP);

7]@FOR(CITIES(J):

8]@SUM(PLANTS(I):SHIP(I,J))>DEM(J));

9]@FOR(PLANTS(I):

10]@SUM(CITIES(J):SHIP(I,J))<CAP(I));

11]DATA:

12]CAP=35,50,40;

13]DEM=45,20,30,30;

14]COST=8,6,10,9,

15]9,12,13,7,

16]14,9,16,5;

17]ENDDATA

END

Lines 1-5 define the SETS needed to generate the objective function and constraints.

In line 2, we create the three power plants (the supply points) and specify that each has a capacity (given in the DATA section). In line 3, we create the four cities (the demand points) and specify that each has a demand (given in the DATA section). The LINK statement in line 4 creates a LINK(I,J) as I runs over all PLANTS and J runs over all CITIES.

Thus, objects LINK(1,1), LINK (1,2), LINK(1,3), LINK(1,4), LINK(2,1), LINK (2,2),

LINK(2,3), LINK(2,4), LINK(3,1), LINK (3,2), LINK(3,3), LINK(3,4) are created and

stored in this order. Attributes with multiple subscripts are stored so that the rightmost subscripts advance most rapidly. Each LINK has two attributes: a per-unit shipping cost [(COST), given in the DATA section] and the amount shipped (SHIP), for which LINGO will solve.

Line 6 creates the objective function. We sum over all links the product of the unit shipping cost and the amount shipped. Using the @FOR and @SUM operators, lines 7–8 generate all demand constraints. They ensure that for each city, the sum of the amount shipped into the city will be at least as large as the city’s demand. Note that the extra parenthesis after SHIP(I,J) in line 8 is to close the @SUM operator, and the extra parenthesis after DEM(J) is to close the @FOR operator. Using the @FOR and @SUM operators, lines 9–10 generate all supply constraints. They ensure that for each plant, the total shipped out of the plant will not exceed the plant’s capacity. Lines 11–17 contain the data needed for the problem. Line 12 defines each plant’s capacity, and line 13 defines each city’s demand. Lines 14–16 contain the unit shipping cost from each plant to each city. These costs correspond to the ordering of the links described previously. ENDDATA ends the data section, and END ends the program. Typing GO will solve the problem.

This program can be used to solve any transportation problem. If, for example, we wanted to solve a problem with 15 supply points and 10 demand points, we would change line 2 to create 15 supply points and line 3 to create 10 demand points. Moving to line 12, we would type in the 15 plant capacities. In line 13, we would type in the demands for the 10 demand points. Then in line 14, we would type in the 150 shipping costs. Observe that the part of the program (lines 6–10) that generates the objective function and constraints remains unchanged! Notice also that our LINGO formulation does not require that the transportation problem be balanced.

Obtaining LINGO Data from an Excel Spreadsheet

Often it is easier to obtain data for a LINGO model from a spreadsheet. For example, shipping costs for a transportation problem may be the end result of many computations. As an example, suppose we have created the capacities, demands, and shipping costs for Powerco.xls the Powerco model in the file Powerco.xls (see Figure 3). We have created capacities in the cell range F9:F11 and named the range Cap. As you probably know, you can name a range of cells in Excel by selecting the range and clicking in the name box in the upper left-hand corner of your spreadsheet. Then type the range name and hit the Enter key. In a similar fashion, name the city demands (in cells B12:E12) with the name Demand and the unit shipping costs (in cells B4:E6) with the name Costs.

FIGURE 3

A B C D E F G H

1 OPTIMAL SOLUTION FOR POWERCO COSTS

2 COSTS CITY 1020

3 PLANT 1 2 3 4

4 1 8 6 10 9

5 2 9 12 13 7

6 3 14 9 16 5

7 SHIPMENTS CITY SHIPPED SUPPLIES

8 PLANT 1 2 3 4

9 1 0 10 25 0 35 <= 35

10 2 45 0 5 0 50 <= 50

11 3 0 10 0 30 40 <= 40

12 RECEIVED 45 20 30 30

13 >= >= >= >=

14 DEMANDS 45 20 30 30

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Using an @OLE statement, LINGO can read from a spreadsheet the values of data that are defined in the Sets portion of a program. The LINGO program (see file Transp-Transpspread.lng spread.lng) needed to read our input data from the Powerco.xls file is shown below.

MODEL:

SETS:

PLANTS/P1,P2,P3/:CAP;

CITIES/C1,C2,C3,C4/:DEM;

LINKS(PLANTS,CITIES):COST,SHIP;

ENDSETS

MIN=@SUM(LINKS:COST*SHIP);

@FOR(CITIES(J):

@SUM(PLANTS(I):SHIP(I,J))>DEM(J));

@FOR(PLANTS(I);

@SUM(CITIES(J):SHIP(I,J))<CAP(I));

DATA:

CAP, DEM, COST=@OLE('C:\MPROG\POWERCO.XLS','Cap','Demand','Costs');

ENDDATA

END

The key statement is

CAP, DEM, COST=@OLE('C:\MPROG\POWERCO.XLS','Cap','Demand','Costs');

This statement reads the defined data sets CAP, DEM, and COSTS from the Powerco.xls spreadsheet. Note that the full path location of our Excel file (enclosed in single quotes) must be given first followed by the spreadsheet range names that contain the needed data.

The range names are paired with the data sets in the order listed. Therefore, CAP values are found in range Cap and so on. The @OLE statement is very powerful, because a spreadsheet will usually greatly simplify the creation of data for a LINGO program.

Spreadsheet Solution of Transportation Problems

Powerco.xls In the file Powerco.xls, we show how easy it is to use the Excel Solver to find the optimal solution to a transportation problem. After entering the plant capacities, city demands, and unit shipping costs as shown, we enter trial values of the units shipped from each

plant to each city in the range B9:E11. Then we proceed as follows:

Step 1 Compute the total amount shipped out of each city by copying from F9 to F10:F11 the formula

`=SUM(B9:E9)`

Step 2 Compute the total received by each city by copying from B12 to C12:E12 the formula

`=SUM(B9:B11)`

Step 3 Compute the total shipping cost in cell F2 with the formula

`=SUMPRODUCT(B9:E11,Costs)`

Note that the `=SUMPRODUCT` function works on rectangles as well as rows or columns of numbers. Also, we have named the range of unit shipping costs (B4:E6) as **COSTS**.

Step 4 We now fill in the Solver window shown in Figure 4. We minimize total shipping costs (F2) by changing units shipped from each plant to each city (B9:E11). We constrain amount received by each city (B12:E12) to be at least each city's demand (range name Demand). We constrain the amount shipped out of each plant (F9:F11) to be at most each plant's capacity (range name Cap). After checking the Assume Nonnegative option and Assume Linear Model option, we obtain the optimal solution shown in Figure 3. Note, of course, that the objective function of the optimal solution found by Excel equals the ob-

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FIGURE 4

jective function value found by LINGO and our hand solution. If the problem had multiple optimal solutions, then it is possible that the values of the shipments found by LINGO, Excel, and our hand solution might be different.

PROBLEMS

Group A

1 A company supplies goods to three customers, who each overtime production is limited to 100 pairs. It costs \$1 per

require 30 units. The company has two warehouses. month to hold a pair of shoes in inventory. Formulate a

Warehouse 1 has 40 units available, and warehouse 2 has 30 balanced transportation problem to minimize the total cost

units available. The costs of shipping 1 unit from warehouse of meeting the next six months of demand on time.

to customer are shown in Table 7. There is a penalty for each

4 Steelco manufactures three types of steel at different

unmet customer unit of demand: With customer 1, a penalty

plants. The time required to manufacture 1 ton of steel

cost of \$90 is incurred; with customer 2, \$80; and with

(regardless of type) and the costs at each plant are shown in

customer 3, \$110. Formulate a balanced transportation

Table 8. Each week, 100 tons of each type of steel (1, 2, and

problem to minimize the sum of shortage and shipping costs.

3) must be produced. Each plant is open 40 hours per week.

2 Referring to Problem 1, suppose that extra units could a Formulate a balanced transportation problem to min-

be purchased and shipped to either warehouse for a total imize the cost of meeting Steelco's weekly requirements.

cost of \$100 per unit and that all customer demand must be b Suppose the time required to produce 1 ton of steel

met. Formulate a balanced transportation problem to depends on the type of steel as well as on the plant at

minimize the sum of purchasing and shipping costs. which it is produced (see Table 9, page 372).

Could a

3 A shoe company forecasts the following demands during transportation problem still be formulated?

the next six months: month 1—200; month 2—260; month 5 A hospital needs to purchase 3 gallons of a perishable

3—240; month 4—340; month 5—190; month 6—150. It medicine for use during the current month and 4 gallons for

costs \$7 to produce a pair of shoes with regular-time labor use during the next month. Because the medicine is

(RT) and \$11 with overtime labor (OT). During each month,

regular production is limited to 200 pairs of shoes, and

TA B L E 8

TA B L E 7 Cost (\$)

Time

To Plant Steel 1 Steel 2 Steel 3 (minutes)

From Customer 1 Customer 2 Customer 3

1 60 40 28 20

Warehouse 1 \$15 \$35 \$25 2 50 30 30 16

Warehouse 2 \$10 \$50 \$40 3 43 20 20 15

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TA B L E 9 TA B L E 12

Time (minutes) To (\$)

Plant Steel 1 Steel 2 Steel 3 From (\$)

1 15 12 15 Field 1 1 2

2 15 15 20 Field 2 2 1

3 10 10 15

TA B L E 13

perishable, it can only be used during the month of purchase. Project (\$)

Two companies (Daisy and Laroach) sell the medicine. The Auditor 1 2 3

medicine is in short supply. Thus, during the next two

months, the hospital is limited to buying at most 5 gallons 1 120 150 190

from each company. The companies charge the prices shown in Table 10. Formulate a balanced transportation model to minimize the cost of purchasing the needed medicine.

6 A bank has two sites at which checks are processed. Site 1 can process 10,000 checks per day, and site 2 can process

8 The Ayatola Oil Company controls two oil fields. Field 1 can produce up to 40 million barrels of oil per day, and checks: vendor, salary, and personal. The processing cost field 2 can produce up to 50 million barrels of oil per day.

per check depends on the site (see Table 11). Each day, At field 1, it costs \$3 to extract and refine a barrel of oil; at 5,000 checks of each type must be processed. Formulate a field 2, the cost is \$2. Ayatola sells oil to two countries:

balanced transportation problem to minimize the daily cost England and Japan. The shipping cost per barrel is shown of processing checks.

in Table 12. Each day, England is willing to buy up to 40

7† The U.S. government is auctioning off oil leases at two million barrels (at \$6 per barrel), and Japan is willing to

sites: 1 and 2. At each site, 100,000 acres of land are to be buy up to 30 million barrels (at \$6.50 per barrel). Formulate

auctioned. Cliff Ewing, Blake Barnes, and Alexis Pickens are a balanced transportation problem to maximize Ayatola's

bidding for the oil. Government rules state that no bidder can profits.

receive more than 40% of the land being auctioned. Cliff has

9 For the examples and problems of this section, discuss

bid \$1,000/acre for site 1 land and \$2,000/acre for site 2 land.

whether it is reasonable to assume that the proportionality

Blake has bid \$900/acre for site 1 land and \$2,200/acre for site assumption holds for the objective function.

2 land. Alexis has bid \$1,100/acre for site 1 land and

\$1,900/acre for site 2 land. Formulate a balanced transportation 10 Touche Young has three auditors. Each can work as

model to maximize the government's revenue. many as 160 hours during the next month, during which

time three projects must be completed. Project 1 will take

130 hours; project 2, 140 hours; and project 3, 160 hours.

The amount per hour that can be billed for assigning each

TA B L E 10 auditor to each project is given in Table 13. Formulate a

balanced transportation problem to maximize total billings

Current Month's Next Month's during the next month.

Price per Price per

Company Gallon (\$) Gallon (\$)

Group B

Daisy 800 720

Laroach 710 750 11† Paperco recycles newsprint, uncoated paper, and

coated paper into recycled newsprint, recycled uncoated

paper, and recycled coated paper. Recycled newsprint can

be produced by processing newsprint or uncoated paper.

Recycled coated paper can be produced by recycling any

TA B L E 11 type of paper. Recycled uncoated paper can be produced by

Site (¢)

processing uncoated paper or coated paper. The process

used to produce recycled newsprint removes 20% of the

Checks 1 2 input's pulp, leaving 80% of the input's pulp for recycled

Vendor 5 3 paper. The process used to produce recycled coated paper

removes 10% of the input's pulp. The process used to

Salary 4 4 produce recycled uncoated paper removes 15% of the input's

Personal 2 5 pulp. The purchasing costs, processing costs, and availability

of each type of paper are shown in Table 14. To meet demand,

† ‡

This problem is based on Jackson (1980). This problem is based on Glassey and Gupta (1974).

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TA B L E 14 Paperco must produce at least 250 tons of recycled newsprint

pulp, at least 300 tons of recycled uncoated paper pulp, and

Purchase Processing

Cost per Ton Cost per Ton at least 150 tons of recycled coated paper pulp. Formulate

of Pulp (\$) of Input (\$) Availability a balanced transportation problem that can be used to

minimize the cost of meeting Paperco's demands.

Newsprint 10 500

12 Explain how each of the following would modify the

Coated paper 19 300

formulation of the Sailco problem as a balanced trans-

Uncoated paper 18 200 portation problem:

NP used for RNP 3 a Suppose demand could be backlogged at a cost of

NP used for RCP 4 \$30/sailboat/month. (Hint: Now it is permissible to ship

UCP used for RNP 4 from, say, month 2 production to month 1 demand.)

UCP used for RUP 1 b If demand for a sailboat is not met on time, the sale

UCP used for RCP 6 is lost and an opportunity cost of \$450 is incurred.

CP used for RUP 5 c Sailboats can be held in inventory for a maximum of two months.

CP used for RCP 3

d At a cost of \$440/sailboat, Sailco can purchase up to 10 sailboats/month from a subcontractor.

7.2 Finding Basic Feasible Solutions for Transportation Problems

Consider a balanced transportation problem with m supply points and n demand points. From (2), we see that such a problem contains $m \times n$ equality constraints. From our experience with the Big M method and the two-phase simplex method, we know it is difficult to find a bfs if all of an LP's constraints are equalities. Fortunately, the special structure of a balanced transportation problem makes it easy for us to find a bfs.

Before describing three methods commonly used to find a bfs to a balanced transportation problem, we need to make the following important observation. If a set of values for the x_{ij} 's satisfies all but one of the constraints of a balanced transportation problem, then the values for the x_{ij} 's will automatically satisfy the other constraint. For example, in the Powerco problem, suppose a set of values for the x_{ij} 's is known to satisfy all the constraints with the exception of the first supply constraint. Then this set of x_{ij} 's must supply $d_1 \times d_2 \times d_3 \times d_4 = 125$ million kwh to cities 1–4 and supply $s_2 \times s_3 \times 125 \times s_1 = 90$ million kwh from plants 2 and 3. Thus, plant 1 must supply $125 \times (125 \times s_1) = 35$ million kwh, so the x_{ij} 's must also satisfy the first supply constraint. The preceding discussion shows that when we solve a balanced transportation problem, we may omit from consideration any one of the problem's constraints and solve an LP having $m \times n - 1$ constraints. We (arbitrarily) assume that the first supply constraint is omitted from consideration.

In trying to find a bfs to the remaining $m \times n - 1$ constraints, you might think that any collection of $m \times n - 1$ variables would yield a basic solution. Unfortunately, this is not the case. For example, consider (3), a balanced transportation problem. (We omit the costs because they are not needed to find a bfs.)

4

(3)

5

3 2 4

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In matrix form, the constraints for this balanced transportation problem may be written as

$x_{11} \ x_{12} \ x_{13} \ x_{21} \ x_{22} \ x_{23}$

x_{11}

1 1 1 0 0 4

x_{12}

0 0 0 1 1 5

x_{13}

1 0 0 1 0 0 3 (3)

x_{21}

0 1 0 0 1 0 2

x22

0 0 1 0 0 1 4

x23

After dropping the first supply constraint, we obtain the following linear system:

$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 & 1 & 1 & 5 \\ 1 & 0 & 0 & 1 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{21} \\ x_{22} \\ x_{23} \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$

x11

0 0 0 1 1 1 x12 5

1 0 0 1 0 0 x13 3

$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 & 1 & 1 & 5 \\ 1 & 0 & 0 & 1 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{21} \\ x_{22} \\ x_{23} \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$

0 1 0 0 1 0 x21 2

0 0 1 0 0 1 x22 4

x23

A basic solution to (3) must have four basic variables. Suppose we try $BV = \{x_{11}, x_{12}, x_{21}, x_{22}\}$. Then

0 0 1 1

1 0 1 0

$B = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

0 1 0 1

0 0 0 0

For $\{x_{11}, x_{12}, x_{21}, x_{22}\}$ to yield a basic solution, it must be possible to use EROs to transform B to I_4 . Because $\text{rank } B = 3$ and EROs do not change the rank of a matrix, there is no way that EROs can be used to transform B into I_4 . Thus, $BV = \{x_{11}, x_{12}, x_{21}, x_{22}\}$ cannot yield a basic solution to (3). Fortunately, the simple concept of a loop may be used to determine whether an arbitrary set of $m - n + 1$ variables yields a basic solution to a balanced transportation problem.

DEFINITION ■ An ordered sequence of at least four different cells is called a loop if

1 Any two consecutive cells lie in either the same row or same column

2 No three consecutive cells lie in the same row or column

3 The last cell in the sequence has a row or column in common with the first cell

in the sequence ■

In the definition of a loop, the first cell is considered to follow the last cell, so the loop may be thought of as a closed path. Here are some examples of the preceding definition:

Figure 5 represents the loop (2, 1)–(2, 4)–(4, 4)–(4, 1). Figure 6 represents the loop

(1, 1)–(1, 2)–(2, 2)–(2, 3)–(4, 3)–(4, 5)–(3, 5)–(3, 1). In Figure 7, the path (1, 1)–(1, 2)–

(2, 3)–(2, 1) does not represent a loop, because (1, 2) and (2, 3) do not lie in the same

row or column. In Figure 8, the path (1, 2)–(1, 3)–(1, 4)–(2, 4)–(2, 2) does not represent

a loop, because (1, 2), (1, 3), and (1, 4) all lie in the same row.

Theorem 1 (which we state without proof) shows why the concept of a loop is important.

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FIGURE 5

FIGURE 6

FIGURE 7

FIGURE 8

THEOREM 1

In a balanced transportation problem with m supply points and n demand points, the cells corresponding to a set of $m \times n - 1$ variables contain no loop if and only if the $m \times n - 1$ variables yield a basic solution.

Theorem 1 follows from the fact that a set of $m \times n - 1$ cells contains no loop if and only if the $m \times n - 1$ columns corresponding to these cells are linearly independent. Because $(1, 1)-(1, 2)-(2, 2)-(2, 1)$ is a loop, Theorem 1 tells us that $\{x_{11}, x_{12}, x_{22}, x_{21}\}$ cannot yield a basic solution for (3). On the other hand, no loop can be formed with the cells $(1, 1)-(1, 2)-(1, 3)-(2, 1)$, so $\{x_{11}, x_{12}, x_{13}, x_{21}\}$ will yield a basic solution to (3).

We are now ready to discuss three methods that can be used to find a basic feasible solution for a balanced transportation problem:

- 1 northwest corner method
- 2 minimum-cost method
- 3 Vogel's method

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Northwest Corner Method for Finding a Basic Feasible Solution

To find a bfs by the northwest corner method, we begin in the upper left (or northwest) corner of the transportation tableau and set x_{11} as large as possible. Clearly, x_{11} can be no larger than the smaller of s_1 and d_1 . If $x_{11} = s_1$, cross out the first row of the transportation tableau; this indicates that no more basic variables will come from row 1. Also change d_1 to $d_1 - s_1$. If $x_{11} = d_1$, cross out the first column of the transportation tableau; this indicates that no more basic variables will come from column 1. Also change s_1 to $s_1 - d_1$. If $x_{11} = s_1 = d_1$, cross out either row 1 or column 1 (but not both). If you cross out row 1, change d_1 to 0; if you cross out column 1, change s_1 to 0.

Continue applying this procedure to the most northwest cell in the tableau that does not lie in a crossed-out row or column. Eventually, you will come to a point where there is only one cell that can be assigned a value. Assign this cell a value equal to its row or column demand, and cross out both the cell's row and column. A basic feasible solution has now been obtained.

We illustrate the use of the northwest corner method by finding a bfs for the balanced transportation problem in Table 15. (We do not list the costs because they are not needed to apply the algorithm.) We indicate the crossing out of a row or column by placing an $\square$ by the row's supply or column's demand.

To begin, we set $x_{11} = \min\{5, 2\} = 2$. Then we cross out column 1 and change s_1 to $5 - 2 = 3$. This yields Table 16. The most northwest remaining variable is x_{12} . We set $x_{12} = \min\{3, 4\} = 3$. Then we cross out row 1 and change d_2 to $4 - 3 = 1$. This yields

Table 17. The most northwest available variable is now x_{22} . We set $x_{22} = \min\{1, 1\} = 1$. Because both the supply and demand corresponding to the cell are equal, we may cross out either row 2 or column 2 (but not both). For no particular reason, we choose to cross out row 2. Then d_2 must be changed to $1 - 1 = 0$. The resulting tableau is Table 18. At the next step, this will lead to a degenerate bfs.

TABLE 15

5
1
3
2 4 2 1

TABLE 16

2 3
1
3
 $\square$ 4 2 1

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TABLE 17

2 3 $\square$
1
3
 $\square$ 1 2 1

TABLE 18

2 3 $\square$
1 $\square$
3
 $\square$ 0 2 1

The most northwest available cell is now x_{32} , so we set $x_{32} = \min\{3, 0\} = 0$. Then we cross out column 2 and change s_3 to $3 - 0 = 3$. The resulting tableau is Table 19. We now set $x_{33} = \min\{3, 2\} = 2$. Then we cross out column 3 and reduce s_3 to $3 - 2 = 1$. The resulting tableau is Table 20. The only available cell is x_{34} . We set $x_{34} = \min\{1, 1\} = 1$. Then we cross out row 3 and column 4. No cells are available, so we are finished. We have obtained the bfs $x_{11} = 2$, $x_{12} = 3$, $x_{22} = 1$, $x_{32} = 0$, $x_{33} = 2$, $x_{34} = 1$.

Why does the northwest corner method yield a bfs? The method ensures that no basic variable will be assigned a negative value (because no right-hand side ever becomes nega-

TABLE 19

2 3 $\square$
1 $\square$

0 3

□ □ 2 1

T A B L E 20

2 3 □

1 □

0 2 1

□ □ □ 1

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tive) and also that each supply and demand constraint is satisfied (because every row and column is eventually crossed out). Thus, the northwest corner method yields a feasible solution. To complete the northwest corner method, $m - n$ rows and columns must be crossed out. The last variable assigned a value results in a row and column being crossed out, so the northwest corner method will assign values to $m - n + 1$ variables. The variables chosen by the northwest corner method cannot form a loop, so Theorem 1 implies that the northwest corner method must yield a bfs.

Minimum-Cost Method for Finding a Basic Feasible Solution

The northwest corner method does not utilize shipping costs, so it can yield an initial bfs that has a very high shipping cost. Then determining an optimal solution may require several pivots. The minimum-cost method uses the shipping costs in an effort to produce a bfs that has a lower total cost. Hopefully, fewer pivots will then be required to find the problem's optimal solution.

To begin the minimum-cost method, find the variable with the smallest shipping cost (call it x_{ij}). Then assign x_{ij} its largest possible value, $\min\{s_i, d_j\}$. As in the northwest corner method, cross out row i or column j and reduce the supply or demand of the noncrossed-out row or column by the value of x_{ij} . Then choose from the cells that do not lie in a crossed-out row or column the cell with the minimum shipping cost and repeat the procedure. Continue until there is only one cell that can be chosen. In this case, cross out both the cell's row and column. Remember that (with the exception of the last variable) if a variable satisfies both a supply and demand constraint, only cross out a row or column, not both. To illustrate the minimum cost method, we find a bfs for the balanced transportation problem in Table 21. The variable with the minimum shipping cost is x_{22} . We set $x_{22} = \min\{10, 8\} = 8$. Then we cross out column 2 and reduce s_2 to $10 - 8 = 2$ (Table 22). We could now choose either x_{11} or x_{21} (both having shipping costs of 2). We arbitrarily choose x_{21} and set $x_{21} = \min\{2, 12\} = 2$. Then we cross out row 2 and change d_1 to $12 - 2 = 10$ (Table 23). Now we set $x_{11} = \min\{5, 10\} = 5$, cross out row 1, and change d_1 to $10 - 5 = 5$ (Table 24). The minimum cost that does not lie in a crossed-out row or column is x_{31} . We set $x_{31} = \min\{15, 5\} = 5$, cross out column 1, and reduce s_3 to $15 - 5 = 10$ (Table 25). Now we set $x_{33} = \min\{10, 4\} = 4$, cross out column 3, and reduce s_3 to $10 - 4 = 6$ (Table 26). The only cell that we can choose is x_{34} . We set $x_{34} = \min\{6, 6\}$ and cross out both row 3 and column 4. We have now obtained the bfs: $x_{11} = 5$, $x_{21} = 2$, $x_{22} = 8$, $x_{31} = 5$, $x_{33} = 4$, and $x_{34} = 6$.

Because the minimum-cost method chooses variables with small shipping costs to be basic variables, you might think that this method would always yield a bfs with a relatively low total shipping cost. The following problem shows how the minimum-cost method can be fooled into choosing a relatively high-cost bfs.

TABLE 21 2 3 5 6

5

2 1 3 5

10

3 8 4 6

15

12 8 4 6

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TABLE 22 2 3 5 6

5

2 1 3 5

8 2

3 8 4 6

15

12 □ 4 6

TABLE 23

2 3 5 6

5

2 1 3 5

2 8 □

3 8 4 6

15

10 □ 4 6

TABLE 24 2 3 5 6

5

2 1 3 5

2 8 □

3 8 4 6

15

5 □ 4 6

TABLE 25

2 3 5 6

5

2 1 3 5

2 8 □

3 8 4 6

5 10

□ □ 4 6

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TA B L E 26 2 3 5 6

5

2 1 3 5

2 8 □

3 8 4 6

5 4 6

□ □ □ 6

TA B L E 27 6 7 8

10

15 80 78

15

15 5 5

If we apply the minimum-cost method to Table 27, we set $x_{11} = 10$ and cross out row 1. This forces us to make x_{22} and x_{23} basic variables, thereby incurring their high shipping costs. Thus, the minimum-cost method will yield a costly bfs. Vogel's method for finding a bfs usually avoids extremely high shipping costs.

Vogel's Method for Finding a Basic Feasible Solution

Begin by computing for each row (and column) a "penalty" equal to the difference between the two smallest costs in the row (column). Next find the row or column with the largest penalty. Choose as the first basic variable the variable in this row or column that has the smallest shipping cost. As described in the northwest corner and minimum-cost methods, make this variable as large as possible, cross out a row or column, and change the supply or demand associated with the basic variable. Now recompute new penalties (using only cells that do not lie in a crossed-out row or column), and repeat the procedure until only one uncrossed cell remains. Set this variable equal to the supply or demand associated with the variable, and cross out the variable's row and column. A bfs has now been obtained.

We illustrate Vogel's method by finding a bfs to Table 28. Column 2 has the largest penalty, so we set $x_{12} = \min\{10, 5\} = 5$. Then we cross out column 2 and reduce s_1 to $10 - 5 = 5$. After recomputing the new penalties (observe that after a column is crossed out, the column penalties will remain unchanged), we obtain Table 29. The largest penalty now occurs in column 3, so we set $x_{13} = \min\{5, 5\}$. We may cross out either row 1 or column 3. We arbitrarily choose to cross out column 3, and we reduce s_1 to $5 - 5 = 0$. Because each row has only one cell that is not crossed out, there are no row penalties.

The resulting tableau is Table 30. Column 1 has the only (and, of course, the largest) penalty. We set $x_{11} = \min\{0, 15\} = 0$, cross out row 1, and change d_1 to $15 - 0 = 15$. The result is Table 31. No penalties can be computed, and the only cell that is not in a crossed-out row or column is x_{21} . Therefore, we set $x_{21} = 15$ and cross out both column 1 and row 2. Our application of Vogel's method is complete, and we have obtained the bfs: $x_{11} = 0$, $x_{12} = 5$, $x_{13} = 5$, and $x_{21} = 15$ (see Table 32).

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TABLE 28 Supply Row Penalty

6 7 8

10 7-6=1

15 80 78

15 78 - 15 = 63

Demand 15 5 5

Column 15 - 6 = 9 80 - 7 = 73 78 - 8 = 70

Penalty

TABLE 29 Supply Row Penalty

6 7 8

5 5 8-6=2

15 80 78

15 78 - 15 = 63

Demand 15 5

Column

Penalty 9 --- 70

TABLE 30 Supply Row Penalty

6 7 8

5 5 0 ---

15 80 78

15 ---

Demand 15 5

Column 9 --- ---

Penalty

TABLE 31 Supply Row Penalty

6 7 8

0 5 5 5 ---

15 80 78

15 ---

Demand 15 5

Column

Penalty --- --- ---

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TABLE 32.6.7.8

0.5510

15.8078

15.15

15.55

Observe that Vogel's method avoids the costly shipments associated with x_{22} and x_{23} . This is because the high shipping costs resulted in large penalties that caused Vogel's method to choose other variables to satisfy the second and third demand constraints. Of the three methods we have discussed for finding a bfs, the northwest corner method requires the least effort, and Vogel's method requires the most effort. Extensive research [Glover et al. (1974)] has shown, however, that when Vogel's method is used to find an initial bfs, it usually takes substantially fewer pivots than if the other two methods had been used. For this reason, the northwest corner and minimum-cost methods are rarely used to find a basic feasible solution to a large transportation problem.

PROBLEMS

Group A

1 Use the northwest corner method to find a bfs for 3 Use Vogel's method to find a bfs for Problems 5 and 6

Problems 1, 2, and 3 of Section 7.1. of Section 7.1.

2 Use the minimum-cost method to find a bfs for Problems 4 How should Vogel's method be modified to solve a

4, 7, and 8 of Section 7.1. (Hint: For a maximization maximization problem? problem, call the minimum-cost method the maximum-profit method or the maximum-revenue method.)

7.3 The Transportation Simplex Method

In this section, we show how the simplex algorithm simplifies when a transportation problem is solved. We begin by discussing the pivoting procedure for a transportation problem. Recall that when the pivot row was used to eliminate the entering basic variable from other constraints and row 0, many multiplications were usually required. In solving a transportation problem, however, pivots require only additions and subtractions.

How to Pivot in a Transportation Problem

By using the following procedure, the pivots for a transportation problem may be performed within the confines of the transportation tableau:

Step 1 Determine (by a criterion to be developed shortly) the variable that should enter the basis.

Step 2 Find the loop (it can be shown that there is only one loop) involving the entering variable and some of the basic variables.

Step 3 Counting only cells in the loop, label those found in step 2 that are an even num-

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ber (0, 2, 4, and so on) of cells away from the entering variable as even cells. Also label those that are an odd number of cells away from the entering variable as odd cells.

Step 4 Find the odd cell whose variable assumes the smallest value. Call this value θ .

The variable corresponding to this odd cell will leave the basis. To perform the pivot, decrease the value of each odd cell by θ and increase the value of each even cell by θ . The values of variables not in the loop remain unchanged. The pivot is now complete. If $\theta = 0$, then the entering variable will equal 0, and an odd variable that has a current value of 0 will leave the basis. In this case, a degenerate bfs existed before and will result after the pivot. If more than one odd cell in the loop equals θ , you may arbitrarily choose one of these odd cells to leave the basis; again, a degenerate bfs will result.

We illustrate the pivoting procedure on the Powerco example. When the northwest corner method is applied to the Powerco example, the bfs in Table 33 is found. For this bfs, the basic variables are $x_{11} = 35$, $x_{21} = 10$, $x_{22} = 20$, $x_{23} = 20$, $x_{33} = 10$, and $x_{34} = 30$.

Suppose we want to find the bfs that would result if x_{14} were entered into the basis.

The loop involving x_{14} and some of the basic variables is

E O E O E O

(1, 4)–(3, 4)–(3, 3)–(2, 3)–(2, 1)–(1, 1)

In this loop, (1, 4), (3, 3), and (2, 1) are the even cells, and (1, 1), (3, 4), and (2, 3) are the odd cells. The odd cell with the smallest value is $x_{23} = 20$. Thus, after the pivot, x_{23} will have left the basis. We now add 20 to each of the even cells and subtract 20 from each of the odd cells. The bfs in Table 34 results. Because each row and column has as many ≤ 20 s as ≥ 20 s, the new solution will satisfy each supply and demand constraint. By choosing the smallest odd variable (x_{23}) to leave the basis, we have ensured that all variables will remain nonnegative. Thus, the new solution is feasible. There is no loop involving the cells (1, 1), (1, 4), (2, 1), (2, 2), (3, 3), and (3, 4), so the new solution is a bfs. After the pivot, the new bfs is $x_{11} = 15$, $x_{14} = 20$, $x_{21} = 30$, $x_{22} = 20$, $x_{33} = 30$, and $x_{34} = 10$, and all other variables equal 0.

TABLE 33

Northwest Corner Basic

Feasible Solution for Powerco

10 20 20 50

10 30 40

45 20 30 30

TABLE 34

New Basic Feasible Solution

After x_{14} Is Pivoted into Basis

10 + 20 20 20 – 20 50

(nonbasic)

10 + 20 30 – 20 40

45 20 30 30

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The preceding illustration of the pivoting procedure makes it clear that each pivot in a transportation problem involves only additions and subtractions. Using this fact, we can show that if all the supplies and demands for a transportation problem are integers, then the transportation problem will have an optimal solution in which all the variables are integers. Begin by observing that, by the northwest corner method, we can find a bfs in which each variable is an integer. Each pivot involves only additions and subtractions, so each bfs obtained by performing the simplex algorithm (including the optimal solution) will assign all variables integer values. The fact that a transportation problem with integer supplies and demands has an optimal integer solution is useful, because it ensures that we need not worry about whether the Divisibility Assumption is justified.

Pricing Out Nonbasic Variables (Based on Chapter 6)

To complete our discussion of the transportation simplex, we now show how to compute row 0 for any bfs. From Section 6.2, we know that for a bfs in which the set of basic variables is BV , the coefficient of the variable x_{ij} (call it c_{ij}) in the tableau's row 0 is given by $c_{ij} - c_{BV}^{-1}a_{ij} = c_{ij}$

where c_{ij} is the objective function coefficient for x_{ij} and a_{ij} is the column for x_{ij} in the original LP (we are assuming that the first supply constraint has been dropped).

Because we are solving a minimization problem, the current bfs will be optimal if all the c_{ij} 's are nonpositive; otherwise, we enter into the basis the variable with the most positive c_{ij} .

After determining c_{BV}^{-1} , we can easily determine c_{ij} . Because the first constraint has been dropped, c_{BV}^{-1} will have $m - n + 1$ elements. We write

$$c_{BV}^{-1} = [u_2 \ u_3 \ \dots \ u_m \ v_1 \ v_2 \ v_n]$$

where $u_2, u_3, \dots, u_m$ are the elements of c_{BV}^{-1} corresponding to the $m - 1$ supply constraints, and $v_1, v_2, \dots, v_n$ are the elements of c_{BV}^{-1} corresponding to the n demand constraints.

To determine c_{BV}^{-1} , we use the fact that in any tableau, each basic variable x_{ij} must have $c_{ij} = 0$. Thus, for each of the $m - n + 1$ variables in BV ,

$$c_{BV}^{-1}a_{ij} = c_{ij} = 0 \quad (4)$$

For a transportation problem, the equations in (4) are very easy to solve. To illustrate the solution of (4), we find c_{BV}^{-1} for (5), by applying the northwest corner method bfs to the Powerco problem.

8 6 10 9

35

35

9 12 13 7

10 20 20 50 (5)

14 9 16 5

10 30 40

45 20 30 30

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For this bfs, $BV = \{x_{11}, x_{21}, x_{22}, x_{23}, x_{33}, x_{34}\}$. Applying (4) we obtain

0

0

1

$c_{11} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 8 \quad v_1 = 8 \quad 0$

0

0

0

1

0

1

$c_{21} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 9 \quad u_2 = v_1 = 9 \quad 0$

0

0

0

1

0

0

$c_{22} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 12 \quad u_2 = v_2 = 12 \quad 0$

1

0

0

1

0

0

$c_{23} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 13 \quad u_2 = v_3 = 13 \quad 0$

0

1

0

0

1

0

$c_{33} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] = 16 \quad u_3 = v_3 = 16 \quad 0$

0

1

0

0

1

0

$$c_{34} = [u_2 \ u_3 \ v_1 \ v_2 \ v_3 \ v_4] \begin{bmatrix} 5 \\ 5 \\ 5 \\ 0 \end{bmatrix}$$

0

0

1

For each basic variable x_{ij} (except those having $i = 1$), we see that (4) reduces to $u_i - v_j = c_{ij}$. If we define $u_1 = 0$, we see that (4) reduces to $u_i - v_j = c_{ij}$ for all basic variables. Thus, to solve for c_{BVB}^{-1} , we must solve the following system of $m - n$ equations: $u_1 = 0$, $u_i - v_j = c_{ij}$ for all basic variables.

For (5), we find c_{BVB}^{-1} by solving

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$$u_1 - u_1 = 0 \quad (6)$$

$$u_1 - v_1 = 8 \quad (7)$$

$$u_2 - v_1 = 9 \quad (8)$$

$$u_2 - v_2 = 12 \quad (9)$$

$$u_2 - v_3 = 13 \quad (10)$$

$$u_3 - v_3 = 16 \quad (11)$$

$$u_3 - v_4 = 5 \quad (12)$$

From (7), $v_1 = 8$. From (8), $u_2 = 1$. Then (9) yields $v_2 = 11$, and (10) yields $v_3 = 12$.

From (11), $u_3 = 4$. Finally, (12) yields $v_4 = 1$. For each nonbasic variable, we now compute $c_{ij} - u_i - v_j = c_{ij}$. We obtain

$$c_{12} = 0 - 0 - 11 = -6 \quad c_{13} = 0 - 0 - 12 = -10 \quad c_{14} = 0 - 0 - 1 = -9 \quad c_{24} = 1 - 1 - 7 = -5$$

$$c_{31} = 4 - 8 - 14 = -18 \quad c_{32} = 4 - 11 - 9 = -6$$

Because c_{32} is the most positive c_{ij} , we would next enter x_{32} into the basis. Each unit of x_{32} that is entered into the basis will decrease Powerco's cost by \$6.

How to Determine the Entering Nonbasic Variable

(Based on Chapter 5)

For readers who have not covered Chapter 6, we now discuss how to determine whether a bfs is optimal, and, if it is not, how to determine which nonbasic variable should enter the basis. Let u_i ($i = 1, 2, \dots, m$) be the shadow price of the i th supply constraint, and let v_j ($j = 1, 2, \dots, n$) be the shadow price of the j th demand constraint. We assume that the first supply constraint has been dropped, so we may set $u_1 = 0$. From the definition of shadow price, if we were to increase the right-hand side of the i th supply and j th demand constraint by 1, the optimal z -value would decrease by $u_i - v_j$. Equivalently, if we were to decrease the right-hand side of the i th supply and j th demand constraint by 1, the optimal z -value would increase by $u_i - v_j$. Now suppose x_{ij} is a nonbasic variable. Should we enter x_{ij} into the basis? Observe that if we increase x_{ij} by 1, costs directly increase by c_{ij} . Also, increasing

x_{ij} by 1 means that one less unit will be shipped from supply point i and one less unit will be shipped to demand point j . This is equivalent to reducing the right-hand sides of the i th supply constraint and j th demand constraint by 1. This will increase z by $-u_i - v_j$. Thus, increasing x_{ij} by 1 will increase z by a total of $c_{ij} - u_i - v_j$. So if $c_{ij} - u_i - v_j \leq 0$ (or $u_i + v_j \geq c_{ij} \geq 0$) for all nonbasic variables, the current bfs will be optimal. If, however, a nonbasic variable x_{ij} has $c_{ij} - u_i - v_j > 0$ (or $u_i + v_j < c_{ij}$), then z can be decreased by $u_i + v_j - c_{ij}$ per unit of x_{ij} by entering x_{ij} into the basis. Thus, we may conclude that if $u_i + v_j \geq c_{ij} \geq 0$ for all nonbasic variables, then the current bfs is optimal. Otherwise, the nonbasic variable with the most positive value of $u_i + v_j - c_{ij}$ should enter the basis. How do we find the u_i 's and v_j 's? The coefficient of a nonbasic variable x_{ij} in row 0 of any tableau is the amount by which a unit increase in x_{ij} will decrease z , so we can conclude that the coefficient of any nonbasic variable (and, it turns out, any basic variable) in row 0 is $u_i + v_j - c_{ij}$. So we may solve for the u_i 's and v_j 's by solving the following system of equations: $u_1 = 0$ and $u_i + v_j - c_{ij} = 0$ for all basic variables.

To illustrate the previous discussion, consider the bfs for the Powerco problem shown in (5).

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8 6 10 9

35

35

9 12 13 7 (5)

10 20 20 50

14 9 16 5

10 30 40

45 20 30 30

We

find the u_i 's and v_j 's by solving

$$u_1 = u_1 = 0 \quad (6)$$

$$u_1 + v_1 = 8 \quad (7)$$

$$u_2 + v_1 = 9 \quad (8)$$

$$u_2 + v_2 = 12 \quad (9)$$

$$u_2 + v_3 = 13 \quad (10)$$

$$u_3 + v_3 = 16 \quad (11)$$

$$u_3 + v_4 = 5 \quad (12)$$

From (7), $v_1 = 8$. From (8), $u_2 = 1$. Then (9) yields $v_2 = 11$, and (10) yields $v_3 = 12$.

From (11), $u_3 = 4$. Finally, (12) yields $v_4 = 1$. For each nonbasic variable, we now compute $c_{ij} - u_i - v_j - c_{ij}$. We obtain

$$c_{12} - u_1 - v_2 - c_{12} = 0 - 11 - 6 - 5 = -32$$

$$c_{14} - u_1 - v_4 - c_{14} = 0 - 1 - 1 - 7 = -9$$

$$c_{31} - u_3 - v_1 - c_{31} = 4 - 8 - 14 - 2 = -20$$

$$c_{32} - u_3 - v_2 - c_{32} = 4 - 11 - 9 - 6 = -22$$

Because c_{32} is the most positive c_{ij} , we would next enter x_{32} into the basis. Each unit of x_{32} that is entered into the basis will decrease Powerco's cost by \$6.

We can now summarize the procedure for using the transportation simplex to solve a transportation (min) problem.

Summary and Illustration

of the Transportation Simplex Method

Step 1 If the problem is unbalanced, balance it.

Step 2 Use one of the methods described in Section 7.2 to find a bfs.

Step 3 Use the fact that $u_1 = 0$ and $u_i + v_j = c_{ij}$ for all basic variables to find the $[u_1 \ u_2 \ \dots \ u_m \ v_1 \ v_2 \ \dots \ v_n]$ for the current bfs.

Step 4 If $u_i + v_j = c_{ij} = 0$ for all nonbasic variables, then the current bfs is optimal. If this is not the case, then we enter the variable with the most positive $u_i + v_j = c_{ij}$ into the basis using the pivoting procedure. This yields a new bfs.

Step 5 Using the new bfs, return to steps 3 and 4.

For a maximization problem, proceed as stated, but replace step 4 by step 4 .

Step 4 If $u_i + v_j = c_{ij} = 0$ for all nonbasic variables, then the current bfs is optimal.

Otherwise, enter the variable with the most negative $u_i + v_j = c_{ij}$ into the basis using the pivoting procedure described earlier.

7.3 The Transportation Simplex Method 387

We illustrate the procedure for solving a transportation problem by solving the Powerco problem. We begin with the bfs (5). We have already determined that x_{32} should enter the basis. As shown in Table 35, the loop involving x_{32} and some of the basic variables is (3, 2)–(3, 3)–(2, 3)–(2, 2). The odd cells in this loop are (3, 3) and (2, 2). Because $x_{33} = 10$ and $x_{22} = 20$, the pivot will decrease the value of x_{33} and x_{22} by 10 and increase the value of x_{32} and x_{23} by 10. The resulting bfs is shown in Table 36. The u_i 's and v_j 's for the new bfs were obtained by solving

$$u_2 + u_1 = 0 \quad u_2 + v_3 = 13$$

$$u_2 + v_2 = 12 \quad u_2 + v_1 = 9$$

$$u_3 + v_4 = 5 \quad u_3 + v_2 = 9$$

$$u_1 + v_1 = 82 \quad u_2 + v_1 = 9$$

In computing $c_{ij} = u_i + v_j - c_{ij}$ for each nonbasic variable, we find that $c_{12} = 5$, $c_{24} = 1$, and $c_{13} = 2$ are the only positive c_{ij} 's. Thus, we next enter x_{12} into the basis. The loop involving x_{12} and some of the basic variables is (1, 2)–(2, 2)–(2, 1)–(1, 1). The odd cells are (2, 2) and (1, 1). Because $x_{22} = 10$ is the smallest entry in an odd cell, we decrease x_{22} and x_{11} by 10 and increase x_{12} and x_{21} by 10. The resulting bfs is shown in Table 37. For this bfs, the u_i 's and v_j 's were determined by solving

$$u_1 + u_1 = 0 \quad u_1 + v_2 = 6$$

$$u_2 + v_1 = 9 \quad u_3 + v_2 = 9$$

$$u_1 + v_1 = 8 \quad u_3 + v_4 = 5$$

$$u_2 + v_3 = 13 \quad u_3 + v_4 = 5$$

In computing c_{ij} for each nonbasic variable, we find that the only positive c_{ij} is $c_{13} =$

2. Thus, x_{13} enters the basis. The loop involving x_{13} and some of the basic variables is

TABLE 35 8 6 10 9

Loop Involving 35

Entering Variable x_{32} 35

9 12 13 7

10 20 20 50

14 9 16 5

10 30 40

45 20 30 30

TABLE 36 v_j j

= 8 11

11 12

12 77

x_{32} Has Entered the Basis,

8 6 10 9

and x_{12} Enters Next

$u_{ii} = 00$ 35

35

9 12 13 7

11 10 10 30 50

14 9 16 5

-2

2 10 30 40

45 20 30 30

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TABLE 37 v_{ij}

= 88 66 12

12 22

x_{12} Has Entered the Basis,

8 6 10 9

and x_{13} Enters Next

$u_{ii} = 00$ 25 10

35

9 12 13 7

11 20 30 50

14 9 16 5

33 10 30 40

45 20 30 30

TABLE 38

vvjj = 66 66 10

10 22

Optimal Tableau for Powerco

8 6 10 9

uu_{ii} = 00 10 25

35

9 12 13 7

33 45 5 50

14 9 16 5

33 10 30 40

45 20 30 30

(1, 3)–(2, 3)–(2, 1)–(1, 1). The odd cells are x_{23} and x_{11} . Because $x_{11} = 25$ is the smallest entry in an odd cell, we decrease x_{23} and x_{11} by 25 and increase x_{13} and x_{21} by 25. The resulting bfs is shown in Table 38. For this bfs, the u_i 's and v_j 's were obtained by solving

$$u_2 - u_1 = 0 \quad u_2 - v_3 = 13$$

$$u_2 - v_1 = 9 \quad u_1 - v_3 = 10$$

$$u_3 - v_4 = 5 \quad u_3 - v_2 = 9$$

$$u_1 - v_2 = 6 \quad u_3 - v_2 = 9$$

The reader should check that for this bfs, all $c_{ij} \geq 0$, so an optimal solution has been obtained. Thus, the optimal solution to the Powerco problem is $x_{12} = 10$, $x_{13} = 25$, $x_{21} = 45$, $x_{23} = 5$, $x_{32} = 10$, $x_{34} = 30$, and

$$z = 6(10) + 10(25) + 9(45) + 13(5) + 9(10) + 5(30) = \$1,020$$

PROBLEMS

Group A

Use the transportation simplex to solve Problems 1–8 in Section 7.1. Begin with the bfs found in Section 7.2.

7.3 The Transportation Simplex Method 389

7.4 Sensitivity Analysis for Transportation Problemst

We have already seen that for a transportation problem, the determination of a bfs and of row 0 for a given set of basic variables, as well as the pivoting procedure, all simplify. It should therefore be no surprise that certain aspects of the sensitivity analysis discussed in Section 6.3 can be simplified. In this section, we discuss the following three aspects of sensitivity analysis for the transportation problem:

Change 1 Changing the objective function coefficient of a nonbasic variable.

Change 2 Changing the objective function coefficient of a basic variable.

Change 3 Increasing a single supply by Δ and a single demand by Δ .

We illustrate three changes using the Powerco problem. Recall from Section 7.3 that the optimal solution for the Powerco problem was $z = \$1,020$; the optimal tableau is Table 39.

Changing the Objective Function Coefficient
of a Nonbasic Variable

As in Section 6.3, changing the objective function coefficient of a nonbasic variable x_{ij} will leave the right-hand side of the optimal tableau unchanged. Thus, the current basis will still be feasible. We are not changing $c_{BVB}[1]$, so the u_i 's and v_j 's remain unchanged. In row 0, only the coefficient of x_{ij} will change. Thus, as long as the coefficient of x_{ij} in the optimal row 0 is nonpositive, the current basis remains optimal.

To illustrate the method, we answer the following question: For what range of values of the cost of shipping 1 million kwh of electricity from plant 1 to city 1 will the current basis remain optimal? Suppose we change c_{11} from 8 to $8 + \Delta$. For what values of Δ will the current basis remain optimal? Now $\Delta c_{11} = u_1 - v_1 - c_{11} = 0 - 6 - (8 + \Delta) = -2 - \Delta$. Thus, the current basis remains optimal for $-2 - \Delta \leq 0$, or $\Delta \geq -2$, and $c_{11} = 8 - 2 = 6$.

Changing the Objective Function Coefficient of a Basic Variable

Because we are changing $c_{BVB}[1]$, the coefficient of each nonbasic variable in row 0 may change, and to determine whether the current basis remains optimal, we must find the new u_i 's and v_j 's and use these values to price out all nonbasic variables. The current basis remains optimal as long as all nonbasic variables price out nonpositive. To illustrate the idea, we determine for the Powerco problem the range of values of the cost of shipping 1 million kwh from plant 1 to city 3 for which the current basis remains optimal.

Suppose we change c_{13} from 10 to $10 + \Delta$. Then the equation $\Delta c_{13} = 0$ changes from $u_1 - v_3 = 10$ to $u_1 - v_3 = 10 + \Delta$. Thus, to find the u_i 's and v_j 's, we must solve the following equations:

$$u_2 - u_1 = 0 \quad u_3 - v_2 = 90 \quad \Delta$$

$$u_2 - v_1 = 9 \quad u_1 - v_3 = 10 \quad \Delta$$

$$u_1 - v_2 = 6 \quad u_3 - v_4 = 50 \quad \Delta$$

$$u_2 - v_3 = 13 \quad u_1 - v_3 = 10 \quad \Delta$$

This section covers topics that may be omitted with no loss of continuity.

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TABLE 39 City 1 City 2 City 3 City 4 Supply

Optimal Tableau for Powerco

$$v_j = 6 \quad 6 \quad 10 \quad 2$$

$$8 \quad 6 \quad 10 \quad 9$$

$$\text{Plant 1 } u_i = 0 \quad 10 \quad 25$$

$$35$$

$$9 \quad 12 \quad 13 \quad 7$$

$$\text{Plant 2 } 3 \quad 45 \quad 5 \quad 50$$

$$14 \quad 9 \quad 16 \quad 5$$

$$\text{Plant 3 } 3 \quad 10 \quad 30 \quad 40$$

$$\text{Demand } 45 \quad 20 \quad 30 \quad 30$$

Solving these equations, we obtain $u_1 = 0$, $v_2 = 6$, $v_3 = 10 + \Delta$, $v_1 = 6 + \Delta$, $u_2 = 3 + \Delta$

$u_3 \leq 3$, and $v_4 \leq 2$.

We now price out each nonbasic variable. The current basis will remain optimal as long as each nonbasic variable has a nonpositive coefficient in row 0.

$$c_{11} - u_1 - v_1 = 8 - 0 - 2 = 6 > 0 \text{ for } x_{11}$$

$$c_{14} - u_1 - v_4 = 9 - 0 - 2 = 7 > 0$$

$$c_{22} - u_2 - v_2 = 12 - 3 - 0 = 9 > 0 \text{ for } x_{22}$$

$$c_{24} - u_2 - v_4 = 7 - 3 - 2 = 2 > 0 \text{ for } x_{24}$$

$$c_{31} - u_3 - v_1 = 14 - 5 - 2 = 7 > 0 \text{ for } x_{31}$$

$$c_{33} - u_3 - v_3 = 16 - 5 - 3 = 8 > 0 \text{ for } x_{33}$$

Thus, the current basis remains optimal for $u_2 \leq 2$, or $8 - 10 + 2 \leq c_{13} \leq 10 + 2 \leq 12$.

Increasing Both Supply s_i and Demand d_j by Δ

Observe that this change maintains a balanced transportation problem. Because the u_i 's and v_j 's may be thought of as the negative of each constraint's shadow prices, we know from (37) of Chapter 6 that if the current basis remains optimal,

$$\text{New } z\text{-value} = \text{old } z\text{-value} + \Delta u_i + \Delta v_j$$

For example, if we increase plant 1's supply and city 2's demand by 1 unit, then (new cost) $= 1,020 + 1(0) + 1(6) = \$1,026$.

We may also find the new values of the decision variables as follows:

1 If x_{ij} is a basic variable in the optimal solution, then increase x_{ij} by Δ .

2 If x_{ij} is a nonbasic variable in the optimal solution, then find the loop involving x_{ij} and some of the basic variables. Find an odd cell in the loop that is in row i . Increase the value of this odd cell by Δ and go around the loop, alternately increasing and then decreasing current basic variables in the loop by Δ .

To illustrate the first situation, suppose we increase s_1 and d_2 by 2. Because x_{12} is a basic variable in the optimal solution, the new optimal solution will be the one shown in Table 40. The new optimal z -value is $1,020 + 2u_1 + 2v_2 = \$1,032$. To illustrate the second situation, suppose we increase both s_1 and d_1 by 1. Because x_{11} is a nonbasic variable in the current optimal solution, we must find the loop involving x_{11} and some of the

7. 4 Sensitivity Analysis for Transportation Problems 391

TA B L E 40 City 1 City 2 City 3 City 4 Supply

Optimal Tableau for Powerco If

$$v_j = 6 \ 6 \ 10 \ 2$$

$$s_1 = 35 \ 2 \ 37 \text{ and}$$

$$d_2 = 20 \ 2 \ 22 \ 8 \ 6 \ 10 \ 9$$

$$\text{Plant 1} = 00$$

$$u_i = 12 \ 25$$

$$37$$

$$9 \ 12 \ 13 \ 7$$

$$\text{Plant 2} = 33 \ 45 \ 5 \ 50$$

14 9 16 5

Plant 3 3 10 30 40

Demand 45 22 30 30

TA B L E 41 City 1 City 2 City 3 City 4 Supply

Optimal Tableau for Powerco If

$v_j = 6 \ 6 \ 10 \ 2$

$s_1 = 35 \ 1 \ 36$ and

$d_1 = 45 \ 1 \ 46 \ 8 \ 6 \ 10 \ 9$

Plant 1 $u_{ii} = 00 \ 10 \ 26$

36

9 12 13 7

Plant 2 33 46 4 50

14 9 16 5

Plant 3 33 10 30 40

Demand 46 20 30 30

basic variables. The loop is (1, 1)–(1, 3)–(2, 3)–(2, 1). The odd cell in the loop and row 1 is x_{13} . Thus, the new optimal solution will be obtained by increasing both x_{13} and x_{21} by 1 and decreasing x_{23} by 1. This yields the optimal solution shown in Table 41. The new optimal z -value is found from (new z -value) $= 1,020 + v_1 - v_1 = \$1,026$. Observe that if both s_1 and d_1 were increased by 6, the current basis would be infeasible. (Why?)

PROBLEMS

Group A

The following problems refer to the Powerco example. a The company's goal is to minimize the cost of meet-

ing customers' demands. Find two optimal bfs for this

1 Determine the range of values of c_{14} for which the transportation problem.

current basis remains optimal.

b Suppose that customer 2's demand increased by one

2 Determine the range of values of c_{34} for which the unit. By how much would costs increase? current basis remains optimal.

3 If s_2 and d_3 are both increased by 3, what is the new TA B L E 42 optimal solution?

To

4 If s_3 and d_3 are both decreased by 2, what is the new optimal solution? From Customer 1 Customer 2 Customer 3 Supply

5 Two plants supply three customers with medical Plant 1 \$55 \$65 \$80 35

supplies. The unit costs of shipping from the plants to the Plant 2 \$10 \$15 \$25 50

customers, along with the supplies and demands, are given Demand 10 10 10

in Table 42.

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7.5 Assignment Problems

Although the transportation simplex appears to be very efficient, there is a certain class of transportation problems, called assignment problems, for which the transportation simplex is often very inefficient. In this section, we define assignment problems and discuss an efficient method that can be used to solve them.

EXAMPLE 4 Machine Assignment Problem

Machineco has four machines and four jobs to be completed. Each machine must be assigned to complete one job. The time required to set up each machine for completing each job is shown in Table 43. Machineco wants to minimize the total setup time needed to complete the four jobs. Use linear programming to solve this problem.

Solution Machineco must determine which machine should be assigned to each job. We define (for $i, j \in 1, 2, 3, 4$)

$x_{ij} = 1$ if machine i is assigned to meet the demands of job j

$x_{ij} = 0$ if machine i is not assigned to meet the demands of job j

Then Machineco's problem may be formulated as

$$\min z = 14x_{11} + 5x_{12} + 8x_{13} + 7x_{14} + 2x_{21} + 12x_{22} + 6x_{23} + 5x_{24}$$

$$\min z = 7x_{31} + 8x_{32} + 3x_{33} + 9x_{34} + 2x_{41} + 4x_{42} + 6x_{43} + 10x_{44}$$

$$\text{s.t. } x_{11} + x_{12} + x_{13} + x_{14} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{21} + x_{22} + x_{23} + x_{24} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{31} + x_{32} + x_{33} + x_{34} = 1 \text{ (Machine constraints)}$$

(13)

$$\text{s.t. } x_{41} + x_{42} + x_{43} + x_{44} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{11} + x_{21} + x_{31} + x_{41} = 1 \text{ (Job constraints)}$$

$$\text{s.t. } x_{12} + x_{22} + x_{32} + x_{42} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{13} + x_{23} + x_{33} + x_{43} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{14} + x_{24} + x_{34} + x_{44} = 1 \text{ (Machine constraints)}$$

$$\text{s.t. } x_{ij} = 0 \text{ or } x_{ij} = 1 \text{ (Machine constraints)}$$

The first four constraints in (13) ensure that each machine is assigned to a job, and the last four ensure that each job is completed. If $x_{ij} = 1$, then the objective function will pick up the time required to set up machine i for job j ; if $x_{ij} = 0$, then the objective function will not pick up the time required.

Ignoring for the moment the $x_{ij} = 0$ or $x_{ij} = 1$ restrictions, we see that Machineco faces a balanced transportation problem in which each supply point has a supply of 1 and each

TABLE 43

Setup Times for Machineco

Time (Hours)

Machine Job 1 Job 2 Job 3 Job 4

1 14 5 8 7

2 2 12 6 5

3 7 8 3 9

4 2 4 6 10

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demand point has a demand of 1. In general, an assignment problem is a balanced transportation problem in which all supplies and demands are equal to 1. Thus, an assignment problem is characterized by knowledge of the cost of assigning each supply point to each demand point. The assignment problem's matrix of costs is its cost matrix.

All the supplies and demands for the Machineco problem (and for any assignment problem) are integers, so our discussion in Section 7.3 implies that all variables in Machineco's optimal solution must be integers. Because the right-hand side of each constraint is equal to 1, each x_{ij} must be a nonnegative integer that is no larger than 1, so each x_{ij} must equal 0 or 1. This means that we can ignore the restrictions that $x_{ij} \geq 0$ or 1 and solve (13) as a balanced transportation problem. By the minimum cost method, we obtain the bfs in Table 44. The current bfs is highly degenerate. (In any bfs to an $m \times m$ assignment problem, there will always be m basic variables that equal 1 and $m - 1$ basic variables that equal 0.)

We find that $c_{43} = 1$ is the only positive c_{ij} . We therefore enter x_{43} into the basis. The loop involving x_{43} and some of the basic variables is $(4, 3)-(1, 3)-(1, 2)-(4, 2)$. The odd variables in the loop are x_{13} and x_{42} . Because $x_{13} = x_{42} = 0$, either x_{13} or x_{42} will leave

TABLE 44 Job 1 Job 2 Job 3 Job 4

Basic Feasible Solution

$v_j = 3 \ 4 \ 8 \ 7$

for Machineco

14 5 8 7

Machine 1 $u_i = 0 \ 1 \ 0 \ 0$

1

2 12 6 5

Machine 2 $u_i = 2 \ 1 \ 1$

7 8 3 9

Machine 3 $u_i = 5 \ 1 \ 1$

2 4 6 10

Machine 4 $u_i = 1 \ 1 \ 0 \ 1$

1 1 1 1

TABLE 45 Job 1 Job 2 Job 3 Job 4

x_{43} Has Entered the Basis

$v_j = 3 \ 5 \ 7 \ 7$

14 5 8 7

Machine 1 $u_i = 0 \ 1 \ 0$

1

2 12 6 5

Machine 2 $\begin{bmatrix} 2 & 1 & 1 \end{bmatrix}$

7 8 3 9

Machine 3 $\begin{bmatrix} 4 & 1 & 1 \end{bmatrix}$

2 4 6 10

Machine 4 $\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \end{bmatrix}$

1 1 1 1

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the basis. We arbitrarily choose x_{13} to leave the basis. After performing the pivot, we obtain the bfs in Table 45. All c_{ij} 's are now nonpositive, so we have obtained an optimal assignment: $x_{12} = 1$, $x_{24} = 1$, $x_{33} = 1$, and $x_{41} = 1$. Thus, machine 1 is assigned to job 2, machine 2 is assigned to job 4, machine 3 is assigned to job 3, and machine 4 is assigned to job 1. A total setup time of $5 + 5 + 3 + 2 = 15$ hours is required.

The Hungarian Method

Looking back at our initial bfs, we see that it was an optimal solution. We did not know that it was optimal, however, until performing one iteration of the transportation simplex. This suggests that the high degree of degeneracy in an assignment problem may cause the transportation simplex to be an inefficient way of solving assignment problems. For this reason (and the fact that the algorithm is even simpler than the transportation simplex), the Hungarian method is usually used to solve assignment (min) problems:

Step 1 Find the minimum element in each row of the $m \times m$ cost matrix. Construct a new matrix by subtracting from each cost the minimum cost in its row. For this new matrix, find the minimum cost in each column. Construct a new matrix (called the reduced cost matrix) by subtracting from each cost the minimum cost in its column.

Step 2 Draw the minimum number of lines (horizontal, vertical, or both) that are needed to cover all the zeros in the reduced cost matrix. If m lines are required, then an optimal solution is available among the covered zeros in the matrix. If fewer than m lines are needed, then proceed to step 3.

Step 3 Find the smallest nonzero element (call its value k) in the reduced cost matrix that is uncovered by the lines drawn in step 2. Now subtract k from each uncovered element of the reduced cost matrix and add k to each element that is covered by two lines. Return to step 2.

REMARKS 1 To solve an assignment problem in which the goal is to maximize the objective function, multiply the profits matrix through by -1 and solve the problem as a minimization problem.

2 If the number of rows and columns in the cost matrix are unequal, then the assignment problem is unbalanced. The Hungarian method may yield an incorrect solution if the problem is unbalanced. Thus, any assignment problem should be balanced (by the addition of one or more dummy points) before it is solved by the Hungarian method.

3 In a large problem, it may not be easy to find the minimum number of lines needed to cover all zeros in the current cost matrix. For a discussion of how to find the minimum number of lines

needed, see Gillett (1976). It can be shown that if j lines are required, then only j “jobs” can be assigned to zero costs in the current matrix. This explains why the algorithm terminates when m lines are required.

Solution of Machineco Example by the Hungarian Method

We illustrate the Hungarian method by solving the Machineco problem (see Table 46).

Step 1 For each row, we subtract the row minimum from each element in the row, obtaining Table 47. We now subtract 2 from each cost in column 4, obtaining Table 48.

Step 2 As shown, lines through row 1, row 3, and column 1 cover all the zeros in the reduced cost matrix. From remark 3, it follows that only three jobs can be assigned to zero costs in the current cost matrix. Fewer than four lines are required to cover all the zeros, so we proceed to step 3.

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TA B L E 46 Row Minimum

Cost Matrix for Machineco 14 5 8 7 5

2 12 6 5 2

7 8 3 9 3

2 4 6 10 2

TA B L E 47

Cost Matrix After Row 9 0 3 2

Minimums Are Subtracted

0 10 4 3

4 5 0 6

0 2 4 8

0

Column Minimum 0 0 2

TA B L E 48

Cost Matrix After Column 9 0 3 0

Minimums Are Subtracted

0 10 4 1

4 5 0 4

0 2 4 6

Step 3 The smallest uncovered element equals 1, so we now subtract 1 from each uncovered element in the reduced cost matrix and add 1 to each twice-covered element. The resulting matrix is Table 49. Four lines are now required to cover all the zeros. Thus, an optimal solution is available. To find an optimal assignment, observe that the only covered 0 in column 3 is x_{33} , so we must have $x_{33} = 1$. Also, the only available covered zero in column 2 is x_{12} , so we set $x_{12} = 1$ and observe that neither row 1 nor column 2 can be used again. Now the only available covered zero in column 4 is x_{24} . Thus, we choose $x_{24} = 1$ (which now

excludes both row 2 and column 4 from further use). Finally, we choose $x_{41} = 1$.

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TABLE 49

Four Lines Required; Optimal 10 0 3 0

Solution Is Available

0 9 3 0

5 5 0 4

0 1 3 5

Thus, we have found the optimal assignment $x_{12} = 1$, $x_{24} = 1$, $x_{33} = 1$, and $x_{41} = 1$.

Of course, this agrees with the result obtained by the transportation simplex.

Intuitive Justification of the Hungarian Method

To give an intuitive explanation of why the Hungarian algorithm works, we need to discuss the following result: If a constant is added to each cost in a row (or column) of a balanced transportation problem, then the optimal solution to the problem is unchanged. To show why the result is true, suppose we add k to each cost in the first row of the Machineco problem. Then

New objective function $=$ old objective function $+ k(x_{11} + x_{12} + x_{13} + x_{14})$

Because any feasible solution to the Machineco problem must have $x_{11} + x_{12} + x_{13} + x_{14} = 1$,

New objective function $=$ old objective function $+ k$

Thus, the optimal solution to the Machineco problem remains unchanged if a constant k is added to each cost in the first row. A similar argument applies to any other row or column.

Step 1 of the Hungarian method consists (for each row and column) of subtracting a constant from each element in the row or column. Thus, step 1 creates a new cost matrix having the same optimal solution as the original problem. Step 3 of the Hungarian method is equivalent (see Problem 7 at the end of this section) to adding k to each cost that lies in a covered row and subtracting k from each cost that lies in an uncovered column (or vice versa). Thus, step 3 creates a new cost matrix with the same optimal solution as the initial assignment problem. Each time step 3 is performed, at least one new zero is created in the cost matrix.

Steps 1 and 3 also ensure that all costs remain nonnegative. Thus, the net effect of steps 1 and 3 of the Hungarian method is to create a sequence of assignment problems (with nonnegative costs) that all have the same optimal solution as the original assignment problem. Now consider an assignment problem in which all costs are nonnegative. Any feasible assignment in which all the x_{ij} 's that equal 1 have zero costs must be optimal for such an assignment problem. Thus, when step 2 indicates that m lines are required to cover all the zeros in the cost matrix, an optimal solution to the original problem has been found.

Computer Solution of Assignment Problems

To solve assignment problems in LINDO, type in the objective function and constraints. Also, many menu-driven programs require the user to input only a list of supply and de-

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mand points (such as jobs and machines, respectively) and a cost matrix. LINGO can also be used to easily solve assignment problems, including the following model to solve the Assign.Ing Machineco example (file Assign.Ing).

MODEL:

1]SETS:

2]MACHINES/1..4/;

3]JOBS/1..4/;

4]LINKS(MACHINES,JOBS):COST,ASSIGN;

5]ENDSETS

6]MIN=@SUM(LINKS:COST*ASSIGN);

7]@FOR(MACHINES(I):

8]@SUM(JOBS(J):ASSIGN(I,J))<1);

9]@FOR(JOBS(J):

10]@SUM(MACHINES(I):ASSIGN(I,J))>1);

11]DATA:

12]COST = 14,5,8,7,

13]2,12,6,5,

14]7,8,3,9,

15]2,4,6,10;

16]ENDDATA

END

Line 2 defines the four supply points (machines), and line 3 defines the four demand points (jobs). In line 4, we define each possible combination of jobs and machines (16 in all) and associate with each combination an assignment cost [for example $COST(1, 2) = 5$] and a variable $ASSIGN(I,J)$. $ASSIGN(I,J)$ equals 1 if machine i is used to perform job j ; it equals 0 otherwise. Line 5 ends the definition of sets.

Line 6 expresses the objective function by summing over all possible (I,J) combinations the product of the assignment cost and $ASSIGN(I,J)$. Lines 7–8 limit each MACHINE to performing at most one job by forcing (for each machine) the sum of $ASSIGN(I,J)$ over all JOBS to be at most 1. Lines 9–10 require that each JOB be completed by forcing (for each job) the sum of $ASSIGN(I,J)$ over all MACHINES to be at least 1. Lines 12–16 input the cost matrix.

Observe that this LINGO program can (with simple editing) be used to solve any assignment problem (even if it is not balanced!). For example, if you had 10 machines available to perform 8 jobs, you would edit line 2 to indicate that there are 10 machines (replace 1..4 with 1..10). Then edit line 3 to indicate that there are 8 jobs. Finally, in line 12, you would type the 80 entries of your cost matrix, following “COST=” and you would be ready to roll!

REMARK 1 From our discussion of the Machineco example, it is unnecessary to force the $ASSIGN(I,J)$ to

equal 0 or 1; this will happen automatically!

PROBLEMS

Group A

1 Five employees are available to perform four jobs. The TA B L E 50

time it takes each person to perform each job is given in

Table 50. Determine the assignment of employees to jobs Time (hours)

that minimizes the total time required to perform the four Person Job 1 Job 2 Job 3 Job 4 jobs.

1 22 18 30 18

2† Doc Councillman is putting together a relay team for 2 18 — 27 22

the 400-meter relay. Each swimmer must swim 100 meters

3 26 20 28 28

of breaststroke, backstroke, butterfly, or freestyle. Doc

believes that each swimmer will attain the times given in 4 16 22 — 14

5 21 — 25 28

This problem is based on Machol (1970). Note: Dashes indicate person cannot do that particular job.

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Table 51. To minimize the team's time for the race, which did not bid on the given job). Person 1 can do only one job,

swimmer should swim which stroke? but persons 2 and 3 can each do as many as two jobs.

3 Tom Cruise, Freddy Prinze Jr., Harrison Ford, and Matt Determine the minimum cost assignment of persons to jobs.

LeBlanc are marooned on a desert island with Jennifer 5 Greydog Bus Company operates buses between Boston

Aniston, Courteney Cox, Gwyneth Paltrow, and Julia and Washington, D.C. A bus trip between these two cities

Roberts. The "compatibility measures" in Table 52 indicate takes 6 hours. Federal law requires that a driver rest for four

how much happiness each couple would experience if they or more hours between trips. A driver's workday consists of

spent all their time together. The happiness earned by a two trips: one from Boston to Washington and one from

couple is proportional to the fraction of time they spend Washington to Boston. Table 54 gives the departure times

together. For example, if Freddie and Gwyneth spend half for the buses. Greydog's goal is to minimize the total

their time together, they earn happiness of $\frac{1}{2} \cdot \frac{1}{2} \cdot (9) = 4.5$. downtime for all drivers. How should Greydog assign crews

a Let x_{ij} be the fraction of time that the i th man spends to trips? Note: It is permissible for a driver's "day" to overlap

with the j th woman. The goal of the eight people is to midnight. For example, a Washington-based driver can be

maximize the total happiness of the people on the is- assigned to the Washington–Boston 3 P.M. trip and the

land. Formulate an LP whose optimal solution will yield Boston–Washington 6 A.M. trip.

the optimal values of the x_{ij} 's. 6 Five male characters (Billie, John, Fish, Glen, and Larry)

b Explain why the optimal solution in part (a) will and five female characters (Ally, Georgia, Jane, Rene, and

have four $x_{ij} \leq 1$ and twelve $x_{ij} \leq 0$. The optimal solu- Nell) from Ally McBeal are marooned on a desert island.

tion requires that each person spend all his or her time The problem is to determine what percentage of time each

with one person of the opposite sex, so this result is of- woman on the island should spend with each man. For

ten referred to as the Marriage Theorem. example, Ally could spend 100% of her time with John or

c Determine the marriage partner for each person. she could “play the field” by spending 20% of her time with

d Do you think the Proportionality Assumption of lin- each man. Table 55 shows a “happiness index” for each

ear programming is valid in this situation? potential pairing of a man and woman. For example, if Larry

and Rene spend all their time together, they earn 8 units of

4 A company is taking bids on four construction jobs. Three happiness for the island.

people have placed bids on the jobs. Their bids (in thousands

a Play matchmaker and determine an allocation of

of dollars) are given in Table 53 (a * indicates that the person

each man and woman's time that earns the maximum to-

tal happiness for the island. Assume that happiness

earned by a couple is proportional to the amount of time

TA B L E 51 they spend together.

Time (seconds) b Explain why the optimal solution to this problem

Swimmer Free Breast Fly Back will, for any matrix of “happiness indices,” always in-
volve each woman spending all her time with one man.

Gary Hall 54 54 51 53

Mark Spitz 51 57 52 52

Jim Montgomery 50 53 54 56

Chet Jastremski 56 54 55 53

TA B L E 54

Departure Departure

Trip Time Trip Time

TA B L E 52

Boston 1 6 A.M. Washington 1 5:30 A.M.

JA CC GP JR

Boston 2 7:30 A.M. Washington 2 9 A.M.

TC 7 5 8 2 Boston 3 11.30 A.M. Washington 3 3 P.M.

FP 7 8 9 4 Boston 4 7 P.M. Washington 4 6:30 P.M.

HF 3 5 7 9 Boston 5 12:30 A.M. Washington 5 12 midnight

ML 5 5 6 7

TA B L E 55

TA B L E 53

Ally Georgia Jane Rene Nell

Job

Billie 8 6 4 7 5

Person 1 2 3 4

John 5 7 6 4 9

1 50 46 42 40 Fish 10 6 5 2 10

2 51 48 44 * Glen 1 0 0 0 0

3 * 47 45 45 Larry 5 7 9 8 6

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c What assumption made in the problem is needed for TA B L E 57

the Marriage Theorem to hold?

Bids

Route Route Route Route

Group B Company 1 2 3 4

7 Any transportation problem can be formulated as an

1 \$4,000 \$5,000 — —

assignment problem. To illustrate the idea, determine an

assignment problem that could be used to find the optimal 2 — \$4,000 — \$4,000

solution to the transportation problem in Table 56. (Hint: 3 \$3,000 — \$2,000 —

You will need five supply and five demand points). 4 — — \$4,000 \$5,000

8 The Chicago board of education is taking bids on the

city's four school bus routes. Four companies have made the

bids in Table 57. b Suppose that each company can be assigned two

a Suppose each bidder can be assigned only one route. routes. Use the assignment method to minimize

Use the assignment method to minimize Chicago's cost Chicago's cost of running the four bus routes. (Hint:

of running the four bus routes. Two supply points will be needed for each company.)

9 Show that step 3 of the Hungarian method is equivalent

to performing the following operations: (1) Add k to each

TA B L E 56 cost that lies in a covered row. (2) Subtract k from each cost

3 1 that lies in an uncovered column.

2 10 Suppose c_{ij} is the smallest cost in row i and column j

of an assignment problem. Must $x_{ij} \leq 1$ in any optimal

2 3 assignment?

3

1 4

7.6 Transshipment Problems

A transportation problem allows only shipments that go directly from a supply point to a demand point. In many situations, shipments are allowed between supply points or between demand points. Sometimes there may also be points (called transshipment points) through which goods can be transshipped on their journey from a supply point to a demand point. Shipping problems with any or all of these characteristics are transshipment problems. Fortunately, the optimal solution to a transshipment problem can be found by solving a transportation problem.

In what follows, we define a supply point to be a point that can send goods to another point but cannot receive goods from any other point. Similarly, a demand point is a point that can receive goods from other points but cannot send goods to any other point. A transshipment point is a point that can both receive goods from other points and send goods to other points. The following example illustrates these definitions (“—” indicates that a shipment is impossible).

EXAMPLE 5 Transshipment

Widgetco manufactures widgets at two factories, one in Memphis and one in Denver. The Memphis factory can produce as many as 150 widgets per day, and the Denver factory can produce as many as 200 widgets per day. Widgets are shipped by air to customers in Los Angeles and Boston. The customers in each city require 130 widgets per day. Because of the deregulation of airfares, Widgetco believes that it may be cheaper to first fly some widgets to New York or Chicago and then fly them to their final destinations. The costs of flying a widget are shown in Table 58. Widgetco wants to minimize the total cost of shipping the required widgets to its customers.

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TABLE 58

Shipping Costs for Transshipments

To (\$)

From Memphis Denver N.Y. Chicago L.A. Boston

Memphis 0 — 8 13 25 28

Denver — 0 15 12 26 25

N.Y. — — 0 6 16 17

Chicago — — 6 0 14 16

L.A. — — — — 0 —

Boston — — — — — 0

In this problem, Memphis and Denver are supply points, with supplies of 150 and 200 widgets per day, respectively. New York and Chicago are transshipment points. Los Angeles and Boston are demand points, each with a demand of 130 widgets per day. A graphical representation of possible shipments is given in Figure 9.

We now describe how the optimal solution to a transshipment problem can be found

by solving a transportation problem. Given a transshipment problem, we create a balanced transportation problem by the following procedure (assume that total supply exceeds total demand):

Step 1 If necessary, add a dummy demand point (with a supply of 0 and a demand equal to the problem's excess supply) to balance the problem. Shipments to the dummy and from a point to itself will, of course, have a zero shipping cost. Let s = total available supply.

Step 2 Construct a transportation tableau as follows: A row in the tableau will be needed for each supply point and transshipment point, and a column will be needed for each demand point and transshipment point. Each supply point will have a supply equal to its original supply, and each demand point will have a demand equal to its original demand. Let s = total available supply. Then each transshipment point will have a supply equal to (point's original supply) $- s$ and a demand equal to (point's original demand) $+ s$. This ensures that any transshipment point that is a net supplier will have a net outflow equal to the point's original supply, and, similarly, a net demander will have a net inflow equal to the point's original demand. Although we don't know how much will be shipped through each transshipment point, we can be sure that the total amount will not exceed s . This explains why we add s to the supply and demand at each transshipment point. By adding the same amounts to the supply and demand, we ensure that the net outflow at each transshipment point will be correct, and we also maintain a balanced transportation tableau.

Los

Memphis New York

Angeles

FIGURE 9

A Transshipment

Problem

Denver Chicago Boston

7. 6 Transshipment Problems 401

For the Widgetco example, this procedure yields the transportation tableau and its optimal solution given in Table 59. Because s = (total supply) $= 150 + 200 = 350$ and (total demand) $= 130 + 130 = 260$, the dummy demand point has a demand of $350 - 260 = 90$. The other supplies and demands in the transportation tableau are obtained by adding $s = 350$ to each transshipment point's supply and demand.

In interpreting the solution to the transportation problem created from a transshipment problem, we simply ignore the shipments to the dummy and from a point to itself. From Table 59, we find that Widgetco should produce 130 widgets at Memphis, ship them to New York, and transship them from New York to Los Angeles. The 130 widgets produced at Denver should be shipped directly to Boston. The net outflow from each city is

Memphis: $220 - 130 = 20 = 220 - 150$

Denver: $220 - 130 = 70 = 220 - 200$

N.Y.: $220 - 130 + 130 = 220 - 0$

Chicago: 350 350 130 220 0

L.A.: 350 350 130 220 0

Boston: 350 350 130 220 0

Dummy: 20 70 130 220 90

A negative net outflow represents an inflow. Observe that each transshipment point (New York and Chicago) has a net outflow of 0; whatever flows into the transshipment point must leave the transshipment point. A graphical representation of the optimal solution to the Widgetco example is given in Figure 10.

Suppose that we modify the Widgetco example and allow shipments between Memphis and Denver. This would make Memphis and Denver transshipment points and would add columns for Memphis and Denver to the Table 59 tableau. The Memphis row in the tableau would now have a supply of 150 350 500, and the Denver row would have

TABLE 59 N.Y. Chicago L.A. Boston Dummy Supply

Representation of 8 13 25 28 0

Transshipment Problem Memphis 130 20 150

as Balanced

Transportation Problem 15 12 26 25 0

Denver

130 70 200

0 6 16 17 0

220 130 350

N.Y.

6 0 14 16 0

350 350

Chicago

Demand

350 350 130 130 90

130 130 Los

Memphis New York

Angeles

FIGURE 10

Optimal Solution

to Widgetco Denver Chicago Boston

130

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a supply of 200 350 550. The new Memphis column would have a demand of 0 350 350, and the new Denver column would have a demand of 0 350 350. Fi-

nally, suppose that shipments between demand points L.A. and Boston were allowed. This

would make L.A. and Boston transshipment points and add rows for L.A. and Boston.

The supply for both the L.A. and Boston rows would be 0 350 350. The demand for

both the L.A. and Boston columns would now be 130 □ 350 □ 480.

PROBLEMS

Group A

1 General Ford produces cars at L.A. and Detroit and has TA B L E 61

a warehouse in Atlanta; the company supplies cars to

customers in Houston and Tampa. The cost of shipping a car To (\$)

between points is given in Table 60 (“—” means that a From Well 1 Well 2 Mobile Galveston N.Y. L.A.

shipment is not allowed). L.A. can produce as many as

Well 1 0 — 10 13 25 28

1,100 cars, and Detroit can produce as many as 2,900 cars.

Houston must receive 2,400 cars, and Tampa must receive Well 2 — 0 15 12 26 25

1,500 cars. Mobile — — 0 6 16 17

a Formulate a balanced transportation problem that Galveston — — 6 0 14 16

can be used to minimize the shipping costs incurred in N.Y. — — — — 0 15

meeting demands at Houston and Tampa. L.A. — — — — 15 0

b Modify the answer to part (a) if shipments between

L.A. and Detroit are not allowed. Note: Dashes indicate shipments that are not allowed.

c Modify the answer to part (a) if shipments between

Houston and Tampa are allowed at a cost of \$5.

2 Sunco Oil produces oil at two wells. Well 1 can produce formulate a transshipment and balanced transportation model

as many as 150,000 barrels per day, and well 2 can produce to minimize the daily cost of transporting and refining the oil

as many as 200,000 barrels per day. It is possible to ship oil requirements of Los Angeles and New York.

directly from the wells to Sunco’s customers in Los Angeles

and New York. Alternatively, Sunco could transport oil to 4 Rework Problem 3 under the assumption that Galveston

the ports of Mobile and Galveston and then ship it by tanker has a refinery capacity of 150,000 barrels per day and Mobile

to New York or Los Angeles. Los Angeles requires 160,000 has one of 180,000 barrels per day.

(Hint: Modify the method

barrels per day, and New York requires 140,000 barrels per used to determine the supply and demand at each

day. The costs of shipping 1,000 barrels between two points transshipment point to incorporate the refinery capacity

are shown in Table 61. Formulate a transshipment model restrictions, but make sure to keep the problem balanced.)

(and equivalent transportation model) that could be used to 5 General Ford has two plants, two warehouses, and three

minimize the transport costs in meeting the oil demands of customers. The locations of these are as follows:

Los Angeles and New York.

Plants: Detroit and Atlanta

3 In Problem 2, assume that before being shipped to Los Warehouses: Denver and New York Angeles or New York, all oil produced at the wells must be Customers: Los Angeles, Chicago, and Philadelphia

refined at either Galveston or Mobile. To refine 1,000 barrels

of oil costs \$12 at Mobile and \$10 at Galveston. Assuming Cars are produced at plants, then shipped to warehouses,

that both Mobile and Galveston have infinite refinery capacity, and finally shipped to customers. Detroit can produce 150

cars per week, and Atlanta can produce 100 cars per week.

Los Angeles requires 80 cars per week; Chicago, 70; and

Philadelphia, 60. It costs \$10,000 to produce a car at each

TA B L E 60 plant, and the cost of shipping a car between two cities is

given in Table 62. Determine how to meet General Ford's

To (\$)

weekly demands at minimum cost.

From L.A. Detroit Atlanta Houston Tampa

L.A. 0 140 100 90 225 Group B

Detroit 145 0 111 110 119 6† A company must meet the following demands for cash

Atlanta 105 115 0 113 78 at the beginning of each of the next six months: month 1,

Houston 89 109 121 0 —

Tampa 210 117 82 — 0

Based on Srinivasan (1974).

7. 6 Transshipment Problems 403

TA B L E 62 TA B L E 63

To (\$) Month of Sale

From Denver New York Bond 1 2 3 4 5 6

Detroit 1,253 637 1 \$0.21 \$0.19 \$0.17 \$0.13 \$0.09 \$0.05

Atlanta 1,398 841 2 \$0.50 \$0.50 \$0.50 \$0.33 \$0.00 \$0.00

To (\$) 3 \$1.00 \$1.00 \$1.00 \$1.00 \$1.00 \$0.00

From Los Angeles Chicago Philadelphia

Denver 1,059 996 1,691

New York 2,786 802 100

used to minimize the cost of meeting the cash demands for the next six months.

b Assume that payment of bills can be made after they are due, but a penalty of 5¢ per month is assessed for each dollar of cash demands that is postponed for one \$200; month 2, \$100; month 3, \$50; month 4, \$80; month month. Assuming all bills must be paid by the end of

5, \$160; month 6, \$140. At the beginning of month 1, the month 6, develop a transshipment model that can be company has \$150 in cash and \$200 worth of bond 1, \$100 used to minimize the cost of paying the next six months' worth of bond 2, and \$400 worth of bond 3. The company bills. (Hint: Transshipment points are needed, in the will have to sell some bonds to meet demands, but a penalty form C_t = cash available at beginning of month t after will be charged for any bonds sold before the end of month bonds for month t have been sold, but before month t 6. The penalties for selling \$1 worth of each bond are as demand is met. Shipments into C_t occur from bond sales shown in Table 63.

and $C_t = 1$. Shipments out of C_t occur to $C_t = 1$ and a Assuming that all bills must be paid on time, for- demands for months 1, 2, t .)
mulate a balanced transportation problem that can be

SUMMARY Notation

m = number of supply points

n = number of demand points

x_{ij} = number of units shipped from supply point i to demand point j

c_{ij} = cost of shipping 1 unit from supply point i to demand point j

s_i = supply at supply point i

d_j = demand at demand point j

c_{ij} = coefficient of x_{ij} in row 0 of a given tableau

a_{ij} = column for x_{ij} in transportation constraints

A transportation problem is balanced if total supply equals total demand. To use the methods of this chapter to solve a transportation problem, the problem must first be balanced by use of a dummy supply or a dummy demand point. A balanced transportation problem may be written as

$i = 1, \dots, m$

$\min \sum_{i,j} c_{ij}x_{ij}$

$i = 1, \dots, m$

$j = 1, \dots, n$

s.t. $\sum_j x_{ij} = s_i$ ($i = 1, 2, \dots, m$)

$j = 1, \dots, n$

(Supply constraints)

$i = 1, \dots, m$

s.t. $\sum_i x_{ij} = d_j$ ($j = 1, 2, \dots, n$)

$i = 1, \dots, m$

(Demand constraints)

$x_{ij} \geq 0$ ($i = 1, 2, \dots, m; j = 1, 2, \dots, n$)

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Finding Basic Feasible Solutions

for Balanced Transportation Problems

We can find a bfs for a balanced transportation problem by the northwest corner method, the minimum-cost method, or Vogel's method. To find a bfs by the northwest corner method, begin in the upper left-hand (or northwest) corner of the transportation tableau and set x_{11} as large as possible. Clearly, x_{11} can be no larger than the smaller of s_1 and d_1 . If $x_{11} = s_1$, then cross out the first row of the transportation tableau; this indicates that no more basic variables will come from row 1 of the tableau. Also change d_1 to $d_1 - s_1$. If $x_{11} = d_1$, then cross out the first column of the transportation tableau and change s_1 to $s_1 - d_1$. If $x_{11} = s_1 = d_1$, cross out either row 1 or column 1 (but not both) of the transportation tableau. If you cross out row 1, change d_1 to 0; if you cross out column 1, change s_1 to 0. Continue applying this procedure to the most northwest cell in the tableau that does not lie in a crossed-out row or column. Eventually, you will come to a point where there is only one cell that can be assigned a value. Assign this cell a value equal to its row or column demand, and cross out both the cell's row and its column. A basic feasible solution has now been obtained.

Finding the Optimal Solution

for a Transportation Problem

Step 1 If the problem is unbalanced, balance it.

Step 2 Use one of the methods described in Section 7.2 to find a bfs.

Step 3 Use the fact that $u_1 = 0$ and $u_i - v_j = c_{ij}$ for all basic variables to find the $[u_1 \ u_2 \ \dots \ u_m \ v_1 \ v_2 \ \dots \ v_n]$ for the current bfs.

Step 4 If $u_i - v_j = c_{ij} \leq 0$ for all nonbasic variables, then the current bfs is optimal. If this is not the case, then we enter the variable with the most positive $u_i - v_j = c_{ij}$ into the basis. To do this, find the loop. Then, counting only cells in the loop, label the even cells. Also label the odd cells. Now find the odd cell whose variable assumes the smallest value, θ . The variable corresponding to this odd cell will leave the basis. To perform the pivot, decrease the value of each odd cell by θ and increase the value of each even cell by θ . The values of variables not in the loop remain unchanged. The pivot is now complete. If $\theta = 0$, then the entering variable will equal 0, and an odd variable that has a current value of 0 will leave the basis. In this case, a degenerate bfs will result. If more than one odd cell in the loop equals θ , you may arbitrarily choose one of these odd cells to leave the basis; again, a degenerate bfs will result. The pivoting yields a new bfs.

Step 5 Using the new bfs, return to steps 3 and 4.

For a maximization problem, proceed as stated, but replace step 4 by step 4'.

Step 4' If $u_i - v_j = c_{ij} \geq 0$ for all nonbasic variables, the current bfs is optimal. Otherwise, enter the variable with the most negative $u_i - v_j = c_{ij}$ into the basis using the pivoting procedure.

Assignment Problems

An assignment problem is a balanced transportation problem in which all supplies and demands equal 1. An $m \times m$ assignment problem may be efficiently solved by the Hungarian method:

7. 1 Summary 405

Step 1 Find the minimum element in each row of the cost matrix. Construct a new matrix by subtracting from each cost the minimum cost in its row. For this new matrix, find the minimum cost in each column. Construct a new matrix (reduced cost matrix) by subtracting from each cost the minimum cost in its column.

Step 2 Cover all the zeros in the reduced cost matrix using the minimum number of lines needed. If m lines are required, then an optimal solution is available among the covered zeros in the matrix. If fewer than m lines are needed, then proceed to step 3.

Step 3 Find the smallest nonzero element (k) in the reduced cost matrix that is uncovered by the lines drawn in step 2. Now subtract k from each uncovered element and add k to each element that is covered by two lines. Return to step 2.

REMARKS 1 To solve an assignment problem in which the goal is to maximize the objective function, multiply the profits matrix through by -1 and solve it as a minimization problem.

2 If the number of rows and columns in the cost matrix are unequal, then the problem is unbalanced. The Hungarian method may yield an incorrect solution if the problem is unbalanced. Thus, any assignment problem should be balanced (by the addition of one or more dummy points) before it is solved by the Hungarian method.

Transshipment Problems

A transshipment problem allows shipment between supply points and between demand points, and it may also contain transshipment points through which goods may be shipped on their way from a supply point to a demand point. Using the following method, a transshipment problem may be transformed into a balanced transportation problem.

Step 1 If necessary, add a dummy demand point (with a supply of 0 and a demand equal to the problem's excess supply) to balance the problem. Shipments to the dummy and from a point to itself will, of course, have a zero shipping cost. Let s = total available supply.

Step 2 Construct a transportation tableau creating a row for each supply point and transshipment point, and a column for each demand point and transshipment point. Each supply point will have a supply equal to its original supply, and each demand point will have a demand equal to its original demand. Let s = total available supply. Then each transshipment point will have a supply equal to (point's original supply) $- s$ and a demand equal to (point's original demand) $+ s$.

Sensitivity Analysis for Transportation Problems

Following the discussion of sensitivity analysis in Chapter 6, we can analyze how a change in a transportation problem affects the problem's optimal solution.

Change 1 Changing the objective function coefficient of a nonbasic variable. As long as the coefficient of x_{ij} in the optimal row 0 is nonpositive, the current basis remains optimal.

Change 2 Changing the objective function coefficient of a basic variable. To see whether the current basis remains optimal, find the new u_i 's and v_j 's and use these values to price out all nonbasic variables. The current basis remains optimal as long as all nonbasic variables have a nonpositive coefficient in row 0.

Change 3 Increasing both supply s_i and demand d_j by Δ .

New z-value Δ old z-value $\Delta \sum u_i \Delta \sum v_j$

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We may find the new values of the decision variables as follows:

1 If x_{ij} is a basic variable in the optimal solution, then increase x_{ij} by Δ .

2 If x_{ij} is a nonbasic variable in the optimal solution, find the loop involving x_{ij} and some of the basic variables. Find an odd cell in the loop that is in row i . Increase the value of this odd cell by Δ and go around the loop, alternately increasing and then decreasing current basic variables in the loop by Δ .

REVIEW PROBLEMS

Group A

1 Televco produces TV picture tubes at three plants. Plant 3 A company must meet the following demands for

1 can produce 50 tubes per week; plant 2, 100 tubes per a product: January, 30 units; February, 30 units; March,

week; and plant 3, 50 tubes per week. Tubes are shipped to 20 units. Demand may be backlogged at a cost of

three customers. The profit earned per tube depends on the \$5/unit/month. All demand must be met by the end of March.

site where the tube was produced and on the customer who Thus, if 1 unit of January demand is met during March, a

purchases the tube (see Table 64). Customer 1 is willing to backlogging cost of 5(2) Δ \$10 is incurred. Monthly

purchase as many as 80 tubes per week; customer 2, as production capacity and unit production cost during each

many as 90; and customer 3, as many as 100. Televco wants month are given in Table 66. A holding cost of \$20/unit is

to find a shipping and production plan that will maximize assessed on the inventory at the end of each month.

profits. a Formulate a balanced transportation problem that

a Formulate a balanced transportation problem that could be used to determine how to minimize the total

can be used to maximize Televco's profits. cost (including backlogging, holding, and production

b Use the northwest corner method to find a bfs to the costs) of meeting demand.

problem. b Use Vogel's method to find a basic feasible solution.

c Use the transportation simplex to find an optimal so- c Use the transportation simplex to determine how to

lution to the problem. meet each month's demand. Make sure to give an inter-

pretation of your optimal solution (for example, 20 units

2 Five workers are available to perform four jobs. The
of month 2 demand is met from month 1 production).
time it takes each worker to perform each job is given in
Table 65. The goal is to assign workers to jobs so as to 4 Appletree Cleaning has five maids. To
complete cleaning
minimize the total time required to perform the four jobs. my house, they must vacuum, clean the
kitchen, clean the
Use the Hungarian method to solve the problem. bathroom, and do general straightening up. The
time it takes
each maid to do each job is shown in Table 67. Each maid

TA B L E 64

TA B L E 66

To (\$)

Production Unit Production

From Customer 1 Customer 2 Customer 3 Month Capacity Cost

Plant 1 75 60 69 January 35 \$400

Plant 2 79 73 68 February 30 \$420

Plant 3 85 76 70 March 35 \$410

TA B L E 67

TA B L E 65

Time (Hours)

Time (Hours)

Clean Clean Straighten

Worker Job 1 Job 2 Job 3 Job 4 Maid Vacuum Kitchen Bathroom Up

1 10 15 10 15 1 6 5 2 1

2 12 8 20 16 2 9 8 7 3

3 12 9 12 18 3 8 5 9 4

4 6 12 15 18 4 7 7 8 3

5 16 12 8 12 5 5 5 6 4

7. 1 Review Problems 407

is assigned one job. Use the Hungarian method to determine TA B L E 70

assignments that minimize the total number of maid-hours

needed to clean my house. No. of Students Distance to (Miles)

District Whites Blacks District 2 District 3

5† Currently, State University can store 200 files on hard

disk, 100 files in computer memory, and 300 files on tape. 1 210 120 3 5

Users want to store 300 word-processing files, 100 2 210 30 — 4

packaged-program files, and 100 data files. Each month a

typical word-processing file is accessed eight times; a typical 3 180 150 — —

packaged-program file, four times; and a typical data file,

two times. When a file is accessed, the time it takes for the

file to be retrieved depends on the type of file and on the storage medium (see Table 68). Formulate a balanced transportation problem that can be used to determine the minimum total distance that students a If the goal is to minimize the total time per month must be bused while still satisfying the Supreme Court's that users spend accessing their files, formulate a bal- requirements. Assume that a student who remains in his or anced transportation problem that can be used to deter- her own district will not be bused. mine where files should be stored.

b Use the minimum cost method to find a bfs. 8 Using the northwest corner method to find a bfs, find

(via the transportation simplex) an optimal solution to the

c Use the transportation simplex to find an optimal transportation (minimization) problem shown in Table 71. solution.

9 Solve the following LP:

6 The Gotham City police have just received three calls for police. Five cars are available. The distance (in city blocks) min $z = 2x_1 + 3x_2 + 4x_3 + 3x_4$ of each car from each call is given in Table 69. Gotham City s.t. $x_1 + x_2 + x_3 + x_4 = 4$ wants to minimize the total distance cars must travel to s.t. $x_1 + x_2 + x_3 + x_4 = 5$ respond to the three police calls. Use the Hungarian method s.t. $x_1 + x_2 + x_3 + x_4 = 3$ to determine which car should respond to which call. s.t. $x_1 + x_2 + x_3 + x_4 = 6$

7 There are three school districts in the town of Busville. min $x_j = 0$ ($j = 1, 2, 3, 4$)

The number of black and white students in each district are shown in Table 70. The Supreme Court requires the schools 10 Find the optimal solution to the balanced transportation

in Busville to be racially balanced. Thus, each school must problem in Table 72 (minimization).

have exactly 300 students, and each school must have the 11 In Problem 10, suppose we increase s_1 to 16 and d_3 to

same number of black students. The distances between 11. The problem is still balanced, and because 31 units

districts are shown in Table 70. (instead of 30 units) must be shipped, one would think that the total shipping costs would be increased. Show that the total shipping cost has actually decreased by \$2, however.

This is called the "more for less" paradox. Explain why

TA B L E 68

increasing both the supply and the demand has decreased

Time (Minutes) cost. Using the theory of shadow prices, explain how one

Storage Word Packaged could have predicted that increasing s_1 and d_3 by 1 would

Medium Processing Program Data decrease total cost by \$2.

Hard disk 5 4 4 12 Use the northwest corner method, the minimum-cost

method, and Vogel's method to find basic feasible solutions

Memory 2 1 1

to the transportation problem in Table 73.

Tape 10 8 6

13 Find the optimal solution to Problem 12.

TA B L E 69 TA B L E 71

Distance (Blocks) 12 14 16

Car Call 1 Call 2 Call 3

60

1 10 11 18

14 13 19

2 6 7 7

3 7 8 5 50

4 5 6 4

17 15 18

5 9 4 7

40

This problem is based on Evans (1984). 40 70 10

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TA B L E 72 TA B L E 74

4 2 4 To (\$)

15 From Dallas Houston N.Y. Chicago

L.A. 300 110 — —

12 8 4

San Diego 420 100 — —

15 Dallas — — 450 550

Houston — — 470 530

10 10 10

TA B L E 73 TA B L E 75

20 11 3 6 Plant Cars Available

5 Atlanta 5,000

Boston 6,000

5 9 10 2 Chicago 4,000

10 L.A. 3,000

18 7 4 1

15

TA B L E 76

3 3 12 12 Warehouse Cars Required

Memphis 6,000

Milwaukee 4,000

14 Oilco has oil fields in San Diego and Los Angeles. The San Diego field can produce 500,000 barrels per day, and N.Y. 4,000 the Los Angeles field can produce 400,000 barrels per day. Denver 2,000 Oil is sent from the fields to a refinery, either in Dallas or San Francisco 2,000 in Houston (assume that each refinery has unlimited capacity). It costs \$700 to refine 100,000 barrels of oil at Dallas and \$900 at Houston. Refined oil is shipped to customers in Chicago and New York. Chicago customers TA B L E 77 require 400,000 barrels per day of refined oil; New York Memphis Milwaukee N.Y. Denver S.F. customers require 300,000. The costs of shipping 100,000 barrels of oil (refined or unrefined) between cities are given Atlanta 371 761 841 1,398 2,496 in Table 74. Formulate a balanced transportation model of Boston 1,296 1,050 206 1,949 3,095 this situation.

Chicago 530 87 802 996 2,142

15 For the Powerco problem, find the range of values of L.A. 1,817 2,012 2,786 1,059 379 c24 for which the current basis remains optimal.

16 For the Powerco problem, find the range of values of c23 for which the current basis remains optimal.

18 During the next three quarters, Airco faces the following

17 A company produces cars in Atlanta, Boston, Chicago, demands for air conditioner compressors: quarter 1—200;

and Los Angeles. The cars are then shipped to warehouses quarter 2—300; quarter 3—100. As many as 240 air

in Memphis, Milwaukee, New York City, Denver, and San compressors can be produced during each quarter. Production

Francisco. The number of cars available at each plant is costs/compressor during each quarter are given in Table 78.

given in Table 75. The cost of holding an air compressor in inventory is

Each warehouse needs to have available the number of \$100/quarter. Demand may be backlogged (as long as it is met

cars given in Table 76. by the end of quarter 3) at a cost of \$60/compressor/quarter.

The distance (in miles) between the cities is given in Formulate the tableau for a balanced transportation problem

Table 77. whose solution tells Airco how to minimize the total cost of

a Assuming that the cost (in dollars) of shipping a car meeting the demands for quarters 1–3. equals the distance between two cities, determine an op-

19 A company is considering hiring people for four types timal shipping schedule.

of jobs. It would like to hire the number of people in Table

b Assuming that the cost (in dollars) of shipping a car 79 for each type of job.

equals the square root of the distance between two cities, Four types of people can be hired by the company. Each

determine an optimal shipping schedule. type is qualified to perform two types of jobs according to

7. 1 Review Problems 409

TA B L E 78 TA B L E 81

Quarter 1 Quarter 2 Quarter 3 Leave Arrive Leave Arrive

Flight Chicago New York Flight New York Chicago

\$200 \$180 \$240

1 6 A.M. 10 A.M. 1 7 A.M. 9 A.M.

2 9 A.M. 1 P.M. 2 8 A.M. 10 A.M.

3 12 noon 4 P.M. 3 10 A.M. 12 noon

TA B L E 79 4 3 P.M. 7 P.M. 4 12 noon 2 P.M.

Job 5 5 P.M. 9 P.M. 5 2 P.M. 4 P.M.

1 2 3 4 6 7 P.M. 11 P.M. 6 4 P.M. 6 P.M.

7 8 P.M. 12 midnight 7 6 P.M. 8 P.M.

Number of people 30 30 40 20

TA B L E 80 York flight with at least 1 hour of downtime between flights.

Braneast wants to schedule the crews to minimize the total

Type of Person downtime. Set up an assignment problem that can be used

1 2 3 4 to accomplish this goal. (Hint: Let $x_{ij} = 1$ if the crew that

flies flight i also flies flight j , and $x_{ij} = 0$ otherwise. If $x_{ij} =$

Jobs qualified for 1 and 3 2 and 3 3 and 4 1 and 4 1, then a cost c_{ij} is incurred, corresponding to the downtime

associated with a crew flying flight i and flight j .) Of course,

some assignments are not possible. Find the flight

assignments that minimize the total downtime. How many

Table 80. A total of 20 Type 1, 30 Type 2, 40 Type 3, and crews should be based in each city?

Assume that at the end

20 Type 4 people have applied for jobs. Formulate a balanced of the day, each crew must be in its home city.

transportation problem whose solution will tell the company

how to maximize the number of employees assigned to 23 A firm producing a single product has three plants and

suitable jobs. (Note: Each person can be assigned to at most four customers. The three plants will produce 3,000, 5,000,

one job.) and 5,000 units, respectively, during the next time period.

The firm has made a commitment to sell 4,000 units to

20 During each of the next two months you can produce customer 1, 3,000 units to customer 2, and at least 3,000

as many as 50 units/month of a product at a cost of \$12/unit units to customer 3. Both customers 3 and 4 also want to

during month 1 and \$15/unit during month 2. The customer buy as many of the remaining units as possible. The profit

is willing to buy as many as 60 units/month during each of associated with shipping a unit from plant i to customer j is

the next two months. The customer will pay \$20/unit during given in Table 82. Formulate a balanced transportation

month 1, and \$16/unit during month 2. It costs \$1/unit to problem that can be used to maximize the company's profit.

hold a unit in inventory for a month. Formulate a balanced

transportation problem whose solution will tell you how to 24 A company can produce as many as 35 units/month.

maximize profit. The demands of its primary customers must be met on time

each month; if it wishes, the company may also sell units to

secondary customers each month. A \$1/unit holding cost is

Group B assessed against each month's ending inventory. The relevant

21† The Carter Caterer Company must have the following data are shown in Table 83. Formulate a balanced

number of clean napkins available at the beginning of each transportation problem that can be used to maximize profits

of the next four days: day 1—15; day 2—12; day 3—18; earned during the next three months.

day 4—6. After being used, a napkin can be cleaned by one 25 My home has four valuable paintings that are up for

of two methods: fast service or slow service. Fast service sale. Four customers are bidding for the paintings. Customer

costs 10¢ per napkin, and a napkin cleaned via fast service 1 is willing to buy two paintings, but each other customer

is available for use the day after it is last used. Slow service is willing to purchase at most one painting. The prices that

costs 6¢ per napkin, and these napkins can be reused two each customer is willing to pay are given in Table 84. Use

days after they are last used. New napkins can be purchased

for a cost of 20¢ per napkin. Formulate a balanced

transportation problem to minimize the cost of meeting the

demand for napkins during the next four days. TA B L E 82

22 Braneast Airlines must staff the daily flights between To Customer (\$)

New York and Chicago shown in Table 81. Each of Braneast's From 1 2 3 4

crews lives in either New York or Chicago. Each day a crew

must fly one New York–Chicago and one Chicago–New Plant 1 65 63 62 64

Plant 2 68 67 65 62

Plant 3 63 60 59 60

This problem is based on Jacobs (1954).

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TA B L E 83 TA B L E 86

Available for To (\$)

Production Primary Secondary Sales

Month Cost/Unit (\$) Demand Demand Price/Unit (\$) From NE NW MW SE SW

1 13 20 15 15 Atlanta 8.41 21.52 6.74 3.00 7.89

2 12 15 20 14 Houston 15.20 13.87 10.67 7.89 3.00

3 13 25 15 16

27† During the month of July, Pittsburgh resident B. Fly must make four round-trip flights between Pittsburgh and

TA B L E 84

Chicago. The dates of the trips are as shown in Table 87.

Bid for (\$) B. Fly must purchase four round-trip tickets. Without a

Customer Painting 1 Painting 2 Painting 3 Painting 4 discounted fare, a round-trip ticket between Pittsburgh and

Chicago costs \$500. If Fly's stay in a city includes a weekend,

1 8 11 — — then he gets a 20% discount on the round-trip fare. If his

2 9 13 12 7 stay in a city is at least 21 days, then he receives a 35%

3 9 — 11 — discount; and if his stay is more than 10 days, then he receives a 30% discount. Of course, only one discount can

4 — — 12 9

be applied toward the purchase of any ticket. Formulate and

solve an assignment problem that minimizes the total cost

of purchasing the four round-trip tickets. (Hint: Let $x_{ij} \in \{0, 1\}$

the Hungarian method to determine how to maximize the if a round-trip ticket is purchased for use on the i th flight

total revenue received from the sale of the paintings. out of Pittsburgh and the j th flight out of Chicago. Also

think about where Fly should buy a ticket if, for example,

26 Powerhouse produces capacitors at three locations: Los Angeles, Chicago, and New York. Capacitors are shipped

from these locations to public utilities in five regions of the

28 Three professors must be assigned to teach six sections

country: northeast (NE), northwest (NW), midwest (MW), of finance. Each professor must teach two sections of

southeast (SE), and southwest (SW). The cost of producing finance, and each has ranked the six time periods during

and shipping a capacitor from each plant to each region of which finance is taught, as shown in Table 88. A ranking of

the country is given in Table 85. Each plant has an annual 10 means that the professor wants to teach that time, and a

production capacity of 100,000 capacitors. Each year, each ranking of 1 means that he or she does not want to teach at

region of the country must receive the following number of that time. Determine an assignment of professors to sections

capacitors: NE, 55,000; NW, 50,000; MW, 60,000; SE, that will maximize the total satisfaction of the professors.

60,000; SW, 45,000. Powerhouse feels shipping costs are 29¢ Three fires have just broken out in New York. Fires 1

too high, and the company is therefore considering building and 2 each require two fire engines, and fire 3 requires three

one or two more production plants. Possible sites are Atlanta fire engines. The “cost” of responding to each fire depends

and Houston. The costs of producing a capacitor and on the time at which the fire engines arrive. Let t_{ij} be the

shipping it to each region of the country are given in Table time (in minutes) when the j th engine arrives at fire i . Then

86. It costs \$3 million (in current dollars) to build a new the cost of responding to each fire is as follows:

plant, and operating each plant incurs a fixed cost (in

addition to variable shipping and production costs) of Fire 1: $6t_{11} + 4t_{12} + 5t_{13}$

\$50,000 per year. A plant at Atlanta or Houston will have Fire 2: $7t_{21} + 3t_{22} + 5t_{23}$

the capacity to produce 100,000 capacitors per year. Fire 3: $9t_{31} + 8t_{32} + 5t_{33}$

Assume that future demand patterns and production costs

Three fire companies can respond to the three fires.

will remain unchanged. If costs are discounted at a rate of

1 Company 1 has three engines available, and companies 2

$11\frac{9}{10}\%$ per year, how can Powerhouse minimize the present

value of all costs associated with meeting current and future demands?

TA B L E 87

Leave Pittsburgh Leave Chicago

TA B L E 85

Monday, July 1 Friday, July 5

To (\$) Tuesday, July 9 Thursday, July 11

From NE NW MW SE SW Monday, July 15 Friday, July 19

L.A. 27.86 4.00 20.54 21.52 13.87 Wednesday, July 24 Thursday, July 25

Chicago 8.02 20.54 2.00 6.74 10.67

N.Y. 2.00 27.86 8.02 8.41 15.20 †

Based on Hansen and Wendell (1982).

Based on Denardo, Rothblum, and Swersey (1988).

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TA B L E 88 TA B L E 89

Professor 9 A.M. 10 A.M. 11 A.M. 1 P.M. 2 P.M. 3 P.M. Company Fire 1 Fire 2 Fire 3

1 8 7 6 5 7 6 1 6 7 9

2 9 9 8 8 4 4 2 5 8 11

3 7 6 9 6 9 9 3 6 9 10

and 3 each have two engines available. The time (in minutes) signing the fire engines. (Hint: Seven demand points

it takes an engine to travel from each company to each fire will be needed.)
 is shown in Table 89. b Would the formulation in part (a) still be valid if the
 a Formulate and solve a transportation problem that cost of fire 1 were 4t11 □ 6t12?
 can be used to minimize the cost associated with as-

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Critical Path Analysis 14.1

14

Critical Path Analysis

LEARNING OBJECTIVES :

After studying this unit you will be able to :

I Explain the generic term network analysis

I Draw diagram & chart representing inter-relationships between various elements of the project and finally establishing critical path

I Identify Dummy Activities

I Identify earliest and latest event times of each event. Identify different types of Float

Help in Decision Making about project times and resource allocation Calculating probability of completing network in a given time

14.1 INTRODUCTION

Planning, Scheduling and Controlling are three important functions of management. Planning involves the formulation of objectives and goals that are subsequently translated into specific plans and projects. Scheduling is concerned about the implementation of activities necessary to achieve the laid down plans. The function of control is to institute a mechanism that can trigger a warning signal if actual performance is deviating (in terms of time, cost and some other measures of effectiveness) from the plan. If such a deviation is unacceptable to the concerned manager, he will be required to take corrective action to bring performance in conformity with the plans. The PERT and CPM models are extremely useful for the purpose of planning, scheduling and controlling the progress and completion of large and complex projects or for carrying out the analysis of these three managerial functions. Before we describe the basic concepts used in the construction and analysis of these models, let us first understand the meaning of a project.

What is a project?

A project can be defined as a set of large number of activities or jobs that are performed in a certain sequence determined logically or technologically and it has to be completed within (i) a specified time, (ii) a specified cost and (iii) meeting the performance standards. Examples of a project from fairly diverse fields are given below:

1. Introducing a new product in the market.
2. Construction of a new bridge over a river or construction of a 25 storied building.

14.2 Advanced Management Accounting

3. Executing a large and complex order on jobbing production.
4. Sending a spacecraft to the mars.

14.2 GENERAL FRAMEWORK OF PERT/CPM

A network is a graphical representation of a project, depicting the flow as well as the sequence of well-defined activities and events. Developed during the 1950s, both CPM (Critical Path Method) and PERT (Programme Evaluation and Review Technique) are network techniques/models. The network approach helps project managers in planning,

scheduling and controlling. As a planning tool it helps the manager to estimate the requirements of resources viz., materials, equipment, manpower, cost and time for each activity or tasks of the project. This approach cannot make decisions by its own. It only provides additional information to executives to facilitate decision making process. Also it does not provide solution to every management problem. It certainly helps in identification of those activities, jobs or events which control the completion of the project.

The working methodology of critical path analysis (CPA) which includes both CPM and PERT, consists of following five steps:

1. Analyse and break down the project in terms of specific activities and/ or events.
2. Determine the interdependence and sequence of specific activities and prepare a network.
3. Assign estimates of time, cost or both to all the activities of the network.
4. Identify the longest or critical path through the network.
5. Monitor, evaluate and control the progress of the project by replanning, rescheduling and reassignment of resources.

The central task in the control aspect of these models is to identify the longest path through the network. The longest path is the critical path because it equals the minimum time required to complete the project. All other paths other than the critical path (i.e. non-critical or slack paths) offer flexibility in scheduling and transferring resources, because they take less time to complete than the critical path.

14.3 ADVANTAGES OF CRITICAL PATH ANALYSIS

There are a number of advantages in using critical path analysis.

1. It allows for a comprehensive view of the entire project. Because of the sequential and concurrent relationships, time scheduling becomes very effective. Identifying the critical activities keeps the executive alert and in a state of preparedness, with alternative plans ready in case these are needed. Breaking down the project into smaller components permits better and closer control.
2. Critical path analysis offers economical and effective system of control based on the principle of management by exception i.e. need for corrective action arises only in exceptional situations and in most of other cases, performance is in conformity with the plans.
3. It is a dynamic tool of management which calls for constant review, a reformulation of

Critical Path Analysis 14.3

the network, and finding the current path of relevance and optimum resources allocation.

14.4 FUNDAMENTALS OF A CPA NETWORK

◆ Activity

An activity is any portion of a project which consumes time or resources and has a definable beginning and ending. For example, "laying of pipe" is an activity requiring the use of resource mainly effort. Activity may involve labour, paper work, contractual negotiations, machinery operations, etc. Commonly used terms synonymous with "activity" are "task" and

"job".

Tail Head

7

1 3

Figures 1 and 2

10

3 4

14

14

11 24

3 5

7

1 4 6

2

Figures 3 to 6

14.4 Advanced Management Accounting

Activities are graphically represented by arrows, usually with description and time estimates written along the arrows. The tail of the arrow portraying an activity represents the starting point of the activity and its head represents its completion. The arrow may be straight slanting, or bent but not broken (see figure-1). The arrow is not a vector and need not be drawn to scale.

◆ Events

The beginning and ending points of an activity or a group of activities are called events. Synonyms of an event are "node" and "connectors". An event is often represented graphically by a numbered circle (see figure-2), although any geometric figure such as square, oval, rectangle etc. will serve the purpose. We shall, however, stick to the most commonly used convention for representing an event viz, the circle. A few examples of events are as follows: (i) Material procured, (ii) Design completed, (iii) Project started, (iv) Bricks laid, etc. All activities in a network must commence from some event. Such events are called the tail events because they are connected to the tail of an activity. These are shown in figure 3. Similarly, all activities in a network must have terminal points called the head event, because it is at the head of an activity. These are shown in figure-4. Figure-5 depicts the tail and head events connected by arrows representing activities i.e. it depicts the dual role of an event. Event 14 is the head event for one activity and tail event for another.

In a network, symbol "i" is used for the tail event (also called preceding event) and "j" for the head event (or succeeding event) of an activity. The activity, then being i-j.

If an event represents the joint completion of more than one activity, it is called a merge event. If an event represents the joint initiation of more than one activity, it is called a burst event.

A network is, then, a graphical representation of a project plan, showing the inter-relationship of the various activities. Networks are also called arrow diagrams (see figure-6). When the results of time estimates and computations have been added to a network, it may be used as a project schedule.

14.4.1 Conventions adopted in drawing networks: There are two conventions normally adopted while drawing networks. In the early stages of network drawing, it is suggested that the conventions should be respected until sufficient experience has been gained to justify dropping them. These conventions are:

(a) Time flows from left to right.

(b) Head events always have a number higher than that of the tail events.

The above stated conventions allow activities to be referred uniquely by their tail and head event numbers, so that "activity 3-4" means only "the activity which starts from event 3 and proceeds to event 4"; it cannot mean "the activity which starts from event 4 and finishes at event 3".

Critical Path Analysis 14.5

14.4.2 Graphical representation of events and activities: Events are represented by numbers within circles. Activities are represented by arrows, the arrow-heads represent the completion of the activities. The length and orientation of the arrow are of no significance whatsoever (chosen only for the convenience of drawing). The activity of leaving place A and walking to place B can equally well be represented by figure-7.

14.4.3 Fundamental properties governing the representation of events and activities: The representation of events and activities is governed by one simple dependency rule which requires that an activity which depends upon another activity is shown to emerge from the head event of the activity upon which it depends and that only dependent activities are drawn in this way. Thus, if activity B depends upon activity A, then the two activities are drawn in figure-8.

The dependency rule gives rise to two fundamental properties of events and activities:

1 2 1

1 2

2

2

1 2

1

Figure 7

A B

1 2 3

Figure 8

14.6 Advanced Management Accounting

1. An event cannot occur until all activities leading to it are complete.

2. No activity can start until its tail event is reached.

The above two properties can be combined into a single one, namely that "no activity may start until all previous activities in the same chain are completed.

14.4.4 Logical sequencing and connection of activities: A project entails several activities. The arrows are arranged to show the plan of logical sequence in which the activities of the project are to be accomplished. The sequence is ascertained for each activity by answering the following three queries viz:

- (i) Which activity or activities must be completed before the start of a particular activity?
- (ii) Which activity or activities should follow this?
- (iii) Which activities can be accomplished simultaneously?

The activity or activities which immediately come before another activity without any intervening activities are called predecessor activities to that activity. The activities which follow another activity without any intervening activities are called successor activities to that activity.

In a project of laying a pipe line, the three activities involved may be trenching, laying pipe and welding pipe. To decide the logical connection between these three activities, it is necessary that they be carried out in series, the reasoning being that the pipe cannot be laid until trenching has been done and welding cannot be undertaken until the pipe has been laid. This way we decide the logical sequencing between different activities.

Errors in logical sequencing: Two types of errors in logic may arise while drawing a network, particularly when it is a complicated one. These are known as looping and dangling.

(1) Looping: Normally in a network, the arrow points from left to right. This convention is to be strictly adhered, as this would avoid illogical looping, as shown wrongly below:

1 2 3 4

Figure 9

Critical Path Analysis 14.7

(2) Dangling: The situation represented by the following diagram is also at fault, since the activity represented by the dangling arrow 9-11 is undertaken with no result.

A

8 9 10

11

Figure 10

To overcome the problem arising due to dangling arrows, following rules may be adopted.

(i) All events, except the first and the last, must have at least one activity entering and one activity leaving them.

(ii) All activities must start and finish with an event.

(3) Duplicate activities: Consider the following figure 11:

A

x Y 8

6 7

B

Figure 11

In the above figure, activities A and B may be called duplicate activities because they have same head event (i.e 6) and the same tail event (i.e.7). One remedy for such a situation is the introduction of a dummy activity.

(4) Dummy activity: It is a hypothetical activity which consumes no resource and time. It is represented by dotted lines and is inserted in the network to clarify activity pattern under the following situations:

(i) It is created to make activities with common starting and finishing events distinguishable.

(ii) to identify and maintain the proper precedence relationship between activities that are not connected by events.

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(iii) to bring all "loose ends" to a single initial and a single terminal event in each network using dummies, if necessary.

For example, problem of duplicate activities in the figure-11 above may be circumvented as shown in figure-12.

A

x Y

6 8

B

7

Figure 12

Figure-13 shows three cases for the following set of dependency relationships:

Activity C is dependent upon both A and B.

Activity D is dependent upon A alone.

1 4 2 B 3 C 4

A C

3 A

B D A

1 D 5

3

2 5

B C

2 3 5

A D

1 4 6

Figure 13

The first portrayal (on top left of figure-13) is clearly wrong since it shows D as dependent upon not only A but also B which is not desired. The other portrayal (ii) is also wrong since A is being shown twice and thus contravenes the fundamental axiom of network that there must be one arrow for each activity. The way out to this dilemma is the representation by means of the dummy activity. In the third portrayal of figure -13, C is dependent upon both A and B (via dummy) whereas D is dependent upon just A.

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14.4.5 Numbering the events: The event numbers in a network should in some respect reflect their logical sequences. When a complicated network has been drawn then the problem of assigning numbers to the events involved in the network arises. A rule devised by D.R. Fulkerson, involving the following steps may be followed to resolve the problem of numbering the events.

- (i) An "initial" event is one which has arrow/arrows coming out of it and none of the arrow entering it. In a network there will be only one such event. Call it "1".
- (ii) Delete all arrows coming out from the event 1. This will give us at least one more "initial event".
- (iii) Number these events as "2, 3...."
- (iv) Delete all emerging arrows from these numbered events which will create new initial events. Then follow step (iii).
- (v) Continue the above steps till last event is obtained which has no arrows coming out of it. Consider the numbering of events in the following figure.

y 4

1 2 5 6 7 8

3

x

Figure 14

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A F A F

B B G

C G H

C

A F

A F

B G

B G

C H

C H

A F

A F

B G

B G

C H

C H

Figure 15

Critical Path Analysis 14.11

Here we proceed from left to right. The event with least x-co-ordinate is assigned the smallest integer, say 1. Other events are assigned progressively higher integers with regard to x-co-ordinate. If two or more events (4 and 5 above) have the same x-co-ordinate, the one towards arrow should have higher number.

Furthermore, it is not necessary, and in fact also not desirable to number the events consecutively. It would be a better scheme to number the events as 10, 20, 30, 40, 50, 60, 70, in the above diagram instead of 1,2,3,4,5,6,7. This affords insertion of more activities and events omitted by oversight or having become necessary in view of certain logic revisions. It was mentioned earlier that it is desirable that all the activity arrows point from left to right. If the arrow is vertical it may point downwards or upwards.

For the sake of presentability it is to be recommended that activities emanating from one event or converging to another may make as great angles between themselves as possible.

A few more conventions are given below:

(i) Keep the arrow to the extreme right. (ii) As far as possible avoid drawing arrows that cross each other. Usually by suitable 'stretching' the network diagram it is possible to avoid this. (iii) Where, however, crossing is unavoidable, bridging may be done. This applies to dummies as well. Draw boldly a big network. Smaller ones are confusing. Use of pencil and rubber is recommended.

Exercise : Depict the following dependency relationships by means of network diagrams. The Alphabets stand for activities.

1. A and B control F; B and C control G.
2. A and B control F; B controls G while C controls G and H.
3. A controls F and G; B controls G; while C controls G and H.
4. A controls F and G; B and C control G with H depending upon C.
5. F and G are controlled by A, G and H are controlled by B with H controlled by B and C.
6. A controls F, G and H; B controls G and H with H controlled by C.

Answer. The required networks are given in figure-15

B E

H

A C F

G

D

Figure 16

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J K L

M

Figure 17

F G

A B C D E

H J L

M

I K

Figure 18

Exercise : Find out the superfluous (unnecessary) dummy activities in the networks below.

14.4.6 Basic steps involved in drawing a CPM/PERT network : Network is defined as a diagram representing the activities and events of a project, their sequence and inter-relationships.

The basic steps involved in drawing a network are:

- (i) Breaking up of the entire project into smaller systems known as tasks.
- (ii) For each task ascertain the activities and events to be performed.
- (iii) For each activity determine the preceding and succeeding activities.
- (iv) For each activity determine or estimate the time and other resources needed.
- (v) Draw a network depicting the assembly of tasks into a project.

14.4.7 Network Construction

Illustration

The activities involved in the computer installation process are detailed below. You are required to draw the network.

Activity Predecessor Activities

- A. Physical Preparation None
- B. Organisational Planning None
- C. Personnel Selection B
- D. Equipment Installation A
- E. Personnel Training C

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- F. Detailed Systems Design C
- G. File Conversion F
- H. Establish Standards and Controls F
- I. Programme Preparation H
- J. Programme Testing I
- K. Parallel Operations D, E, G, J.
- L. Finalise Systems Documentation I
- M. Follow up. K, L

A A B < C (ii)

NONE < A
 (i)
 NONE < B B
 B
 C
 D
 D
 A A C < E (iv)
 A<D C<F
 B (iii) B
 C C
 E
 D J F
 F<G
 F<H
 A (v)
 E G I
 H<I
 B C F H I <J
 K D, E, G and
 J < K (vi)
 D J
 E G
 A I
 B C F H
 Figure 19

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Solution

The network is shown in figure 19. The construction has been carried out step by step which is explained below. The symbol '<' is to be read as "precedes" in these figures.

(i) Since A and B are preceded by no activity, they are shown emanating from the start event. We do not put arrows and the head events for A and B because they have been drawn as such only tentatively in the following step (s) their position may have to be changed.

(ii) Here C is shown as dependent upon B. Again we neither show arrow and nor the head event of C because its position is tentative.

(iii) D is shown as dependent upon A.

(iv) We show here C preceding E and F.

(v) F is shown preceding G and H, H precedes I and I precedes J.

(vi) K is shown as dependent upon D,E,G and J, the latter four therefore, are made to

converge on the same head event.

(vii) L is shown as dependent upon I and M as dependent on K and L. Note carefully that K and L are the terminal activities and, therefore, converge on the last event.

About this point the student is cautioned since the beginners do tend to show the last activities dangling and do not converge them to the common last event.

Of course, the network would be constructed in just one diagram (as shown in Figure-19).

We drew so many for explanation only. However, for a stylish presentation it is desirable that this one diagram is suitably sketched i.e. faired up in another diagram. The student may number the events as an exercise.

M

K I < L

L < M

L

D J

E G

A I

B C F H

Figure 19 contd.

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Example : Construct the network for the following activity data:

Activity Preceded by

A None

B None

C A

D B,C

E D

F E

G B,C

H F

I F,G

J H,I

K B

L F,G,K

NONE < A C

A

NONE < B

B

A A < C

B

D

E F

C

D

 $A B < D C$ $C < D D < F$ $B A E < F$

B

H

E F

D

 $B, C < G$

C

 $F < H$

G

A

B Figure 20

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H I

 $E F < H, I$

F

 $G < I$

D

C G

A

B

H

I J

 $I, H < J$

E F

D

C G

A

B

J

0 1

 $B < K$ $F, G, K < L$

- .

L

+ /

* K

Figure 20 (Contd.)

(The students may remember the events as an exercise)

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Solution: The network is shown in figure-20 step by step with dependency relationships that are incorporated at each step which the student should, by now, be able to follow.

Please carefully review the steps where dummies are incorporated.

14.4.8 Concurrent activities: Activities may not always be discrete i.e. they may be done in part allowing the subsequent activities to commence before the preceding activity is fully completed. Activities of this kind are to be frequently encountered in batch production. If, for example, a batch of 50 spindles is to be processed on two machines obviously it is not necessary to process all the items of the batch on the first machine and then transfer these to the next machine. A few items processed on the first machine may be transferred to the second machine before completion of the entire batch on the first machine. Since this is a matter of great practical importance we shall dwell upon it at a greater length. Such simultaneous or concurrent activities are to be encountered in sewage work e.g., trenching, laying pipe, welding pipe and back filling, all going on simultaneously with suitable lags on construction work.

Illustration

A batch of 4 axles is to be proceeded on the following three machines in this sequence:

Lathe (L), Milling (M) and Grinding (G)

Instead of first working on these 4 axles on lathe and then on milling and finally on grinding in sequence, it is desired to process the first axle on the lathe and as and when it is processed, it is taken up on milling and the 2nd axle on the lathe, and so on. In other words, each of the three activities L,M and G have been quartered for the sake of concurrent operations. You are required to draw the network.

Solution

The dependency relationships are sorted out hereunder:

Quartered Activity Preceded by

L1 None

L2 L1

L3 L2

L4 L3

M1 L1

M2 L 2, M 1

M3 L 3, M 2

M4 L 4, M 3

G1 M1

G2 M 2, G 1

G3 M 3, G 2

G4 M 4,G 3

The network is now constructed below. (The student is urged to draw his own network. Mere understanding of our solution is not enough.)

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1 2 4 8

L1 L2 L3 L4

3 5 6 9 10 12

M1 M2 M3 M4

7 11 13 14

G1 G2 G3 G4

Figure 21

(N.B.: The concurrent activities so drawn are known: as 'ladders' in the network jargon.)

Illustration

Draw a network diagram for the following data.

Task Immediate Predecessor

A —

B —

C B

D B

E B

F E

G A, D,C.

Solution

The required network is given below.

4

A G

C

B 2 D

1 3 6

E

F

5

Figure 22

Critical Path Analysis 14.19

Illustration

Draw a network diagram for the following data.

Activity Preceding activities

A —

B A

C A
 D B
 E A
 F B, E
 G C
 H D, F
 I G
 J H, I

Solution: The required network is given below:

5
 E F
 1 A B D H J
 2 4 6 8 9
 C I
 3 G
 7

Figure 23

14.5 CRITICAL PATH ANALYSIS

You have already been familiarised with the logic of the critical path analysis by way of introduction of this chapter.

The purpose of the analysis is two-fold: (i) to find the critical path, i.e. the sequence of activities with the longest duration. Once it is found it is marked in bold sequence of arrows on the network. For a simple network as of figure -24 the various sequences can be enumerated and the durations of activities encompassed by them simply added, to find the critical sequence. As stated earlier, one could indeed end up with more than one critical sequence; and (ii) to find the float associated with each non-critical activity.

♦ Systematic analysis: The enumeration method would be too cumbersome computationally for any real life project, Even the computer would be hard pressed to proceed this way. There is a systematic way that cuts short the analysis time to manageable proportions.

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It is accomplished by performing the following steps:

1. Calculate the time schedule for each activity. This represents the time by which an activity must begin and the time before which it must be completed. The time schedule data for each activity includes, the earliest start, earliest finish, the latest start, latest finish, and finally the float, which is the spare time associated with an activity.
2. Calculate the time schedule for the completion of entire project. This represents the scheduled date for the completion of the entire project and the probability of completing the project on or before the deadline.
3. Identify the critical activities. These activities are the ones which must be started and completed on schedule, or else the project is likely to be delayed.

4. Determine the critical path for the network. This path represents the critical activities which must be closely followed in order to complete the project on time.

14.5.1 Scheduling computations: After preparing the network diagram, we wish to know how long it will take to complete the project and also to identify the activities in the network that are to be placed under strict control. The basic scheduling computations involve a forward and then a backward pass through the network. The process of tracing the network from START to END is called forward pass, and from END to START is called backward pass. Based on a specified occurrence time for the initial network event, the forward pass computations give the earliest start and finish times for each activity. By the specification of the latest allowable occurrence time for the terminal network event, the backward pass computation will give the latest allowable start and finish times for each activity.

14.5.2 Forward pass computations: As stated above, the purpose of the forward pass is to compute the earliest start (EST) and finish time (EFT) for each activity. The EST time indicates the earliest time that a given activity can be scheduled. Earliest finish time for an activity indicates the time by which the activity can be completed, at the earliest. To compute these time estimates, we will first of all compute the earliest allowable occurrence time for various events of the network.

It is a convention to keep the earliest allowable occurrence time of the START event as zero. To understand, how this time estimate for other events is computed, let us consider the following network diagram.

$$E_3 = 14$$

$$3$$

$$8 \ 6$$

$$6 \ 16$$

$$1 \ 2 \ 0 \ 5 \ 6 \ E_6 = 52$$

$$E_1 = 0 \ 10 \ 20 \ E_5 = 36$$

$$E_2 = 6$$

$$4$$

$$E_4 = 16$$

Figure 24

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♦ Earliest allowable occurrence time of an event or Earliest Event Time :

In the network shown above, event 1 stands for the beginning of the activity 1 - 2 and we can say that it occurs at the time zero i.e. $E_1 = 0$. Event 2 stands for the finish of the activity 1-2 thus event 2 can occur at the earliest time E_2 which is computed as

$$E_2 = 0 + D_{12} = 0 + 6 = 6$$

where D_{12} stands for the duration of activity 1 - 2

Event 3 stands for finish of the activity 2-3 and its earliest time is

$$E_3 = E_2 + D_{23} = 6 + 8 = 14$$

The event 4 can occur either at the end of the activity 3 - 4 or at the finish of activity 2 - 4. In this case, there will be two time estimates as follows:

$$E_4 = E_3 + D_{34} = 14 + 0 = 14$$

$$E_4 = E_2 + D_{24} = 6 + 10 = 16$$

In case two or more time estimates exist for a particular event, then the time estimate with maximum value is retained as the earliest event time and other values are discarded. This maximum value represents the completion of all the activities ending at the event under consideration. In the above example, the earliest event time for event 4 will be 16.

A general rule can also be given here for determining the earliest event time as below:

$$E_j = \text{Max} (E_i + D_{ij})$$

where E_j is the earliest time for event j , E_i is the earliest time for event i , and D_{ij} is the duration of the activity $i-j$.

◆ Earliest start and earliest finish times of an activity

After computing the earliest event time of various events, one can easily compute the earliest start and finish times of all the activities on the network. The earliest start time of an activity is given by the earliest allowable occurrence time of the tail event of that activity. Thus, in our example, the earliest start time of the activity 1-2 will be given by the earliest time of the event 1 i.e. it will be 0. The earliest start time for the activities 2 - 3 and 2 - 4 will be given by the earliest time of event 2 which is equal to 6. The earliest time for the activities 3-4 and 3-5 will be 14 which is the earliest time for the event 3 and so on.

The earliest finish time of an activity will be simply equal to the earliest start time of the activity plus the duration of that activity. Hence, in our example, earliest finish time of activity 1-2 will be $0+6 = 6$, for activity 2-3, it will be $6+8 = 14$ and for activity 2-4, it will be $6+10 = 16$ and so on.

The complete computations for all the activities are shown in columns (3) and (4) of table -1 given on the next page.

(*Note: In various networks we have omitted subscripts for E_i 's and L_j 's)

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14.5.3 Backward pass computations: The purpose of the backward pass is to compute the latest start and finish times for each activity. These computations are precisely a "mirror image" of the forward pass computations. The term "latest allowable occurrence time of an event" (denoted by L_i) is used in the sense that the project terminal event must occur on or before some arbitrary scheduled time. Thus, the backward pass computations are started rolling back by arbitrarily specifying the latest allowable occurrence time for the project terminal event. If no schedule date for the completion of the project is specified, then the convention of setting the latest allowable time for the terminal event equal to its earliest time, determined in the forward pass, is usually followed, i.e. $L = E^*$ for the terminal event of the project. This convention is called the zero slack convention. Following this, one can also interpret the latest allowable activity finish time (LFT) as the time to which the completion of an activity can be delayed without directly causing any increase in the total time to complete the project.

To explain the computation, let us again consider the network diagram in figure-24. The terminal event is 6 so we set $L_6 = E_6 = 52$ and we start rolling back. The latest allowable occurrence time for the events 5 and 4 are $L_5 = 52 - 16 = 36$ and $L_4 = 36 - 20 = 16$ respectively. It may be noted here that we can roll back to event 3 via activity 3-5 as well as activity 4-3. So there are two latest allowable occurrence times for the event 3 as given below;

$$L_3 = L_4 - D_{34} = 16 - 0 = 16$$

$$L_3 = L_5 - D_{35} = 36 - 6 = 30$$

We retain the minimum value as the latest occurrence time for the event 3 and ignore other values. Therefore, the latest allowable occurrence time for the event 3 is 16. Similarly

$$L_2 = L_3 - D_{23} = 16 - 8 = 8$$

$$\text{and } L_2 = L_4 - D_{24} = 16 - 10 = 6$$

The latest occurrence time for the event 2 is thus 6 and the latest occurrence time for the event 1 is equal to its earliest time i.e. zero.

In general, the latest allowable occurrence time of an event can be calculated by selecting an appropriate formula among the following two:

$$L_i = L_j - D_{ij}$$

$$\text{or } L_i = \text{minimum } (L_j - D_{ij})$$

The second formula is used for the event having two or more latest allowable occurrence time estimates.

♦ Latest start and latest finish times of an activity

After computing latest allowable occurrence time for various events, one can compute the latest start and latest finish times of an activity. The latest finish time of an activity is equal to the latest allowable occurrence time of the head event of that activity.

$$\text{i.e. } LFT(i - j) = L_j$$

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The latest start time of an activity is equal to its latest finish time minus its duration.

$$\text{i.e. } LST(i - j) = LFT(i - j) - D_{ij}$$

These computations are shown in column (4) and column (6) of table-1 given below:

Table 1

START FINISH

Activity Duration Earliest time Latest time Earliest time Latest Time

(1) (2) (3) (4) (5) (6)

1-2 6 0 0 6 6

2-3 8 6 8 14 16

2-4 10 6 6 16 16

3-4 0 14 16 14 16

3-5 6 14 30 20 36

4-5 20 16 16 36 36

5-6 16 36 36 52 52

14.5.4 The critical path determination: After having computed various time estimates,

we are now interested in finding the critical path of the network. A network will consist of a number of paths. A path is a continuous series of activities through the network that leads from the initial event (or node) of the network to its terminal event. For finding the critical path, we list out all possible paths through a network along with their duration. In the network under consideration, various paths have been listed below.

Path Length (in days)

1-2-3-5-6 36

1-2-4-5-6 52

1-2-3-4-5-6 50

♦ Critical path: A path in a project network is called critical if it is the longest path. The activities lying on the critical path are called the critical activities.

In the above example, the path 1-2-4-5-6 with the longest duration of 52 days is the critical path and the activities 1-2, 2-4, 4-5 and 5-6 are the critical activities.

14.5.5 Calculation of Floats: It may be observed that for every critical activity in a network, the earliest start and latest start time are the same. This is so since the critical activities cannot be scheduled later than their earliest schedule time without delaying the total project duration, they do not have any flexibility in scheduling. However, non-critical activities do have some flexibility i.e. these activities can be delayed for some time without affecting the project duration. This flexibility is termed as slack in case of an event and as float in case of an activity.

(*Note: In various networks we have omitted subscripts for E_i s and L_j s)

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Some people do not make any distinction between a slack and a float.

♦ Slack time for an event

The slack time or slack of an event in a network is the difference between the latest event time and the earliest event time. Mathematically it may be calculated using the formula $L_i - E_i$, where L_i is the latest allowable occurrence time and E_i is the earliest allowable occurrence time of an event i .

♦ Total float of an activity: The total activity float is equal to the difference between the earliest and latest allowable start or finish times for the activity in question. Thus, for an activity $(i-j)$, the total float is given by

$$TF_{ij} = LST - EST \text{ or } TF_{ij} = LFT - EFT$$

In other words, it is the difference between the maximum time available for the activity and the actual time it takes to complete. Thus, total float indicates the amount of time by which the actual completion of an activity can exceed its earliest expected completion time without causing any delay in the project duration.

♦ Free Float: It is defined as that portion of the total float within which an activity can be manipulated without affecting the float of the succeeding activities. It can be determined by subtracting the head event slack from the total float of an activity.

$$\text{i.e. } FF_{ij} = TF_{ij} - (\text{slack of event } j)$$

The free float indicates the value by which an activity in question can be delayed beyond

the earliest starting point without affecting the earliest start, and therefore, the total float of the activities following it.

♦ Independent float : It is defined as that portion of the total float within which an activity can be delayed for start without affecting float of the preceding activities. It is computed by subtracting the tail event slack from the free float of an activity.

$$IF_{ij} = FF_{ij} - (\text{slack of event } i)$$

The independent float is always either equal to or less than the free float of an activity. If a negative value is obtained, the independent float is taken to be zero .

♦ Interfering float: Utilization of the float of an activity can affect the float of subsequent activities in the network. Thus, interfering float can be defined as that part of the total float which causes a reduction in the float of the successor activities. In other words, it can be defined as the difference between the latest finish time of the activity under consideration and the earliest start time of the following activity, or zero, whichever is larger. Thus, interfering float refers to that portion of the activity float which cannot be consumed without affecting adversely the float of the subsequent activity or activities.

Illustration

Activity Duration

1-2 4 days

1-3 12 days

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1-4 10 days

2-4 8 days

2-5 6 days

3-6 8 days

4-6 10 days

5-7 10 days

6-7 0 days

6-8 8 days

7-8 10 days

8-9 6 days

With the help of the activities given above draw a network. Determine its critical path, earliest start time, earliest finish time, latest start time, latest finish time, total float, free float and independent float.

Solution

The network based on the activities given in the example is as follows:

$$E=4$$

$$L=4 \quad E = 10$$

$$6 \quad L = 12$$

$$2 \quad 5$$

$$4 \quad 10$$

$$8 \quad 7 \quad E = 22$$

$L = 22$
 10 10 0 10
 $E=0$ 1 6
 4 16 8 9
 $L=0$ 8 $E = 38$
 $E = 12$ $E = 22$ $E = 32$
 $L = 32$ $L = 38$
 $12 L = 12$ 8 $L = 22$
 $3 E = 12$
 $L = 14$

Figure - 25

The time estimates and floats are calculated as below:

Job or Duration EST EFT LST LFT Slack of
 activity of an (= E_i) ($E_i + D_{ij}$) ($L_j - D_{ij}$) (= L_j) tail head Total float Free float Independent
 activity event event (LST- (Total Float (Free float-
 D_{ij} ($L_i - E_i$) ($L_j - E_j$) EST) -Slack of slack of
 head event) tail event)

1-2 4 0 4 0 4 0 0 0 0 0
 1-3 12 0 12 2 14 0 2 2 0 0
 1-4 10 0 10 2 12 0 0 2 2 2

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2-4 8 4 12 4 12 0 0 0 0 0
 2-5 6 4 10 6 12 0 2 2 0 0
 3-6 8 12 20 14 22 2 0 2 2 0
 4-6 10 12 22 12 22 0 0 0 0 0
 5-7 10 10 20 12 22 2 0 2 2 0
 6-7 0 22 22 22 22 0 0 0 0 0
 6-8 8 22 30 24 32 0 0 2 2 2
 7-8 10 22 32 22 32 0 0 0 0 0
 8-9 6 32 38 32 38 0 0 0 0 0

Critical path is represented by 1-2-4-6-7-8-9.

(Note: In this table we have given detailed calculations merely for explanation.)

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Illustration

Analyse the network below for the critical path and for different floats

5
 2 6
 $E=3$ $E = 19$
 $L=3$ $L = 19$
 3 10 6 1

$E=0$ 1 3 15 $E = 17$

$L=0$ 4 4 $L = 18$

$E = 13$

$L = 13$

14 1

4 $E = 14$

$L = 17$

Figure 26

Network Analysis Table

Activity Duration Start Finish Float

E L E L Total FreeInd. Interfering

1-2 3 0 0 3 3 0 0 0 0

1-3 4 0 9 4 13 9 9 9 0

1-4 14 0 3 14 17 3 0 0 3

2-3 10 3 3 13 13 0 0 0 0

2-6 5 3 14 8 19 11 11 11 0

3-5 4 13 14 17 18 1 0 0 1

3-6 6 13 13 19 19 0 0 0 0

4-5 1 14 17 15 18 3 2 0 1

5-6 1 17 18 18 19 1 1 0 0

(N.B.: Activities must always be arranged in the i-j sequence.)

Explanation: Consider activity 1 - 3. The slack of head event 3 is $13 - 13 = 0$. Therefore, free float = $9 - 0 = 9$. Likewise slack of tail event 1 is $0 - 0 = 0$. Therefore independent float = 9.

Illustration

Analyse the network below for the critical path and calculate total float, free float and independent float for each activity.

14.28 Advanced Management Accounting

$E = 43$ $E = 63$ $E = 83$

$E = 36$ $L = 43$ $L = 63$ $L = 83$

$L = 41$

4 8 9 10

A/36 D/2 K/20 M/20

0 J/9

1 2 3 E/1 L/8

B/4 C/2

$E=0$ $E=4$ F/15 7

G/9

$L=0$ $E=6$ $E = 34$

$L=4$ $L=6$ $L = 34$

5

I/9

E = 21 H/4 Critical Path: 2-3-5-6-7-8-9-10

L = 21

6 E = 25

L = 25

Figure - 27

Solution

Network Analysis Table

i-j D Start Finish Floats

E L E L T F I

1- 2 4 0 0 4 4 0 0 0

1-4 36 0 5 36 41 5 0 0

2-3 2 4 4 6 6 0 0 0

3-5 15 6 6 21 21 0 0 0

3-8 10 6 33 16 43 27 27 27

4-8 2 36 41 38 43 5 5 0

5-6 4 21 21 25 25 0 0 0

5-8 9 21 34 30 43 13 13 13

6-7 9 25 25 34 34 0 0 0

7-8 9 34 34 43 43 0 0 0

7-9 8 34 55 42 63 21 21 21

8-9 20 43 43 63 63 0 0 0

9-10 20 63 63 83 83 0 0 0

Critical Path : BCHIJKM = 83 weeks.

Illustration

A small assembly plant assembles PCs through 9 interlinked stages according to following precedence / process:

Critical Path Analysis 14.29

Stage Duration

From to (Hours)

1-2 4

1-3 12

1-4 10

2-4 8

2-5 6

3-6 8

4-6 10

5-7 10

6-7 0

6-8 8

7-8 10

8-9 6

- (i) Draw an arrow diagram (network) representing the above assembly work.
- (ii) Tabulate earliest start, earliest finish, latest start and latest finish time for all the stages.
- (iii) Find the critical path and the assembly duration.
- (iv) Tabulate total float, free float and independent float.

Solution

- (i) The network diagram representing the given assembly work is as follows:

E=4 6 E = 10

> 5

L=4 2 L = 12

10 >

4 7

8

E=0 0 10

L=0

10 10 8 6

1 > 4 > 6 > 8 > 9

E = 12 E = 22 E = 32 E = 38

L = 12 L = 22 L = 32

12 L = 38

8

3 E = 12

L = 14

Figure 28

14.30 Advanced Management Accounting

- (ii) The earliest start, earliest finish, latest start and latest finish times for all the stages are calculated in the table given below:

Stage or Duration Earliest Earliest Latest Latest Floats

Activity (Hours) Start Finish Start Finish Total Free Independent

(EST) (EFT) (LST) (LFT) (LST-EST) (Total Float (Free Float)

or -Slack of - Slack of

(LFT-EFT) Head event) tail event)

(1) (2) (3) (4) (5) (6) (7) (8) (9)

1-2 4 0 4 0 4 0 0 0

1-3 12 0 12 2 14 2 0 0

1-4 10 0 10 2 12 2 2 2

2-4 8 4 12 4 12 0 0 0

2-5 6 4 10 6 12 2 0 0

3-6 8 12 20 14 22 2 2 0

4-6 10 22 12 22 0 0 0

5-7 10 12 20 12 22 2 2 0

6-7 0 10 22 22 22 0 0 0

6-8 8 22 30 24 32 2 2 2

7-8 10 22 32 22 32 0 0 0

8-9 6 22 38 32 38 0 0 0

(iii) The critical path for the above network is given by 1 - 2 - 4 - 6 - 7 - 8 - 9. The assembly duration is 38 hours.

(iv) The total float, free float and independent float are given by columns (7), (8) and (9) respectively of the above table.

14.6 Distinction between PERT & CPM

PERT CPM

PERT is used for non-repetitive Jobs like planning the CPM is used for non-Repetitive Job assembly of the space platform like building a house

It is a probabilistic Model It is a deterministic Model

It is event oriented It is activity oriented

It incorporates Statistical Analysis It does not incorporate Statistical Analysis

It serve as useful control device It is difficult to use CPM as a control device

Critical Path Analysis 14.31

SUMMARY :

- Two most commonly used network Models
 - ü Critical Path Method(CPM)
 - ü Programme Evaluation Review Technique(PERT)
- Construction of Network diagrams
- Calculation of Critical Path and Project Completion time together with Floats, earliest start and finishing times
- Application of technique to crash cost problems
- Critical path should be longest route through the network
- Resource schedule problems and use of bar charts
- Identifying the project duration
- Cheapest critical activities to crash and not reduce project completion time by reducing non-critical activities

SELF-EXAMINATION QUESTIONS

1. Define a project and briefly explain the four common implications which characterise a project and state the five steps of the working methodology of critical path analysis.
2. Distinguish between PERT and CPM.
3. What kinds of decision making situation may be analysed by using CPM technique.
4. Draw a network corresponding to the following information. Obtain the early and late start

and completion times, and determine the critical activities.

Activity Duration

1-2 4 days

1-3 6 days

2-6 8 days

3-4 7 days

3-5 4 days

4-6 6 days

5-6 5 days

5-7 19 days

6-7 10 days

14.32 Advanced Management Accounting

5. A work project consists of twelve activities labelled A through L. Upon being asked to specify the order in which the jobs had to be done, the managers answered as follows :

A, B and C are the first activities of the project and can start simultaneously and immediately;

A and B precede D while B precedes E, F and H. Activities F and C precede G while E and

H precede I and J. C, D, F and J precede K which in turn, precedes L. Further I, G and L are

the terminal activities of the project.

The completion times of the various activities are listed here :

Activity Time (days)

A 6

B 4

C 10

D 1

E 6

F 3

G 14

H 1

I 9

J 2

K 7

L 5

(a) Draw a network diagram corresponding to this project.

(b) Obtain the lengths of all the paths and determine critical path.

ANSWER TO SELF-EXAMINATION QUESTIONS

5. The network diagram is shown in Figure below :

E 4

H

2 6

I

B F

C G

1 5 9

A J L

D K

3 7 8

Figure 29

Critical Path Analysis 14.33

The various paths and their lengths are as follows :

Path Length

1-3-7-8-9 19

1-2-3-7-8-9 17

1-2-4-5-7-8-9 19

1-2-5-7-8-9 24* Critical

1-2-5-9 19

1-2-4-5-9 14

1-2-6-7-8-9 19

1-2-6-9 21

1-6-7-8-9 22

1-6-9 24* Critical

ESD.36 System Project Management

Lecture 1

- Class Introduction

Instructor(s)

Prof. Olivier de Weck

Dr. James Lyneis, Prof. Dan Braha

Today's Agenda

□ Welcome and Introductions

□ Definitions – Initial Discussion

□ Course Objectives

□ Schedule

□ Term Project, Homeworks

□ Questions?

2

Introductions

□ Olivier de Weck

□ Dipl. Ing. Industrial Engineering – ETH Zurich '93

□ 1993-1997 Engineering Program Manager Swiss

F/A-18 Project, McDonnell Douglas, St. Louis

- S.M. '99 Ph.D. '01 Aerospace Systems – MIT
- Associate Professor – dual appointment AA and ESD, Executive Director Production in Innovation Economy (PIE) Study
- Research:
- Systems Engineering for Changeability and Commonality
- <http://strategic.mit.edu>
- Space Logistics
- <http://spacelogistics.mit.edu>

3

Introductions

- James Lyneis
- S.B.s MIT – EE and System Dynamics ('71)
- PhD – Univ. of Michigan (Mgt. Science; '74)
- Sloan Faculty '74-'78; Senior Lecturer '98-Present
- Professor of the Practice, WPI '02-Present
- Consultant with Pugh-Roberts Associates, '78-'02, working on many project models

4

Introductions

- Dan Braha
- Professor at University of Massachusetts
- Sabbatical at MIT ESD in AY 2012/13
- Affiliation with New England Complex Systems Institute (NECSI)
- Specializes in Complexity research, with application to complex projects and human organizations

The Structure and Diffusion Dynamics of Large Scale Organizational Networks Social Network Analysis of ProductDesign and Development Organizational Networks

5

Introductions

- Let's go around the room & remote sites
- Name

- Company and Job (present or past)
- One observation on project management

6

Project Definition

- A Project is a set of tasks that
- Are related to each other
- Have a specific objective to be completed

within certain specifications

- Have defined start and end dates
- Have funding limits
- Consume resources

7

The “Iron Triangle”

Cost Scope

Project

- Why “iron” triangle?

Schedule □ Risk if all three are
constrained !

8

System Definition

- A System is a set of physical or virtual objects
whose interrelationships enable desired
function(s).

- more than the sum of its parts
- Undesired (emergent) functions often exist
- System complexity scales with the number of
objects as well as the type and number of
interconnections between them
- Instantaneously available functions, versus
“lifecycle” properties (scalability, flexibility,
robustness ...)

- A Product is a “System” sold for profit

9

Example System: F/A-18 Aircraft

- Clean sheet design ~ 1978 F/A-18 A/B
- Re-designs: C/D (1987), E/F (1999)
- Hardware, Software, Humans ...

☐ What is inside the system boundary?

10

Project Management

☐ Project Management comprises a body of methods and tools that facilitate the achievement of project objectives

☐ Within time

☐ Within cost

☐ Within scope

☐ At the desired performance/specification level

☐ While effectively and efficiently utilizing resources

☐ While carefully managing risks and opportunities

11

Research and Development

R D

Research, Technology Product/System

Development Development

☐ Unstructured methods ☐ Structured methods

☐ Difficult to plan ☐ Generally planned

☐ Predictable

☐ Unpredictable

Our focus is on

downstream

development.

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+ Discussion Point 1:

- Why is complex (S)PM hard?

☐ Main obstacles to easy project success:

☐ Poorly defined project objectives or shifting system requirements

☐ What do you think? ...

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Concept Question 1 (DEMO)

☐ A project is

☐ A - ongoing management of facilities

☐ B - a finite undertaking with a specific goal

☐ C - task-based design

☐ D - doomed to failure from the outset

☐ E – all of the above

☐ F – none of the above

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Relationship w/other SDM core classes

☐ System Architecture (ESD.34) is about the “DNA” of the ARTIFACTS themselves – atomic unit: object

☐ Concept, form, function, decomposition ...

☐ Systems Engineering (ESD.33) is about the PROCESSES to understand and design systems – atomic unit: process

☐ QFD, DOE, Requirements Analysis and Verification, ...

☐ Integrating the Lean Enterprise (ESD.61J) is about the PEOPLE and ORGANIZATIONS – atomic unit: person

☐ Principles of lean manufacturing, organizational models

☐ System Project Management (ESD.36) is about how to best utilize resources to implement a set of objectives – atomic unit: task

☐ CPM, DSM, System Dynamics

15

Task as an Object-Process-Diagram

System

Inputs n r Elements

o Task m

Resources Executing Deliverables

p q

Time

Personnel Facilities

Money (\$)

Consumables Using OPM Nomenclature

16

Ref: OPM, D. Dori, 2002

Project = set of related tasks

m

o A n

r

m

q

p

B n

r

m

q D

n

r

q

F m

Deliverable from A

is an input to C B and D use the p q

same people

r

m

C n r F requires deliverables

From D and E

p q E m

p

C and E need access

to the same facility

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+ System Project Management

ESD.36 Framework

Project

Preparation Project Project

Planning Monitoring

Project

Control

Enterprise has

chosen what product Project

or system to develop Adaptation Project

Completion

Next

Project Project

Learning

Start

18

- Course Objectives

□ Introduce advanced methods and tools of Project

Management in a development context

□ CPM/PERT, Critical Chain, Design Structure Matrix

□ System Dynamics

□ Earned Value Management

- Understand how methods work (strengths, limitations)
- Industry Examples
- Gain appreciation for organizational and human aspects
- Case Studies
- Managing International Projects, Portfolios of Projects ...
- Learn from each other
- Class Discussions
- Project Assignments
- □ Improve development projects in your career/firm

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+ Class

Schedule

(next 7 weeks, refer to syllabus)

Business Trip 20

Readings

- Required Readings
- NO Paper Class Reader Packet
- Read ahead of lecture ~ 1-2 papers/chapters per session
- Check reading assignments in the syllabus
- Next:
- MIT Press Book Chapter: Introduction to PM
- ABCs of the Critical Path Method (1963)
- Optional Readings
- Textbooks
- Available at MIT Library (Dewey)
- Purchase only if you think useful beyond class
- (e.g. MIT COOP, amazon.com etc)

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Draft Textbook

- “Successfully Designing and Managing Complex Projects”
- de Weck, Lyneis
- MIT Press, draft in development
- textbook to support SDM core class ESD.36
- current draft ~ 300 pages

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Case Studies

- Case 1: Civil Engineering Project
- BAE Systems: DIA Baggage Handling System
- HBS-9-396-311
- or “live case” BP Wind Farm Development

- Case 2: Aerospace Project
- Raise and Fall of Iridium (or Mission to Mars HBS-9-603-083)
- HBS-9-601-040
- Presented by Prof. Joel Schindall
- Case 3: Software Project
- Microsoft .Net (and/or Microsoft Office 2000)
- HBS-9-602-086
- Presented by Prof. Paulo Gomes
- Case 4: Voted by Students

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Project Assignment

- Apply Design Structure Matrix (DSM) method, generally at your sponsor company site
- System Dynamics Project (incl. simulation)
- Survey of Methods & Tools in company
- Analyze Success or Failure of a significant Past Product/System Development Project
- Work in teams of 4 (nominally)
- 1-page project proposals due on 9/18
- Get approval by 9/25
- Project Update due on: 10/25
- Final presentation in class on December 4 or 6

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Previous Project Examples

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
36 37 38 39 40 41

Marketing & Competition Research x 1 1

Cost Risk Analysis to Bid 1 x 1

Customer Requirement Negotiation 1 1 x

DSM

Technical Screening for Qualification 1 1 1 x

Project Authorization & Kick off 1 1 1 x

Failure Effect Mode Analysis x

Control of Engineering Changes

Project

Design for Catalyst Volume & Emissions 1 1 1 1 1

CAE for Catalyst Volume & Emission 1 1 1 x 1

Design for Tuning Volume, Pressure & Flow 1 1 1 x 1 1

CAE for Tuning Volume, Pressure & Flow 1 1 1 x 1

Design for Legal Noise and Sound Quality 1 1 1 x 1 1

CAE for Legal Noise and Sound Quality 1 1 1 x 1

Design for Durability Target 1 1 1 x 1 1 1 1

CAE for Durability Target 1 1 1 x 1 1 1

Cost Analysis for Manufacturing 1 1 1 1 x 1

Exhaust

Prototype for DV Test & Pre-prod Build 1 1 1 1 1 1 1 1 1 x

Design Verification Test 1 1 x

Testing for Emission 1 1 x

Testing for Tuning Volume, Pressure & Flow 1 1 x

System

Testing for Legal Noise and Sound Quality 1 1 x

Sub-tier Supplier DV Bench Testing 1 1 1 1 1 x

Vehicular Proving Ground Testing 1 1 1 1 1 1 1 1 1 1 1 1 x

Data Analysis and Editing 1 x

Design Sub-System Durability Testing

Design Release

Release of Accurate 3D Geometry

1 1

1

1

1

1

1

1

1

1 1 x

x

x

1

Issues Sales Authority to Release (SAR) 1 1 x

Creation and Release of all 2D drawings 1 1 1 x

Mfg Facility Lineup & Tooling Process 1 1 x 1 1

Mfg Cell Setup and Fine Tuning 1 1 1 x 1 1 1

PSO & "Run-at-Rate" Demonstration 1 1 1 1 x

Quality Control for Manufacturing 1 1 1 1 1 x

Mfg Process Evaluation 1 1 1 1 1 1 x

Product Verification Testing 1 1 1 1 1 1 1 x

Sub-tier Supplier PV Testing 1 1 1 1 1 1 1 1 x

Production Part Approval Process 1 1 1 1 x

Production 1 1 1 x

Warranty Cost Analysis 1 1 1 1 1 1 1 1 1 1 1 x

Customer Satisfaction Evaluation 1 1 1 1 1 1 1 1 1 x

- Understand Iterations
- Reduce Expected Project Duration

25

Understanding the sources of change --

External factors; management responses; side effects

Out-of-Sequence

Work, Worksite

Congestion,

Coordination

Fatigue, Problems,

Burnout Morale Problems

Average P eo p le P ro d u ctivity Q u ality

Employee Skill

and Quality P ro g ress

W o rk W o rk

Overtime To Be Done R e a lly D o n e

Scope Customer

Growth Changes Schedule

Know n U n d is c o v e r e d

Acceleration

R e w o rk R e w o rk

R ew o rk

D isco very

Hiring

Apparent

Progress

26

+ Understand the dynamics of a cancelled
project at NASA

Baseline Plan Staffing

FY2000 Replan

700

Actuals Through FY02 & Projections FY02-
FY06 (FY2000 Re-plan)

600 Model for Baseline

Project

Cancelled

500

September

2002
400
Staff - EP
300
200
FY00 Re-plan
First Launch
100 Baseline First
Hypergolic Launch
Maintenance
0 Facility Set
Operational
M1-3
M4-6
M7-9
M10-12
M13-15
M16-18
M19-21
M22-24
M25-27
M28-30
M31-33
M34-36
M37-39
M40-42
M43-45
M46-48
M49-51
M52-54
M55-57
M58-60
M61-63
M64-66
M67-69
M70-72
M73-75
M76-78
M79-81
M82-84
M85-87

M88-90

M91-93

M94-96

M97-99

M100-102

M103-105

M106-108

M109-111

M112-114

M115-117

M118-120

M121-123

M124-126

M127-129

M130-132

FY1996 FY1997 FY1998 FY1999 FY2000 FY2001 FY2002 FY2003 FY2004 FY2005 FY2006

Sep 4, 2008

Time 27

Discussion Point 2: Failure?

- ☐ What can lead to projects failing ?
- ☐ What is success/failure?
- ☐ Project Manager is unqualified and overwhelmed.
- ☐ What else...

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Homework Assignments

- ☐ 6 Individual assignments,
- but can cooperate
- (acknowledge !)

- ☐ Don't spend more than
- ~10-15 hours per HW !

Electrical CityCar

Design Project

HW1: Critical Path and Network of Tasks

HW2: Design Structure Matrix and Iterations

HW3: System Dynamics – Initial Model

HW4: Budgeting and Earned Value

HW5: System Dynamics – Brook's Law

HW6: Project Organizational Design

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Grading

- Homeworks 60%
- Project Assignment 25%
- Active Participation 15%
- Total 100%
- Each HW counts 10%
- All project team members receive same project grade, work together
- People in this class do get A,B,C ...even F ... you want a good grade? ... you have to earn it !

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The Course Site

- Getting Started

- SDM students should add themselves to the class list (if not already there)
- Non-SDM students get permission from instructors
- Contact TA via email to be added
- Course number is ESD.36
- Make sure you go to the Fall 2012 version

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The Course Site

- Main features

- Syllabus (under > Materials > General)
- Calendar
- Schedule
- Handouts
- Go to "Materials" (store all shared files)
- will be posted before each lecture
- Homework
- Submit assignments through the course site
- Receive comments and grades
- FAQ / Forums
- Forum will be used for various topics

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- Remember

- Read the syllabus!
- Answers many of your questions
- Take a look at what's already uploaded

- Reading for next class
- Sign up on the course site if needed
- Non-SDM students need permission of instructor to enroll.
- Use your name cards every class.
- PLEASE ... okay, pay attention
- Laptops needed to answer concept questions
- IM TA's during lecture from remote sites if problems surface

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MIT OpenCourseWare

<http://ocw.mit.edu>

ESD.□□ 6\VWHP□3URMHFW□0DQDJHPHQW

Fall 2012

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Module B

Transportation and

Assignment Solution

Methods

B-1

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B-2 Module B Transportation and Assignment Solution Methods

Solution of the Transportation Model

The following example was used in Chapter 6 of the text to demonstrate the formulation of the transportation model. Wheat is harvested in the Midwest and stored in grain elevators in three different cities—Kansas City, Omaha, and Des Moines. These grain elevators supply three flour mills, located in Chicago, St. Louis, and Cincinnati. Grain is shipped to the mills in railroad cars, each of which is capable of holding one ton of wheat. Each grain elevator is able to supply the following number of tons (i.e., railroad cars) of wheat to the mills on a monthly basis:

Grain Elevator Supply

1. Kansas City 150
2. Omaha 175
3. Des Moines 275

Total 600 tons

Each mill demands the following number of tons of wheat per month.

Mill Demand

- A. Chicago 200

B. St. Louis 100
 C. Cincinnati 300
 Total 600 tons

The cost of transporting one ton of wheat from each grain elevator (source) to each mill (destination) differs according to the distance and rail system. These costs are shown in the following table. For example, the cost of shipping one ton of wheat from the grain elevator at Omaha to the mill at Chicago is \$7.

Mill

Grain Elevator A. Chicago B. St. Louis C. Cincinnati

1. Kansas City \$6 \$8 \$10
2. Omaha 7 11 11
3. Des Moines 4 5 12

The problem is to determine how many tons of wheat to transport from each grain elevator to each mill on a monthly basis in order to minimize the total cost of transportation. The linear programming model for this problem is formulated in the equations that follow: minimize $Z = \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C}$ subject to

$$x_{1A} + x_{1B} + x_{1C} = 150$$

$$x_{2A} + x_{2B} + x_{2C} = 175$$

$$x_{3A} + x_{3B} + x_{3C} = 275$$

$$x_{1A} + x_{2A} + x_{3A} = 200$$

$$x_{1B} + x_{2B} + x_{3B} = 100$$

$$x_{1C} + x_{2C} + x_{3C} = 300$$

$$x_{ij} \geq 0$$

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Solution of the Transportation Model B-3

In this model the decision variables, x_{ij} , represent the number of tons of wheat transported from each grain elevator, i (where $i = 1, 2, 3$), to each mill, j (where $j = A, B, C$). The objective function represents the total transportation cost for each route. Each term in the objective function reflects the cost of the tonnage transported for one route. For example, if 20 tons are transported from elevator 1 to mill A, the cost of \$6 is multiplied by $x_{1A}(=20)$, which equals \$120.

The first three constraints in the linear programming model represent the supply at each elevator; the last three constraints represent the demand at each mill. As an example, consider the first supply constraint, $x_{1A} + x_{1B} + x_{1C} = 150$. This constraint represents the tons of wheat transported from Kansas City to all three mills: Chicago (x_{1A}), St. Louis (x_{1B}), and Cincinnati (x_{1C}). The amount transported from Kansas City is limited to the 150 tons available. Note that this constraint (as well as all others) is an equation ($=$) rather than a ... inequality because all the tons of wheat available will be needed to meet the total demand of 600 tons. In other words, the three mills demand 600 total tons, which is the

exact amount that can be supplied by the three grain elevators. Thus, all that can be supplied will be, in order to meet demand. This type of model, in which supply exactly equals demand, is referred to as a balanced transportation model. The balanced model will be used to demonstrate the solution of a transportation problem.

Transportation problems are Transportation models are solved manually within the context of a tableau, as

solved manually within a tableau in the simplex method. The tableau for our wheat transportation model is shown in

format. Table B-1.

Table B-1

To

The Transportation Tableau

From A B C Supply

6 8 10

1 150

7 11 11

2 175

4 5 12

3 275

Demand 200 100 300 600

Each cell in a transportation tableau Each cell in the tableau represents the amount transported from one source to one des-

is analogous to a decision variable tination. Thus, the amount placed in each cell is the value of a decision variable for that cell.

that indicates the amount allocated For example, the cell at the intersection of row 1 and column A represents the decision vari-

from a source to a destination.

able x_{1A}. The smaller box within each cell contains the unit transportation cost for that

The supply and demand values route. For example, in cell 1A the value, \$6, is the cost of transporting one ton of wheat

along the outside rim of a tableau from Kansas City to Chicago. Along the outer rim of the tableau are the supply and demand

are called rim requirements. constraint quantity values, which are referred to as rim requirements.

The two methods for solving a transportation model are the stepping-stone method and the modified distribution method (also known as MODI). In applying the simplex method,

Transportation models do not start

an initial solution had to be established in the initial simplex tableau. This same condition

at the origin where all decision must be met in solving a transportation model. In a transportation model, an initial feasi-

variables equal zero; they must be ble solution can be found by several alternative methods, including the northwest corner

given an initial feasible solution. method, the minimum cell cost method, and Vogel's approximation model.

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B-4 Module B Transportation and Assignment Solution Methods

The Northwest Corner Method

In the northwest corner method With the northwest corner method, an initial allocation is made to the cell in the upper

the largest possible allocation is left-hand corner of the tableau (i.e., the "northwest corner"). The amount allocated is the

made to the cell in the upper left- most possible, subject to the supply and demand constraints for that cell. In our example,

hand corner of the tableau,

we first allocate as much as possible to cell 1A (the northwest corner). This amount is 150

followed by allocations to adjacent

feasible cells.

tons, since that is the maximum that can be supplied by grain elevator 1 at Kansas City, even though 200 tons are demanded by mill A at Chicago. This initial allocation is shown in Table B-2.

We next allocate to a cell adjacent to cell 1A, in this case either cell 2A or cell 1B.

However, cell 1B no longer represents a feasible allocation, because the total tonnage of wheat available at source 1 (i.e., 150 tons) has already been allocated. Thus, cell 2A represents the only feasible alternative, and as much as possible is allocated to this cell. The amount allocated at 2A can be either 175 tons, the supply available from source 2 (Omaha), or 50 tons, the amount now demanded at destination A. (Recall that 150 of the 200 tons demanded at A have already been supplied.) Because 50 tons is the most constrained amount, it is allocated to cell 2A, as shown in Table B-2.

Table B-2

To

The Initial NW Corner Solution

From A B C Supply

6 8 10

1 150 150

7 11 11

2 50 100 25 175

4 5 12

3 275 275

Demand 200 100 300 600

The third allocation is made in the same way as the second allocation. The only feasible cell adjacent to cell 2A is cell 2B. The most that can be allocated is either 100 tons (the amount demanded at mill B) or 125 tons (175 tons minus the 50 tons allocated to cell 2A).

The smaller (most constrained) amount, 100 tons, is allocated to cell 2B, as shown in Table B-2.

The initial solution is complete The fourth allocation is 25 tons to cell 2C, and the fifth allocation is 275 tons to cell 3C,

when all rim requirements are both of which are shown in Table B-2. Notice that all of the row and column allocations

satisfied. add up to the appropriate rim requirements.

The transportation cost of this solution is computed by substituting the cell allocations (i.e., the amounts transported),

$$x_{1A} = 150$$

$$x_{2A} = 50$$

$$x_{2B} = 100$$

$$x_{2C} = 25$$

$$x_{3C} = 275$$

into the objective function:

$$\begin{aligned} Z &= \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C} \\ &= 6(150) + 8(0) + 10(0) + 7(50) + 11(100) + 11(25) + 4(0) + 5(0) + 12(275) \\ &= \$5,925 \end{aligned}$$

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Solution of the Transportation Model B-5

The steps of the northwest corner method are summarized here:

1. Allocate as much as possible to the cell in the upper left-hand corner, subject to the supply and demand constraints.
2. Allocate as much as possible to the next adjacent feasible cell.
3. Repeat step 2 until all rim requirements have been met.

The Minimum Cell Cost Method

In the minimum cell cost method With the minimum cell cost method, the basic logic is to allocate to the cells with the low-

as much as possible is allocated to est costs. The initial allocation is made to the cell in the tableau having the lowest cost. In

the cell with the minimum cost. the transportation tableau for our example problem, cell 3A has the minimum cost of \$4.

As much as possible is allocated to this cell; the choice is either 200 tons or 275 tons. Even though 275 tons could be supplied to cell 3A, the most we can allocate is 200 tons, since only 200 tons are demanded. This allocation is shown in Table B-3.

Table B-3

To

The Initial Minimum Cell From A B C Supply

Cost Allocation

6 8 10

1 150

7 11 11

2 175

4 5 12

3 200 275

Demand 200 100 300 600

Notice that all the remaining cells in column A have now been eliminated, because all the wheat demanded at destination A, Chicago, has now been supplied by source 3, Des Moines.

The next allocation is made to the cell that has the minimum cost and also is feasible. This is cell 3B, which has a cost of \$5. The most that can be allocated is 75 tons (275 tons minus the 200 tons already supplied). This allocation is shown in Table B-4.

Table B-4

To

The Second Minimum Cell From A B C Supply

Cost Allocation

6 8 10

1 150

7 11 11

2 175

4 5 12

3 200 75 275

Demand 200 100 300 600

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B-6 Module B Transportation and Assignment Solution Methods

The third allocation is made to cell 1B, which has the minimum cost of \$8. (Notice that cells with lower costs, such as 1A and 2A, are not considered because they were previously ruled out as infeasible.) The amount allocated is 25 tons. The fourth allocation of 125 tons is made to cell 1C, and the last allocation of 175 tons is made to cell 2C. These allocations, which complete the initial minimum cell cost solution, are shown in Table B-5.

Table B-5

To

The Initial Solution From A B C Supply

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The minimum cell cost method The total cost of this initial solution is \$4,550, as compared with a total cost of \$5,925 for

will provide a solution with a the initial northwest corner solution. It is not a coincidence that a lower total cost is derived

lower cost than the northwest using the minimum cell cost method; it is a logical occurrence. The northwest corner

corner solution because it

considers cost in the

method does not consider cost at all in making allocations—the minimum cell cost

allocation process. method does. It is therefore quite natural that a lower initial cost will be attained using the

latter method. Thus, the initial solution achieved by using the minimum cell cost method is usually better in that, because it has a lower cost, it is closer to the optimal solution; fewer subsequent iterations will be required to achieve the optimal solution.

The specific steps of the minimum cell cost method are summarized next:

1. Allocate as much as possible to the feasible cell with the minimum transportation cost, and adjust the rim requirements.
2. Repeat step 1 until all rim requirements have been met.

Vogel's Approximation Model

The third method for determining an initial solution, Vogel's approximation model (also

A penalty cost is the difference called VAM), is based on the concept of penalty cost or regret. If a decision maker incor-

between the largest and rectly chooses from several alternative courses of action, a penalty may be suffered (and the

next largest cell cost in a row decision maker may regret the decision that was made). In a transportation problem, the

(or column).

courses of action are the alternative routes, and a wrong decision is allocating to a cell that does not contain the lowest cost.

VAM allocates as much as possible In the VAM method, the first step is to develop a penalty cost for each source and desti-

to the minimum cost cell in the nation. For example, consider column A in Table B-6. Destination A, Chicago, can be sup-

row or column with the largest plied by Kansas City, Omaha, and Des Moines. The best decision would be to supply

penalty cost.

Chicago from source 3 because cell 3A has the minimum cost of \$4. If a wrong decision was made and the next higher cost of \$6 was selected at cell 1A, a "penalty" of \$2 per ton would result (i.e., $\$6 - \$4 = \$2$). This demonstrates how the penalty cost is determined for each row and column of the tableau. The general rule for computing a penalty cost is to subtract the minimum cell cost from the next higher cell cost in each row and column. The penalty costs for our example are shown at the right and at the bottom of Table B-6.

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Solution of the Transportation Model B-7

Table B-6

To

The VAM Penalty Costs From A B C Supply

6 8 10

2

1 150

7 11 11

4

2 175

4 5 12

1

3 275

Demand 200 100 300 600

2 3 1

The initial allocation in the VAM method is made in the row or column that has the highest penalty cost. In Table B-6, row 2 has the highest penalty cost of \$4. We allocate as much as possible to the feasible cell in this row with the minimum cost. In row 2, cell 2A has the lowest cost of \$7, and the most that can be allocated to cell 2A is 175 tons. With this allocation the greatest penalty cost of \$4 has been avoided because the best course of action has been selected. The allocation is shown in Table B-7.

Table B-7

To

The Initial VAM Allocation From A B C Supply

6 8 10

2

1 150

7 11 11

2 175 175

4 5 12

1

3 275

Demand 200 100 300 600

2 3 2

After each VAM cell allocation, After the initial allocation is made, all the penalty costs must be recomputed. In some

all row and column penalty costs cases the penalty costs will change; in other cases they will not change. For example, the

are recomputed. penalty cost for column C in Table B-7 changed from \$1 to \$2 (because cell 2C is no longer

considered in computing penalty cost), and the penalty cost in row 2 was eliminated altogether (because no more allocations are possible for that row).

Next, we repeat the previous step and allocate to the row or column with the highest penalty cost, which is now column B with a penalty cost of \$3 (see Table B-7). The cell in column B with the lowest cost is 3B, and we allocate as much as possible to this cell, 100 tons. This allocation is shown in Table B-8.

Note that all penalty costs have been recomputed in Table B-8. Since the highest penalty cost is now \$8 for row 3 and since cell 3A has the minimum cost of \$4, we allocate 25 tons to this cell, as shown in Table B-9.

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B-8 Module B Transportation and Assignment Solution Methods

Table B-8

To

The Second VAM Allocation From A B C Supply

6 8 10

4

1 150

7 11 11

2 175 175

4 5 12

8

3 100 275

Demand 200 100 300 600

2 2

Table B-9

To

The Third VAM Allocation From A B C Supply

6 8 10

1 150

7 11 11

2 175 175

4 5 12

3 25 100 275

Demand 200 100 300 600

2

Table B-9 also shows the recomputed penalty costs after the third allocation. Notice that

by now only column C has a penalty cost. Rows 1 and 3 have only one feasible cell, so a penalty does not exist for these rows. Thus, the last two allocations are made to column C. First, 150 tons are allocated to cell 1C because it has the lowest cell cost. This leaves only cell 3C as a feasible possibility, so 150 tons are allocated to this cell. Both of these allocations are shown in Table B-10.

Table B-10

To

The Initial VAM Solution From A B C Supply

6 8 10

1 150 150

7 11 11

2 175 175

4 5 12

3 25 100 150 275

Demand 200 100 300 600

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Solution of the Transportation Model B-9

VAM and minimum cell cost both The total cost of this initial Vogel's approximation model solution is \$5,125, which is

provide better initial solutions not as high as the northwest corner initial solution of \$5,925. It is also not as low as the

than the northwest corner method. minimum cell cost solution of \$4,550. Like the minimum cell cost method, VAM typi-

cally results in a lower cost for the initial solution than does the northwest corner method.

The steps of Vogel's approximation model can be summarized in the following list:

1. Determine the penalty cost for each row and column by subtracting the lowest cell cost in the row or column from the next lowest cell cost in the same row or column.
2. Select the row or column with the highest penalty cost (breaking ties arbitrarily or choosing the lowest-cost cell).
3. Allocate as much as possible to the feasible cell with the lowest transportation cost in the row or column with the highest penalty cost.
4. Repeat steps 1, 2, and 3 until all rim requirements have been met.

The Stepping-Stone Solution Method

Once an initial basic feasible solution has been determined by any of the previous three methods, the next step is to solve the model for the optimal (i.e., minimum total cost) solu-

Once an initial solution is derived, tion. There are two basic solution methods: the stepping-stone solution method and the

the problem must be solved using modified distribution method (MODI). The stepping-stone solution method will be

either the stepping-stone method demonstrated first. Because the initial solution obtained by the minimum cell cost method or the modified distribution had the lowest total cost of the three initial solutions, we will use it as the starting solution. method (MODI).

Table B-11 repeats the initial solution that was developed from the minimum cell cost method.

Table B-11

To

The Minimum Cell From A B C Supply

Cost Solution

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The stepping-stone method The basic solution principle in a transportation problem is to determine whether a

determines whether there is a cell transportation route not at present being used (i.e., an empty cell) would result in a

with no allocation that would lower total cost if it were used. For example, Table B-11 shows four empty cells (1A, 2A,

reduce cost if used.

2B, and 3C) representing unused routes. Our first step in the stepping-stone method is to evaluate these empty cells to see whether the use of any of them would reduce total cost.

If we find such a route, then we will allocate as much as possible to it.

First, let us consider allocating one ton of wheat to cell 1A. If one ton is allocated to cell 1A, cost will be increased by \$6—the transportation cost for cell 1A. However, by allocating one ton to cell 1A, we increase the supply in row 1 to 151 tons, as shown in Table B-12.

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B-10 Module B Transportation and Assignment Solution Methods

Table B-12

To

The Allocation of One Ton to From A B C Supply

Cell 1A

+1 6 8 10

1 25 125 150 151

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The constraints of the problem cannot be violated, and feasibility must be maintained.

If we add one ton to cell 1A, we must subtract one ton from another allocation along that row. Cell 1B is a logical candidate because it contains 25 tons. By subtracting one ton from cell 1B, we now have 150 tons in row 1, and we have satisfied the supply constraint again. At the same time, subtracting one ton from cell 1B has reduced total cost by \$8.

However, by subtracting one ton from cell 1B, we now have only 99 tons allocated to column B, where 100 tons are demanded, as shown in Table B-13. To compensate for this constraint violation, one ton must be added to a cell that already has an allocation. Since cell 3B has 75 tons, we will add one ton to this cell, which again satisfies the demand constraint of 100 tons.

Table B-13

To

The Subtraction of One Ton From A B C Supply

from Cell 1B

+1 6 -1 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

99

A requirement of this solution method is that units can be added to and subtracted from only those cells that already have allocations. That is why one ton was added to cell 3B and not to cell 2B. It is from this requirement that the method derives its name. The process of adding and subtracting units from allocated cells is analogous to crossing a pond by stepping on stones (i.e., only allocated-to cells).

By allocating one extra ton to cell 3B we have increased cost by \$5, the transportation cost for that cell. However, we have also increased the supply in row 3 to 276 tons, a violation of the supply constraint for this source. As before, this violation can be remedied by subtracting one ton from cell 3A, which contains an allocation of 200 tons. This satisfies the supply constraint again for row 3, and it also reduces the total cost by \$4, the transportation cost for cell 3A. These allocations and deletions are shown in Table B-14.

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Solution of the Transportation Model B-11

Table B-14

To

The Addition of One Ton to Cell From A B C Supply

3B and the Subtraction of One

Ton from Cell 3A +1 6 -1 8 10

1 25 125 150

7 11 11

2 175 175

-1 4 +1 5 12

3 200 75 275

Demand 200 100 300 600

Notice in Table B-14 that by subtracting one ton from cell 3A, we did not violate the demand constraint for column A, since we previously added one ton to cell 1A.

Now let us review the increases and reductions in costs resulting from this process. We initially increased cost by \$6 at cell 1A, then reduced cost by \$8 at cell 1B, then increased cost by \$5 at cell 3B, and, finally, reduced cost by \$4 at cell 3A:

1A : 1B : 3B : 3A

+ \$6 - 8 + 5 - 4 = - \$1

An empty cell that will reduce cost In other words, for each ton allocated to cell 1A (a route not at present used), total cost

is a potential entering variable. will be reduced by \$1. This indicates that the initial solution is not optimal because a lower

cost can be achieved by allocating additional tons of wheat to cell 1A (i.e., cell 1A is analogous to a pivot column in the simplex method). Our goal is to determine the cell or entering "variable" that will reduce cost the most. Another variable (empty cell) may result in an even greater decrease in cost than cell 1A. If such a cell exists, it will be selected as the entering variable; if not, cell 1A will be selected. To identify the appropriate entering variable, the remaining empty cells must be tested as cell 1A was.

To evaluate the cost reduction Before testing the remaining empty cells, let us identify a few of the general characteris-

potential of an empty cell, a closed tics of the stepping-stone process. First, we always start with an empty cell and form a closed

path connecting used cells to the path of cells that now have allocations. In developing the path, it is possible to skip over

empty cell is identified.

both unused and used cells. In any row or column there can be only one addition and one subtraction. (For example, in row 1, wheat is added at cell 1A and is subtracted at cell 1B.)

Let us test cell 2A to see if it results in a cost reduction. The stepping-stone closed path for cell 2A is shown in Table B-15. Notice that the path for cell 2A is slightly more complex

Table B-15

To

The Stepping-Stone Path for From A B C Supply

Cell 2A

6 - 8 + 10

1 25 125 150

+ 7 11 - 11

2 175 175

- 4 + 5 12

3 200 75 275

Demand 200 100 300 600

2A : 2C : 1C : 1B : 3B : 3A

+ \$7 - 11 + 10 - 8 + 5 - 4 = - \$1

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B-12 Module B Transportation and Assignment Solution Methods

than the path for cell 1A. Notice also that the path crosses itself at one point, which is perfectly acceptable. An allocation to cell 2A will reduce cost by \$1, as shown in the computation in Table B-15. Thus, we have located another possible entering variable, although it is no better than cell 1A.

The remaining stepping-stone paths and the resulting computations for cells 2B and 3C are shown in Tables B-16 and B-17, respectively.

Table B-16

To

The Stepping-Stone Path for From A B C Supply

Cell 2B

6 + 8 - 10

1 25 125 150

7 - 11 + 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

2B : 2C : 1C : 1B

+ \$11 - 11 + 10 - 8 = + \$2

Table B-17

To

The Stepping-Stone Path for From A B C Supply

Cell 3C

6 + 8 - 10

1 25 125 150

7 11 11

2 175 175

4 - 5 + 12

3 200 75 275

Demand 200 100 300 600

3C : 1C : 1B : 3B

+ \$12 - 10 + 8 - 5 = + \$5

After all empty cells are evaluated, Notice that after all four unused routes are evaluated, there is a tie for the entering vari-

the one with the greatest able between cells 1A and 2A. Both show a reduction in cost of \$1 per ton allocated to that

cost reduction potential is the route. The tie can be broken arbitrarily. We will select cell 1A (i.e., x1A) to enter the solution.

entering variable.

Because the total cost of the model will be reduced by \$1 for each ton we can reallocate to cell 1A, we naturally want to reallocate as much as possible. To determine how much to allocate, we need to look at the path for cell 1A again, as shown in Table B-18.

When reallocating units to the The stepping-stone path in Table B-18 shows that tons of wheat must be subtracted at

entering variable (cell), the cells 1B and 3A to meet the rim requirements and thus satisfy the model constraints.

amount is the minimum Because we cannot subtract more than is available in a cell, we are limited by the 25 tons in

amount subtracted on the

cell 1B. In other words, if we allocate more than 25 tons to cell 1A, then we must subtract stepping-stone path.

more than 25 tons from 1B, which is impossible because only 25 tons are available.

Therefore, 25 tons is the amount we reallocate to cell 1A according to our path. That is, 25 tons are added to 1A, subtracted from 1B, added to 3B, and subtracted from 3A. This reallocation is shown in Table B-19.

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Solution of the Transportation Model B-13

Table B-18

To

The Stepping-Stone Path for From A B C Supply

Cell 1A

+ 6 - 8 10

1 25 125 150

7 11 11

2 175 175

- 4 + 5 12

3 200 75 275

Demand 200 100 300 600

Table B-19

To

The Second Iteration of the From A B C Supply

Stepping-Stone Method

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 175 100 275

Demand 200 100 300 600

The process culminating in Table B-19 represents one iteration of the stepping-stone method. We selected x_{1A} as the entering variable, and it turned out that x_{1B} was the leaving variable (because it now has a value of zero in Table B-19). Thus, at each iteration one variable enters and one leaves (just as in the simplex method).

Now we must check to see whether the solution shown in Table B-19 is, in fact, optimal.

We do this by plotting the paths for the unused routes (i.e., empty cells 2A, 1B, 2B, and 3C) that are shown in Table B-19. These paths are shown in Tables B-20 through B-23.

Table B-20

To

The Stepping-Stone Path for From A B C Supply

Cell 2A

- 6 8 + 10

1 25 125 150

+ 7 11 - 11

2 175 175

4 5 12

3 175 100 275

Demand 200 100 300 600

2A : 2C : 1C : 1A

+ \$7 - 11 + 10 - 6 = \$0

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B-14 Module B Transportation and Assignment Solution Methods

Table B-21

To

The Stepping-Stone Path for From A B C Supply

Cell 1B

- 6 + 8 10

1 25 125 150

7 11 11

2 175 175

+ 4 - 5 12

3 175 100 275

Demand 200 100 300 600

1B : 3B : 3A : 1A

+ \$8 - 5 + 4 - 6 = + \$1

Table B-22

To

The Stepping-Stone Path for From A B C Supply

Cell 2B

- 6 8 + 10

1 25 125 150

7 + 11 - 11

2 175 175

+ 4 - 5 12

3 175 100 275

Demand 200 100 300 600

2B : 3B : 3A : 1A : 1C : 2C

+ \$11 - 5 + 4 - 6 + 10 - 11 = + \$3

Table B-23

To

The Stepping-Stone Path for

From A B C Supply

Cell 3C

+ 6 8 - 10

1 25 125 150

7 11 11

2 175 175

- 4 5 + 12

3 175 100 275

Demand 200 100 300 600

3C : 3A : 1A : 1C

+ \$12 - 4 + 6 - 10 = + \$4

The stepping-stone process is Our evaluation of the four paths indicates no cost reductions; therefore, the solution

repeated until none of the shown in Table B-19 is optimal. The solution and total minimum cost are

empty cells will reduce cost

(i.e., an optimal solution). $x_{1A} = 25$ tons $x_{2C} = 175$ tons $x_{3A} = 175$ tons

$x_{1C} = 125$ tons $x_{3B} = 100$ tons

$Z = \$6(25) + 8(0) + 10(125) + 7(0) + 11(0) + 11(175) + 4(175) + 5(100) + 12(0)$
 $= \$4,525$

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Solution of the Transportation Model B-15

Multiple optimal solutions However, notice in Table B-20 that the path for cell 2A resulted in a cost change of \$0. In

occur when an empty cell has a other words, allocating to this cell would neither increase nor decrease total cost. This situ-

cost change of zero and all other ation indicates that the problem has multiple optimal solutions.

Thus, x_{2A} could be

empty cells are positive.

entered into the solution and there would not be a change in the total minimum cost of

An alternative optimal solution is

\$4,525. To identify the alternative solution, we would allocate as much as possible to cell 2A,

determined by allocating to the

empty cell a zero cost change. which in this case is 25 tons of wheat. The alternative solution is shown in Table B-24.

Table B-24 To

The Alternative Optimal From A B C Supply

Solution

6 8 10

1 150 150

7 11 11

2 25 150 175

4 5 12

3 175 100 275

Demand 200 100 300 600

The solution in Table B-24 also results in a total minimum cost of \$4,525. The steps of the stepping-stone method are summarized here:

1. Determine the stepping-stone paths and cost changes for each empty cell in the tableau.
2. Allocate as much as possible to the empty cell with the greatest net decrease in cost.
3. Repeat steps 1 and 2 until all empty cells have positive cost changes that indicate an optimal solution.

The Modified Distribution Method

The modified distribution method (MODI) is basically a modified version of the stepping-stone method. However, in the MODI method the individual cell cost changes are deter-

MODI is a modified version of
 mined mathematically, without identifying all the stepping-stone paths for the empty cells.
 the stepping-stone method in To demonstrate MODI, we will again use the initial solution obtained
 by the minimum
 which math equations replace the cell cost method. The tableau for the initial solution with the
 modifications required by
 stepping-stone paths. MODI is shown in Table B-25.

Table B-25 v_j $v_A = v_B = v_C =$

The Minimum Cell Cost

To

Initial Solution

u_i From A B C Supply

6 8 10

$u_1 = 1$ 25 125 150

7 11 11

$u_2 = 2$ 175 175

4 5 12

$u_3 = 3$ 200 75 275

Demand 200 100 300 600

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B-16 Module B Transportation and Assignment Solution Methods

The extra left-hand column with the u_i symbols and the extra top row with the v_j symbols represent column and row values that must be computed in MODI. These values are computed for all cells with allocations by using the following formula:

$$u_i + v_j = c_{ij}$$

The value c_{ij} is the unit transportation cost for cell ij . For example, the formula for cell 1B is

$$u_1 + v_B = c_{1B}$$

and, since $c_{1B} = 8$,

$$u_1 + v_B = 8$$

The formulas for the remaining cells that presently contain allocations are

$$x_{1C}: u_1 + v_C = 10$$

$$x_{2C}: u_2 + v_C = 11$$

$$x_{3A}: u_3 + v_A = 4$$

$$x_{3B}: u_3 + v_B = 5$$

Now there are five equations with six unknowns. To solve these equations, it is necessary to assign only one of the unknowns a value of zero. Thus, if we let $u_1 = 0$, we can solve for all remaining u_i and v_j values.

$$x_{1B}: u_1 + v_B = 8$$

$$0 + v_B = 8$$

$$v_B = 8$$

$$x1C: u1 + vC = 10$$

$$0 + vC = 10$$

$$vC = 10$$

$$x2C: u2 + vC = 11$$

$$u2 + 10 = 11$$

$$u2 = 1$$

$$x3B: u3 + vB = 5$$

$$u3 + 8 = 5$$

$$u3 = -3$$

$$x3A: u3 + vA = 4$$

$$-3 + vA = 4$$

$$vA = 7$$

Notice that the equation for cell 3B had to be solved before the cell 3A equation could be solved. Now all the u_i and v_j values can be substituted into the tableau, as shown in Table B-26.

Table B-26 $vA = 7$ $vB = 8$ $vC = 10$

The Initial Solution with All u_i

To

and v_j Values

u_i From A B C Supply

6 8 10

$u1 = 0$ 1 25 125 150

7 11 11

$u2 = 1$ 2 175 175

4 5 12

$u3 = -3$ 3 200 75 275

Demand 200 100 300 600

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Solution of the Transportation Model B-17

Next, we use the following formula to evaluate all empty cells:

$$c_{ij} - u_i - v_j = k_{ij}$$

where k_{ij} equals the cost increase or decrease that would occur by allocating to a cell.

For the empty cells in Table B-26, the formula yields the following values:

$$x1A: k1A = c1A - u1 - vA = 6 - 0 - 7 = -1$$

$$x2A: k2A = c2A - u2 - vA = 7 - 1 - 7 = -1$$

$$x2B: k2B = c2B - u2 - vB = 11 - 1 - 8 = +2$$

$$x3C: k3C = c3C - u3 - vC = 12 - (-3) - 10 = +5$$

Each MODI allocation replicates These calculations indicate that either cell 1A or cell 2A will decrease cost by \$1 per allo-

the stepping-stone allocation. cated ton. Notice that those are exactly the same cost changes for all four empty cells as

were computed in the stepping-stone method. That is, the same information is obtained by evaluating the paths in the stepping-stone method and by using the mathematical formulas of the MODI.

After each allocation to an empty We can select either cell 1A or 2A to allocate to because they are tied at -1. If cell 1A is

cell, the u_i and v_j values must be selected as the entering nonbasic variable, then the stepping-stone path for that cell must

recomputed. be determined so that we know how much to reallocate. This is the same path previously

identified in Table B-18. Reallocating along this path results in the tableau shown in

Table B-27 (and previously shown in Table B-19).

Table B-27 $v_A = v_B = v_C =$

The Second Iteration of the

To

MODI Solution Method

u_i From A B C Supply

6 8 10

$u_1 = 1$ 25 125 150

7 11 11

$u_2 = 2$ 175 175

4 5 12

$u_3 = 3$ 175 100 275

Demand 200 100 300 600

The u_i and v_j values for Table B-27 must now be recomputed using our formula for the allocated-to cells:

x1A: $u_1 + v_A = 6$

$0 + v_A = 6$

$v_A = 6$

x1C: $u_1 + v_C = 10$

$0 + v_C = 10$

$v_C = 10$

x2C: $u_2 + v_C = 11$

$u_2 + 10 = 11$

$u_2 = 1$

x3A: $u_3 + v_A = 4$

$u_3 + 6 = 4$

$u_3 = -2$

x3B: $u_3 + v_B = 5$

$-2 + v_B = 5$

$v_B = 7$

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B-18 Module B Transportation and Assignment Solution Methods

These new u_i and v_j values are shown in Table B-28.

Table B-28 $v_A = 6$ $v_B = 7$ $v_C = 10$

The New u_i and v_j Values for

To

the Second Iteration

u_i From A B C Supply

6 8 10

$u_1 = 0$ 1 25 125 150

7 11 11

$u_2 = 1$ 2 175 175

4 5 12

$u_3 = -2$ 3 175 100 275

Demand 200 100 300 600

The cost changes for the empty cells are now computed using the formula

$c_{ij} - u_i - v_j = k_{ij}$:

x1B: $k_{1B} = c_{1B} - u_1 - v_B = 8 - 0 - 7 = +1$

x2A: $k_{2A} = c_{2A} - u_2 - v_A = 7 - 1 - 6 = 0$

x2B: $k_{2B} = c_{2B} - u_2 - v_B = 11 - 1 - 7 = +3$

x3C: $k_{3C} = c_{3C} - u_3 - v_C = 12 - (-2) - 10 = +4$

Because none of these values is negative, the solution shown in Table B-28 is optimal.

However, as in the stepping-stone method, cell 2A with a zero cost change indicates multiple optimal solutions.

The steps of the modified distribution method can be summarized as follows:

1. Develop an initial solution using one of the three methods available.
2. Compute u_i and v_j values for each row and column by applying the formula $u_i + v_j = c_{ij}$ to each cell that has an allocation.
3. Compute the cost change, k_{ij} , for each empty cell using $c_{ij} - u_i - v_j = k_{ij}$.
4. Allocate as much as possible to the empty cell that will result in the greatest net decrease in cost (most negative k_{ij}). Allocate according to the stepping-stone path for the selected cell.
5. Repeat steps 2 through 4 until all k_{ij} values are positive or zero.

The Unbalanced Transportation Model

Thus far, the methods for determining an initial solution and an optimal solution have been demonstrated within the context of a balanced transportation model. Realistically, however, an unbalanced problem is a more likely occurrence. Consider our example of transporting wheat. By changing the demand at Cincinnati to 350 tons, we create a situation in which total demand is 650 tons and total supply is 600 tons.

When demand exceeds supply, To compensate for this difference in the transportation tableau, a “dummy” row is added

a dummy row is added to the tableau, as shown in Table B-29. The dummy row is assigned a supply of 50 tons

tableau. to balance the model. The additional 50 tons demanded, which cannot be supplied, will be allocated to a cell in the dummy row. The transportation costs for the cells in the dummy row are zero because the tons allocated to these cells are not amounts really transported but the amounts by which demand was not met. These dummy cells are, in effect, slack variables.

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Solution of the Transportation Model B-19

Table B-29

To

An Unbalanced Model From A B C Supply

(Demand 7 Supply)

6 8 10

1 150

7 11 11

2 175

4 5 12

3 275

0 0 0

Dummy 50

Demand 200 100 350 650

When a supply exceeds demand, Now consider our example with the supply at Des Moines increased to 375 tons. This

a dummy column is added to the increases total supply to 700 tons, while total demand remains at 600 tons. To compensate for

tableau. this imbalance, we add a dummy column instead of a dummy row, as shown in Table B-30.

Table B-30

To

An Unbalanced Model From A B C Dummy Supply

(Supply 7 Demand)

6 8 10 0

1 150

7 11 11 0

2 175

4 5 12 0

3 375

Demand 200 100 300 100 700

The addition of a dummy row or a dummy column has no effect on the initial solution

methods or on the methods for determining an optimal solution. The dummy row or column cells are treated the same as any other tableau cell. For example, in the minimum cell cost method, three cells would be tied for the minimum cost cell, each with a cost of zero. In this case (or any time there is a tie between cells) the tie would be broken arbitrarily.

Degeneracy

In all the tableaus showing a solution to the wheat transportation problem, the following condition was met:

$m \text{ rows} + n \text{ columns} - 1 = \text{the number of cells with allocations}$

For example, in any of the balanced tableaus for wheat transportation, the number of rows was three (i.e., $m = 3$) and the number of columns was three (i.e., $n = 3$); thus,

In a transportation tableau

$3 + 3 - 1 = 5$ cells with allocations.

with m rows and n columns, there These tableaus always had five cells with allocations; thus, our condition for normal

must be $m + n - 1$ cells with solution was met. When this condition is not met and fewer than $m + n - 1$ cells have

allocations; if not, it is degenerate. allocations, the tableau is said to be degenerate.

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B-20 Module B Transportation and Assignment Solution Methods

Consider the wheat transportation example with the supply values changed to the amounts shown in Table B-31. The initial solution shown in this tableau was developed using the minimum cell cost method.

Table B-31

To

The Minimum Cell Cost Initial

From A B C Supply

Solution

6 8 10

1 100 50 150

7 11 11

2 250 250

4 5 12

3 200 200

Demand 200 100 300 600

The tableau shown in Table B-31 does not meet the condition

$m + n - 1 = \text{the number of cells with allocations}$

$3 + 3 - 1 = 5$ cells

In a degenerate tableau, not all of because there are only four cells with allocations. The difficulty resulting from a degenerate

the stepping-stone paths or MODI solution is that neither the stepping-stone method nor MODI will work unless the preceding condition is met (there is an appropriate number of cells with allocations). When the tableau is degenerate, a closed path cannot be completed for all cells in the stepping-stone method, and not all the $u_i + v_j = c_{ij}$ computations can be completed in MODI. For example, a closed path cannot be determined for cell 1A in Table B-31.

To rectify a degenerate tableau, an empty cell must be artificially designated as a cell with an allocation. Cell 1A in Table B-32 is designated arbitrarily as a cell with artificial allocation of zero. (However, any symbol, such as $\square$, could be used to signify the artificial allocation.) This indicates that this cell will be treated as a cell with an allocation in determining stepping-stone paths or MODI formulas, although there is no real allocation in this cell. Notice that the location of 0 was arbitrary because there is no general rule for allocating the artificial cell. Allocating zero to a cell does not guarantee that all the stepping-stone paths can be determined.

Table B-32

To

The Initial Solution From A B C Supply

6 8 10

1 0 100 50 150

7 11 11

2 250 250

4 5 12

3 200 200

Demand 200 100 300 600

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Solution of the Transportation Model B-21

For example, if zero had been allocated to cell 2B instead of to cell 1A, none of the stepping-stone paths could have been determined, even though technically the tableau would no longer be degenerate. In such a case, the zero must be reallocated to another cell and all paths determined again. This process must be repeated until an artificial allocation has been made that will enable the determination of all paths. In most cases, however, there is more than one possible cell to which such an allocation can be made.

The stepping-stone paths and cost changes for this tableau follow:

2A 2C 1C 1A

$$x_{2A}: 7 - 11 + 10 - 6 = 0$$

2B 2C 1C 1B

$$x_{2B}: 11 - 11 + 10 - 8 = +2$$

3B 1B 1A 3A

x3B: $5 - 8 + 6 - 4 = -1$

3C 1C 1A 3A

x3C: $12 - 10 + 6 - 4 = +4$

Because cell 3B shows a \$1 decrease in cost for every ton of wheat allocated to it, we will allocate 100 tons to cell 3B. This results in the tableau shown in Table B-33.

Table B-33

To

The Second Stepping-Stone From A B C Supply

Iteration

6 8 10

1 100 50 150

7 11 11

2 250 250

4 5 12

3 100 100 200

Demand 200 100 300 600

A normal problem can become Notice that the solution in Table B-33 now meets the condition $m + n - 1 = 5$. Thus,

degenerate at any iteration and in applying the stepping-stone method (or MODI) to this tableau, it is not necessary to

vice versa. make an artificial allocation to an empty cell. It is quite possible to begin the solution process

with a normal tableau and have it become degenerate or begin with a degenerate tableau and have it become normal. If it had been indicated that the cell with the zero should have units subtracted from it, no actual units could have been subtracted. In that case the zero would have been moved to the cell that represents the entering variable. (The solution shown in Table B-33 is optimal; however, multiple optimal solutions exist at cell 2A.)

Prohibited Routes

Sometimes one or more of the routes in the transportation model are prohibited. That is, units cannot be transported from a particular source to a particular destination. When this situation occurs, we must make sure that no units in the optimal solution are allocated to the cell representing this route. In our study of the simplex tableau, we learned that assigning a large relative cost or a coefficient of M to a variable would keep it out of the final solution. This same principle can be used in a transportation model for a prohibited route. A

large cost such as M so that its value of M is assigned as the transportation cost for a cell that represents a prohibited

route. Thus, when the prohibited cell is evaluated, it will always contain a large positive cost

change of M , which will keep it from being selected as an entering variable.

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B-22 Module B Transportation and Assignment Solution Methods

Solution of the Assignment Model

An assignment model is a special The assignment model is a special form of a linear programming model that is similar to

form of the transportation model the transportation model. There are differences, however. In the assignment model, the

in which all supply and supply at each source and the demand at each destination are limited to one unit each.

demand values equal one.

The following example from the text will be used to demonstrate the assignment model and its special solution method. The Atlantic Coast Conference has four basketball games on a particular night. The conference office wants to assign four teams of officials to the four games in a way that will minimize the total distance traveled by the officials. The distances in miles for each team of officials to each game location are shown in Table B-34.

Table B-34

Game Sites

The Travel Distances to Each

Game for Each Team of Officials RALEIGH ATLANTA DURHAM CLEMSON

Officials

A 210 90 180 160

B 100 70 130 200

C 175 105 140 170

D 80 65 105 120

The supply is always one team of officials, and the demand is for only one team of officials at each game. Table B-34 is already in the proper form for the assignment.

An opportunity cost table is The first step in the assignment method of solution is to develop an opportunity cost

developed by first subtracting table. We accomplish this by first subtracting the minimum value in each row from every

the minimum value in each value in the row. These computations are referred to as row reductions. We applied a similar

row from all other row values and

principle in the VAM method when we determined penalty costs. In other words, the best

then repeating this process for

each column. course of action is determined for each row, and the penalty or “lost opportunity” is devel-

oped for all other row values. The row reductions for this example are shown in Table B-35.

Table B-35

Game Sites

The Assignment Tableau with

Row Reductions Officials RALEIGH ATLANTA DURHAM CLEMSON

A 120 0 90 70
 B 30 0 60 130
 C 70 0 35 65
 D 15 0 40 55

Next, the minimum value in each column is subtracted from all column values. These computations are called column reductions and are shown in Table B-36, which represents the completed opportunity cost table for our example. Assignments can be made in this table wherever a zero is present. For example, team A can be assigned to Atlanta. An optimal solution results when each of the four teams can be uniquely assigned to a different game.

Table B-36

Game Sites

The Tableau with

Column Reductions Officials RALEIGH ATLANTA DURHAM CLEMSON

A 105 0 55 15
 B 15 0 25 75
 C 55 0 0 10
 D 0 0 5 0

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Solution of the Assignment Model B-23

Assignments are made to locations Notice in Table B-36 that the assignment of team A to Atlanta means that no other team

with zeros in the opportunity cost can be assigned to that game. Once this assignment is made, the zero in row B is infeasible,

table. which indicates that there is not a unique optimal assignment for team B. Therefore, Table B-36 does not contain an optimal solution.

An optimal solution occurs when A test to determine whether four unique assignments exist in Table B-36 is to draw the

the number of independent unique minimum number of horizontal or vertical lines necessary to cross out all zeros through

assignments equals the number of the rows and columns of the table. For example, Table B-37 shows that three lines are

rows or columns.

required to cross out all zeros.

Table B-37

Game Sites

The Opportunity Cost Table

with the Line Test Officials RALEIGH ATLANTA DURHAM CLEMSON

A 105 0 55 15
 B 15 0 25 75
 C 55 0 0 10
 D 0 0 5 0

If the number of unique The three lines indicate that there are only three unique assignments, whereas four are assignments is less than the required for an optimal solution. (Note that even if the three lines could have been drawn number of rows (or columns), differently, the subsequent solution method would not be affected.) Next, subtract the minimum value that is not crossed out from all values not crossed out. Then, add this minimum value to those cells where two lines intersect. The minimum value not crossed out in Table B-37 is 15. The second iteration for this model with the appropriate changes is shown in Table B-38.

Table B-38

Game Sites

The Second Iteration

Officials RALEIGH ATLANTA DURHAM CLEMSON

A 90 0 40 0

B 0 0 10 60

C 55 15 0 10

D 0 15 5 0

In a line test, all zeros are crossed No matter how the lines are drawn in Table B-38, at least four are required to cross out out by horizontal and vertical all the zeros. This indicates that four unique assignments can be made and that an optimal lines; the minimum uncrossed solution has been reached. Now let us make the assignments from Table B-38.

value is subtracted from all

First, team A can be assigned to either the Atlanta game or the Clemson game. We will uncrossed values and added to cell values where two lines cross.

assign team A to Atlanta first. This means that team A cannot be assigned to any other game, and no other team can be assigned to Atlanta. Therefore, row A and the Atlanta column can be eliminated. Next, team B is assigned to Raleigh. (Team B cannot be assigned to Atlanta, which has already been eliminated.) The third assignment is of team C to the Durham game. This leaves team D for the Clemson game. These assignments and their respective distances (from Table B-34) are summarized as follows:

Assignment Distance

Team A : Atlanta 90

Team B : Raleigh 100

Team C : Durham 140

Team D : Clemson 120

450 miles

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B-24 Module B Transportation and Assignment Solution Methods

Now let us go back and make the initial assignment of team A to Clemson (the alternative assignment we did not initially make). This will result in the following set of assignments:

Assignment Distance

Team A : Clemson 160

Team B : Atlanta 70

Team C : Durham 140

Team D : Raleigh __80

450 miles

These two assignments represent multiple optimal solutions for our example problem.

Both assignments will result in the officials traveling a minimum total distance of 450 miles.

When supply exceeds demands, a Like a transportation problem, an assignment model can be unbalanced when supply

dummy column is added to the exceeds demand or demand exceeds supply. For example, assume that, instead of four

assignment tableau. teams of officials, there are five teams to be assigned to the four games. In this case

a dummy column is added to the assignment tableau to balance the model, as shown in Table B-39.

Table B-39 Game Sites

An Unbalanced Assignment

Tableau with a Dummy Column Officials RALEIGH ATLANTA DURHAM CLEMSON DUMMY

A 210 90 180 160 0

B 100 70 130 200 0

C 175 105 140 170 0

D 80 65 105 120 0

E 95 115 120 100 0

When demand exceeds supply, a In solving this model, one team of officials would be assigned to the dummy column. If

dummy row is added to the there were five games and only four teams of officials, a dummy row would be added

assignment tableau. instead of a dummy column. The addition of a dummy row or column does not affect the

solution method.

A prohibited assignment is given Prohibited assignments are also possible in an assignment problem, just as prohibited

a large relative cost of M so that it routes can occur in a transportation model. In the transportation model, an M value was

will never be selected. assigned as a large cost for the cell representing the prohibited route. This same method is

used for a prohibited assignment. A value of M is placed in the cell that represents the prohibited assignment.

The steps of the assignment solution method are summarized here:

1. Perform row reductions by subtracting the minimum value in each row from all row values.
2. Perform column reductions by subtracting the minimum value in each column from all column values.
3. In the completed opportunity cost table, cross out all zeros, using the minimum number of horizontal or vertical lines.
4. If fewer than m lines are required (where m = the number of rows or columns), subtract the minimum uncrossed value from all uncrossed values, and add this same minimum value to all cells where two lines intersect. Leave all other values unchanged, and repeat step 3.
5. If m lines are required, the tableau contains the optimal solution and m unique assignments can be made. If fewer than m lines are required, repeat step 4.

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Problems B-25

Problems

1. Green Valley Mills produces carpet at plants in St. Louis and Richmond. The carpet is then shipped

to two outlets located in Chicago and Atlanta. The cost per ton of shipping carpet from each of the two plants to the two warehouses is as follows:

To

From Chicago Atlanta

St. Louis \$40 \$65

Richmond 70 30

The plant at St. Louis can supply 250 tons of carpet per week; the plant at Richmond can supply 400

tons per week. The Chicago outlet has a demand of 300 tons per week, and the outlet at Atlanta demands 350 tons per week. The company wants to know the number of tons of carpet to ship from each plant to each outlet in order to minimize the total shipping cost. Solve this transportation problem.

2. A transportation problem involves the following costs, supply, and demand:

To

From 1 2 3 4 Supply

1 \$500 \$750 \$300 \$450 12

2 650 800 400 600 17

3 400 700 500 550 11

Demand 10 10 10 10

- a. Find the initial solution using the northwest corner method, the minimum cell cost method, and Vogel's approximation model. Compute total cost for each.
- b. Using the VAM initial solution, find the optimal solution using the modified distribution method (MODI).

3. Consider the following transportation tableau and solution:

To

From A B C Supply

12 10 6

1 600 600

4 15 3

2 400 400

9 7 M

3 300 300

11 8 6

4 500 300 800

0 0 0

Dummy 200 200

Demand 900 500 900 2,300

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B-26 Module B Transportation and Assignment Solution Methods

- a. Is this a balanced or an unbalanced transportation problem? Explain.
 - b. Is this solution degenerate? Explain. If it is degenerate, show how it would be put into proper form.
 - c. Is there a prohibited route in this problem?
 - d. Compute the total cost of this solution.
 - e. What is the value of x_{2B} in this solution?
4. Solve the following transportation problem:

To

From 1 2 3 Supply

1 \$ 40 \$ 10 \$ 20 800

2 15 20 10 500

3 20 25 30 600

Demand 1,050 500 650

5. Given a transportation problem with the following costs, supply, and demand, find the initial solution using the minimum cell cost method and Vogel's approximation model. Is the VAM solution optimal?

To

From 1 2 3 Supply

A \$ 6 \$ 7 \$ 4 100

B 5 3 6 180

C 8 5 7 200

Demand 135 175 170

6. Consider the following transportation problem:

To

From 1 2 3 Supply

A \$ 6 9 M 130

B 12 3 5 70

C 4 8 11 100

Demand 80 110 60

a. Find the initial solution by using VAM and then solve it using the stepping-stone method.

b. Formulate this problem as a general linear programming model.

7. Solve the following linear programming problem:

minimize $Z = 3x_{11} + 12x_{12} + 8x_{13} + 10x_{21} + 5x_{22} + 6x_{23} + 6x_{31} + 7x_{32} + 10x_{33}$

subject to

$x_{11} + x_{12} + x_{13} = 90$ $x_{11} + x_{21} + x_{31} \leq 70$

$x_{21} + x_{22} + x_{23} = 30$ $x_{12} + x_{22} + x_{32} \leq 110$

$x_{31} + x_{32} + x_{33} = 100$ $x_{13} + x_{23} + x_{33} \leq 80$

$x_{ij} \geq 0$

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Problems B-27

8. Consider the following transportation problem:

To

From 1 2 3 Supply

A \$ 6 \$ 9 \$ 7 130

B 12 3 5 70

C 4 11 11 100

Demand 80 110 60

a. Find the initial solution using the minimum cell cost method.

b. Solve using the stepping-stone method.

9. Steel mills in three cities produce the following amounts of steel:

Location Weekly Production (tons)

A. Bethlehem 150

B. Birmingham 210

C. Gary 320

680

These mills supply steel to four cities where manufacturing plants have the following demand:

Location Weekly Demand (tons)

1. Detroit 130

2. St. Louis 70

3. Chicago 180

4. Norfolk 240

620

Shipping costs per ton of steel are as follows:

To

From 1 2 3 4

A \$14 9 16 18

B 11 8 7 16

C 16 12 10 22

Because of a truckers' strike, shipments are at present prohibited from Birmingham to Chicago.

a. Set up a transportation tableau for this problem and determine the initial solution. Identify the method used to find the initial solution.

b. Solve this problem using MODI.

c. Are there multiple optimal solutions? Explain. If so, identify them.

d. Formulate this problem as a general linear programming model.

10. In Problem 9, what would be the effect on the optimal solution of a reduction in production capacity at the Gary mill from 320 tons to 290 tons per week?

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B-28 Module B Transportation and Assignment Solution Methods

11. Coal is mined and processed at the following four mines in Kentucky, West Virginia, and Virginia:

Location Capacity (tons)

A. Cabin Creek 90

B. Surry 50

C. Old Fort 80

D. McCoy __60

280

These mines supply the following amount of coal to utility power plants in three cities:

Plant Demand (tons)

1. Richmond 120

2. Winston-Salem 100

3. Durham 110

330

The railroad shipping costs (\$1,000s) per ton of coal are shown in the following table. Because of railroad construction, shipments are now prohibited from Cabin Creek to Richmond:

To

From 1 2 3

A \$ 7 \$10 \$ 5

B 12 9 4

C 7 3 11

D 9 5 7

- Set up the transportation tableau for this problem, determine the initial solution using VAM, and compute total cost.
- Solve using MODI.
- Are there multiple optimal solutions? Explain. If there are alternative solutions, identify them.
- Formulate this problem as a linear programming model.

12. Oranges are grown, picked, and then stored in warehouses in Tampa, Miami, and Fresno. These warehouses supply oranges to markets in New York, Philadelphia, Chicago, and Boston. The following table shows the shipping costs per truckload (\$100s), supply, and demand. Because of an agreement between distributors, shipments are prohibited from Miami to Chicago.

To

From New York Philadelphia Chicago Boston Supply

Tampa \$ 9 \$ 14 \$ 12 \$ 17 200

Miami 11 10 6 10 200

Fresno 12 8 15 7 200

Demand 130 170 100 150

- Set up the transportation tableau for this problem and determine the initial solution using the minimum cell cost method.
- Solve using MODI.
- Are there multiple optimal solutions? Explain. If so, identify them.
- Formulate this problem as a linear programming model.

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Problems B-29

13. A manufacturing firm produces diesel engines in four cities—Phoenix, Seattle, St. Louis, and Detroit. The company is able to produce the following numbers of engines per month:

Plant Production

- Phoenix 5
- Seattle 25
- St. Louis 20
- Detroit 25

Three trucking firms purchase the following numbers of engines for their plants in three cities:

Firm Demand

- Greensboro 10
- Charlotte 20
- Louisville 15

The transportation costs per engine (\$100s) from sources to destinations are shown in the follow-

ing table. However, the Charlotte firm will not accept engines made in Seattle, and the Louisville firm will not accept engines from Detroit; therefore, these routes are prohibited.

To

From A B C

1 \$ 7 \$ 8 \$ 5

2 6 10 6

3 10 4 5

4 3 9 11

a. Set up the transportation tableau for this problem. Find the initial solution using VAM.

b. Solve for the optimal solution using the stepping-stone method. Compute the total minimum cost.

c. Formulate this problem as a linear programming model.

14. The Interstate Truck Rental firm has accumulated extra trucks at three of its truck leasing outlets,

as shown in the following table:

Extra

Leasing Outlet Trucks

1. Atlanta 70

2. St. Louis 115

3. Greensboro 60

Total 245

The firm also has four outlets with shortages of rental trucks, as follows:

Truck

Leasing Outlet Shortage

A. New Orleans 80

B. Cincinnati 50

C. Louisville 90

D. Pittsburgh 25

Total 245

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B-30 Module B Transportation and Assignment Solution Methods

The firm wants to transfer trucks from those outlets with extras to those with shortages at the minimum total cost. The following costs of transporting these trucks from city to city have been determined:

To

From A B C D

1 \$ 70 80 45 90

2 120 40 30 75

3 110 60 70 80

a. Find the initial solution using the minimum cell cost method.

b. Solve using the stepping-stone method.

15. The Sholtz Beer Company has breweries in two cities; the breweries can supply the following numbers of barrels of draft beer to the company's distributors each month:

Brewery Monthly Supply (bbl)

A. Tampa 3,500

B. St. Louis 5,000

Total 8,500

The distributors, which are spread throughout six states, have the following total monthly demand:

Distributor Monthly Demand (bbl)

1. Tennessee 1,600

2. Georgia 1,800

3. North Carolina 1,500

4. South Carolina 950

5. Kentucky 1,250

6. Virginia 1,400

Total 8,500

The company must pay the following shipping costs per barrel:

To

From 1 2 3 4 5 6

A \$0.50 0.35 0.60 0.45 0.80 0.75

B 0.25 0.65 0.40 0.55 0.20 0.65

a. Find the initial solution using VAM.

b. Solve using the stepping-stone method.

16. In Problem 15, the Sholtz Beer Company management has negotiated a new shipping contract with

a trucking firm between its Tampa brewery and its distributor in Kentucky that reduces the shipping cost per barrel from \$0.80 per barrel to \$0.65 per barrel. How will this cost change affect the optimal solution?

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Problems B-31

17. Computers Unlimited sells microcomputers to universities and colleges on the East Coast and ships

them from three distribution warehouses. The firm is able to supply the following numbers of microcomputers to the universities by the beginning of the academic year:

Distribution Supply

Warehouse (microcomputers)

1. Richmond 420

2. Atlanta 610

3. Washington, D.C. 340

Total 1,370

Four universities have ordered microcomputers that must be delivered and installed by the beginning of the academic year:

Demand

University (microcomputers)

A. Tech 520

B. A and M 250

C. State 400

D. Central 380

Total 1,550

The shipping and installation costs per microcomputer from each distributor to each university are as follows:

To

From A B C D

1 \$22 17 30 18

2 15 35 20 25

3 28 21 16 14

a. Find the initial solution using VAM.

b. Solve using MODI.

18. In Problem 17, Computers Unlimited wants to better meet demand at the four universities it supplies. It is considering two alternatives: (1) expand its warehouse at Richmond to a capacity of 600

at a cost equivalent to an additional \$6 in handling and shipping per unit or (2) purchase a new warehouse in Charlotte that can supply 300 units with shipping costs of \$19 to Tech, \$26 to A and M, \$22 to State, and \$16 to Central. Which alternative should management select based solely on transportation costs (i.e., no capital costs)?

19. Computers Unlimited in Problem 17 has determined that when it is unable to meet the demand for

microcomputers at the universities it supplies, the universities tend to purchase microcomputers elsewhere in the future. Thus, the firm has estimated a shortage cost for each microcomputer demanded but not supplied that reflects the loss of future sales and goodwill. These costs for each university are as follows:

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University Cost/Microcomputer

A. Tech \$40

B. A and M 65

C. State 25

D. Central 50

Solve Problem 17 with these shortage costs included. Compute the total transportation cost and the total shortage cost.

20. A severe winter ice storm has swept across North Carolina and Virginia, followed by over a foot of

snow and frigid, single-digit temperatures. These weather conditions have resulted in numerous downed power lines and power outages in the region causing dangerous conditions for much of the population. Local utility companies have been overwhelmed and have requested assistance from unaffected utility companies across the Southeast. The following table shows the number of utility trucks with crews available from five different companies in Georgia, South Carolina, and Florida; the demand for crews in seven different areas that local companies cannot get to; and the weekly cost (\$1,000s) of a crew going to a specific area (based on the visiting company's normal charges, the distance the crew has to come, and living expenses in an area):

Area (Cost = \$1,000s)

Crews

Crew NC-E NC-SW NC-P NC-W VA-SW VA-C VA-T Available

GA-1 15.2 14.3 13.9 13.5 14.7 16.5 18.7 12

GA-2 12.8 11.3 10.6 12.0 12.7 13.2 15.6 10

SC-1 12.4 10.8 9.4 11.3 13.1 12.8 14.5 14

FL-1 18.2 19.4 18.2 17.9 20.5 20.7 22.7 15

FL-2 19.3 20.2 19.5 20.2 21.2 21.3 23.5 12

Crews

Needed 9 7 6 8 10 9 7

Determine the number of crews that should be sent from each utility to each affected area that will minimize total costs.

21. A large manufacturing company is closing three of its existing plants and intends to transfer some

of its more skilled employees to three plants that will remain open. The number of employees available for transfer from each closing plant is as follows:

Closing Plant Transferable Employees

1 60

2 105

3 70

Total 235

The following number of employees can be accommodated at the three plants remaining open:

Open Plants Employees Demanded

A 45

B 90

C 35

Total 170

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Problems B-33

Each transferred employee will increase product output per day at each plant as shown in the following table. The company wants to transfer employees so as to ensure the maximum increase in product output.

To

From A B C

1 5 8 6

2 10 9 12

3 7 6 8

a. Find the initial solution using VAM.

b. Solve using MODI.

22. The Sav-Us Rental Car Agency has six lots in Nashville, and it wants to have a certain number of cars available at each lot at the beginning of each day for local rental. The agency would like a model it could quickly solve at the end of each day that would tell it how to redistribute the cars among the six lots in the minimum total time. The times required to travel between the six lots are as follows:

To (minutes)

From 1 2 3 4 5 6

1 — 12 17 18 10 20

2 14 — 10 19 16 15

3 14 10 — 12 8 9

4 8 16 14 — 12 15

5 11 21 16 18 — 10

6 24 12 9 17 15 —

The agency would like the following numbers of cars at each lot at the end of the day. Also shown are the numbers of available cars at each lot at the end of a particular day. Determine the optimal reallocation of rental cars using any initial solution approach and any solution method.

Lot

Cars 1 2 3 4 5 6

Available 37 20 14 26 40 28

Desired 30 25 20 40 30 20

23. Bayville has built a new elementary school so that the town now has a total of four schools—Addison, Beeks, Canfield, and Daley. Each has a capacity of 400 students. The school wants to assign children to schools so that their travel time by bus is as short as possible. The school has partitioned the town into five districts conforming to population density—north, south, east, west, and central. The average bus travel time from each district to each school is shown as follows:

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Travel Time (min.)

Student

District Addison Beeks Canfield Daley Population

North 12 23 35 17 250

South 26 15 21 27 340

East 18 20 22 31 310

West 29 24 35 10 210

Central 15 10 23 16 290

Determine the number of children that should be assigned from each district to each school to minimize total student travel time.

24. In Problem 23, the school board has determined that it does not want any one school to be more

crowded than any other school. It would like to assign students from each district to each school so that enrollments are evenly balanced among the four schools. However, the school board is concerned that this might significantly increase travel time. Determine the number of students to be assigned from each district to each school so that school enrollments are evenly balanced. Does this

new solution appear to result in a significant increase in travel time per student?

25. The Easy Time Grocery chain operates in major metropolitan areas on the eastern seaboard. The

stores have a “no-frills” approach, with low overhead and high volume. They generally buy their stock in volume at low prices. However, in some cases they actually buy stock at stores in other areas

and ship it in. They can do this because of high prices in the cities they operate in compared with costs in other locations. One example is baby food. Easy Time purchases baby food at stores in Albany, Binghamton, Claremont, Dover, and Edison and then trucks it to six stores in and around New York City. The stores in the outlying areas know what Easy Time is up to, so they limit the number of cases of baby food Easy Time can purchase. The following table shows the profit Easy Time makes per case of baby food based on where the chain purchases it and at which store it's sold, plus the available baby food per week at purchase locations and the shelf space available at each Easy Time store per week:

Easy Time Store

Purchase

Location 1 2 3 4 5 6 Supply

Albany 9 8 11 12 7 8 26

Binghamton 10 10 8 6 9 7 40

Claremont 8 6 6 5 7 4 20

Dover 4 6 9 5 8 10 40

Edison 12 10 8 9 6 7 45

Demand 25 15 30 18 27 35

Determine where Easy Time should purchase baby food and how the food should be distributed in order to maximize profit. Use any initial solution approach and any solution method.

26. Suppose that in Problem 25 Easy Time can purchase all the baby food it needs from a New York City distributor at a price that will result in a profit of \$9 per case at stores 1, 3, and 4, \$8 per case at

stores 2 and 6, and \$7 per case at store 5. Should Easy Time purchase all, none, or some of its baby food from the distributor rather than purchasing it at other stores and trucking it in?

27. In Problem 25, if Easy Time could arrange to purchase more baby food from one of the outlying locations, which should it be, how many additional cases could be purchased, and how much would this increase profit?

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Problems B-35

28. The Roadnet Transport Company has expanded its shipping capacity by purchasing 90 trailer trucks from a competitor that went bankrupt. The company subsequently located 30 of the purchased trucks at each of its shipping warehouses in Charlotte, Memphis, and Louisville. The company makes shipments from each of these warehouses to terminals in St. Louis, Atlanta, and New York. Each truck is capable of making one shipment per week. The terminal managers have indicated their capacity of extra shipments. The manager at St. Louis can accommodate 40 additional trucks per week, the manager at Atlanta can accommodate 60 additional trucks, and the manager at

New York can accommodate 50 additional trucks. The company makes the following profit per truckload shipment from each warehouse to each terminal. The profits differ as a result of differences in products shipped, shipping costs, and transport rates:

Terminal

Warehouse St. Louis Atlanta New York

Charlotte \$1,800 \$2,100 \$1,600

Memphis 1,000 700 900

Louisville 1,400 800 2,200

Determine how many trucks to assign to each route (i.e., warehouse to terminal) in order to maximize profit.

29. During the Gulf War, Operation Desert Storm required large amounts of military matériel and supplies to be shipped daily from supply depots in the United States to bases in the Middle East. The critical factor in the movement of these supplies was speed. The following table shows the number of planeloads of supplies available each day from each of six supply depots and the number of daily loads demanded at each of five bases. (Each planeload is approximately equal in tonnage.) Also included are the transport hours per plane, including loading and fueling, actual flight time, and unloading and refueling:

Military Base

Supply

Depot A B C D E Supply

1 36 40 32 43 29 7
 2 28 27 29 40 38 10
 3 34 35 41 29 31 8
 4 41 42 35 27 36 8
 5 25 28 40 34 38 9
 6 31 30 43 38 40 6
 Demand 9 6 12 8 10

Determine the optimal daily flight schedule that will minimize total transport time.

30. PM Computer Services produces personal computers from component parts it buys on the open market. The company can produce a maximum of 300 personal computers per month. PM wants to determine its production schedule for the first six months of the new year. The cost to produce a personal computer in January will be \$1,200. However, PM knows the cost of component parts will decline each month such that the overall cost to produce a PC will be 5% less each month. The cost of holding a computer in inventory is \$15 per unit per month. Following is the demand for the company's computers each month:

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Month Demand Month Demand

January 180 April 210

February 260 May 400

March 340 June 320

Determine a production schedule for PM that will minimize total cost.

31. In Problem 30, suppose the demand for personal computers increased each month as follows:

Month Demand

January 410

February 320

March 500

April 620

May 430

June 380

In addition to the regular production capacity of 300 units per month, PM Computer Services can also produce an additional 200 computers per month using overtime. Overtime production adds 20% to the cost of a personal computer.

Determine a production schedule for PM that will minimize total cost.

32. National Foods Company has five plants where it processes and packages fruits and vegetables. It has suppliers in six cities in California, Texas, Alabama, and Florida. The company has owned and operated its own trucking system in the past for transporting fruits and vegetables from its suppliers to its plants. However, it is now considering outsourcing all its shipping to outside trucking firms and getting rid of its own trucks. It currently spends \$245,000 per month to

operate its own trucking system. It has determined monthly shipping costs (in \$1,000s per ton) using outside shippers from each of its suppliers to each of its plants as shown in the following table:

Processing Plants (\$1,000s per ton)

Suppliers Denver St. Paul Louisville Akron Topeka Supply (tons)

Sacramento 3.7 4.6 4.9 5.5 4.3 18

Bakersfield 3.4 5.1 4.4 5.9 5.2 15

San Antonio 3.3 4.1 3.7 2.9 2.6 10

Montgomery 1.9 4.2 2.7 5.4 3.9 12

Jacksonville 6.1 5.1 3.8 2.5 4.1 20

Ocala 6.6 4.8 3.5 3.6 4.5 15

Demand (tons) 20 15 15 15 20 90

Should National Foods continue to operate its own shipping network or sell its trucks and out-source its shipping to independent trucking firms?

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Problems B-37

33. In Problem 32, National Foods would like to know what the effect would be on the optimal solution and the company's decision regarding its shipping if it negotiates with its suppliers in Sacramento, Jacksonville, and Ocala to increase their capacity to 25 tons per month? What would be the effect of negotiating instead with its suppliers at San Antonio and Montgomery to increase their capacity to 25 tons each?

34. Orient Express is a global distribution company that transports its clients' products to customers in

Hong Kong, Singapore, and Taipei. All the products Orient Express ships are stored at three distribution centers, one in Los Angeles, one in Savannah, and one in Galveston. For the coming month the company has 450 containers of computer components available at the Los Angeles center, 600 containers available at Savannah, and 350 containers available in Galveston. The company has orders for 600 containers from Hong Kong, 500 containers from Singapore, and 500 containers from Taipei. The shipping costs per container from each U.S. port to each of the overseas ports are shown in the following table:

Overseas Port

U.S. Distribution

Center Hong Kong Singapore Taipei

Los Angeles \$300 \$210 \$340

Savannah 490 520 610

Galveston 360 320 500

The Orient Express, as the overseas broker for its U.S. customers, is responsible for unfulfilled orders, and it incurs stiff penalty costs from overseas customers if it does not meet an order. The Hong Kong customers charge a penalty cost of \$800 per container for unfulfilled demand, Singapore customers charge a penalty cost of \$920 per container, and Taipei customers charge

\$1,100 per container. Formulate and solve a transportation model to determine the shipments from each U.S. distribution center to each overseas port that will minimize shipping costs. Indicate what portion of the total cost is a result of penalties.

35. Binford Tools manufactures garden tools. It uses inventory, overtime, and subcontracting to absorb

demand fluctuations. Expected demand, regular and overtime production capacity, and subcontracting capacity are provided in the following table for the next four quarters for its basic line of steel garden tools:

Regular Overtime Subcontracting

Quarter Demand Capacity Capacity Capacity

1 9,000 9,000 1,000 3,000

2 12,000 10,000 1,500 3,000

3 16,000 12,000 2,000 3,000

4 19,000 12,000 2,000 3,000

The regular production cost per unit is \$20, the overtime cost per unit is \$25, the cost to subcontract a unit is \$27, and the inventory carrying cost is \$2 per unit. The company has 300 units in inventory at the beginning of the year.

Determine the optimal production schedule for the four quarters that will minimize total costs.

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36. Solve the following linear programming problem:

minimize $Z = 18x_{11} + 30x_{12} + 20x_{13} + 18x_{14} + 25x_{21} + 27x_{22} + 22x_{23}$

$+ 16x_{24} + 30x_{31} + 26x_{32} + 19x_{33} + 32x_{34} + 40x_{41} + 36x_{42}$

$+ 27x_{43} + 29x_{44} + 30x_{51} + 26x_{52} + 18x_{53} + 24x_{54}$

subject to

$x_{11} + x_{12} + x_{13} + x_{14} \leq 1$

$x_{21} + x_{22} + x_{23} + x_{24} \leq 1$

$x_{31} + x_{32} + x_{33} + x_{34} \leq 1$

$x_{41} + x_{42} + x_{43} + x_{44} \leq 1$

$x_{51} + x_{52} + x_{53} + x_{54} \leq 1$

$x_{11} + x_{21} + x_{31} + x_{41} + x_{51} = 1$

$x_{12} + x_{22} + x_{32} + x_{42} + x_{52} = 1$

$x_{13} + x_{23} + x_{33} + x_{43} + x_{53} = 1$

$x_{14} + x_{24} + x_{34} + x_{44} + x_{54} = 1$

$x_{ij} \geq 0$

37. A plant has four operators to be assigned to four machines. The time (minutes) required by each

worker to produce a product on each machine is shown in the following table. Determine the optimal assignment and compute total minimum time.

Machine

Operator A B C D

1 10 12 9 11

2 5 10 7 8

3 12 14 13 11

4 8 15 11 9

38. A shop has four machinists to be assigned to four machines. The hourly cost of having each machine operated by each machinist is as follows:

Machine

Machinist A B C D

1 \$12 11 8 14

2 10 9 10 8

3 14 8 7 11

4 6 8 10 9

However, because he does not have enough experience, machinist 3 cannot operate machine B.

- Determine the optimal assignment and compute total minimum cost.
- Formulate this problem as a general linear programming model.

39. The Omega pharmaceutical firm has five salespersons, whom the firm wants to assign to five sales regions. Given their various previous contacts, the salespersons are able to cover the

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Problems B-39

regions in different amounts of time. The amount of time (days) required by each salesperson to cover each city is shown in the following table. Which salesperson should be assigned to each region to minimize total time? Identify the optimal assignments and compute total minimum time.

Region

Salesperson A B C D E

1 17 10 15 16 20

2 12 9 16 9 14

3 11 16 14 15 12

4 14 10 10 18 17

5 13 12 9 15 11

40. The Bunker Manufacturing firm has five employees and six machines and wants to assign the employees to the machines to minimize cost. A cost table showing the cost incurred by each employee on each machine follows. Because of union rules regarding departmental transfers, employee 3 cannot be assigned to machine E and employee 4 cannot be assigned to machine B. Solve this problem, indicate the optimal assignment, and compute total minimum cost.

Machine

Employee A B C D E F

1 \$12 \$ 7 \$20 \$14 \$ 8 \$10

2 10 14 13 20 9 11

3 5 3 6 9 7 10

4 9 11 7 16 9 10

5 10 6 14 8 10 12

41. Given the following cost table for an assignment problem, determine the optimal assignment and compute total minimum cost. Identify all alternative solutions if there are multiple optimal solutions.

Machine

Operator A B C D

1 \$10 \$2 \$ 8 \$ 6

2 9 5 11 9

3 12 7 14 14

4 3 1 4 2

42. An electronics firm produces electronic components, which it supplies to various electrical manufacturers. Quality control records indicate that different employees produce different numbers of defective items. The average number of defects produced by each employee for each of six components

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is given in the following table. Determine the optimal assignment that will minimize the total average number of defects produced by the firm per month.

Component

Employee A B C D E F

1 30 24 16 26 30 22

2 22 28 14 30 20 13

3 18 16 25 14 12 22

4 14 22 18 23 21 30

5 25 18 14 16 16 28

6 32 14 10 14 18 20

43. A dispatcher for the Citywide Taxi Company has six taxicabs at different locations and five customers who have called for service. The mileage from each taxi's present location to each customer is shown in the following table. Determine the optimal assignment(s) that will minimize the total mileage traveled.

Customer

Cab 1 2 3 4 5

A 7 2 4 10 7

B 5 1 5 6 6
 C 8 7 6 5 5
 D 2 5 2 4 5
 E 3 3 5 8 4
 F 6 2 4 3 4

44. The Southeastern Conference has nine basketball officials who must be assigned to three conference games, three to each game. The conference office wants to assign the officials so that the total distance they travel will be minimized. The distance (in miles) each official would travel to each game is given in the following table:

Game

Official Athens Columbia Nashville

1 165 90 130
 2 75 210 320
 3 180 170 140
 4 220 80 60
 5 410 140 80
 6 150 170 190
 7 170 110 150
 8 105 125 160
 9 240 200 155

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Problems B-41

- a. Should this problem be solved by the transportation method or by the assignment method? Explain.
- b. Determine the optimal assignment(s) that will minimize the total distance traveled by the officials.

45. In Problem 44, officials 2 and 8 had a recent confrontation with one of the coaches in the game in

Athens. They were forced to eject the coach after several technical fouls. The conference office decided that it would not be a good idea to have these two officials work the Athens game so soon after this confrontation, so they decided that officials 2 and 8 will not be assigned to the Athens game. How will this affect the optimal solution to this problem?

46. State University has planned six special catered events for the November Saturday of its homecom-

ing football game. The events include an alumni brunch, a parents' brunch, a booster club luncheon, a postgame party for season ticket holders, a lettermen's dinner, and a fund-raising dinner for major contributors. The university wants to use local catering firms as well as the university catering service to cater these events and it has asked the caterers to bid on each event. The bids (in

\$1,000s) based on menu guidelines for the events prepared by the university are shown in the following table:

Event	Alumni	Parents’	Booster	Postgame	Lettermen’s	Contributor’s
Caterer	Brunch	Brunch	Club	Lunch	Party	Dinner
Al’s	\$12.6	\$10.3	\$14.0	\$19.5	\$25.0	\$30.0
Bon Apetît	14.5	13.0	16.5	17.0	22.5	32.0
Custom	13.0	14.0	17.6	21.5	23.0	35.0
Divine	11.5	12.6	13.0	18.7	26.2	33.5
Epicurean	10.8	11.9	12.9	17.5	21.9	28.5
Fouchés	13.5	13.5	15.5	22.3	24.5	36.0
University	12.5	14.3	16.0	22.0	26.7	34.0

The Bon Apetît, Custom, and University caterers can handle two events, whereas the other four caterers can handle only one. The university is confident all the caterers will do a high-quality job, so it wants to select the caterers for the events that will result in the lowest total cost. Determine the optimal selection of caterers that will minimize total cost.

47. A university department head has five instructors to be assigned to four different courses. All the instructors have taught the courses in the past and have been evaluated by the students. The rating for each instructor for each course is given in the following table (a perfect score is 100). The department head wants to know the optimal assignment of instructors to courses that will maximize the overall average evaluation. The instructor who is not assigned to teach a course will be assigned to grade exams. Solve this problem using the assignment method.

Course	Instructor	A	B	C	D
1	80	75	90	85	
2	95	90	90	97	
3	85	95	88	91	
4	93	91	80	84	
5	91	92	93	88	

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48. The coach of the women’s swim team at State University is preparing for the conference swim meet and must choose the four swimmers she will assign to the 800-meter medley relay team. The medley relay consists of four strokes—the backstroke, breaststroke, butterfly, and freestyle. The coach has computed the average times (in minutes) each of her top six swimmers has achieved in each of the four strokes for 200 meters in previous swim meets during the season as follows:

Stroke (min)	Swimmer 1	Swimmer 2	Swimmer 3	Swimmer 4	Swimmer 5	Swimmer 6
--------------	-----------	-----------	-----------	-----------	-----------	-----------

Swimmer Backstroke Breaststroke Butterfly Freestyle

Annie 2.56 3.07 2.90 2.26

Beth 2.63 3.01 3.12 2.35

Carla 2.71 2.95 2.96 2.29

Debbie 2.60 2.87 3.08 2.41

Erin 2.68 2.97 3.16 2.25

Fay 2.75 3.10 2.93 2.38

Determine the medley relay team and its total expected relay time for the coach.

49. Biggio's Department Store has six employees available to assign to four departments in the store—

home furnishings, china, appliances, and jewelry. Most of the six employees have worked in each of the four departments on several occasions in the past and have demonstrated that they perform better in some departments than in others. The average daily sales for each of the six employees in each of the four departments is shown in the following table.

Department Sales (\$)

Home

Employee Furnishings China Appliances Jewelry

1 340 160 610 290

2 560 370 520 450

3 270 — 350 420

4 360 220 630 150

5 450 190 570 310

6 280 320 490 360

Employee 3 has not worked in the china department before, so the manager does not want to assign

this employee to china.

Determine which employee to assign to each department and indicate the total expected daily sales.

50. The Vanguard Publishing Company has eight college students it hires as salespeople to sell encyclo-

pedias during the summer. The company desires to allocate them to three sales territories.

Territory

1 requires three salespeople, and territories 2 and 3 require two salespeople each. It is estimated that

each salesperson will be able to generate the amounts of dollar sales per day in each of the three ter-

ritories as given in the following table. The company desires to allocate the salespeople to the three territories so that sales will be maximized. Solve this problem using any method to determine the initial solution and any solution method. Compute the maximum total sales per day.

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Problems B-43

Territory

Salesperson 1 2 3

A \$110 \$150 \$130

B 90 120 80

C 205 160 175

D 125 100 115

E 140 105 150

F 100 140 120

G 180 210 160

H 110 120 90

51. Carolina Airlines, a small commuter airline in North Carolina, has six flight attendants whom it wants to assign to six monthly flight schedules in a way that will minimize the number of nights they will be away from their homes. The numbers of nights each attendant must be away from home with each schedule are given in the following table. Identify the optimal assignments that will minimize the total number of nights the attendants will be away from home.

Schedule

Attendant A B C D E F

1 7 4 5 10 5 7

2 4 5 4 12 7 5

3 9 9 10 7 10 7

4 11 6 7 5 9 9

5 5 8 5 10 7 5

6 10 12 10 9 9 9

52. The football coaching staff at Tech focuses its recruiting on several key states, including Georgia, Florida, Virginia, Pennsylvania, New York, and New Jersey. The staff includes seven assistant coaches, two of whom are responsible for Florida, a high school talent-rich state, whereas one coach is assigned to each of the other five states. The staff has been together for a long time and at one time or another all the coaches have recruited all of the states. The head coach has accumulated some data on the past success rate (i.e., percentage of targeted recruits signed) for each coach in each state as shown in the following table:

State

Coach GA FL VA PA NY NJ

Allen 62 56 65 71 55 63

Bush 65 70 63 81 75 72

Crumb 46 53 62 55 64 50

Doyle 58 66 70 67 71 49

Evans 77 73 69 80 80 74

Fouch 68 73 72 80 78 57

Goins 72 60 74 72 62 61

Determine the optimal assignment of coaches to recruiting regions that will maximize the overall success rate and indicate the average percentage success rate for the staff with this assignment.

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ESD.36 System & Project Management

Lecture 2

- Critical Path Method (CPM)

Instructor(s)

Prof. Olivier de Weck

Today's Agenda

- Overview of PM methods and tools

- CPM 101

- Critical Paths, Slack

- Task "Crashing" and Cost

- Conclusions and Class Discussions

- Introduce HW3

- ESD.36 SPM 2

History of Project Management

- Big Projects since antiquity

- Pyramids (Egypt), Great Wall (China)

- Enormous workforce, but little documented evidence of formal project management

- Formal Project Management

- Henry Gantt (1861-1919) □ bar chart 1910

- 1957 Sputnik Crisis □ revival of "scientific management"

- Polaris (1958) □ Project Evaluation and Review

Technique (PERT)

- DuPont Company (1960) □ Critical Path Method (CPM)

- 1960's NASA projects: Mercury, Gemini, Apollo

- Work Breakdown Structures (WBS)

- Cost and Schedule Tracking, Configuration Management

- ESD.36 SPM 3

Comments about early PM

- Project decomposition necessary due to complexity

- Resource allocation and workload smoothing

- Schedule urgency .."before the decade is out"

JFK

- Circumstances
 - Complex Relations between Government and Contractors
 - “Shielded” from Society, Competition, Regulations
 - Cold War Pressures for Nuclear Power, Space Race ..
 - Other Innovations
 - Project Manager as a central figure
 - Beginnings of Matrix Organization
 - “Earned Value” – adopted by USAF (1963)
 - Professionalization since 1969
 - Diffusion into other industries: computers, automotive ...
 - Project Management Institute (PMI) founded – PMBOK
 - ISO 10006:2003 Quality in Project Management
 - Recent criticism about PM standards as “bureaucratic”
- ESD.36 SPM 4

Fundamental Approaches

- How to represent task relationships?
- Network-based (graph theory) methods
- CPM, PERT,

a b

- Task is a node or an arc d e

c

- Matrix-based methods
- DSM - Tasks are columns and rows x x
- Interrelationships are off-diagonal entries

xx

x x

x x

- System Dynamics x x
- Feedback loops, causal relationships
- Stocks and flows simulation
- Tasks that are done or waiting to be done

PEOPLE PRODUCTIVITY

are stocks – “amount of work” WORK WORK
WORK

- Doing project work causes a “flow”

TO BE

BEING DONE DONE

DONE

- ESD.36 SPM 5

Gantt Charts

- Attributed to Henry Gantt – most popular PM tool (80%)
- Used to plan big shipbuilding projects (cargo ships WWI)
- Graphical way of showing task durations, project schedule
- Does not explicitly show relationships between tasks
- Limited use for project tracking
- Easy to understand

today calendar

completion

Gantt Chart Builder System (Excel) 1.6

08 Sep'03 22 Sep'03 06 Oct'03

Bus Unit Project % Start Finish 08/09 15/09 22/09 29/09 06/10 13/10

Project "XYZ" 30 9-Sep-03 6-Oct-03

Mkt Customer Clinic 100 09-Sep-2003 12-Sep-2003

milestone

Sys Requirements Definition 100 11-Sep-2003 15-Sep-2003

Eng Parts Design 50 15-Sep-2003 22-Sep-2003

Sys Design Review 0 23-Sep-2003 23-Sep-2003

Mfg Manufacturing 0 24-Sep-2003 05-Oct-2003

Sys Product Release 0 06-Oct-2003 06-Oct-2003

tasks actual planned

- ESD.36 SPM D 6

CPM 101

- Represent a project (set of task) as a network using graph theory
- Capture task durations
- Capture task logic (dependencies)

(Parallel)

Task ID (Series) Expected

Duration B,8

A,5 B,8 D,3

A,5

C,2

"B can only start after A

is completed" "B and C do not depend on each other"

- ESD.36 SPM 7

CPM Assumptions

- Project consists of a collection of well defined tasks (jobs)

- Project ends when all jobs completed
- Jobs may be started and stopped independently of each other within a given sequence (no “continuous-flow” processes)
- Jobs are ordered □ “technological sequence”

- ESD.36 SPM 8

Task Representations

- Tasks as Nodes of a Graph
- Circles

ES EF

- Boxes

A,5 ID:A Dur:5

LS LF

- Tasks as Arcs of a Graph
- Tasks are uni-directional arrows
- Nodes now represent “states” of a project
- Kelley-Walker form

A, 5

broken fixed

- ESD.36 SPM 9

Concept Question 1

What in project management is named after Henry Gantt (1861-1919)?

- A new method for managing budgets in shipyards
- A matrix representation of the task relationships known as the Gantt-Matrix
- A horizontal bar chart known as the Gantt-Chart
- Gantt-teams, a new form of organizational structure
- Gantt was a mechanical engineer and did not care about projects

- ESD.36 SPM 10

Work Breakdown Structure

- Used to create the task (job) list
- Tree-decomposition of project tasks
- WBS identifies “terminal elements”
- The key starting point for project planning
- Required by US Govt as part of SOW

□ Can be activity-oriented or deliverable- oriented

□ Use “sticky-notes” method early on

□ Carl L. Pritchard. Nuts and Bolts Series 1: How to Build a Work Breakdown Structure. ISBN 1890367125

Job A Job B Job X Job G

- ESD.36 SPM 11

WBS – Painting a Room

□ 1 Prepare materials

□ 1.1 Buy paint

□ 1.2 Buy brushes/rollers

□ 1.3 Buy wallpaper remover

□ 2. Prepare room

□ 2.1 Remove old wallpaper

□ 2.2 Remove detachable decorations

□ 2.3 Cover floor with old newspapers

□ 2.4 Cover electrical outlets/switches with tape

□ 2.5 Cover furniture with sheets

□ 3. Paint the room

□ 4. Clean up the room

□ 4.1 Dispose or store left over paint

□ 4.2 Clean brushes/rollers

□ 4.3 Dispose of old newspapers

□ 4.4 Remove covers Source: <http://www.wikipedia.org>

- ESD.36 SPM 12

+ WBS of MIT/NASA Space

- Logistics Project

IP S C M & L A

1 . 0 P r o j e c t 2 . 0 T e r r e s t r i a l 3 . 0 S p a c e L o g i s t i c s 4 . 0 E x p l o r a t i o n 5 . 0 I n t e r p l a n e t a r y 6 . 0 P r o j e c t

M a n a g e m e n t A n a l o g i e s N e t w o r k M o d e l S u p p l y / D e m a n d S C A r c h T r a d e I n t e g r a t i o n a n d

(T a s k 1) (T a s k 2) M o d e l i n g S t u d i e s D i s s e m i n a t i o n

(T a s k 3) (T a s k 4)

1 . 1 M I T P r o j e c t 2 . 1 L o w D e n s i t y E q u i p m e n t 3 . 1 M o d e l F o r m u l a t i o n 4 . 1 S u p p l y / D e m a n d 5 . 1 T r a d e S t u d i e s 6 . 1 N A S A W o r k s h o p I

M a n a g e m e n t S C M L e s s o n s L e a r n e d D a t a M o d e l i n g P l a n n i n g a n d S e t u p

2 . 2 L o n g - R a n g e M i l i t a r y 6 . 2 N A S A W o r k s h o p I I

1 . 1 . 1 M I T P r o j e c t S C M L e s s o n s L e a r n e d 3 . 1 . 1 M I T S e g m e n t 4 . 1 . 1 M I T S e g m e n t 5 . 1 . 1 M I T S e g m e n t

Manager/PI

2.3 Remote Environments 6.3 Internal Integration

1.1.2 MIT Travel SCM Lessons Learned 3.1.2 JPL Segment 4.1.2 JPL Segment 5.1.2 JPL Segment

2.4 ISS SCM

1.1.3 MIT Resource Lessons Learned 3.2 Model 4.1.3 USA Segment 5.2 Trade Studies 6.3.1 MIT Segment

Administration Development Execution

2.5 Shuttle/ISS

1.1.4 Reserves Lessons Learned 4.2 Data 5.3 Trade Studies 6.3.2 JPL Segment

3.2.1 MIT Segment Collection Reporting

1.2 JPL Project 6.3.3 USA Segment

Management 3.2.2 JPL Segment 4.2.1 JPL Segment

6.4 Public Education

1.2.1 JPL Project 3.3 Model 4.2.2 USA Segment and Outreach

Manager/Co-PI Validation

1.2.2 JPL Travel 4.3 Data 6.4.1 MIT Segment

3.3.1 MIT Segment Validation

1.2.3 JPL Resource 6.4.2 JPL Segment

Administration 3.3.2 JPL Segment 4.3.1 JPL Segment

1.2.4 Reserves 6.4.3 USA Segment

3.3.3 USA Segment 4.3.2 USA Segment

1.3 USA Project 6.5 Space SCM

Management Executive Short Course

1.3.1 USA Project

MIT (incl. PSI): cyan

Manager/Co-PI

JPL: gray

1.3.2 USA Travel USA: green

Color code WBS by

Organization in charge

1.3.3 USA Resource

Administration

- ESD.36 SPM 13

1.3.4 Reserves

Discussion Point 1

□ Why is it difficult to come up with a good WBS (task list, task structure) in a complex project?

□ Not all tasks known ahead of time if completely new product/system

□ Others...?

- ESD.36 SPM 14

WBS Guidelines

□ No more than 100-200 terminal elements, if more □ use subprojects

□ Can be up to 3-4 Levels deep

□ Not more than 5-9 jobs at one level

□ Human cognitive “bandwidth” only 3 bits=23=8

□ Short term memory for most people 5-9 items

□ Poorer planning if “too-fine grained” – dilution of attention

□ The more tasks there are, the more intricate dependencies there will be to keep track of

□ Jobs should be of similar size/complexity

□ Manageable chunks □ sense of progress

□ Level of graininess □ very difficult to answer

- ESD.36 SPM 15

Task List

□ List all tasks in a table with

□ Identifying symbol (tag, ID number)

□ Task description

□ Immediate prerequisite jobs

□ Expected task duration

□ Arrange jobs in “technological order”

□ No job appears in the list until all its predecessors have been listed

□ Iterations are NOT allowed □ “cycle error”

□ Job a precedes b precedes c precedes a

□ We will discuss iterations a lot in this class !!!

- ESD.36 SPM 16

Simple Example: Job List

□ Two Parts X and Y: Manufacture and Assembly

Job # Description Immediate Time

Predecessors [min]

A Start 0

B Get materials for X A 10

C Get materials for Y A 20

D Turn X on lathe B,C 30
 E Turn Y on lathe B,C 20
 F Polish Y E 40
 G Assemble X and Y D,F 20
 H Finish G 0

- ESD.36 SPM 17

Project Graph

- Each job is drawn on a graph as a circle*
- Connect each job with immediate predecessor(s) – unidirectional arrows “→”
- Jobs with no predecessor connect to “Start”
- Jobs with no successors connect to “Finish”
- “Start” and “Finish” are pseudo-jobs of length 0
- A finite number of “arrow paths” from “Start” to “Finish” will be the result
- Total time of each path is the sum of job times
- Path with the longest total time □ “critical path”
- There can be multiple critical paths □ minimum time to complete project

* or other symbol, see before

- ESD.36 SPM 18

Project Graph

C,20 E,20 F,40

critical path

A,0 G,20 H,0

Start Finish

B,10 D,30

4 unique paths: A,C,E,F,G,H; A,C,D,G,H; A,B,D,G,H; A,B,E,F,G,H

100 70 60 90

- ESD.36 SPM 19

Critical Path

- CP is the “bottleneck route”
- Shortening or lengthening tasks on the critical path directly affects project finish
- Duration of “non-critical” tasks is irrelevant
- “Crashing” all jobs is ineffective, focus on the few % of jobs that are on the CP
- “Crashing” tasks can shift the CP to a different task

- Shortening tasks – technical and economical challenge
- How can it be done?
- Previously non-critical tasks can become critical
- Lengthening of non-critical tasks can also shift the critical path (see HW1).

- ESD.36 SPM 20

Discussion Point 2

- What is the usefulness of knowing the CP in a project?
- Tells which task to shorten to finish project earlier.
- Others ...?

- ESD.36 SPM 21

Critical Path Algorithm

- For large projects there are many paths
- Need a algorithm to identify the CP efficiently
- Develop information about each task in context of the overall project
- Times
- Start time (S)
- For each job: Earliest Start (ES)
- Earliest start time of a job if all its predecessors start at ES
- Job duration: t
- Earliest Finish (EF) = $(ES) + t$
- Finish time (F) – earliest finish time of the overall project
- Show algorithm using project graph

- ESD.36 SPM 22

CP Algorithm

1. Mark the value of S to left and right of Start
2. Consider any new unmarked job, all of whose predecessors have been marked. Mark to the left of the new job the largest number to the right of its immediate predecessors: (ES)
3. Add to ES the job time t and mark result to the right (EF)
4. Stop when Finish has been reached

- ESD.36 SPM 23

CP Algorithm - Graphical

critical path

E,20 F,40

0 C,20 20 20 40 40 80

Finish

S=0 A,0 0 G,20 H,0

80 100 100 F=100

Start

0 B,10 10 20 D,30 50

- ESD.36 SPM 24

Concept Question 2

Possible

A project starts with (A,5). Task (B,10)

can start after A is completed. This is Answers

also true for task (E,5). Task (C,8) □20

depends only on (B,10), while task (F,10)

□22

depends on both (B,10) and (E,5). Task

(D,5) is the last task in the project and it □25

can start once (C,8) and (F,10) have □27

been finished.

□30

The Earliest Finish (EF) time for the □35

whole project is:

□40

- ESD.36 SPM 25

Latest Start and Finish Times

□ Set target finish time for project: $T \geq F$

□ Usually target is a specific calendar date,
e.g. October 1, 2007

□ When is the latest date the project can be
started?

□ Late Finish (LF) - latest time a job can be
finished, without delaying the project
beyond its target time (T)

□ Late Start: $LS = LF - t$

- ESD.36 SPM 26

Determine LF and LS

□ Work from the end of the project: T

1. Mark value of T to left and right of Finish

2. Consider any new unmarked job, all of

whose successors have been marked - mark
 to the right the smallest LS time marked to
 the left of any of its immediate successors
 3. Subtract from this number, LF, the job time t
 and mark result to the left of the job: LS
 4. Continue upstream until Start has been
 reached, then stop
 - ESD.36 SPM 27

LS and LF : Project Graph

0 20 20 40 40 80

0 20 20 40 40 80

C,20 E,20 F,40

Finish

S=0 A,0 G,20 H,0

Start 80 100 100 100

80 100 100 100

0 0

0 0 0 B,10 10 D,30

Legend

0 10 20 50 early ES EF

10 20 50 80 late LS LF

- ESD.36 SPM 28

Slack

- Some tasks have $ES=LS$ --> no slack
- Total Slack of a task $TS=LS-ES$
- Maximum amount of time a task may be delayed
beyond its early start without delaying project
completion
- Slack time is precious ... managerial freedom, don't
squander it unnecessarily
- e.g. resource, work load smoothing
- When $T=F$ then all critical tasks have $TS=0$
- At least one path from Start->Finish with critical jobs
only
- When $T>F$, then all critical jobs have $TS=T-F$

- ESD.36 SPM 29

Project Graph - Slack

0 20 20 40 40 80

0 20 20 40 40 80

C,20 E,20 F,40

Finish

TS=0 TS=0 TS=0

TS=0 TS=0

S=0 A,0 G,20 H,0

Start 80 100 100 100

TS=10 TS=30 80 100 100 100

0 0

0 0 0 B,10 10 D,30

Legend

0 10 20 50 early ES EF

10 20 50 80 late LS LF

- ESD.36 SPM 30

Task Times Detail - Task i

ES(i) LS(i) EF(i) LF(i)

Duration t(i)

Total Slack Duration t(i)

j is the immediate

TS(i) successor of i with

the smallest ES

FS(i) ES(j) $j > i$

Free Slack

□ Free Slack (FS) is the amount a job can be delayed without delaying the Early Start (ES) of any other job.

FS ≤ TS always

- ESD.36 SPM 31

Main CPM Errors

□ Estimated job times are wrong

□ Predecessor relationships may contain cycles □ “cycle error”

□ List of prerequisites contains more than the immediate predecessors, e.g. a□b, b□c and a,b□c

□ Overlooked some predecessor relationships

□ Some predecessor relationships may be listed that are spurious

□ and Some tasks/jobs may be missing !!!

- ESD.36 SPM 32

Gradual Refinement of CPM

- Job Times
- Given rough time estimates construct CPM chart
- Re-estimate times for CP and those with very small TS
- Iterate until the critical path is stable
- Focus attention on a subset of tasks
- Predecessor Relationships
- Check algorithmically for cycle errors and pre-predecessor errors
- Cancel all except immediate predecessor relationships
- Wrong or Missing Facts
- Cannot be detected by computers!

- ESD.36 SPM 33

Sample CPM (2007)

A&D High Tech Case, Online Store Project, 60 tasks, HW2, 2007

- ESD.36 SPM 34

Crashing Tasks

- What is we want to speed up project completion?
- Options
- Work overtime
- Put more resources on the critical tasks
- Parallelize tasks that are really serial (later in class)
- Cost of speedup?
- Is there a net savings resulting from reduction in overall project time?

- ESD.36 SPM 35

Cost Calculations

- Can compute project costs if cost of each job is included in the task data
- (Potentially) shorten crew jobs by adding personnel or working overtime with existing personnel
- Speedup carries price tag: "normal time", "crash time"
- Assign some critical jobs to their "crash time"
- Direct costs will increase as we "crash" critical tasks
- Indirect (fixed, overhead) costs will decrease as the overall project duration decreases - "standing army phenomenon"
- Minimize the sum of fixed and direct costs

- ESD.36 SPM 36

Typical Cost Pattern

Project Total Costs

Costs [\$]

“natural pace”

savings

Cum Fixed Costs

Direct Costs

crash Total Project

A B

Time [days]

- ESD.36 SPM 37

CPM Judgment

□ +

□ Focuses attention on a subset of critical tasks

□ Determine effect of shortening/lengthening tasks

□ Evaluate costs of a “crash” program

□ -

□ Doesn’t capture task iterations, in fact ...

□ Prohibits iterations = “cycle error”

□ Treats task durations as deterministic

- ESD.36 SPM 38

Summary

□ CPM is useful, despite criticism, to identify the critical path - focus on a subset of the project

□ Slack (TS and FS) is precious

□ apply flexibility to smooth resource/schedules

□ PERT treats task times as probabilistic

□ Next lecture

□ Selective “crashing” of critical tasks can reduce (or increase) total project cost

□ CPM does not allow for task iterations

- ESD.36 SPM 39

Class Frustrations

□ Poor examples set by project managers

□ Perception of PMs as bureaucratic “box-checkers” and “ankle-biters”

Why?

□ Traditional Project Management ...

□ Doesn’t acknowledge the existence of iterations

- Is inflexible, “changing the plan” considered a failure
- Does not think of projects in a probabilistic sense
- “Hostage” to existing project management software
- In a reactive mode – no “early warning” systems

- ESD.36 SPM 40

HW1 Introduction

- You are Project Manager for the new CityCar development project
- Plan the project
- Task list
- Create project graph
- Critical path
- Slack times
- Re-planning after change
- “managerial”-type questions

- ESD.36 SPM 41

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Paper 5 of 5 — CSC 425 note

Level 400 • Second Semester • Handout

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Module B

Transportation and

Assignment Solution

Methods

B-1

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B-2 Module B Transportation and Assignment Solution Methods

Solution of the Transportation Model

The following example was used in Chapter 6 of the text to demonstrate the formulation of the transportation model. Wheat is harvested in the Midwest and stored in grain elevators in three different cities—Kansas City, Omaha, and Des Moines. These grain elevators supply three flour mills, located in Chicago, St. Louis, and Cincinnati. Grain is shipped to the mills in railroad cars, each of which is capable of holding one ton of wheat. Each grain elevator is able to supply the following number of tons (i.e., railroad cars) of wheat to the mills on a monthly basis:

Grain Elevator Supply

1. Kansas City 150
2. Omaha 175
3. Des Moines 275

Total 600 tons

Each mill demands the following number of tons of wheat per month.

Mill Demand

- A. Chicago 200
- B. St. Louis 100
- C. Cincinnati 300

Total 600 tons

The cost of transporting one ton of wheat from each grain elevator (source) to each mill (destination) differs according to the distance and rail system. These costs are shown in the following table. For example, the cost of shipping one ton of wheat from the grain elevator at Omaha to the mill at Chicago is \$7.

Mill

Grain Elevator A. Chicago B. St. Louis C. Cincinnati

1. Kansas City \$6 \$ 8 \$10
2. Omaha 7 11 11
3. Des Moines 4 5 12

The problem is to determine how many tons of wheat to transport from each grain elevator to each mill on a monthly basis in order to minimize the total cost of transportation. The linear programming model for this problem is formulated in the equations that follow: minimize $Z = \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C}$ subject to

$$x_{1A} + x_{1B} + x_{1C} = 150$$

$$x_{2A} + x_{2B} + x_{2C} = 175$$

$$x_{3A} + x_{3B} + x_{3C} = 275$$

$$x_{1A} + x_{2A} + x_{3A} = 200$$

$$x_{1B} + x_{2B} + x_{3B} = 100$$

$$x_{1C} + x_{2C} + x_{3C} = 300$$

$$x_{ij} \geq 0$$

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Solution of the Transportation Model B-3

In this model the decision variables, x_{ij} , represent the number of tons of wheat transported from each grain elevator, i (where $i = 1, 2, 3$), to each mill, j (where $j = A, B, C$). The objective function represents the total transportation cost for each route. Each term in the objective function reflects the cost of the tonnage transported for one route. For example, if 20 tons are transported from elevator 1 to mill A, the cost of \$6 is multiplied by $x_{1A}(=20)$, which equals \$120.

The first three constraints in the linear programming model represent the supply at each elevator; the last three constraints represent the demand at each mill. As an example, consider the first supply constraint, $x_{1A} + x_{1B} + x_{1C} = 150$. This constraint represents the tons of wheat transported from Kansas City to all three mills: Chicago (x_{1A}), St. Louis (x_{1B}), and Cincinnati (x_{1C}). The amount transported from Kansas City is limited to the 150 tons available. Note that this constraint (as well as all others) is an equation (=) rather than a ... inequality because all the tons of wheat available will be needed to meet the total demand of 600 tons. In other words, the three mills demand 600 total tons, which is the exact amount that can be supplied by the three grain elevators. Thus, all that can be supplied will be, in order to meet demand. This type of model, in which supply exactly equals demand, is referred to as a balanced transportation model. The balanced model will be used to demonstrate the solution of a transportation problem.

Transportation problems are Transportation models are solved manually within the context of a tableau, as

solved manually within a tableau in the simplex method. The tableau for our wheat transportation model is shown in

format. Table B-1.

Table B-1

To

The Transportation Tableau

From A B C Supply

6 8 10

1 150

7 11 11

2 175

4 5 12

3 275

Demand 200 100 300 600

Each cell in a transportation tableau Each cell in the tableau represents the amount transported from one source to one des-

is analogous to a decision variable tination. Thus, the amount placed in each cell is the value of a decision variable for that cell.

that indicates the amount allocated For example, the cell at the intersection of row 1 and column A represents the decision vari-

from a source to a destination.

able x_{1A}. The smaller box within each cell contains the unit transportation cost for that

The supply and demand values route. For example, in cell 1A the value, \$6, is the cost of transporting one ton of wheat

along the outside rim of a tableau from Kansas City to Chicago. Along the outer rim of the tableau are the supply and demand

are called rim requirements. constraint quantity values, which are referred to as rim requirements.

The two methods for solving a transportation model are the stepping-stone method and

the modified distribution method (also known as MODI). In applying the simplex method,

Transportation models do not start

an initial solution had to be established in the initial simplex tableau. This same condition

at the origin where all decision must be met in solving a transportation model. In a transportation model, an initial feasi-

variables equal zero; they must be ble solution can be found by several alternative methods, including the northwest corner

given an initial feasible solution. method, the minimum cell cost method, and Vogel's approximation model.

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B-4 Module B Transportation and Assignment Solution Methods

The Northwest Corner Method

In the northwest corner method With the northwest corner method, an initial allocation is made to the cell in the upper

the largest possible allocation is left-hand corner of the tableau (i.e., the "northwest corner"). The amount allocated is the

made to the cell in the upper left- most possible, subject to the supply and demand constraints for that cell. In our example,

hand corner of the tableau,

we first allocate as much as possible to cell 1A (the northwest corner). This amount is 150

followed by allocations to adjacent feasible cells.

tons, since that is the maximum that can be supplied by grain elevator 1 at Kansas City, even though 200 tons are demanded by mill A at Chicago. This initial allocation is shown in Table B-2.

We next allocate to a cell adjacent to cell 1A, in this case either cell 2A or cell 1B.

However, cell 1B no longer represents a feasible allocation, because the total tonnage of wheat available at source 1 (i.e., 150 tons) has already been allocated. Thus, cell 2A represents the only feasible alternative, and as much as possible is allocated to this cell. The amount allocated at 2A can be either 175 tons, the supply available from source 2 (Omaha), or 50 tons, the amount now demanded at destination A. (Recall that 150 of the 200 tons demanded at A have already been supplied.) Because 50 tons is the most constrained amount, it is allocated to cell 2A, as shown in Table B-2.

Table B-2

To

The Initial NW Corner Solution

From A B C Supply

6 8 10

1 150 150

7 11 11

2 50 100 25 175

4 5 12

3 275 275

Demand 200 100 300 600

The third allocation is made in the same way as the second allocation. The only feasible cell adjacent to cell 2A is cell 2B. The most that can be allocated is either 100 tons (the amount demanded at mill B) or 125 tons (175 tons minus the 50 tons allocated to cell 2A). The smaller (most constrained) amount, 100 tons, is allocated to cell 2B, as shown in Table B-2.

The initial solution is complete The fourth allocation is 25 tons to cell 2C, and the fifth allocation is 275 tons to cell 3C,

when all rim requirements are both of which are shown in Table B-2. Notice that all of the row and column allocations

satisfied. add up to the appropriate rim requirements.

The transportation cost of this solution is computed by substituting the cell allocations (i.e., the amounts transported),

$$x_{1A} = 150$$

$$x_{2A} = 50$$

$$x_{2B} = 100$$

$$x_{2C} = 25$$

$$x_{3C} = 275$$

into the objective function:

$$\begin{aligned} Z &= \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C} \\ &= 6(150) + 8(0) + 10(0) + 7(50) + 11(100) + 11(25) + 4(0) + 5(0) + 12(275) \\ &= \$5,925 \end{aligned}$$

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Solution of the Transportation Model B-5

The steps of the northwest corner method are summarized here:

1. Allocate as much as possible to the cell in the upper left-hand corner, subject to the supply and demand constraints.
2. Allocate as much as possible to the next adjacent feasible cell.
3. Repeat step 2 until all rim requirements have been met.

The Minimum Cell Cost Method

In the minimum cell cost method With the minimum cell cost method, the basic logic is to allocate to the cells with the low-

as much as possible is allocated to est costs. The initial allocation is made to the cell in the tableau having the lowest cost. In

the cell with the minimum cost. the transportation tableau for our example problem, cell 3A has the minimum cost of \$4.

As much as possible is allocated to this cell; the choice is either 200 tons or 275 tons. Even though 275 tons could be supplied to cell 3A, the most we can allocate is 200 tons, since only 200 tons are demanded. This allocation is shown in Table B-3.

Table B-3

To

The Initial Minimum Cell From A B C Supply

Cost Allocation

6 8 10

1 150

7 11 11

2 175

4 5 12

3 200 275

Demand 200 100 300 600

Notice that all the remaining cells in column A have now been eliminated, because all the wheat demanded at destination A, Chicago, has now been supplied by source 3, Des Moines.

The next allocation is made to the cell that has the minimum cost and also is feasible.

This is cell 3B, which has a cost of \$5. The most that can be allocated is 75 tons (275 tons minus the 200 tons already supplied). This allocation is shown in Table B-4.

Table B-4

To

The Second Minimum Cell From A B C Supply

Cost Allocation

6 8 10

1 150

7 11 11

2 175

4 5 12

3 200 75 275

Demand 200 100 300 600

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B-6 Module B Transportation and Assignment Solution Methods

The third allocation is made to cell 1B, which has the minimum cost of \$8. (Notice that cells with lower costs, such as 1A and 2A, are not considered because they were previously ruled out as infeasible.) The amount allocated is 25 tons. The fourth allocation of 125 tons is made to cell 1C, and the last allocation of 175 tons is made to cell 2C. These allocations, which complete the initial minimum cell cost solution, are shown in Table B-5.

Table B-5

To

The Initial Solution From A B C Supply

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The minimum cell cost method The total cost of this initial solution is \$4,550, as compared with a total cost of \$5,925 for

will provide a solution with a the initial northwest corner solution. It is not a coincidence that a lower total cost is derived

lower cost than the northwest using the minimum cell cost method; it is a logical occurrence. The northwest corner

corner solution because it

considers cost in the

method does not consider cost at all in making allocations—the minimum cell cost

allocation process. method does. It is therefore quite natural that a lower initial cost will be attained using the

latter method. Thus, the initial solution achieved by using the minimum cell cost method is usually better in that, because it has a lower cost, it is closer to the optimal solution; fewer subsequent iterations will be required to achieve the optimal solution.

The specific steps of the minimum cell cost method are summarized next:

1. Allocate as much as possible to the feasible cell with the minimum transportation cost, and adjust the rim requirements.
2. Repeat step 1 until all rim requirements have been met.

Vogel's Approximation Model

The third method for determining an initial solution, Vogel's approximation model (also

A penalty cost is the difference called VAM), is based on the concept of penalty cost or regret. If a decision maker incor-

between the largest and rectly chooses from several alternative courses of action, a penalty may be suffered (and the

next largest cell cost in a row decision maker may regret the decision that was made). In a transportation problem, the

(or column).

courses of action are the alternative routes, and a wrong decision is allocating to a cell that does not contain the lowest cost.

VAM allocates as much as possible In the VAM method, the first step is to develop a penalty cost for each source and desti-

to the minimum cost cell in the nation. For example, consider column A in Table B-6. Destination A, Chicago, can be sup-

row or column with the largest plied by Kansas City, Omaha, and Des Moines. The best decision would be to supply

penalty cost.

Chicago from source 3 because cell 3A has the minimum cost of \$4. If a wrong decision was made and the next higher cost of \$6 was selected at cell 1A, a "penalty" of \$2 per ton would result (i.e., $\$6 - \$4 = \$2$). This demonstrates how the penalty cost is determined for each row and column of the tableau. The general rule for computing a penalty cost is to subtract the minimum cell cost from the next higher cell cost in each row and column. The penalty costs for our example are shown at the right and at the bottom of Table B-6.

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Solution of the Transportation Model B-7

Table B-6

To

The VAM Penalty Costs From A B C Supply

6 8 10

2

1 150

7 11 11

4

2 175

4 5 12

1

3 275

Demand 200 100 300 600

2 3 1

The initial allocation in the VAM method is made in the row or column that has the highest penalty cost. In Table B-6, row 2 has the highest penalty cost of \$4. We allocate as much as possible to the feasible cell in this row with the minimum cost. In row 2, cell 2A has the lowest cost of \$7, and the most that can be allocated to cell 2A is 175 tons. With this allocation the greatest penalty cost of \$4 has been avoided because the best course of action has been selected. The allocation is shown in Table B-7.

Table B-7

To

The Initial VAM Allocation From A B C Supply

6 8 10

2

1 150

7 11 11

2 175 175

4 5 12

1

3 275

Demand 200 100 300 600

2 3 2

After each VAM cell allocation, After the initial allocation is made, all the penalty costs must be recomputed. In some

all row and column penalty costs cases the penalty costs will change; in other cases they will not change. For example, the

are recomputed. penalty cost for column C in Table B-7 changed from \$1 to \$2 (because cell 2C is no longer

considered in computing penalty cost), and the penalty cost in row 2 was eliminated altogether (because no more allocations are possible for that row).

Next, we repeat the previous step and allocate to the row or column with the highest penalty cost, which is now column B with a penalty cost of \$3 (see Table B-7). The cell in column B with the lowest cost is 3B, and we allocate as much as possible to this cell, 100 tons. This allocation is shown in Table B-8.

Note that all penalty costs have been recomputed in Table B-8. Since the highest penalty cost is now \$8 for row 3 and since cell 3A has the minimum cost of \$4, we allocate 25 tons to this cell, as shown in Table B-9.

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B-8 Module B Transportation and Assignment Solution Methods

Table B-8

To

The Second VAM Allocation From A B C Supply

6 8 10

4

1 150

7 11 11

2 175 175

4 5 12

8

3 100 275

Demand 200 100 300 600

2 2

Table B-9

To

The Third VAM Allocation From A B C Supply

6 8 10

1 150

7 11 11

2 175 175

4 5 12

3 25 100 275

Demand 200 100 300 600

2

Table B-9 also shows the recomputed penalty costs after the third allocation. Notice that by now only column C has a penalty cost. Rows 1 and 3 have only one feasible cell, so a penalty does not exist for these rows. Thus, the last two allocations are made to column C. First, 150 tons are allocated to cell 1C because it has the lowest cell cost. This leaves only cell 3C as a feasible possibility, so 150 tons are allocated to this cell. Both of these allocations are shown in Table B-10.

Table B-10

To

The Initial VAM Solution From A B C Supply

6 8 10

1 150 150

7 11 11

2 175 175

4 5 12

3 25 100 150 275

Demand 200 100 300 600

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Solution of the Transportation Model B-9

VAM and minimum cell cost both The total cost of this initial Vogel's approximation model solution is \$5,125, which is

provide better initial solutions not as high as the northwest corner initial solution of \$5,925. It is also not as low as the

than the northwest corner method. minimum cell cost solution of \$4,550. Like the minimum cell cost method, VAM typi-

cally results in a lower cost for the initial solution than does the northwest corner method.

The steps of Vogel's approximation model can be summarized in the following list:

1. Determine the penalty cost for each row and column by subtracting the lowest cell cost in the row or column from the next lowest cell cost in the same row or column.
2. Select the row or column with the highest penalty cost (breaking ties arbitrarily or choosing the lowest-cost cell).
3. Allocate as much as possible to the feasible cell with the lowest transportation cost in the row or column with the highest penalty cost.
4. Repeat steps 1, 2, and 3 until all rim requirements have been met.

The Stepping-Stone Solution Method

Once an initial basic feasible solution has been determined by any of the previous three methods, the next step is to solve the model for the optimal (i.e., minimum total cost) solu-

Once an initial solution is derived, tion. There are two basic solution methods: the stepping-stone solution method and the

the problem must be solved using modified distribution method (MODI). The stepping-stone solution method will be

either the stepping-stone method demonstrated first. Because the initial solution obtained by the minimum cell cost method

or the modified distribution

had the lowest total cost of the three initial solutions, we will use it as the starting solution. method (MODI).

Table B-11 repeats the initial solution that was developed from the minimum cell cost method.

Table B-11

To

The Minimum Cell From A B C Supply

Cost Solution

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The stepping-stone method The basic solution principle in a transportation problem is to determine whether a

determines whether there is a cell transportation route not at present being used (i.e., an empty cell) would result in a

with no allocation that would lower total cost if it were used. For example, Table B-11 shows four empty cells (1A, 2A,

reduce cost if used.

2B, and 3C) representing unused routes. Our first step in the stepping-stone method is to evaluate these empty cells to see whether the use of any of them would reduce total cost.

If we find such a route, then we will allocate as much as possible to it.

First, let us consider allocating one ton of wheat to cell 1A. If one ton is allocated to cell 1A, cost will be increased by \$6—the transportation cost for cell 1A. However, by allocating one ton to cell 1A, we increase the supply in row 1 to 151 tons, as shown in Table B-12.

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B-10 Module B Transportation and Assignment Solution Methods

Table B-12

To

The Allocation of One Ton to From A B C Supply

Cell 1A

+1 6 8 10

1 25 125 150 151

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

The constraints of the problem cannot be violated, and feasibility must be maintained.

If we add one ton to cell 1A, we must subtract one ton from another allocation along that row. Cell 1B is a logical candidate because it contains 25 tons. By subtracting one ton from cell 1B, we now have 150 tons in row 1, and we have satisfied the supply constraint again. At the same time, subtracting one ton from cell 1B has reduced total cost by \$8.

However, by subtracting one ton from cell 1B, we now have only 99 tons allocated to column B, where 100 tons are demanded, as shown in Table B-13. To compensate for this con-

straint violation, one ton must be added to a cell that already has an allocation. Since cell 3B has 75 tons, we will add one ton to this cell, which again satisfies the demand constraint of 100 tons.

Table B-13

To

The Subtraction of One Ton From A B C Supply

from Cell 1B

+1 6 -1 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

99

A requirement of this solution method is that units can be added to and subtracted from only those cells that already have allocations. That is why one ton was added to cell 3B and not to cell 2B. It is from this requirement that the method derives its name. The process of adding and subtracting units from allocated cells is analogous to crossing a pond by stepping on stones (i.e., only allocated-to cells).

By allocating one extra ton to cell 3B we have increased cost by \$5, the transportation cost for that cell. However, we have also increased the supply in row 3 to 276 tons, a violation of the supply constraint for this source. As before, this violation can be remedied by subtracting one ton from cell 3A, which contains an allocation of 200 tons. This satisfies the supply constraint again for row 3, and it also reduces the total cost by \$4, the transportation cost for cell 3A. These allocations and deletions are shown in Table B-14.

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Solution of the Transportation Model B-11

Table B-14

To

The Addition of One Ton to Cell From A B C Supply

3B and the Subtraction of One

Ton from Cell 3A +1 6 -1 8 10

1 25 125 150

7 11 11

2 175 175

-1 4 +1 5 12

3 200 75 275

Demand 200 100 300 600

Notice in Table B-14 that by subtracting one ton from cell 3A, we did not violate the demand constraint for column A, since we previously added one ton to cell 1A.

Now let us review the increases and reductions in costs resulting from this process. We initially increased cost by \$6 at cell 1A, then reduced cost by \$8 at cell 1B, then increased cost by \$5 at cell 3B, and, finally, reduced cost by \$4 at cell 3A:

$$1A : 1B : 3B : 3A \\ + \$6 - 8 + 5 - 4 = - \$1$$

An empty cell that will reduce cost In other words, for each ton allocated to cell 1A (a route not at present used), total cost

is a potential entering variable. will be reduced by \$1. This indicates that the initial solution is not optimal because a lower

cost can be achieved by allocating additional tons of wheat to cell 1A (i.e., cell 1A is analogous to a pivot column in the simplex method). Our goal is to determine the cell or entering “variable” that will reduce cost the most. Another variable (empty cell) may result in an even greater decrease in cost than cell 1A. If such a cell exists, it will be selected as the entering variable; if not, cell 1A will be selected. To identify the appropriate entering variable, the remaining empty cells must be tested as cell 1A was.

To evaluate the cost reduction Before testing the remaining empty cells, let us identify a few of the general characteris-

potential of an empty cell, a closed tics of the stepping-stone process. First, we always start with an empty cell and form a closed

path connecting used cells to the path of cells that now have allocations. In developing the path, it is possible to skip over

empty cell is identified.

both unused and used cells. In any row or column there can be only one addition and one subtraction. (For example, in row 1, wheat is added at cell 1A and is subtracted at cell 1B.)

Let us test cell 2A to see if it results in a cost reduction. The stepping-stone closed path for cell 2A is shown in Table B-15. Notice that the path for cell 2A is slightly more complex

Table B-15

To

The Stepping-Stone Path for From A B C Supply

Cell 2A

$$6 - 8 + 10$$

$$1 \ 25 \ 125 \ 150$$

$$+ 7 \ 11 - 11$$

$$2 \ 175 \ 175$$

$$- 4 + 5 \ 12$$

$$3 \ 200 \ 75 \ 275$$

$$\text{Demand } 200 \ 100 \ 300 \ 600$$

$$2A : 2C : 1C : 1B : 3B : 3A$$

$$+ \$7 - 11 + 10 - 8 + 5 - 4 = - \$1$$

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B-12 Module B Transportation and Assignment Solution Methods

than the path for cell 1A. Notice also that the path crosses itself at one point, which is perfectly acceptable. An allocation to cell 2A will reduce cost by \$1, as shown in the computation in Table B-15. Thus, we have located another possible entering variable, although it is no better than cell 1A.

The remaining stepping-stone paths and the resulting computations for cells 2B and 3C are shown in Tables B-16 and B-17, respectively.

Table B-16

To

The Stepping-Stone Path for From A B C Supply

Cell 2B

6 + 8 - 10

1 25 125 150

7 - 11 + 11

2 175 175

4 5 12

3 200 75 275

Demand 200 100 300 600

2B : 2C : 1C : 1B

+ \$11 - 11 + 10 - 8 = + \$2

Table B-17

To

The Stepping-Stone Path for From A B C Supply

Cell 3C

6 + 8 - 10

1 25 125 150

7 11 11

2 175 175

4 - 5 + 12

3 200 75 275

Demand 200 100 300 600

3C : 1C : 1B : 3B

+ \$12 - 10 + 8 - 5 = + \$5

After all empty cells are evaluated, Notice that after all four unused routes are evaluated, there is a tie for the entering vari-

the one with the greatest able between cells 1A and 2A. Both show a reduction in cost of \$1 per ton allocated to that

cost reduction potential is the route. The tie can be broken arbitrarily. We will select cell 1A (i.e., x1A) to enter the solution.

entering variable.

Because the total cost of the model will be reduced by \$1 for each ton we can reallocate to cell 1A, we naturally want to reallocate as much as possible. To determine how much to allocate, we need to look at the path for cell 1A again, as shown in Table B-18.

When reallocating units to the The stepping-stone path in Table B-18 shows that tons of wheat must be subtracted at

entering variable (cell), the cells 1B and 3A to meet the rim requirements and thus satisfy the model constraints.

amount is the minimum Because we cannot subtract more than is available in a cell, we are limited by the 25 tons in

amount subtracted on the

cell 1B. In other words, if we allocate more than 25 tons to cell 1A, then we must subtract stepping-stone path.

more than 25 tons from 1B, which is impossible because only 25 tons are available.

Therefore, 25 tons is the amount we reallocate to cell 1A according to our path. That is, 25 tons are added to 1A, subtracted from 1B, added to 3B, and subtracted from 3A. This reallocation is shown in Table B-19.

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Solution of the Transportation Model B-13

Table B-18

To

The Stepping-Stone Path for From A B C Supply

Cell 1A

+ 6 - 8 10

1 25 125 150

7 11 11

2 175 175

- 4 + 5 12

3 200 75 275

Demand 200 100 300 600

Table B-19

To

The Second Iteration of the From A B C Supply

Stepping-Stone Method

6 8 10

1 25 125 150

7 11 11

2 175 175

4 5 12

3 175 100 275

Demand 200 100 300 600

The process culminating in Table B-19 represents one iteration of the stepping-stone method. We selected x_{1A} as the entering variable, and it turned out that x_{1B} was the leaving variable (because it now has a value of zero in Table B-19). Thus, at each iteration one variable enters and one leaves (just as in the simplex method).

Now we must check to see whether the solution shown in Table B-19 is, in fact, optimal.

We do this by plotting the paths for the unused routes (i.e., empty cells 2A, 1B, 2B, and 3C) that are shown in Table B-19. These paths are shown in Tables B-20 through B-23.

Table B-20

To

The Stepping-Stone Path for From A B C Supply

Cell 2A

- 6 8 + 10

1 25 125 150

+ 7 11 - 11

2 175 175

4 5 12

3 175 100 275

Demand 200 100 300 600

2A : 2C : 1C : 1A

+ \$7 - 11 + 10 - 6 = \$0

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Table B-21

To

The Stepping-Stone Path for From A B C Supply

Cell 1B

- 6 + 8 10

1 25 125 150

7 11 11

2 175 175

+ 4 - 5 12

3 175 100 275

Demand 200 100 300 600

1B : 3B : 3A : 1A

+ \$8 - 5 + 4 - 6 = + \$1

Table B-22

To

The Stepping-Stone Path for From A B C Supply

Cell 2B

- 6 8 + 10

1 25 125 150

7 + 11 - 11

2 175 175

+ 4 - 5 12

3 175 100 275

Demand 200 100 300 600

2B : 3B : 3A : 1A : 1C : 2C

+ \$11 - 5 + 4 - 6 + 10 - 11 = + \$3

Table B-23

To

The Stepping-Stone Path for

From A B C Supply

Cell 3C

+ 6 8 - 10

1 25 125 150

7 11 11

2 175 175

- 4 5 + 12

3 175 100 275

Demand 200 100 300 600

3C : 3A : 1A : 1C

+ \$12 - 4 + 6 - 10 = + \$4

The stepping-stone process is Our evaluation of the four paths indicates no cost reductions; therefore, the solution

repeated until none of the shown in Table B-19 is optimal. The solution and total minimum cost are empty cells will reduce cost

(i.e., an optimal solution). x 1A = 25 tons x 2C = 175 tons x 3A = 175 tons

x 1C = 125 tons x 3B = 100 tons

$Z = \$6(25) + 8(0) + 10(125) + 7(0) + 11(0) + 11(175) + 4(175) + 5(100) + 12(0)$

= \$4,525

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Solution of the Transportation Model B-15

Multiple optimal solutions However, notice in Table B-20 that the path for cell 2A resulted in a cost change of \$0. In

occur when an empty cell has a other words, allocating to this cell would neither increase nor decrease total cost. This situ-

cost change of zero and all other ation indicates that the problem has multiple optimal solutions. Thus, x2A could be

empty cells are positive.

entered into the solution and there would not be a change in the total minimum cost of

An alternative optimal solution is

\$4,525. To identify the alternative solution, we would allocate as much as possible to cell 2A,

determined by allocating to the

empty cell a zero cost change. which in this case is 25 tons of wheat. The alternative solution is shown in Table B-24.

Table B-24 To

The Alternative Optimal From A B C Supply

Solution

6 8 10

1 150 150

7 11 11

2 25 150 175

4 5 12

3 175 100 275

Demand 200 100 300 600

The solution in Table B-24 also results in a total minimum cost of \$4,525. The steps of the stepping-stone method are summarized here:

1. Determine the stepping-stone paths and cost changes for each empty cell in the tableau.
2. Allocate as much as possible to the empty cell with the greatest net decrease in cost.
3. Repeat steps 1 and 2 until all empty cells have positive cost changes that indicate an optimal solution.

The Modified Distribution Method

The modified distribution method (MODI) is basically a modified version of the stepping-stone method. However, in the MODI method the individual cell cost changes are deter-

MODI is a modified version of

mined mathematically, without identifying all the stepping-stone paths for the empty cells.

the stepping-stone method in To demonstrate MODI, we will again use the initial solution obtained by the minimum

which math equations replace the cell cost method. The tableau for the initial solution with the modifications required by

stepping-stone paths. MODI is shown in Table B-25.

Table B-25 $v_j - v_A = v_B = v_C =$

The Minimum Cell Cost

To

Initial Solution

u_i From A B C Supply

6 8 10

$$u_1 = 1 \ 25 \ 125 \ 150$$

$$7 \ 11 \ 11$$

$$u_2 = 2 \ 175 \ 175$$

$$4 \ 5 \ 12$$

$$u_3 = 3 \ 200 \ 75 \ 275$$

$$\text{Demand } 200 \ 100 \ 300 \ 600$$

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B-16 Module B Transportation and Assignment Solution Methods

The extra left-hand column with the u_i symbols and the extra top row with the v_j symbols represent column and row values that must be computed in MODI. These values are computed for all cells with allocations by using the following formula:

$$u_i + v_j = c_{ij}$$

The value c_{ij} is the unit transportation cost for cell ij . For example, the formula for cell 1B is

$$u_1 + v_B = c_{1B}$$

and, since $c_{1B} = 8$,

$$u_1 + v_B = 8$$

The formulas for the remaining cells that presently contain allocations are

$$x_{1C}: u_1 + v_C = 10$$

$$x_{2C}: u_2 + v_C = 11$$

$$x_{3A}: u_3 + v_A = 4$$

$$x_{3B}: u_3 + v_B = 5$$

Now there are five equations with six unknowns. To solve these equations, it is necessary to assign only one of the unknowns a value of zero. Thus, if we let $u_1 = 0$, we can solve for all remaining u_i and v_j values.

$$x_{1B}: u_1 + v_B = 8$$

$$0 + v_B = 8$$

$$v_B = 8$$

$$x_{1C}: u_1 + v_C = 10$$

$$0 + v_C = 10$$

$$v_C = 10$$

$$x_{2C}: u_2 + v_C = 11$$

$$u_2 + 10 = 11$$

$$u_2 = 1$$

$$x_{3B}: u_3 + v_B = 5$$

$$u_3 + 8 = 5$$

$$u_3 = -3$$

$$x_{3A}: u_3 + v_A = 4$$

$$-3 + v_A = 4$$

$$v_A = 7$$

Notice that the equation for cell 3B had to be solved before the cell 3A equation could be

solved. Now all the u_i and v_j values can be substituted into the tableau, as shown in Table B-26.

Table B-26 $v_A = 7$ $v_B = 8$ $v_C = 10$

The Initial Solution with All u_i

To

and v_j Values

u_i From A B C Supply

6 8 10

$u_1 = 0$ 1 25 125 150

7 11 11

$u_2 = 1$ 2 175 175

4 5 12

$u_3 = -3$ 3 200 75 275

Demand 200 100 300 600

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Solution of the Transportation Model B-17

Next, we use the following formula to evaluate all empty cells:

$$c_{ij} - u_i - v_j = k_{ij}$$

where k_{ij} equals the cost increase or decrease that would occur by allocating to a cell.

For the empty cells in Table B-26, the formula yields the following values:

$$x_{1A}: k_{1A} = c_{1A} - u_1 - v_A = 6 - 0 - 7 = -1$$

$$x_{2A}: k_{2A} = c_{2A} - u_2 - v_A = 7 - 1 - 7 = -1$$

$$x_{2B}: k_{2B} = c_{2B} - u_2 - v_B = 11 - 1 - 8 = +2$$

$$x_{3C}: k_{3C} = c_{3C} - u_3 - v_C = 12 - (-3) - 10 = +5$$

Each MODI allocation replicates These calculations indicate that either cell 1A or cell 2A will decrease cost by \$1 per allo-

the stepping-stone allocation. cated ton. Notice that those are exactly the same cost changes for all four empty cells as

were computed in the stepping-stone method. That is, the same information is obtained by evaluating the paths in the stepping-stone method and by using the mathematical formulas of the MODI.

After each allocation to an empty We can select either cell 1A or 2A to allocate to because they are tied at -1. If cell 1A is

cell, the u_i and v_j values must be selected as the entering nonbasic variable, then the stepping-stone path for that cell must

recomputed. be determined so that we know how much to reallocate. This is the same path previously

identified in Table B-18. Reallocating along this path results in the tableau shown in

Table B-27 (and previously shown in Table B-19).

Table B-27 $v_A = v_B = v_C =$

The Second Iteration of the

To

MODI Solution Method

ui From A B C Supply

6 8 10

$u_1 = 1 \ 25 \ 125 \ 150$

7 11 11

$u_2 = 2 \ 175 \ 175$

4 5 12

$u_3 = 3 \ 175 \ 100 \ 275$

Demand 200 100 300 600

The u_i and v_j values for Table B-27 must now be recomputed using our formula for the allocated-to cells:

x1A: $u_1 + v_A = 6$

$0 + v_A = 6$

$v_A = 6$

x1C: $u_1 + v_C = 10$

$0 + v_C = 10$

$v_C = 10$

x2C: $u_2 + v_C = 11$

$u_2 + 10 = 11$

$u_2 = 1$

x3A: $u_3 + v_A = 4$

$u_3 + 6 = 4$

$u_3 = -2$

x3B: $u_3 + v_B = 5$

$-2 + v_B = 5$

$v_B = 7$

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B-18 Module B Transportation and Assignment Solution Methods

These new u_i and v_j values are shown in Table B-28.

Table B-28 $v_A = 6 \ v_B = 7 \ v_C = 10$

The New u_i and v_j Values for

To

the Second Iteration

ui From A B C Supply

6 8 10

$u_1 = 0 \ 1 \ 25 \ 125 \ 150$

7 11 11

$u_2 = 1 \ 2 \ 175 \ 175$

4 5 12

$$u_3 = -2 \ 3 \ 175 \ 100 \ 275$$

Demand 200 100 300 600

The cost changes for the empty cells are now computed using the formula

$$c_{ij} - u_i - v_j = k_{ij}:$$

$$x_{1B}: k_{1B} = c_{1B} - u_1 - v_B = 8 - 0 - 7 = +1$$

$$x_{2A}: k_{2A} = c_{2A} - u_2 - v_A = 7 - 1 - 6 = 0$$

$$x_{2B}: k_{2B} = c_{2B} - u_2 - v_B = 11 - 1 - 7 = +3$$

$$x_{3C}: k_{3C} = c_{3C} - u_3 - v_C = 12 - (-2) - 10 = +4$$

Because none of these values is negative, the solution shown in Table B-28 is optimal.

However, as in the stepping-stone method, cell 2A with a zero cost change indicates multiple optimal solutions.

The steps of the modified distribution method can be summarized as follows:

1. Develop an initial solution using one of the three methods available.
2. Compute u_i and v_j values for each row and column by applying the formula $u_i + v_j = c_{ij}$ to each cell that has an allocation.
3. Compute the cost change, k_{ij} , for each empty cell using $c_{ij} - u_i - v_j = k_{ij}$.
4. Allocate as much as possible to the empty cell that will result in the greatest net decrease in cost (most negative k_{ij}). Allocate according to the stepping-stone path for the selected cell.
5. Repeat steps 2 through 4 until all k_{ij} values are positive or zero.

The Unbalanced Transportation Model

Thus far, the methods for determining an initial solution and an optimal solution have been demonstrated within the context of a balanced transportation model. Realistically, however, an unbalanced problem is a more likely occurrence. Consider our example of transporting wheat. By changing the demand at Cincinnati to 350 tons, we create a situation in which total demand is 650 tons and total supply is 600 tons.

When demand exceeds supply, To compensate for this difference in the transportation tableau, a “dummy” row is added

a dummy row is added to the to the tableau, as shown in Table B-29. The dummy row is assigned a supply of 50 tons

tableau. to balance the model. The additional 50 tons demanded, which cannot be supplied, will be allocated to a cell in the dummy row. The transportation costs for the cells in the dummy row are zero because the tons allocated to these cells are not amounts really transported but the amounts by which demand was not met. These dummy cells are, in effect, slack variables.

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Solution of the Transportation Model B-19

Table B-29

To

An Unbalanced Model From A B C Supply

(Demand 7 Supply)

6 8 10

1 150

7 11 11

2 175

4 5 12

3 275

0 0 0

Dummy 50

Demand 200 100 350 650

When a supply exceeds demand, Now consider our example with the supply at Des Moines increased to 375 tons. This

a dummy column is added to the increases total supply to 700 tons, while total demand remains at 600 tons. To compensate for

tableau. this imbalance, we add a dummy column instead of a dummy row, as shown in Table B-30.

Table B-30

To

An Unbalanced Model From A B C Dummy Supply

(Supply 7 Demand)

6 8 10 0

1 150

7 11 11 0

2 175

4 5 12 0

3 375

Demand 200 100 300 100 700

The addition of a dummy row or a dummy column has no effect on the initial solution methods or on the methods for determining an optimal solution. The dummy row or column cells are treated the same as any other tableau cell. For example, in the minimum cell cost method, three cells would be tied for the minimum cost cell, each with a cost of zero. In this case (or any time there is a tie between cells) the tie would be broken arbitrarily.

Degeneracy

In all the tableaus showing a solution to the wheat transportation problem, the following condition was met:

$m \text{ rows} + n \text{ columns} - 1 = \text{the number of cells with allocations}$

For example, in any of the balanced tableaus for wheat transportation, the number of rows was three (i.e., $m = 3$) and the number of columns was three (i.e., $n = 3$); thus,

In a transportation tableau

$3 + 3 - 1 = 5$ cells with allocations.

with m rows and n columns, there These tableaus always had five cells with allocations; thus, our condition for normal

must be $m + n - 1$ cells with solution was met. When this condition is not met and fewer than $m + n - 1$ cells have

allocations; if not, it is degenerate. allocations, the tableau is said to be degenerate.

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B-20 Module B Transportation and Assignment Solution Methods

Consider the wheat transportation example with the supply values changed to the amounts shown in Table B-31. The initial solution shown in this tableau was developed using the minimum cell cost method.

Table B-31

To

The Minimum Cell Cost Initial

From A B C Supply

Solution

6 8 10

1 100 50 150

7 11 11

2 250 250

4 5 12

3 200 200

Demand 200 100 300 600

The tableau shown in Table B-31 does not meet the condition

$m + n - 1 =$ the number of cells with allocations

$3 + 3 - 1 = 5$ cells

In a degenerate tableau, not all of because there are only four cells with allocations. The difficulty resulting from a degenerate

the stepping-stone paths or MODI solution is that neither the stepping-stone method nor MODI will work unless the preced-

equations can be developed. ing condition is met (there is an appropriate number of cells with allocations). When the

tableau is degenerate, a closed path cannot be completed for all cells in the stepping-stone method, and not all the $u_i + v_j = c_{ij}$ computations can be completed in MODI. For example, a closed path cannot be determined for cell 1A in Table B-31.

To rectify a degenerate tableau, an To create a closed path, one of the empty cells must be artificially designated as a cell

empty cell must artificially be with an allocation. Cell 1A in Table B-32 is designated arbitrarily as a cell with artificial

treated as an occupied cell. allocation of zero. (However, any symbol, such as $\square$, could be used to signify the artificial

allocation.) This indicates that this cell will be treated as a cell with an allocation in determining stepping-stone paths or MODI formulas, although there is no real allocation in this cell. Notice that the location of 0 was arbitrary because there is no general rule for allocat-

ing the artificial cell. Allocating zero to a cell does not guarantee that all the stepping-stone paths can be determined.

Table B-32

To

The Initial Solution From A B C Supply

6 8 10

1 0 100 50 150

7 11 11

2 250 250

4 5 12

3 200 200

Demand 200 100 300 600

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Solution of the Transportation Model B-21

For example, if zero had been allocated to cell 2B instead of to cell 1A, none of the stepping-stone paths could have been determined, even though technically the tableau would no longer be degenerate. In such a case, the zero must be reallocated to another cell and all paths determined again. This process must be repeated until an artificial allocation has been made that will enable the determination of all paths. In most cases, however, there is more than one possible cell to which such an allocation can be made.

The stepping-stone paths and cost changes for this tableau follow:

2A 2C 1C 1A

x2A: $7 - 11 + 10 - 6 = 0$

2B 2C 1C 1B

x2B: $11 - 11 + 10 - 8 = +2$

3B 1B 1A 3A

x3B: $5 - 8 + 6 - 4 = -1$

3C 1C 1A 3A

x3C: $12 - 10 + 6 - 4 = +4$

Because cell 3B shows a \$1 decrease in cost for every ton of wheat allocated to it, we will allocate 100 tons to cell 3B. This results in the tableau shown in Table B-33.

Table B-33

To

The Second Stepping-Stone From A B C Supply

Iteration

6 8 10

1 100 50 150

7 11 11

2 250 250

4 5 12

3 100 100 200

Demand 200 100 300 600

A normal problem can become Notice that the solution in Table B-33 now meets the condition $m + n - 1 = 5$. Thus,

degenerate at any iteration and in applying the stepping-stone method (or MODI) to this tableau, it is not necessary to

vice versa. make an artificial allocation to an empty cell. It is quite possible to begin the solution process

with a normal tableau and have it become degenerate or begin with a degenerate tableau and

have it become normal. If it had been indicated that the cell with the zero should have units

subtracted from it, no actual units could have been subtracted. In that case the zero would

have been moved to the cell that represents the entering variable. (The solution shown in

Table B-33 is optimal; however, multiple optimal solutions exist at cell 2A.)

Prohibited Routes

Sometimes one or more of the routes in the transportation model are prohibited. That is,

units cannot be transported from a particular source to a particular destination. When this

situation occurs, we must make sure that no units in the optimal solution are allocated to

the cell representing this route. In our study of the simplex tableau, we learned that assign-

ing a large relative cost or a coefficient of M to a variable would keep it out of the final solu-

A prohibited route is assigned tion. This same principle can be used in a transportation model for a prohibited route. A

a large cost such as M so that its value of M is assigned as the transportation cost for a cell that represents a prohibited

will never receive an allocation. route. Thus, when the prohibited cell is evaluated, it will always contain a large positive cost

change of M , which will keep it from being selected as an entering variable.

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B-22 Module B Transportation and Assignment Solution Methods

Solution of the Assignment Model

An assignment model is a special The assignment model is a special form of a linear programming model that is similar to

form of the transportation model the transportation model. There are differences, however. In the assignment model, the

in which all supply and supply at each source and the demand at each destination are limited to one unit each.

demand values equal one.

The following example from the text will be used to demonstrate the assignment model

and its special solution method. The Atlantic Coast Conference has four basketball games

on a particular night. The conference office wants to assign four teams of officials to the

four games in a way that will minimize the total distance traveled by the officials. The dis-

tances in miles for each team of officials to each game location are shown in Table B-34.

Table B-34

Game Sites

The Travel Distances to Each

Game for Each Team of Officials RALEIGH ATLANTA DURHAM CLEMSON

Officials

A 210 90 180 160

B 100 70 130 200

C 175 105 140 170

D 80 65 105 120

The supply is always one team of officials, and the demand is for only one team of officials at each game. Table B-34 is already in the proper form for the assignment.

An opportunity cost table is The first step in the assignment method of solution is to develop an opportunity cost

developed by first subtracting table. We accomplish this by first subtracting the minimum value in each row from every

the minimum value in each value in the row. These computations are referred to as row reductions. We applied a similar

row from all other row values and

principle in the VAM method when we determined penalty costs. In other words, the best

then repeating this process for

each column. course of action is determined for each row, and the penalty or “lost opportunity” is developed for all other row values. The row reductions for this example are shown in Table B-35.

Table B-35

Game Sites

The Assignment Tableau with

Row Reductions Officials RALEIGH ATLANTA DURHAM CLEMSON

A 120 0 90 70

B 30 0 60 130

C 70 0 35 65

D 15 0 40 55

Next, the minimum value in each column is subtracted from all column values. These computations are called column reductions and are shown in Table B-36, which represents the completed opportunity cost table for our example. Assignments can be made in this table wherever a zero is present. For example, team A can be assigned to Atlanta. An optimal solution results when each of the four teams can be uniquely assigned to a different game.

Table B-36

Game Sites

The Tableau with

Column Reductions Officials RALEIGH ATLANTA DURHAM CLEMSON

A 105 0 55 15

B 15 0 25 75

C 55 0 0 10

D 0 0 5 0

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Solution of the Assignment Model B-23

Assignments are made to locations Notice in Table B-36 that the assignment of team A to Atlanta means that no other team

with zeros in the opportunity cost can be assigned to that game. Once this assignment is made, the zero in row B is infeasible,

table. which indicates that there is not a unique optimal assignment for team B. Therefore,

Table B-36 does not contain an optimal solution.

An optimal solution occurs when A test to determine whether four unique assignments exist in Table B-36 is to draw the

the number of independent unique minimum number of horizontal or vertical lines necessary to cross out all zeros through

assignments equals the number of the rows and columns of the table. For example, Table B-37 shows that three lines are

rows or columns.

required to cross out all zeros.

Table B-37

Game Sites

The Opportunity Cost Table

with the Line Test Officials RALEIGH ATLANTA DURHAM CLEMSON

A 105 0 55 15

B 15 0 25 75

C 55 0 0 10

D 0 0 5 0

If the number of unique The three lines indicate that there are only three unique assignments, whereas four are

assignments is less than the required for an optimal solution. (Note that even if the three lines could have been drawn

number of rows (or columns), differently, the subsequent solution method would not be affected.)

Next, subtract the min-

a line test must be used.

imum value that is not crossed out from all values not crossed out. Then, add this mini-

imum value to those cells where two lines intersect. The minimum value not crossed out in

Table B-37 is 15. The second iteration for this model with the appropriate changes is shown

in Table B-38.

Table B-38

Game Sites

The Second Iteration

Officials RALEIGH ATLANTA DURHAM CLEMSON

A 90 0 40 0

B 0 0 10 60

C 55 15 0 10

D 0 15 5 0

In a line test, all zeros are crossed No matter how the lines are drawn in Table B-38, at least four are required to cross out

out by horizontal and vertical all the zeros. This indicates that four unique assignments can be made and that an optimal

lines; the minimum uncrossed solution has been reached. Now let us make the assignments from Table B-38.

value is subtracted from all

First, team A can be assigned to either the Atlanta game or the Clemson game. We will uncrossed values and added to

cell values where two lines cross.

assign team A to Atlanta first. This means that team A cannot be assigned to any other game, and no other team can be assigned to Atlanta. Therefore, row A and the Atlanta column can be eliminated. Next, team B is assigned to Raleigh. (Team B cannot be assigned to Atlanta, which has already been eliminated.) The third assignment is of team C to the Durham game. This leaves team D for the Clemson game. These assignments and their respective distances (from Table B-34) are summarized as follows:

Assignment Distance

Team A : Atlanta 90

Team B : Raleigh 100

Team C : Durham 140

Team D : Clemson _120

450 miles

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B-24 Module B Transportation and Assignment Solution Methods

Now let us go back and make the initial assignment of team A to Clemson (the alternative assignment we did not initially make). This will result in the following set of assignments:

Assignment Distance

Team A : Clemson 160

Team B : Atlanta 70

Team C : Durham 140

Team D : Raleigh __80

450 miles

These two assignments represent multiple optimal solutions for our example problem. Both assignments will result in the officials traveling a minimum total distance of 450 miles.

When supply exceeds demands, a Like a transportation problem, an assignment model can be unbalanced when supply

dummy column is added to the exceeds demand or demand exceeds supply. For example, assume that, instead of four

assignment tableau. teams of officials, there are five teams to be assigned to the four games. In this case

a dummy column is added to the assignment tableau to balance the model, as shown in Table B-39.

Table B-39 Game Sites

An Unbalanced Assignment

Tableau with a Dummy Column Officials RALEIGH ATLANTA DURHAM CLEMSON DUMMY

A 210 90 180 160 0

B 100 70 130 200 0

C 175 105 140 170 0

D 80 65 105 120 0

E 95 115 120 100 0

When demand exceeds supply, a In solving this model, one team of officials would be assigned to the dummy column. If

dummy row is added to the there were five games and only four teams of officials, a dummy row would be added

assignment tableau. instead of a dummy column. The addition of a dummy row or column does not affect the

solution method.

A prohibited assignment is given Prohibited assignments are also possible in an assignment problem, just as prohibited

a large relative cost of M so that it routes can occur in a transportation model. In the transportation model, an M value was

will never be selected. assigned as a large cost for the cell representing the prohibited route. This same method is

used for a prohibited assignment. A value of M is placed in the cell that represents the prohibited assignment.

The steps of the assignment solution method are summarized here:

1. Perform row reductions by subtracting the minimum value in each row from all row values.
2. Perform column reductions by subtracting the minimum value in each column from all column values.
3. In the completed opportunity cost table, cross out all zeros, using the minimum number of horizontal or vertical lines.
4. If fewer than m lines are required (where m = the number of rows or columns), subtract the minimum uncrossed value from all uncrossed values, and add this same minimum value to all cells where two lines intersect. Leave all other values unchanged, and repeat step 3.
5. If m lines are required, the tableau contains the optimal solution and m unique

assignments can be made. If fewer than m lines are required, repeat step 4.

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Problems B-25

Problems

1. Green Valley Mills produces carpet at plants in St. Louis and Richmond. The carpet is then shipped

to two outlets located in Chicago and Atlanta. The cost per ton of shipping carpet from each of the two plants to the two warehouses is as follows:

To

From Chicago Atlanta

St. Louis \$40 \$65

Richmond 70 30

The plant at St. Louis can supply 250 tons of carpet per week; the plant at Richmond can supply 400

tons per week. The Chicago outlet has a demand of 300 tons per week, and the outlet at Atlanta demands 350 tons per week. The company wants to know the number of tons of carpet to ship from each plant to each outlet in order to minimize the total shipping cost. Solve this transportation problem.

2. A transportation problem involves the following costs, supply, and demand:

To

From 1 2 3 4 Supply

1 \$500 \$750 \$300 \$450 12

2 650 800 400 600 17

3 400 700 500 550 11

Demand 10 10 10 10

a. Find the initial solution using the northwest corner method, the minimum cell cost method, and Vogel's approximation model. Compute total cost for each.

b. Using the VAM initial solution, find the optimal solution using the modified distribution method (MODI).

3. Consider the following transportation tableau and solution:

To

From A B C Supply

12 10 6

1 600 600

4 15 3

2 400 400

9 7 M

3 300 300

11 8 6

4 500 300 800

0 0 0

Dummy 200 200

Demand 900 500 900 2,300

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B-26 Module B Transportation and Assignment Solution Methods

- a. Is this a balanced or an unbalanced transportation problem? Explain.
 - b. Is this solution degenerate? Explain. If it is degenerate, show how it would be put into proper form.
 - c. Is there a prohibited route in this problem?
 - d. Compute the total cost of this solution.
 - e. What is the value of x_{2B} in this solution?
4. Solve the following transportation problem:

To

From 1 2 3 Supply

1 \$ 40 \$ 10 \$ 20 800

2 15 20 10 500

3 20 25 30 600

Demand 1,050 500 650

5. Given a transportation problem with the following costs, supply, and demand, find the initial solution using the minimum cell cost method and Vogel's approximation model. Is the VAM solution optimal?

To

From 1 2 3 Supply

A \$ 6 \$ 7 \$ 4 100

B 5 3 6 180

C 8 5 7 200

Demand 135 175 170

6. Consider the following transportation problem:

To

From 1 2 3 Supply

A \$ 6 9 M 130

B 12 3 5 70

C 4 8 11 100

Demand 80 110 60

- a. Find the initial solution by using VAM and then solve it using the stepping-stone method.
- b. Formulate this problem as a general linear programming model.

7. Solve the following linear programming problem:

minimize $Z = 3x_{11} + 12x_{12} + 8x_{13} + 10x_{21} + 5x_{22} + 6x_{23} + 6x_{31} + 7x_{32} + 10x_{33}$

subject to

$$x_{11} + x_{12} + x_{13} = 90 \quad x_{11} + x_{21} + x_{31} \dots 70$$

$$x_{21} + x_{22} + x_{23} = 30 \quad x_{12} + x_{22} + x_{32} \dots 110$$

$$x_{31} + x_{32} + x_{33} = 100 \quad x_{13} + x_{23} + x_{33} \dots 80$$

$$x_{ij} \geq 0$$

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Problems B-27

8. Consider the following transportation problem:

To

From 1 2 3 Supply

A \$ 6 \$ 9 \$ 7 130

B 12 3 5 70

C 4 11 11 100

Demand 80 110 60

a. Find the initial solution using the minimum cell cost method.

b. Solve using the stepping-stone method.

9. Steel mills in three cities produce the following amounts of steel:

Location Weekly Production (tons)

A. Bethlehem 150

B. Birmingham 210

C. Gary 320

680

These mills supply steel to four cities where manufacturing plants have the following demand:

Location Weekly Demand (tons)

1. Detroit 130

2. St. Louis 70

3. Chicago 180

4. Norfolk 240

620

Shipping costs per ton of steel are as follows:

To

From 1 2 3 4

A \$14 9 16 18

B 11 8 7 16

C 16 12 10 22

Because of a truckers' strike, shipments are at present prohibited from Birmingham to Chicago.

a. Set up a transportation tableau for this problem and determine the initial solution. Identify the method used to find the initial solution.

- b. Solve this problem using MODI.
 - c. Are there multiple optimal solutions? Explain. If so, identify them.
 - d. Formulate this problem as a general linear programming model.
10. In Problem 9, what would be the effect on the optimal solution of a reduction in production capacity at the Gary mill from 320 tons to 290 tons per week?

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B-28 Module B Transportation and Assignment Solution Methods

11. Coal is mined and processed at the following four mines in Kentucky, West Virginia, and Virginia:

Location Capacity (tons)

- A. Cabin Creek 90
- B. Surry 50
- C. Old Fort 80
- D. McCoy __60

280

These mines supply the following amount of coal to utility power plants in three cities:

Plant Demand (tons)

- 1. Richmond 120
- 2. Winston-Salem 100
- 3. Durham 110

330

The railroad shipping costs (\$1,000s) per ton of coal are shown in the following table. Because of railroad construction, shipments are now prohibited from Cabin Creek to Richmond:

To

From 1 2 3

A \$ 7 \$10 \$ 5

B 12 9 4

C 7 3 11

D 9 5 7

- a. Set up the transportation tableau for this problem, determine the initial solution using VAM, and compute total cost.
 - b. Solve using MODI.
 - c. Are there multiple optimal solutions? Explain. If there are alternative solutions, identify them.
 - d. Formulate this problem as a linear programming model.
12. Oranges are grown, picked, and then stored in warehouses in Tampa, Miami, and Fresno. These warehouses supply oranges to markets in New York, Philadelphia, Chicago, and Boston. The following table shows the shipping costs per truckload (\$100s), supply, and demand. Because of an agreement between distributors, shipments are prohibited from Miami to Chicago.

To

From New York Philadelphia Chicago Boston Supply

Tampa \$ 9 \$ 14 \$ 12 \$ 17 200

Miami 11 10 6 10 200

Fresno 12 8 15 7 200

Demand 130 170 100 150

- Set up the transportation tableau for this problem and determine the initial solution using the minimum cell cost method.
- Solve using MODI.
- Are there multiple optimal solutions? Explain. If so, identify them.
- Formulate this problem as a linear programming model.

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Problems B-29

13. A manufacturing firm produces diesel engines in four cities—Phoenix, Seattle, St. Louis, and Detroit. The company is able to produce the following numbers of engines per month:

Plant Production

- Phoenix 5
- Seattle 25
- St. Louis 20
- Detroit 25

Three trucking firms purchase the following numbers of engines for their plants in three cities:

Firm Demand

- Greensboro 10
- Charlotte 20
- Louisville 15

The transportation costs per engine (\$100s) from sources to destinations are shown in the following table. However, the Charlotte firm will not accept engines made in Seattle, and the Louisville firm will not accept engines from Detroit; therefore, these routes are prohibited.

To

From A B C

1 \$ 7 \$ 8 \$ 5

2 6 10 6

3 10 4 5

4 3 9 11

- Set up the transportation tableau for this problem. Find the initial solution using VAM.
 - Solve for the optimal solution using the stepping-stone method. Compute the total minimum cost.
 - Formulate this problem as a linear programming model.
14. The Interstate Truck Rental firm has accumulated extra trucks at three of its truck leasing outlets,

as shown in the following table:

Extra

Leasing Outlet Trucks

1. Atlanta 70
2. St. Louis 115
3. Greensboro 60
- Total 245

The firm also has four outlets with shortages of rental trucks, as follows:

Truck

Leasing Outlet Shortage

- A. New Orleans 80
- B. Cincinnati 50
- C. Louisville 90
- D. Pittsburgh 25
- Total 245

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B-30 Module B Transportation and Assignment Solution Methods

The firm wants to transfer trucks from those outlets with extras to those with shortages at the minimum total cost. The following costs of transporting these trucks from city to city have been determined:

To

From A B C D

- 1 \$ 70 80 45 90
- 2 120 40 30 75
- 3 110 60 70 80

- a. Find the initial solution using the minimum cell cost method.
- b. Solve using the stepping-stone method.

15. The Sholtz Beer Company has breweries in two cities; the breweries can supply the following numbers of barrels of draft beer to the company's distributors each month:

Brewery Monthly Supply (bbl)

- A. Tampa 3,500
- B. St. Louis 5,000
- Total 8,500

The distributors, which are spread throughout six states, have the following total monthly demand:

Distributor Monthly Demand (bbl)

1. Tennessee 1,600
2. Georgia 1,800
3. North Carolina 1,500
4. South Carolina 950

5. Kentucky 1,250

6. Virginia 1,400

Total 8,500

The company must pay the following shipping costs per barrel:

To

From 1 2 3 4 5 6

A \$0.50 0.35 0.60 0.45 0.80 0.75

B 0.25 0.65 0.40 0.55 0.20 0.65

a. Find the initial solution using VAM.

b. Solve using the stepping-stone method.

16. In Problem 15, the Sholtz Beer Company management has negotiated a new shipping contract with

a trucking firm between its Tampa brewery and its distributor in Kentucky that reduces the shipping cost per barrel from \$0.80 per barrel to \$0.65 per barrel. How will this cost change affect the optimal solution?

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Problems B-31

17. Computers Unlimited sells microcomputers to universities and colleges on the East Coast and ships

them from three distribution warehouses. The firm is able to supply the following numbers of microcomputers to the universities by the beginning of the academic year:

Distribution Supply

Warehouse (microcomputers)

1. Richmond 420

2. Atlanta 610

3. Washington, D.C. 340

Total 1,370

Four universities have ordered microcomputers that must be delivered and installed by the beginning of the academic year:

Demand

University (microcomputers)

A. Tech 520

B. A and M 250

C. State 400

D. Central 380

Total 1,550

The shipping and installation costs per microcomputer from each distributor to each university are

as follows:

To

From A B C D

1 \$22 17 30 18

2 15 35 20 25

3 28 21 16 14

a. Find the initial solution using VAM.

b. Solve using MODI.

18. In Problem 17, Computers Unlimited wants to better meet demand at the four universities it sup-

plies. It is considering two alternatives: (1) expand its warehouse at Richmond to a capacity of 600 at a cost equivalent to an additional \$6 in handling and shipping per unit or (2) purchase a new warehouse in Charlotte that can supply 300 units with shipping costs of \$19 to Tech, \$26 to A and M, \$22 to State, and \$16 to Central. Which alternative should management select based solely on transportation costs (i.e., no capital costs)?

19. Computers Unlimited in Problem 17 has determined that when it is unable to meet the demand for

microcomputers at the universities it supplies, the universities tend to purchase microcomputers elsewhere in the future. Thus, the firm has estimated a shortage cost for each microcomputer demanded but not supplied that reflects the loss of future sales and goodwill. These costs for each university are as follows:

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B-32 Module B Transportation and Assignment Solution Methods

University Cost/Microcomputer

A. Tech \$40

B. A and M 65

C. State 25

D. Central 50

Solve Problem 17 with these shortage costs included. Compute the total transportation cost and the total shortage cost.

20. A severe winter ice storm has swept across North Carolina and Virginia, followed by over a foot of

snow and frigid, single-digit temperatures. These weather conditions have resulted in numerous downed power lines and power outages in the region causing dangerous conditions for much of the population. Local utility companies have been overwhelmed and have requested assistance from unaffected utility companies across the Southeast. The following table shows the number of utility trucks with crews available from five different companies in Georgia, South Carolina, and Florida; the demand for crews in seven different areas that local companies cannot get to; and the weekly cost (\$1,000s) of a crew going to a specific area (based on the visiting company's normal charges, the distance the crew has to come, and living expenses in an area):

Area (Cost = \$1,000s)

Crews

Crew NC-E NC-SW NC-P NC-W VA-SW VA-C VA-T Available

GA-1 15.2 14.3 13.9 13.5 14.7 16.5 18.7 12

GA-2 12.8 11.3 10.6 12.0 12.7 13.2 15.6 10

SC-1 12.4 10.8 9.4 11.3 13.1 12.8 14.5 14

FL-1 18.2 19.4 18.2 17.9 20.5 20.7 22.7 15

FL-2 19.3 20.2 19.5 20.2 21.2 21.3 23.5 12

Crews

Needed 9 7 6 8 10 9 7

Determine the number of crews that should be sent from each utility to each affected area that will minimize total costs.

21. A large manufacturing company is closing three of its existing plants and intends to transfer some

of its more skilled employees to three plants that will remain open. The number of employees available for transfer from each closing plant is as follows:

Closing Plant Transferable Employees

1 60

2 105

3 70

Total 235

The following number of employees can be accommodated at the three plants remaining open:

Open Plants Employees Demanded

A 45

B 90

C 35

Total 170

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Problems B-33

Each transferred employee will increase product output per day at each plant as shown in the following table. The company wants to transfer employees so as to ensure the maximum increase in product output.

To

From A B C

1 5 8 6

2 10 9 12

3 7 6 8

a. Find the initial solution using VAM.

b. Solve using MODI.

22. The Sav-Us Rental Car Agency has six lots in Nashville, and it wants to have a certain number of

cars available at each lot at the beginning of each day for local rental. The agency would like a model it could quickly solve at the end of each day that would tell it how to redistribute the cars among the six lots in the minimum total time. The times required to travel between the six lots are as follows:

To (minutes)

From 1 2 3 4 5 6

1 — 12 17 18 10 20

2 14 — 10 19 16 15

3 14 10 — 12 8 9

4 8 16 14 — 12 15

5 11 21 16 18 — 10

6 24 12 9 17 15 —

The agency would like the following numbers of cars at each lot at the end of the day. Also shown are the numbers of available cars at each lot at the end of a particular day. Determine the optimal reallocation of rental cars using any initial solution approach and any solution method.

Lot

Cars 1 2 3 4 5 6

Available 37 20 14 26 40 28

Desired 30 25 20 40 30 20

23. Bayville has built a new elementary school so that the town now has a total of four schools—Addison, Beeks, Canfield, and Daley. Each has a capacity of 400 students. The school wants to assign children to schools so that their travel time by bus is as short as possible. The school has partitioned the town into five districts conforming to population density—north, south, east, west, and central. The average bus travel time from each district to each school is shown as follows:

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B-34 Module B Transportation and Assignment Solution Methods

Travel Time (min.)

Student

District Addison Beeks Canfield Daley Population

North 12 23 35 17 250

South 26 15 21 27 340

East 18 20 22 31 310

West 29 24 35 10 210

Central 15 10 23 16 290

Determine the number of children that should be assigned from each district to each school to minimize total student travel time.

24. In Problem 23, the school board has determined that it does not want any one school to be more

crowded than any other school. It would like to assign students from each district to each school so

that enrollments are evenly balanced among the four schools. However, the school board is concerned that this might significantly increase travel time. Determine the number of students to be assigned from each district to each school so that school enrollments are evenly balanced. Does this

new solution appear to result in a significant increase in travel time per student?

25. The Easy Time Grocery chain operates in major metropolitan areas on the eastern seaboard. The

stores have a “no-frills” approach, with low overhead and high volume. They generally buy their stock in volume at low prices. However, in some cases they actually buy stock at stores in other areas

and ship it in. They can do this because of high prices in the cities they operate in compared with costs in other locations. One example is baby food. Easy Time purchases baby food at stores in Albany, Binghamton, Claremont, Dover, and Edison and then trucks it to six stores in and around New York City. The stores in the outlying areas know what Easy Time is up to, so they limit the number of cases of baby food Easy Time can purchase. The following table shows the profit Easy Time makes per case of baby food based on where the chain purchases it and at which store it’s sold, plus the available baby food per week at purchase locations and the shelf space available at each Easy Time store per week:

Easy Time Store

Purchase

Location 1 2 3 4 5 6 Supply

Albany 9 8 11 12 7 8 26

Binghamton 10 10 8 6 9 7 40

Claremont 8 6 6 5 7 4 20

Dover 4 6 9 5 8 10 40

Edison 12 10 8 9 6 7 45

Demand 25 15 30 18 27 35

Determine where Easy Time should purchase baby food and how the food should be distributed in order to maximize profit. Use any initial solution approach and any solution method.

26. Suppose that in Problem 25 Easy Time can purchase all the baby food it needs from a New York City distributor at a price that will result in a profit of \$9 per case at stores 1, 3, and 4, \$8 per case at

stores 2 and 6, and \$7 per case at store 5. Should Easy Time purchase all, none, or some of its baby food from the distributor rather than purchasing it at other stores and trucking it in?

27. In Problem 25, if Easy Time could arrange to purchase more baby food from one of the outlying locations, which should it be, how many additional cases could be purchased, and how much would this increase profit?

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Problems B-35

28. The Roadnet Transport Company has expanded its shipping capacity by purchasing 90 trailer trucks from a competitor that went bankrupt. The company subsequently located 30 of the pur-

chased trucks at each of its shipping warehouses in Charlotte, Memphis, and Louisville. The company makes shipments from each of these warehouses to terminals in St. Louis, Atlanta, and New York. Each truck is capable of making one shipment per week. The terminal managers have indicated their capacity of extra shipments. The manager at St. Louis can accommodate 40 additional trucks per week, the manager at Atlanta can accommodate 60 additional trucks, and the manager at

New York can accommodate 50 additional trucks. The company makes the following profit per truckload shipment from each warehouse to each terminal. The profits differ as a result of differences in products shipped, shipping costs, and transport rates:

Terminal

Warehouse St. Louis Atlanta New York

Charlotte \$1,800 \$2,100 \$1,600

Memphis 1,000 700 900

Louisville 1,400 800 2,200

Determine how many trucks to assign to each route (i.e., warehouse to terminal) in order to maximize profit.

29. During the Gulf War, Operation Desert Storm required large amounts of military matériel and supplies to be shipped daily from supply depots in the United States to bases in the Middle East. The critical factor in the movement of these supplies was speed. The following table shows the number of planeloads of supplies available each day from each of six supply depots and the number of daily loads demanded at each of five bases. (Each planeload is approximately equal in tonnage.) Also included are the transport hours per plane, including loading and fueling, actual flight time, and unloading and refueling:

Military Base

Supply

Depot A B C D E Supply

1 36 40 32 43 29 7

2 28 27 29 40 38 10

3 34 35 41 29 31 8

4 41 42 35 27 36 8

5 25 28 40 34 38 9

6 31 30 43 38 40 6

Demand 9 6 12 8 10

Determine the optimal daily flight schedule that will minimize total transport time.

30. PM Computer Services produces personal computers from component parts it buys on the open market. The company can produce a maximum of 300 personal computers per month. PM wants to determine its production schedule for the first six months of the new year. The cost to produce a personal computer in January will be \$1,200. However, PM knows the cost of component parts will decline each month such that the overall cost to produce a PC will be 5% less each month. The cost of holding a computer in inventory is \$15 per unit per month. Following is the demand for the

company's computers each month:

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B-36 Module B Transportation and Assignment Solution Methods

Month Demand Month Demand

January 180 April 210

February 260 May 400

March 340 June 320

Determine a production schedule for PM that will minimize total cost.

31. In Problem 30, suppose the demand for personal computers increased each month as follows:

Month Demand

January 410

February 320

March 500

April 620

May 430

June 380

In addition to the regular production capacity of 300 units per month, PM Computer Services can also produce an additional 200 computers per month using overtime. Overtime production adds 20% to the cost of a personal computer.

Determine a production schedule for PM that will minimize total cost.

32. National Foods Company has five plants where it processes and packages fruits and vegetables. It has suppliers in six cities in California, Texas, Alabama, and Florida. The company has owned and operated its own trucking system in the past for transporting fruits and vegetables from its suppliers to its plants. However, it is now considering outsourcing all its shipping to outside trucking firms and getting rid of its own trucks. It currently spends \$245,000 per month to operate its own trucking system. It has determined monthly shipping costs (in \$1,000s per ton) using outside shippers from each of its suppliers to each of its plants as shown in the following table:

Processing Plants (\$1,000s per ton)

Suppliers Denver St. Paul Louisville Akron Topeka Supply (tons)

Sacramento 3.7 4.6 4.9 5.5 4.3 18

Bakersfield 3.4 5.1 4.4 5.9 5.2 15

San Antonio 3.3 4.1 3.7 2.9 2.6 10

Montgomery 1.9 4.2 2.7 5.4 3.9 12

Jacksonville 6.1 5.1 3.8 2.5 4.1 20

Ocala 6.6 4.8 3.5 3.6 4.5 15

Demand (tons) 20 15 15 15 20 90

Should National Foods continue to operate its own shipping network or sell its trucks and out-source its shipping to independent trucking firms?

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Problems B-37

33. In Problem 32, National Foods would like to know what the effect would be on the optimal solution and the company's decision regarding its shipping if it negotiates with its suppliers in Sacramento, Jacksonville, and Ocala to increase their capacity to 25 tons per month? What would be the effect of negotiating instead with its suppliers at San Antonio and Montgomery to increase their capacity to 25 tons each?

34. Orient Express is a global distribution company that transports its clients' products to customers in

Hong Kong, Singapore, and Taipei. All the products Orient Express ships are stored at three distribution centers, one in Los Angeles, one in Savannah, and one in Galveston. For the coming month the company has 450 containers of computer components available at the Los Angeles center, 600 containers available at Savannah, and 350 containers available in Galveston. The company has orders for 600 containers from Hong Kong, 500 containers from Singapore, and 500 containers from Taipei. The shipping costs per container from each U.S. port to each of the overseas ports are shown in the following table:

Overseas Port

U.S. Distribution

Center Hong Kong Singapore Taipei

Los Angeles \$300 \$210 \$340

Savannah 490 520 610

Galveston 360 320 500

The Orient Express, as the overseas broker for its U.S. customers, is responsible for unfulfilled orders, and it incurs stiff penalty costs from overseas customers if it does not meet an order. The Hong Kong customers charge a penalty cost of \$800 per container for unfulfilled demand, Singapore customers charge a penalty cost of \$920 per container, and Taipei customers charge \$1,100 per container. Formulate and solve a transportation model to determine the shipments from each U.S. distribution center to each overseas port that will minimize shipping costs. Indicate what portion of the total cost is a result of penalties.

35. Binford Tools manufactures garden tools. It uses inventory, overtime, and subcontracting to absorb

demand fluctuations. Expected demand, regular and overtime production capacity, and subcontracting capacity are provided in the following table for the next four quarters for its basic line of steel garden tools:

Regular Overtime Subcontracting

Quarter Demand Capacity Capacity Capacity

1 9,000 9,000 1,000 3,000

2 12,000 10,000 1,500 3,000

3 16,000 12,000 2,000 3,000

4 19,000 12,000 2,000 3,000

The regular production cost per unit is \$20, the overtime cost per unit is \$25, the cost to subcontract a unit is \$27, and the inventory carrying cost is \$2 per unit. The company has 300 units in inventory at the beginning of the year.

Determine the optimal production schedule for the four quarters that will minimize total costs.

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B-38 Module B Transportation and Assignment Solution Methods

36. Solve the following linear programming problem:

$$\begin{aligned} \text{minimize } Z = & 18x_{11} + 30x_{12} + 20x_{13} + 18x_{14} + 25x_{21} + 27x_{22} + 22x_{23} \\ & + 16x_{24} + 30x_{31} + 26x_{32} + 19x_{33} + 32x_{34} + 40x_{41} + 36x_{42} \\ & + 27x_{43} + 29x_{44} + 30x_{51} + 26x_{52} + 18x_{53} + 24x_{54} \end{aligned}$$

subject to

$$x_{11} + x_{12} + x_{13} + x_{14} \leq 1$$

$$x_{21} + x_{22} + x_{23} + x_{24} \leq 1$$

$$x_{31} + x_{32} + x_{33} + x_{34} \leq 1$$

$$x_{41} + x_{42} + x_{43} + x_{44} \leq 1$$

$$x_{51} + x_{52} + x_{53} + x_{54} \leq 1$$

$$x_{11} + x_{21} + x_{31} + x_{41} + x_{51} = 1$$

$$x_{12} + x_{22} + x_{32} + x_{42} + x_{52} = 1$$

$$x_{13} + x_{23} + x_{33} + x_{43} + x_{53} = 1$$

$$x_{14} + x_{24} + x_{34} + x_{44} + x_{54} = 1$$

$$x_{ij} \geq 0$$

37. A plant has four operators to be assigned to four machines. The time (minutes) required by each

worker to produce a product on each machine is shown in the following table. Determine the optimal assignment and compute total minimum time.

Machine

Operator A B C D

1 10 12 9 11

2 5 10 7 8

3 12 14 13 11

4 8 15 11 9

38. A shop has four machinists to be assigned to four machines. The hourly cost of having each machine operated by each machinist is as follows:

Machine

Machinist A B C D

1 \$12 11 8 14

2 10 9 10 8

3 14 8 7 11

4 6 8 10 9

However, because he does not have enough experience, machinist 3 cannot operate machine B.

- Determine the optimal assignment and compute total minimum cost.
- Formulate this problem as a general linear programming model.

39. The Omega pharmaceutical firm has five salespersons, whom the firm wants to assign to five sales regions. Given their various previous contacts, the salespersons are able to cover the

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Problems B-39

regions in different amounts of time. The amount of time (days) required by each salesperson to cover each city is shown in the following table. Which salesperson should be assigned to each region to minimize total time? Identify the optimal assignments and compute total minimum time.

Region

Salesperson A B C D E

1 17 10 15 16 20

2 12 9 16 9 14

3 11 16 14 15 12

4 14 10 10 18 17

5 13 12 9 15 11

40. The Bunker Manufacturing firm has five employees and six machines and wants to assign the employees to the machines to minimize cost. A cost table showing the cost incurred by each employee on each machine follows. Because of union rules regarding departmental transfers, employee 3 cannot be assigned to machine E and employee 4 cannot be assigned to machine B. Solve this problem, indicate the optimal assignment, and compute total minimum cost.

Machine

Employee A B C D E F

1 \$12 \$ 7 \$20 \$14 \$ 8 \$10

2 10 14 13 20 9 11

3 5 3 6 9 7 10

4 9 11 7 16 9 10

5 10 6 14 8 10 12

41. Given the following cost table for an assignment problem, determine the optimal assignment and compute total minimum cost. Identify all alternative solutions if there are multiple optimal solutions.

Machine

Operator A B C D

1 \$10 \$2 \$ 8 \$ 6

2 9 5 11 9

3 12 7 14 14

4 3 1 4 2

42. An electronics firm produces electronic components, which it supplies to various electrical manufacturers. Quality control records indicate that different employees produce different numbers of defective items. The average number of defects produced by each employee for each of six components

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B-40 Module B Transportation and Assignment Solution Methods

is given in the following table. Determine the optimal assignment that will minimize the total average number of defects produced by the firm per month.

Component

Employee A B C D E F

1 30 24 16 26 30 22

2 22 28 14 30 20 13

3 18 16 25 14 12 22

4 14 22 18 23 21 30

5 25 18 14 16 16 28

6 32 14 10 14 18 20

43. A dispatcher for the Citywide Taxi Company has six taxicabs at different locations and five customers who have called for service. The mileage from each taxi's present location to each customer is shown in the following table. Determine the optimal assignment(s) that will minimize the total mileage traveled.

Customer

Cab 1 2 3 4 5

A 7 2 4 10 7

B 5 1 5 6 6

C 8 7 6 5 5

D 2 5 2 4 5

E 3 3 5 8 4

F 6 2 4 3 4

44. The Southeastern Conference has nine basketball officials who must be assigned to three conference games, three to each game. The conference office wants to assign the officials so that the total distance they travel will be minimized. The distance (in miles) each official would travel to each game is given in the following table:

Game

Official Athens Columbia Nashville

1 165 90 130

2 75 210 320

3 180 170 140

4 220 80 60

5 410 140 80

6 150 170 190

7 170 110 150

8 105 125 160

9 240 200 155

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Problems B-41

a. Should this problem be solved by the transportation method or by the assignment method? Explain.

b. Determine the optimal assignment(s) that will minimize the total distance traveled by the officials.

45. In Problem 44, officials 2 and 8 had a recent confrontation with one of the coaches in the game in

Athens. They were forced to eject the coach after several technical fouls. The conference office decided that it would not be a good idea to have these two officials work the Athens game so soon after this confrontation, so they decided that officials 2 and 8 will not be assigned to the Athens game. How will this affect the optimal solution to this problem?

46. State University has planned six special catered events for the November Saturday of its homecom-

ing football game. The events include an alumni brunch, a parents' brunch, a booster club luncheon, a postgame party for season ticket holders, a lettermen's dinner, and a fund-raising dinner for major contributors. The university wants to use local catering firms as well as the university catering service to cater these events and it has asked the caterers to bid on each event. The bids (in

\$1,000s) based on menu guidelines for the events prepared by the university are shown in the following table:

Event

Alumni Parents' Booster Postgame Lettermen's Contributor's

Caterer Brunch Brunch Club Lunch Party Dinner Dinner

Al's \$12.6 \$10.3 \$14.0 \$19.5 \$25.0 \$30.0

Bon Ap  t  t 14.5 13.0 16.5 17.0 22.5 32.0

Custom 13.0 14.0 17.6 21.5 23.0 35.0

Divine 11.5 12.6 13.0 18.7 26.2 33.5

Epicurean 10.8 11.9 12.9 17.5 21.9 28.5

Fouch  s 13.5 13.5 15.5 22.3 24.5 36.0

University 12.5 14.3 16.0 22.0 26.7 34.0

The Bon Ap  t  t, Custom, and University caterers can handle two events, whereas the other four caterers can handle only one. The university is confident all the caterers will do a high-quality job, so it wants to select the caterers for the events that will result in the lowest total cost.

Determine the optimal selection of caterers that will minimize total cost.

47. A university department head has five instructors to be assigned to four different courses. All the

instructors have taught the courses in the past and have been evaluated by the students. The rating

for each instructor for each course is given in the following table (a perfect score is 100). The department head wants to know the optimal assignment of instructors to courses that will maximize the overall average evaluation. The instructor who is not assigned to teach a course will be assigned to grade exams. Solve this problem using the assignment method.

Course

Instructor A B C D

1 80 75 90 85

2 95 90 90 97

3 85 95 88 91

4 93 91 80 84

5 91 92 93 88

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B-42 Module B Transportation and Assignment Solution Methods

48. The coach of the women's swim team at State University is preparing for the conference swim meet

and must choose the four swimmers she will assign to the 800-meter medley relay team. The medley relay consists of four strokes—the backstroke, breaststroke, butterfly, and freestyle. The coach has computed the average times (in minutes) each of her top six swimmers has achieved in each of the four strokes for 200 meters in previous swim meets during the season as follows:

Stroke (min)

Swimmer Backstroke Breaststroke Butterfly Freestyle

Annie 2.56 3.07 2.90 2.26

Beth 2.63 3.01 3.12 2.35

Carla 2.71 2.95 2.96 2.29

Debbie 2.60 2.87 3.08 2.41

Erin 2.68 2.97 3.16 2.25

Fay 2.75 3.10 2.93 2.38

Determine the medley relay team and its total expected relay time for the coach.

49. Biggio's Department Store has six employees available to assign to four departments in the store—

home furnishings, china, appliances, and jewelry. Most of the six employees have worked in each of the four departments on several occasions in the past and have demonstrated that they perform better in some departments than in others. The average daily sales for each of the six employees in each of the four departments is shown in the following table.

Department Sales (\$)

Home

Employee Furnishings China Appliances Jewelry

1 340 160 610 290

2 560 370 520 450

3 270 — 350 420

4 360 220 630 150

5 450 190 570 310

6 280 320 490 360

Employee 3 has not worked in the china department before, so the manager does not want to assign

this employee to china.

Determine which employee to assign to each department and indicate the total expected daily sales.

50. The Vanguard Publishing Company has eight college students it hires as salespeople to sell encyclo-

pedias during the summer. The company desires to allocate them to three sales territories.

Territory

1 requires three salespeople, and territories 2 and 3 require two salespeople each. It is estimated that

each salesperson will be able to generate the amounts of dollar sales per day in each of the three ter-

ritories as given in the following table. The company desires to allocate the salespeople to the three territories so that sales will be maximized. Solve this problem using any method to determine the initial solution and any solution method. Compute the maximum total sales per day.

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Problems B-43

Territory

Salesperson 1 2 3

A \$110 \$150 \$130

B 90 120 80

C 205 160 175

D 125 100 115

E 140 105 150

F 100 140 120

G 180 210 160

H 110 120 90

51. Carolina Airlines, a small commuter airline in North Carolina, has six flight attendants whom it wants to assign to six monthly flight schedules in a way that will minimize the number of nights they will be away from their homes. The numbers of nights each attendant must be away from home with each schedule are given in the following table. Identify the optimal assignments that will minimize the total number of nights the attendants will be away from home.

Schedule

Attendant A B C D E F

1 7 4 5 10 5 7

2 4 5 4 12 7 5

3 9 9 10 7 10 7

4 11 6 7 5 9 9

5 5 8 5 10 7 5

6 10 12 10 9 9 9

52. The football coaching staff at Tech focuses its recruiting on several key states, including Georgia,

Florida, Virginia, Pennsylvania, New York, and New Jersey. The staff includes seven assistant coaches, two of whom are responsible for Florida, a high school talent-rich state, whereas one coach is assigned to each of the other five states. The staff has been together for a long time and at one time or another all the coaches have recruited all of the states. The head coach has accumulated

some data on the past success rate (i.e., percentage of targeted recruits signed) for each coach in each state as shown in the following table:

State

Coach GA FL VA PA NY NJ

Allen 62 56 65 71 55 63

Bush 65 70 63 81 75 72

Crumb 46 53 62 55 64 50

Doyle 58 66 70 67 71 49

Evans 77 73 69 80 80 74

Fouch 68 73 72 80 78 57

Goins 72 60 74 72 62 61

Determine the optimal assignment of coaches to recruiting regions that will maximize the overall success rate and indicate the average percentage success rate for the staff with this assignment.

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ESD.36 System & Project Management

Lecture 2

- Critical Path Method (CPM)

Instructor(s)

Prof. Olivier de Weck

Today's Agenda

□ Overview of PM methods and tools

□ CPM 101

□ Critical Paths, Slack

- Task “Crashing” and Cost
- Conclusions and Class Discussions
- Introduce HW3

- ESD.36 SPM 2

History of Project Management

- Big Projects since antiquity
- Pyramids (Egypt), Great Wall (China)
- Enormous workforce, but little documented evidence of formal project management
- Formal Project Management
- Henry Gantt (1861-1919) □ bar chart 1910
- 1957 Sputnik Crisis □ revival of “scientific management”
- Polaris (1958) □ Project Evaluation and Review

Technique (PERT)

- DuPont Company (1960) □ Critical Path Method (CPM)
- 1960’s NASA projects: Mercury, Gemini, Apollo
- Work Breakdown Structures (WBS)
- Cost and Schedule Tracking, Configuration Management

- ESD.36 SPM 3

Comments about early PM

- Project decomposition necessary due to complexity
- Resource allocation and workload smoothing
- Schedule urgency ..”before the decade is out”

JFK

- Circumstances
- Complex Relations between Government and Contractors
- “Shielded” from Society, Competition, Regulations
- Cold War Pressures for Nuclear Power, Space Race ..
- Other Innovations
- Project Manager as a central figure
- Beginnings of Matrix Organization
- “Earned Value” – adopted by USAF (1963)
- Professionalization since 1969
- Diffusion into other industries: computers, automotive ...
- Project Management Institute (PMI) founded – PMBOK
- ISO 10006:2003 Quality in Project Management
- Recent criticism about PM standards as “bureaucratic”

- ESD.36 SPM 4

Fundamental Approaches

- How to represent task relationships?
- Network-based (graph theory) methods
- CPM, PERT,
- a b
- Task is a node or an arc d e
- c
- Matrix-based methods
- DSM - Tasks are columns and rows x x
- Interrelationships are off-diagonal entries
- xx
- x x
- x x
- System Dynamics x x
- Feedback loops, causal relationships
- Stocks and flows simulation

□ Tasks that are done or waiting to be done
PEOPLE PRODUCTIVITY

are stocks – “amount of work” WORK WORK
WORK

- Doing project work causes a “flow”
- TO BE
- BEING DONE DONE
- DONE

- ESD.36 SPM 5

Gantt Charts

- Attributed to Henry Gantt – most popular PM tool (80%)
- Used to plan big shipbuilding projects (cargo ships WWI)
- Graphical way of showing task durations, project schedule
- Does not explicitly show relationships between tasks
- Limited use for project tracking
- Easy to understand

today calendar

completion

Gantt Chart Builder System (Excel) 1.6

08 Sep'03 22 Sep'03 06 Oct'03

Bus Unit Project % Start Finish 08/09 15/09 22/09 29/09 06/10 13/10

Project "XYZ" 30 9-Sep-03 6-Oct-03

Mkt Customer Clinic 100 09-Sep-2003 12-Sep-2003

milestone

Sys Requirements Definition 100 11-Sep-2003 15-Sep-2003

Eng Parts Design 50 15-Sep-2003 22-Sep-2003

Sys Design Review 0 23-Sep-2003 23-Sep-2003

Mfg Manufacturing 0 24-Sep-2003 05-Oct-2003

Sys Product Release 0 06-Oct-2003 06-Oct-2003

tasks actual planned

- ESD.36 SPM D 6

CPM 101

□ Represent a project (set of task) as a network using graph theory

□ Capture task durations

□ Capture task logic (dependencies)

(Parallel)

Task ID (Series) Expected

Duration B,8

A,5 B,8 D,3

A,5

C,2

“B can only start after A

is completed” “B and C do not depend on each other”

- ESD.36 SPM 7

CPM Assumptions

□ Project consists of a collection of well defined tasks (jobs)

□ Project ends when all jobs completed

□ Jobs may be started and stopped independently of each other within a given sequence (no “continuous-flow” processes)

□ Jobs are ordered □ “technological sequence”

- ESD.36 SPM 8

Task Representations

□ Tasks as Nodes of a Graph

□ Circles

ES EF

□ Boxes

A,5 ID:A Dur:5

LS LF

□ Tasks as Arcs of a Graph

□ Tasks are uni-directional arrows

☐ Nodes now represent “states” of a project

☐ Kelley-Walker form

A, 5

broken fixed

- ESD.36 SPM 9

Concept Question 1

What in project management is named after Henry Gantt (1861-1919)?

☐ A new method for managing budgets in shipyards

☐ A matrix representation of the task relationships known as the Gantt-Matrix

☐ A horizontal bar chart known as the Gantt-Chart

☐ Gantt-teams, a new form of organizational structure

☐ Gantt was a mechanical engineer and did not care about projects

- ESD.36 SPM 10

Work Breakdown Structure

☐ Used to create the task (job) list

☐ Tree-decomposition of project tasks

☐ WBS identifies “terminal elements”

☐ The key starting point for project planning

☐ Required by US Govt as part of SOW

☐ Can be activity-oriented or deliverable- oriented

☐ Use “sticky-notes” method early on

☐ Carl L. Pritchard. Nuts and Bolts Series 1: How to Build a Work Breakdown Structure. ISBN 1890367125

Job A Job B Job X Job G

- ESD.36 SPM 11

WBS – Painting a Room

☐ 1 Prepare materials

☐ 1.1 Buy paint

☐ 1.2 Buy brushes/rollers

☐ 1.3 Buy wallpaper remover

☐ 2. Prepare room

☐ 2.1 Remove old wallpaper

☐ 2.2 Remove detachable decorations

- 2.3 Cover floor with old newspapers
- 2.4 Cover electrical outlets/switches with tape
- 2.5 Cover furniture with sheets
- 3. Paint the room
- 4. Clean up the room
- 4.1 Dispose or store left over paint
- 4.2 Clean brushes/rollers
- 4.3 Dispose of old newspapers
- 4.4 Remove covers Source: <http://www.wikipedia.org>

- ESD.36 SPM 12

+ WBS of MIT/NASA Space

- Logistics Project

IP S C M & L A

1 . 0 P r o j e c t 2 . 0 T e r r e s t r i a l 3 . 0 S p a c e L o g i s t i c s 4 . 0 E x p l o r a t i o n 5 . 0 I n t e r p l a n e t a r y 6 . 0 P r o j e c t

M a n a g e m e n t A n a l o g i e s N e t w o r k M o d e l S u p p l y / D e m a n d S C A r c h T r a d e I n t e g r a t i o n a n d

(T a s k 1) (T a s k 2) M o d e l i n g S t u d i e s D i s s e m i n a t i o n

(T a s k 3) (T a s k 4)

1 . 1 M I T P r o j e c t 2 . 1 L o w D e n s i t y E q u i p m e n t 3 . 1 M o d e l F o r m u l a t i o n 4 . 1 S u p p l y / D e m a n d 5 . 1 T r a d e S t u d i e s 6 . 1 N A S A W o r k s h o p I

M a n a g e m e n t S C M L e s s o n s L e a r n e d D a t a M o d e l i n g P l a n n i n g a n d S e t u p

2 . 2 L o n g - R a n g e M i l i t a r y 6 . 2 N A S A W o r k s h o p I I

1 . 1 . 1 M I T P r o j e c t S C M L e s s o n s L e a r n e d 3 . 1 . 1 M I T S e g m e n t 4 . 1 . 1 M I T S e g m e n t 5 . 1 . 1 M I T S e g m e n t

M a n a g e r / P I

2 . 3 R e m o t e E n v i r o n m e n t s 6 . 3 I n t e r n a l I n t e g r a t i o n

1 . 1 . 2 M I T T r a v e l S C M L e s s o n s L e a r n e d 3 . 1 . 2 J P L S e g m e n t 4 . 1 . 2 J P L S e g m e n t 5 . 1 . 2 J P L S e g m e n t

2 . 4 I S S S C M

1 . 1 . 3 M I T R e s o u r c e L e s s o n s L e a r n e d 3 . 2 M o d e l 4 . 1 . 3 U S A S e g m e n t 5 . 2 T r a d e S t u d i e s 6 . 3 . 1 M I T S e g m e n t

A d m i n i s t r a t i o n D e v e l o p m e n t E x e c u t i o n

2 . 5 S h u t t l e / I S S

1 . 1 . 4 R e s e r v e s L e s s o n s L e a r n e d 4 . 2 D a t a 5 . 3 T r a d e S t u d i e s 6 . 3 . 2 J P L S e g m e n t

3 . 2 . 1 M I T S e g m e n t C o l l e c t i o n R e p o r t i n g

1 . 2 J P L P r o j e c t 6 . 3 . 3 U S A S e g m e n t

M a n a g e m e n t 3 . 2 . 2 J P L S e g m e n t 4 . 2 . 1 J P L S e g m e n t

6 . 4 P u b l i c E d u c a t i o n

1 . 2 . 1 J P L P r o j e c t 3 . 3 M o d e l 4 . 2 . 2 U S A S e g m e n t a n d O u t r e a c h

Manager / Co-PI Validation

1.2.2 JPL Travel 4.3 Data 6.4.1 MIT Segment

3.3.1 MIT Segment Validation

1.2.3 JPL Resource 6.4.2 JPL Segment

Administration 3.3.2 JPL Segment 4.3.1 JPL Segment

1.2.4 Reserves 6.4.3 USA Segment

3.3.3 USA Segment 4.3.2 USA Segment

1.3 USA Project 6.5 Space SCM

Management Executive Short Course

1.3.1 USA Project

MIT (incl. PSI): cyan

Manager / Co-PI

JPL: gray

1.3.2 USA Travel USA: green

Color code WBS by

Organization in charge

1.3.3 USA Resource

Administration

- ESD.36 SPM 13

1.3.4 Reserves

Discussion Point 1

□ Why is it difficult to come up with a good WBS (task list, task structure) in a complex project?

□ Not all tasks known ahead of time if completely new product/system

□ Others...?

- ESD.36 SPM 14

WBS Guidelines

□ No more than 100-200 terminal elements, if more □ use subprojects

□ Can be up to 3-4 Levels deep

□ Not more than 5-9 jobs at one level

□ Human cognitive “bandwidth” only 3 bits=23=8

□ Short term memory for most people 5-9 items

□ Poorer planning if “too-fine grained” – dilution of attention

□ The more tasks there are, the more intricate dependencies there will be to keep track of

- Jobs should be of similar size/complexity
 - Manageable chunks □ sense of progress
 - Level of graininess □ very difficult to answer
- ESD.36 SPM 15

Task List

- List all tasks in a table with
 - Identifying symbol (tag, ID number)
 - Task description
 - Immediate prerequisite jobs
 - Expected task duration
 - Arrange jobs in “technological order”
 - No job appears in the list until all its predecessors have been listed
 - Iterations are NOT allowed □ “cycle error”
 - Job a precedes b precedes c precedes a
 - We will discuss iterations a lot in this class !!!
- ESD.36 SPM 16

Simple Example: Job List

- Two Parts X and Y: Manufacture and Assembly

Job # Description Immediate Time

Predecessors [min]

A Start 0

B Get materials for X A 10

C Get materials for Y A 20

D Turn X on lathe B,C 30

E Turn Y on lathe B,C 20

F Polish Y E 40

G Assemble X and Y D,F 20

H Finish G 0

- ESD.36 SPM 17

Project Graph

- Each job is drawn on a graph as a circle*
- Connect each job with immediate predecessor(s) – unidirectional arrows “→”
- Jobs with no predecessor connect to “Start”
- Jobs with no successors connect to “Finish”
- “Start” and “Finish” are pseudo-jobs of length 0
- A finite number of “arrow paths” from “Start” to “Finish” will be the result

- Total time of each path is the sum of job times
- Path with the longest total time □ “critical path”
- There can be multiple critical paths □ minimum time to complete project

* or other symbol, see before

- ESD.36 SPM 18

Project Graph

C,20 E,20 F,40

critical path

A,0 G,20 H,0

Start Finish

B,10 D,30

4 unique paths: A,C,E,F,G,H; A,C,D,G,H; A,B,D,G,H; A,B,E,F,G,H

100 70 60 90

- ESD.36 SPM 19

Critical Path

- CP is the “bottleneck route”
- Shortening or lengthening tasks on the critical path directly affects project finish
- Duration of “non-critical” tasks is irrelevant
- “Crashing” all jobs is ineffective, focus on the few % of jobs that are on the CP
- “Crashing” tasks can shift the CP to a different task
- Shortening tasks – technical and economical challenge
- How can it be done?
- Previously non-critical tasks can become critical
- Lengthening of non-critical tasks can also shift the critical path (see HW1).

- ESD.36 SPM 20

Discussion Point 2

- What is the usefulness of knowing the CP in a project?
- Tells which task to shorten to finish project earlier.
- Others ...?

- ESD.36 SPM 21

Critical Path Algorithm

- For large projects there are many paths

- Need a algorithm to identify the CP efficiently
- Develop information about each task in context of the overall project
- Times
- Start time (S)
- For each job: Earliest Start (ES)
- Earliest start time of a job if all its predecessors start at ES
- Job duration: t
- Earliest Finish (EF)=(ES)+t
- Finish time (F) – earliest finish time of the overall project
- Show algorithm using project graph

- ESD.36 SPM 22

CP Algorithm

1. Mark the value of S to left and right of Start
2. Consider any new unmarked job, all of whose predecessors have been marked. Mark to the left of the new job the largest number to the right of its immediate predecessors: (ES)
3. Add to ES the job time t and mark result to the right (EF)
4. Stop when Finish has been reached

- ESD.36 SPM 23

CP Algorithm - Graphical

critical path

E,20 F,40

0 C,20 20 20 40 40 80

Finish

S=0 A,0 0 G,20 H,0

80 100 100 F=100

Start

0 B,10 10 20 D,30 50

- ESD.36 SPM 24

Concept Question 2

Possible

A project starts with (A,5). Task (B,10) can start after A is completed. This is Answers also true for task (E,5). Task (C,8) □20 depends only on (B,10), while task (F,10) □22

depends on both (B,10) and (E,5). Task (D,5) is the last task in the project and it can start once (C,8) and (F,10) have been finished.

30

The Earliest Finish (EF) time for the whole project is:

40

- ESD.36 SPM 25

Latest Start and Finish Times

Set target finish time for project: $T \geq F$

Usually target is a specific calendar date, e.g. October 1, 2007

When is the latest date the project can be started?

Late Finish (LF) - latest time a job can be finished, without delaying the project beyond its target time (T)

Late Start: $LS = LF - t$

- ESD.36 SPM 26

Determine LF and LS

Work from the end of the project: T

1. Mark value of T to left and right of Finish
2. Consider any new unmarked job, all of whose successors have been marked - mark to the right the smallest LS time marked to the left of any of its immediate successors
3. Subtract from this number, LF, the job time t and mark result to the left of the job: LS
4. Continue upstream until Start has been reached, then stop

- ESD.36 SPM 27

LS and LF : Project Graph

0 20 20 40 40 80

0 20 20 40 40 80

C,20 E,20 F,40

Finish

S=0 A,0 G,20 H,0

Start 80 100 100 100

80 100 100 100

0 0

0 0 0 B,10 10 D,30

Legend

0 10 20 50 early ES EF

10 20 50 80 late LS LF

- ESD.36 SPM 28

Slack

□ Some tasks have $ES=LS$ --> no slack

□ Total Slack of a task $TS=LS-ES$

□ Maximum amount of time a task may be delayed

beyond its early start without delaying project completion

□ Slack time is precious ... managerial freedom, don't squander it unnecessarily

□ e.g. resource, work load smoothing

□ When $T=F$ then all critical tasks have $TS=0$

□ At least one path from Start->Finish with critical jobs only

□ When $T>F$, then all critical jobs have $TS=T-F$

- ESD.36 SPM 29

Project Graph - Slack

0 20 20 40 40 80

0 20 20 40 40 80

C,20 E,20 F,40

Finish

$TS=0$ $TS=0$ $TS=0$

$TS=0$ $TS=0$

$S=0$ A,0 G,20 H,0

Start 80 100 100 100

$TS=10$ $TS=30$ 80 100 100 100

0 0

0 0 0 B,10 10 D,30

Legend

0 10 20 50 early ES EF

10 20 50 80 late LS LF

- ESD.36 SPM 30

Task Times Detail - Task i

$ES(i)$ $LS(i)$ $EF(i)$ $LF(i)$

Duration $t(i)$

Total Slack Duration $t(i)$

j is the immediate

TS(i) successor of i with

the smallest ES

$FS(i) ES(j) j > i$

Free Slack

□ Free Slack (FS) is the amount a job can be delayed without delaying the Early Start (ES) of any other job.

$FS \leq TS$ always

- ESD.36 SPM 31

Main CPM Errors

□ Estimated job times are wrong

□ Predecessor relationships may contain cycles □ “cycle error”

□ List of prerequisites contains more than the immediate predecessors, e.g. $a \rightarrow b$, $b \rightarrow c$ and $a, b \rightarrow c$

□ Overlooked some predecessor relationships

□ Some predecessor relationships may be listed that are spurious

□ and Some tasks/jobs may be missing !!!

- ESD.36 SPM 32

Gradual Refinement of CPM

□ Job Times

□ Given rough time estimates construct CPM chart

□ Re-estimate times for CP and those with very small TS

□ Iterate until the critical path is stable

□ Focus attention on a subset of tasks

□ Predecessor Relationships

□ Check algorithmically for cycle errors and pre-predecessor errors

□ Cancel all except immediate predecessor relationships

□ Wrong or Missing Facts

□ Cannot be detected by computers!

- ESD.36 SPM 33

Sample CPM (2007)

A&D High Tech Case, Online Store Project, 60 tasks, HW2, 2007

- ESD.36 SPM 34

Crashing Tasks

- What is we want to speed up project completion?
- Options
- Work overtime
- Put more resources on the critical tasks
- Parallelize tasks that are really serial (later in class)
- Cost of speedup?
- Is there a net savings resulting from reduction in overall project time?

- ESD.36 SPM 35

Cost Calculations

- Can compute project costs if cost of each job is included in the task data
- (Potentially) shorten crew jobs by adding personnel or working overtime with existing personnel
- Speedup carries price tag: “normal time”, “crash time”
- Assign some critical jobs to their “crash time”
- Direct costs will increase as we “crash” critical tasks
- Indirect (fixed, overhead) costs will decrease as the overall project duration decreases - “standing army phenomenon”
- Minimize the sum of fixed and direct costs

- ESD.36 SPM 36

Typical Cost Pattern

Project Total Costs

Costs [\$]

“natural pace”

savings

Cum Fixed Costs

Direct Costs

crash Total Project

A B

Time [days]

- ESD.36 SPM 37

CPM Judgment

- +
- Focuses attention on a subset of critical tasks
- Determine effect of shortening/lengthening tasks

- Evaluate costs of a "crash" program
 - -
 - Doesn't capture task iterations, in fact ...
 - Prohibits iterations = "cycle error"
 - Treats task durations as deterministic
- ESD.36 SPM 38

Summary

- CPM is useful, despite criticism, to identify the critical path - focus on a subset of the project
 - Slack (TS and FS) is precious
 - apply flexibility to smooth resource/schedules
 - PERT treats task times as probabilistic
 - Next lecture
 - Selective "crashing" of critical tasks can reduce (or increase) total project cost
 - CPM does not allow for task iterations
- ESD.36 SPM 39

Class Frustrations

- Poor examples set by project managers
- Perception of PMs as bureaucratic "box-checkers" and "ankle-biters"

Why?

- Traditional Project Management ...
 - Doesn't acknowledge the existence of iterations
 - Is inflexible, "changing the plan" considered a failure
 - Does not think of projects in a probabilistic sense
 - "Hostage" to existing project management software
 - In a reactive mode - no "early warning" systems
- ESD.36 SPM 40

HW1 Introduction

- You are Project Manager for the new CityCar development project
- Plan the project
- Task list
- Create project graph
- Critical path
- Slack times
- Re-planning after change

□ “managerial”-type questions

- ESD.36 SPM 41

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Fall 2012

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ESD.36 System Project Management

Lecture 1

- Class Introduction

Instructor(s)

Prof. Olivier de Weck

Dr. James Lyneis, Prof. Dan Braha

Today’s Agenda

□ Welcome and Introductions

□ Definitions – Initial Discussion

□ Course Objectives

□ Schedule

□ Term Project, Homeworks

□ Questions?

2

Introductions

□ Olivier de Weck

□ Dipl. Ing. Industrial Engineering – ETH Zurich ’93

□ 1993-1997 Engineering Program Manager Swiss

F/A-18 Project, McDonnell Douglas, St. Louis

□ S.M. ’99 Ph.D. ’01 Aerospace Systems – MIT

□ Associate Professor – dual appointment AA and

ESD, Executive Director Production in Innovation

Economy (PIE) Study

□ Research:

□ Systems Engineering for Changeability
and Commonality

□ <http://strategic.mit.edu>

□ Space Logistics

□ <http://spacelogistics.mit.edu>

3

Introductions

- James Lyneis
- S.B.s MIT – EE and System Dynamics ('71)
- PhD – Univ. of Michigan (Mgt. Science; '74)
- Sloan Faculty '74-'78; Senior Lecturer '98-Present
- Professor of the Practice, WPI '02-Present
- Consultant with Pugh-Roberts Associates, '78-'02, working on many project models

4

Introductions

- Dan Braha
- Professor at University of Massachusetts
- Sabbatical at MIT ESD in AY 2012/13
- Affiliation with New England Complex Systems Institute (NECSI)
- Specializes in Complexity research, with application to complex projects and human organizations

The Structure and Diffusion Dynamics
of Large Scale Organizational

Networks Social Network Analysis of
ProductDesign and Development
Organizational Networks

5

Introductions

- Let's go around the room & remote sites
- Name
- Company and Job (present or past)
- One observation on project management

6

Project Definition

- A Project is a set of tasks that
- Are related to each other
- Have a specific objective to be completed within certain specifications
- Have defined start and end dates
- Have funding limits

□ Consume resources

7

The “Iron Triangle”

Cost Scope

Project

□ Why “iron” triangle?

Schedule □ Risk if all three are
constrained !

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System Definition

□ A System is a set of physical or virtual objects
whose interrelationships enable desired
function(s).

□ more than the sum of its parts

□ Undesired (emergent) functions often exist

□ System complexity scales with the number of
objects as well as the type and number of
interconnections between them

□ Instantaneously available functions, versus
“lifecycle” properties (scalability, flexibility,
robustness ...)

□ A Product is a “System” sold for profit

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Example System: F/A-18 Aircraft

□ Clean sheet design ~ 1978 F/A-18 A/B

□ Re-designs: C/D (1987), E/F (1999)

□ Hardware, Software, Humans ...

□ What is inside the system boundary?

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Project Management

□ Project Management comprises a body of
methods and tools that facilitate the achievement of
project objectives

□ Within time

□ Within cost

□ Within scope

□ At the desired performance/specification level

□ While effectively and efficiently utilizing resources

☐ While carefully managing risks and opportunities

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Research and Development

R D

Research, Technology Product/System

Development Development

☐ Unstructured methods ☐ Structured methods

☐ Difficult to plan ☐ Generally planned

☐ Predictable

☐ Unpredictable

Our focus is on

downstream

development.

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+ Discussion Point 1:

- Why is complex (S)PM hard?

☐ Main obstacles to easy project success:

☐ Poorly defined project objectives or shifting system requirements

☐ What do you think? ...

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Concept Question 1 (DEMO)

☐ A project is

☐ A - ongoing management of facilities

☐ B - a finite undertaking with a specific goal

☐ C - task-based design

☐ D - doomed to failure from the outset

☐ E - all of the above

☐ F - none of the above

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Relationship w/other SDM core classes

☐ System Architecture (ESD.34) is about the “DNA” of the ARTIFACTS themselves – atomic unit: object

☐ Concept, form, function, decomposition ...

☐ Systems Engineering (ESD.33) is about the PROCESSES to understand and design systems – atomic unit: process

☐ QFD, DOE, Requirements Analysis and Verification, ...

☐ Integrating the Lean Enterprise (ESD.61J) is about the

PEOPLE and ORGANIZATIONS – atomic unit: person

□ Principles of lean manufacturing, organizational models

□ System Project Management (ESD.36) is about how to best utilize resources to implement a set of objectives – atomic unit: task

□ CPM, DSM, System Dynamics

15

Task as an Object-Process-Diagram

System

Inputs n r Elements

o Task m

Resources Executing Deliverables

p q

Time

Personnel Facilities

Money (\$)

Consumables Using OPM Nomenclature

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Ref: OPM, D. Dori, 2002

Project = set of related tasks

m

o A n

r

m

q

p

B n

r

m

q D

n

r

q

F m

Deliverable from A

is an input to C B and D use the p q

same people

r

m

C n r F requires deliverables

From D and E

p q E m

p

C and E need access

to the same facility

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+ System Project Management

ESD.36 Framework

Project

Preparation Project Project

Planning Monitoring

Project

Control

Enterprise has

chosen what product Project

or system to develop Adaptation Project

Completion

Next

Project Project

Learning

Start

18

- Course Objectives

□ Introduce advanced methods and tools of Project Management in a development context

□ CPM/PERT, Critical Chain, Design Structure Matrix

□ System Dynamics

□ Earned Value Management

□ Understand how methods work (strengths, limitations)

□ Industry Examples

□ Gain appreciation for organizational and human aspects

□ Case Studies

□ Managing International Projects, Portfolios of Projects ...

□ Learn from each other

□ Class Discussions

□ Project Assignments

□ □ Improve development projects in your career/firm

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+ Class

Schedule

(next 7 weeks, refer to syllabus)

Business Trip 20

Readings

- Required Readings
- NO Paper Class Reader Packet
- Read ahead of lecture ~ 1-2 papers/chapters per session
- Check reading assignments in the syllabus
- Next:
- MIT Press Book Chapter: Introduction to PM
- ABCs of the Critical Path Method (1963)
- Optional Readings
- Textbooks
- Available at MIT Library (Dewey)
- Purchase only if you think useful beyond class
- (e.g. MIT COOP, amazon.com etc)

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Draft Textbook

- “Successfully Designing and Managing Complex Projects”
- de Weck, Lyneis
- MIT Press, draft in development
- textbook to support SDM core class ESD.36
- current draft ~ 300 pages

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Case Studies

- Case 1: Civil Engineering Project
- BAE Systems: DIA Baggage Handling System
- HBS-9-396-311
- or “live case” BP Wind Farm Development
- Case 2: Aerospace Project
- Raise and Fall of Iridium (or Mission to Mars HBS-9-603-083)
- HBS-9-601-040
- Presented by Prof. Joel Schindall
- Case 3: Software Project
- Microsoft .Net (and/or Microsoft Office 2000)
- HBS-9-602-086
- Presented by Prof. Paulo Gomes
- Case 4: Voted by Students

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Project Assignment

- Apply Design Structure Matrix (DSM) method, generally at your sponsor company site
- System Dynamics Project (incl. simulation)
- Survey of Methods & Tools in company
- Analyze Success or Failure of a significant Past Product/System Development Project
- Work in teams of 4 (nominally)
- 1-page project proposals due on 9/18
- Get approval by 9/25
- Project Update due on: 10/25
- Final presentation in class on December 4 or 6

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Previous Project Examples

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
36 37 38 39 40 41

Marketing & Competition Research x 1 1

Cost Risk Analysis to Bid 1 x 1

Customer Requirement Negotiation 1 1 x

DSM

Technical Screening for Qualification 1 1 1 x

Project Authorization & Kick off 1 1 1 x

Failure Effect Mode Analysis x

Control of Engineering Changes

Project

Design for Catalyst Volume & Emissions 1 1 1 1 1

CAE for Catalyst Volume & Emission 1 1 1 x 1

Design for Tuning Volume, Pressure & Flow 1 1 1 x 1 1

CAE for Tuning Volume, Pressure & Flow 1 1 1 x 1

Design for Legal Noise and Sound Quality 1 1 1 x 1 1

CAE for Legal Noise and Sound Quality 1 1 1 x 1

Design for Durability Target 1 1 1 x 1 1 1

CAE for Durability Target 1 1 1 x 1 1 1

Cost Analysis for Manufacturing 1 1 1 1 x 1

Exhaust

Prototype for DV Test & Pre-prod Build 1 1 1 1 1 1 1 1 x

Design Verification Test 1 1 x

Testing for Emission 1 1 x

Testing for Tuning Volume, Pressure & Flow 1 1 x

System

Testing for Legal Noise and Sound Quality 1 1 x

Sub-tier Supplier DV Bench Testing 1 1 1 1 1 x
 Vehicular Proving Ground Testing 1 1 1 1 1 1 1 1 1 1 1 1 x
 Data Analysis and Editing 1 x
 Design Sub-System Durability Testing
 Design Release
 Release of Accurate 3D Geometry
 1 1
 1
 1
 1
 1
 1
 1
 1
 1
 1 1 x
 x
 x
 1
 Issues Sales Authority to Release (SAR) 1 1 x
 Creation and Release of all 2D drawings 1 1 1 x
 Mfg Facility Lineup & Tooling Process 1 1 x 1 1
 Mfg Cell Setup and Fine Tuning 1 1 1 x 1 1 1
 PSO & "Run-at-Rate" Demonstration 1 1 1 1 x
 Quality Control for Manufacturing 1 1 1 1 1 x
 Mfg Process Evaluation 1 1 1 1 1 1 x
 Product Verification Testing 1 1 1 1 1 1 1 x
 Sub-tier Supplier PV Testing 1 1 1 1 1 1 1 1 x
 Production Part Approval Process 1 1 1 1 x
 Production 1 1 1 x
 Warranty Cost Analysis 1 1 1 1 1 1 1 1 1 1 x
 Customer Satisfaction Evaluation 1 1 1 1 1 1 1 1 1 x
 - Understand Iterations
 - Reduce Expected Project Duration
 25
 Understanding the sources of change --
 External factors; management responses; side effects
 Out-of-Sequence
 Work, Worksite
 Congestion,
 Coordination

Fatigue, Problems,
Burnout Morale Problems
Average P eo p le P ro d u ctivity Q u ality
Employee Skill
and Quality P ro g ress
W o rk W o rk
Overtime To Be Done R e a lly D o n e
Scope Customer
Growth Changes Schedule
Know n U n d is c o v e r e d
Acceleration
R e w o rk R e w o rk
R ew o rk
D isco very
Hiring
Apparent
Progress
26
+ Understand the dynamics of a cancelled
project at NASA
Baseline Plan Staffing
FY2000 Replan
700
Actuals Through FY02 & Projections FY02-
FY06 (FY2000 Re-plan)
600 Model for Baseline
Project
Cancelled
500
September
2002
400
Staff - EP
300
200
FY00 Re-plan
First Launch
100 Baseline First
Hypergolic Launch
Maintenance

0 Facility Set

Operational

M1-3

M4-6

M7-9

M10-12

M13-15

M16-18

M19-21

M22-24

M25-27

M28-30

M31-33

M34-36

M37-39

M40-42

M43-45

M46-48

M49-51

M52-54

M55-57

M58-60

M61-63

M64-66

M67-69

M70-72

M73-75

M76-78

M79-81

M82-84

M85-87

M88-90

M91-93

M94-96

M97-99

M100-102

M103-105

M106-108

M109-111

M112-114

M115-117

M118-120

M121-123

M124-126

M127-129

M130-132

FY1996 FY1997 FY1998 FY1999 FY2000 FY2001 FY2002 FY2003 FY2004 FY2005 FY2006

Sep 4, 2008

Time 27

Discussion Point 2: Failure?

- ☐ What can lead to projects failing ?
- ☐ What is success/failure?
- ☐ Project Manager is unqualified and overwhelmed.
- ☐ What else...

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Homework Assignments

- ☐ 6 Individual assignments,
- but can cooperate
- (acknowledge !)
- ☐ Don't spend more than
- ~10-15 hours per HW !

Electrical CityCar

Design Project

HW1: Critical Path and Network of Tasks

HW2: Design Structure Matrix and Iterations

HW3: System Dynamics – Initial Model

HW4: Budgeting and Earned Value

HW5: System Dynamics – Brook's Law

HW6: Project Organizational Design

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Grading

- ☐ Homeworks 60%
- ☐ Project Assignment 25%
- ☐ Active Participation 15%
- ☐ Total 100%
- ☐ Each HW counts 10%
- ☐ All project team members receive same project
- grade, work together
- ☐ People in this class do get A,B,C ...even F ... you want
- a good grade? ... you have to earn it !

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The Course Site

- Getting Started

- SDM students should add themselves to the class list (if not already there)
- Non-SDM students get permission from instructors
- Contact TA via email to be added
- Course number is ESD.36
- Make sure you go to the Fall 2012 version

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The Course Site

- Main features

- Syllabus (under > Materials > General)
- Calendar
- Schedule
- Handouts
- Go to “Materials” (store all shared files)
- will be posted before each lecture
- Homework
- Submit assignments through the course site
- Receive comments and grades
- FAQ / Forums
- Forum will be used for various topics

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- Remember

- Read the syllabus!
- Answers many of your questions
- Take a look at what’s already uploaded
- Reading for next class
- Sign up on the course site if needed
- Non-SDM students need permission of instructor to enroll.
- Use your name cards every class.
- PLEASE ... okay, pay attention
- Laptops needed to answer concept questions
- IM TA’s during lecture from remote sites if problems surface

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Critical Path Analysis 14.1

14

Critical Path Analysis

LEARNING OBJECTIVES :

After studying this unit you will be able to :

I Explain the generic term network analysis

I Draw diagram & chart representing inter-relationships between various elements of the project and finally establishing critical path

I Identify Dummy Activities

I Identify earliest and latest event times of each event. Identify different types of Float

Help in Decision Making about project times and resource allocation Calculating probability of completing network in a given time

14.1 INTRODUCTION

Planning, Scheduling and Controlling are three important functions of management. Planning involves the formulation of objectives and goals that are subsequently translated into specific plans and projects. Scheduling is concerned about the implementation of activities necessary to achieve the laid down plans. The function of control is to institute a mechanism that can trigger a warning signal if actual performance is deviating (in terms of time, cost and some other measures of effectiveness) from the plan. If such a deviation is unacceptable to the concerned manager, he will be required to take corrective action to bring performance in conformity with the plans. The PERT and CPM models are extremely useful for the purpose of planning, scheduling and controlling the progress and completion of large and complex projects or for carrying out the analysis of these three managerial functions. Before we describe the basic concepts used in the construction and analysis of these models, let us first understand the meaning of a project.

What is a project?

A project can be defined as a set of large number of activities or jobs that are performed in a certain sequence determined logically or technologically and it has to be completed within (i) a specified time, (ii) a specified cost and (iii) meeting the performance standards. Examples of a project from fairly diverse fields are given below:

1. Introducing a new product in the market.
2. Construction of a new bridge over a river or construction of a 25 storied building.

14.2 Advanced Management Accounting

3. Executing a large and complex order on jobbing production.
4. Sending a spacecraft to the mars.

14.2 GENERAL FRAMEWORK OF PERT/CPM

A network is a graphical representation of a project, depicting the flow as well as the sequence of well-defined activities and events. Developed during the 1950s, both CPM (Critical Path Method) and PERT (Programme Evaluation and Review Technique) are network techniques/models. The network approach helps project managers in planning, scheduling and controlling. As a planning tool it helps the manager to estimate the requirements of resources viz., materials, equipment, manpower, cost and time for each activity or tasks of the project. This approach cannot make decisions by its own. It only provides additional information to executives to facilitate decision making process. Also it does not provide solution to every management problem. It certainly helps in identification of those activities, jobs or events which control the completion of the project.

The working methodology of critical path analysis (CPA) which includes both CPM and PERT, consists of following five steps:

1. Analyse and break down the project in terms of specific activities and/ or events.
2. Determine the interdependence and sequence of specific activities and prepare a network.
3. Assign estimates of time, cost or both to all the activities of the network.
4. Identify the longest or critical path through the network.
5. Monitor, evaluate and control the progress of the project by replanning, rescheduling and reassignment of resources.

The central task in the control aspect of these models is to identify the longest path through the network. The longest path is the critical path because it equals the minimum time required to complete the project. All other paths other than the critical path (i.e. non-critical or slack paths) offer flexibility in scheduling and transferring resources, because they take less time to complete than the critical path.

14.3 ADVANTAGES OF CRITICAL PATH ANALYSIS

There are a number of advantages in using critical path analysis.

1. It allows for a comprehensive view of the entire project. Because of the sequential and concurrent relationships, time scheduling becomes very effective. Identifying the critical activities keeps the executive alert and in a state of preparedness, with alternative plans ready in case these are needed. Breaking down the project into smaller components permits better and closer control.
2. Critical path analysis offers economical and effective system of control based on the principle of management by exception i.e. need for corrective action arises only in exceptional situations and in most of other cases, performance is in conformity with the plans.
3. It is a dynamic tool of management which calls for constant review, a reformulation of

Critical Path Analysis 14.3

the network, and finding the current path of relevance and optimum resources allocation.

14.4 FUNDAMENTALS OF A CPA NETWORK

◆ Activity

An activity is any portion of a project which consumes time or resources and has a definable beginning and ending. For example, "laying of pipe" is an activity requiring the use of resource mainly effort. Activity may involve labour, paper work, contractual negotiations, machinery operations, etc. Commonly used terms synonymous with "activity" are "task" and "job".

Tail Head

7

1 3

Figures 1 and 2

10

3 4

14

14

11 24

3 5

7

1 4 6

2

Figures 3 to 6

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Activities are graphically represented by arrows, usually with description and time estimates written along the arrows. The tail of the arrow portraying an activity represents the starting point of the activity and its head represents its completion. The arrow may be straight slanting, or bent but not broken (see figure-1). The arrow is not a vector and need not be drawn to scale.

◆ Events

The beginning and ending points of an activity or a group of activities are called events. Synonyms of an event are "node" and "connectors". An event is often represented graphically by a numbered circle (see figure-2), although any geometric figure such as square, oval, rectangle etc. will serve the purpose. We shall, however, stick to the most commonly used convention for representing an event viz, the circle. A few examples of events are as follows: (i) Material procured, (ii) Design completed, (iii) Project started, (iv) Bricks laid, etc. All activities in a network must commence from some event. Such events are called the tail events because they are connected to the tail of an activity. These are shown in figure 3. Similarly, all activities in a network must have terminal points called the head event, because it is at the head of an activity. These are shown in figure-4. Figure-5 depicts the tail and head events connected by arrows representing activities i.e. it depicts the dual role of an event. Event 14 is the head event for one activity and tail event for another.

In a network, symbol "i" is used for the tail event (also called preceding event) and "j" for the head event (or succeeding event) of an activity. The activity, then being i-j.

If an event represents the joint completion of more than one activity, it is called a merge event. If an event represents the joint initiation of more than one activity, it is called a burst event.

A network is, then, a graphical representation of a project plan, showing the inter-relationship of the various activities. Networks are also called arrow diagrams (see figure-6). When the results of time estimates and computations have been added to a network, it may be used as a project schedule.

14.4.1 Conventions adopted in drawing networks: There are two conventions normally adopted while drawing networks. In the early stages of network drawing, it is suggested that the conventions should be respected until sufficient experience has been gained to justify dropping them. These conventions are:

(a) Time flows from left to right.

(b) Head events always have a number higher than that of the tail events.

The above stated conventions allow activities to be referred uniquely by their tail and head event numbers, so that "activity 3-4" means only "the activity which starts from event 3 and proceeds to event 4"; it cannot mean "the activity which starts from event 4 and finishes at event 3".

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14.4.2 Graphical representation of events and activities: Events are represented by numbers within circles. Activities are represented by arrows, the arrow-heads represent the completion of the activities. The length and orientation of the arrow are of no significance whatsoever (chosen only for the convenience of drawing). The activity of leaving place A and walking to place B can equally well be represented by figure-7.

14.4.3 Fundamental properties governing the representation of events and activities: The representation of events and activities is governed by one simple dependency rule which requires that an activity which depends upon another activity is shown to emerge from the head event of the activity upon which it depends and that only dependent activities are drawn in this way. Thus, if activity B depends upon activity A, then the two activities are drawn in figure-8.

The dependency rule gives rise to two fundamental properties of events and activities:

1 2 1

1 2

2

2

1 2

1

Figure 7

A B

1 2 3

Figure 8

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1. An event cannot occur until all activities leading to it are complete.
2. No activity can start until its tail event is reached.

The above two properties can be combined into a single one, namely that "no activity may start until all previous activities in the same chain are completed.

14.4.4 Logical sequencing and connection of activities: A project entails several activities. The arrows are arranged to show the plan of logical sequence in which the activities of the project are to be accomplished. The sequence is ascertained for each activity by answering the following three queries viz:

- (i) Which activity or activities must be completed before the start of a particular activity?
- (ii) Which activity or activities should follow this?
- (iii) Which activities can be accomplished simultaneously?

The activity or activities which immediately come before another activity without any intervening activities are called predecessor activities to that activity. The activities which follow another activity without any intervening activities are called successor activities to that activity.

In a project of laying a pipe line, the three activities involved may be trenching, laying pipe and welding pipe. To decide the logical connection between these three activities, it is necessary that they be carried out in series, the reasoning being that the pipe cannot be laid until trenching has been done and welding cannot be undertaken until the pipe has been laid. This way we decide the logical sequencing between different activities.

Errors in logical sequencing: Two types of errors in logic may arise while drawing a network, particularly when it is a complicated one. These are known as looping and dangling.

(1) Looping: Normally in a network, the arrow points from left to right. This convention is to be strictly adhered, as this would avoid illogical looping, as shown wrongly below:

1 2 3 4

Figure 9

Critical Path Analysis 14.7

(2) Dangling: The situation represented by the following diagram is also at fault, since the activity represented by the dangling arrow 9-11 is undertaken with no result.

A

8 9 10

11

Figure 10

To overcome the problem arising due to dangling arrows, following rules may be adopted.

- (i) All events, except the first and the last, must have at least one activity entering

and one activity leaving them.

(ii) All activities must start and finish with an event.

(3) Duplicate activities: Consider the following figure 11:

A

x Y 8

6 7

B

Figure 11

In the above figure, activities A and B may be called duplicate activities because they have same head event (i.e 6) and the same tail event (i.e.7). One remedy for such a situation is the introduction of a dummy activity.

(4) Dummy activity: It is a hypothetical activity which consumes no resource and time. It is represented by dotted lines and is inserted in the network to clarify activity pattern under the following situations:

(i) It is created to make activities with common starting and finishing events distinguishable.

(ii) to identify and maintain the proper precedence relationship between activities that are not connected by events.

14.8 Advanced Management Accounting

(iii) to bring all "loose ends" to a single initial and a single terminal event in each network using dummies, if necessary.

For example, problem of duplicate activities in the figure-11 above may be circumvented as shown in figure-12.

A

x Y

6 8

B

7

Figure 12

Figure-13 shows three cases for the following set of dependency relationships:

Activity C is dependent upon both A and B.

Activity D is dependent upon A alone.

1 4 2 B 3 C 4

A C

3 A

B D A

1 D 5

3

2 5

B C

2 3 5

A D

1 4 6

Figure 13

The first portrayal (on top left of figure-13) is clearly wrong since it shows D as dependent upon not only A but also B which is not desired. The other portrayal (ii) is also wrong since A is being shown twice and thus contravenes the fundamental axiom of network that there must be one arrow for each activity. The way out to this dilemma is the representation by means of the dummy activity. In the third portrayal of figure -13, C is dependent upon both A and B (via dummy) whereas D is dependent upon just A.

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14.4.5 Numbering the events: The event numbers in a network should in some respect reflect their logical sequences. When a complicated network has been drawn then the problem of assigning numbers to the events involved in the network arises. A rule devised by D.R. Fulkerson, involving the following steps may be followed to resolve the problem of numbering the events.

- (i) An "initial" event is one which has arrow/arrows coming out of it and none of the arrow entering it. In a network there will be only one such event. Call it "1".
- (ii) Delete all arrows coming out from the event 1. This will give us at least one more "initial event".
- (iii) Number these events as "2, 3...."
- (iv) Delete all emerging arrows from these numbered events which will create new initial events. Then follow step (iii).
- (v) Continue the above steps till last event is obtained which has no arrows coming out of it. Consider the numbering of events in the following figure.

y 4

1 2 5 6 7 8

3

x

Figure 14

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A F A F

B B G

C G H

C

A F

A F

B G

B G

C H

C H

A F

A F

B G

B G

C H

C H

Figure 15

Critical Path Analysis 14.11

Here we proceed from left to right. The event with least x-co-ordinate is assigned the smallest integer, say 1. Other events are assigned progressively higher integers with regard to x-co-ordinate. If two or more events (4 and 5 above) have the same x-co-ordinate, the one towards arrow should have higher number.

Furthermore, it is not necessary, and in fact also not desirable to number the events consecutively. It would be a better scheme to number the events as 10, 20, 30, 40, 50, 60, 70, in the above diagram instead of 1,2,3,4,5,6,7. This affords insertion of more activities and events omitted by oversight or having become necessary in view of certain logic revisions. It was mentioned earlier that it is desirable that all the activity arrows point from left to right. If the arrow is vertical it may point downwards or upwards.

For the sake of presentability it is to be recommended that activities emanating from one event or converging to another may make as great angles between themselves as possible.

A few more conventions are given below:

(i) Keep the arrow to the extreme right. (ii) As far as possible avoid drawing arrows that cross each other. Usually by suitable 'stretching' the network diagram it is possible to avoid this. (iii) Where, however, crossing is unavoidable, bridging may be done. This applies to dummies as well. Draw boldly a big network. Smaller ones are confusing. Use of pencil and rubber is recommended.

Exercise : Depict the following dependency relationships by means of network diagrams.

The Alphabets stand for activities.

1. A and B control F; B and C control G.
2. A and B control F; B controls G while C controls G and H.
3. A controls F and G; B controls G; while C controls G and H.
4. A controls F and G; B and C control G with H depending upon C.
5. F and G are controlled by A, G and H are controlled by B with H controlled by B and C.
6. A controls F, G and H; B controls G and H with H controlled by C.

Answer. The required networks are given in figure-15

B E

H

A C F

G

D

Figure 16

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J K L

M

Figure 17

F G

A B C D E

H J L

M

I K

Figure 18

Exercise : Find out the superfluous (unnecessary) dummy activities in the networks below.

14.4.6 Basic steps involved in drawing a CPM/PERT network : Network is defined as a diagram representing the activities and events of a project, their sequence and inter-relationships.

The basic steps involved in drawing a network are:

- (i) Breaking up of the entire project into smaller systems known as tasks.
- (ii) For each task ascertain the activities and events to be performed.
- (iii) For each activity determine the preceding and succeeding activities.
- (iv) For each activity determine or estimate the time and other resources needed.
- (v) Draw a network depicting the assembly of tasks into a project.

14.4.7 Network Construction

Illustration

The activities involved in the computer installation process are detailed below. You are required to draw the network.

Activity Predecessor Activities

- A. Physical Preparation None
- B. Organisational Planning None
- C. Personnel Selection B
- D. Equipment Installation A
- E. Personnel Training C

Critical Path Analysis 14.13

- F. Detailed Systems Design C
- G. File Conversion F
- H. Establish Standards and Controls F
- I. Programme Preparation H
- J. Programme Testing I

K. Parallel Operations D, E, G, J.

L. Finalise Systems Documentation I

M. Follow up. K, L

A A B < C (ii)

NONE < A

(i)

NONE < B B

B

C

D

D

A A C < E (iv)

A<D C<F

B (iii) B

C C

E

D J F

F<G

F<H

A (v)

E G I

H<I

B C F H I <J

K D, E, G and

J < K (vi)

D J

E G

A I

B C F H

Figure 19

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Solution

The network is shown in figure 19. The construction has been carried out step by step which is explained below. The symbol '<' is to be read as "precedes" in these figures.

(i) Since A and B are preceded by no activity, they are shown emanating from the start event. We do not put arrows and the head events for A and B because they have been drawn as such only tentatively in the following step (s) their position may have to be changed.

(ii) Here C is shown as dependent upon B. Again we neither show arrow and nor the head event of C because its position is tentative.

- (iii) D is shown as dependent upon A.
- (iv) We show here C preceding E and F.
- (v) F is shown preceding G and H, H precedes I and I precedes J.
- (vi) K is shown as dependent upon D,E,G and J, the latter four therefore, are made to converge on the same head event.
- (vii) L is shown as dependent upon I and M as dependent on K and L. Note carefully that K and L are the terminal activities and, therefore, converge on the last event.

About this point the student is cautioned since the beginners do tend to show the last activities dangling and do not converge them to the common last event.

Of course, the network would be constructed in just one diagram (as shown in Figure-19).

We drew so many for explanation only. However, for a stylish presentation it is desirable that this one diagram is suitably sketched i.e. faired up in another diagram. The student may number the events as an exercise.

M

K I < L

L < M

L

D J

E G

A I

B C F H

Figure 19 contd.

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Example : Construct the network for the following activity data:

Activity Preceded by

A None

B None

C A

D B,C

E D

F E

G B,C

H F

I F,G

J H,I

K B

L F,G,K

NONE < A C

A

NONE < B

B

A A < C

B

D

E F

C

D

A B < D C

C < D D < F

B A E < F

B

H

E F

D

B, C < G

C

F < H

G

A

B Figure 20

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H I

E F < H, I

F

G < I

D

C G

A

B

H

I J

I, H < J

E F

D

C G

A

B

J

0 1

$B < K$

$F, G, K < L$

- .

L

+ /

* K

Figure 20 (Contd.)

(The students may remember the events as an exercise)

Critical Path Analysis 14.17

Solution: The network is shown in figure-20 step by step with dependency relationships that are incorporated at each step which the student should, by now, be able to follow.

Please carefully review the steps where dummies are incorporated.

14.4.8 Concurrent activities: Activities may not always be discrete i.e. they may be done in part allowing the subsequent activities to commence before the preceding activity is fully completed. Activities of this kind are to be frequently encountered in batch production. If, for example, a batch of 50 spindles is to be processed on two machines obviously it is not necessary to process all the items of the batch on the first machine and then transfer these to the next machine. A few items processed on the first machine may be transferred to the second machine before completion of the entire batch on the first machine. Since this is a matter of great practical importance we shall dwell upon it at a greater length. Such simultaneous or concurrent activities are to be encountered in sewage work e.g., trenching, laying pipe, welding pipe and back filling, all going on simultaneously with suitable lags on construction work.

Illustration

A batch of 4 axles is to be proceeded on the following three machines in this sequence:

Lathe (L), Milling (M) and Grinding (G)

Instead of first working on these 4 axles on lathe and then on milling and finally on grinding in sequence, it is desired to process the first axle on the lathe and as and when it is processed, it is taken up on milling and the 2nd axle on the lathe, and so on. In other words, each of the three activities L, M and G have been quartered for the sake of concurrent operations. You are required to draw the network.

Solution

The dependency relationships are sorted out hereunder:

Quartered Activity Preceded by

L1 None

L2 L1

L3 L2

L4 L3

M1 L1

M2 L 2, M 1

M3 L 3, M 2

M4 L 4, M 3

G1 M1

G2 M 2, G 1

G3 M 3, G 2

G4 M 4,G 3

The network is now constructed below. (The student is urged to draw his own network. Mere understanding of our solution is not enough.)

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1 2 4 8

L1 L2 L3 L4

3 5 6 9 10 12

M1 M2 M3 M4

7 11 13 14

G1 G2 G3 G4

Figure 21

(N.B.: The concurrent activities so drawn are known: as 'ladders' in the network jargon.)

Illustration

Draw a network diagram for the following data.

Task Immediate Predecessor

A —

B —

C B

D B

E B

F E

G A, D,C.

Solution

The required network is given below.

4

A G

C

B 2 D

1 3 6

E

F

5

Figure 22

Critical Path Analysis 14.19

Illustration

Draw a network diagram for the following data.

Activity Preceding activities

A —

B A

C A

D B

E A

F B, E

G C

H D, F

I G

J H, I

Solution: The required network is given below:

5

E F

1 A B D H J

2 4 6 8 9

C I

3 G

7

Figure 23

14.5 CRITICAL PATH ANALYSIS

You have already been familiarised with the logic of the critical path analysis by way of introduction of this chapter.

The purpose of the analysis is two-fold: (i) to find the critical path, i.e. the sequence of activities with the longest duration. Once it is found it is marked in bold sequence of arrows on the network. For a simple network as of figure -24 the various sequences can be enumerated and the durations of activities encompassed by them simply added, to find the critical sequence. As stated earlier, one could indeed end up with more than one critical sequence; and (ii) to find the float associated with each non-critical activity.

♦ Systematic analysis: The enumeration method would be too cumbersome computationally for any real life project, Even the computer would be hard pressed to proceed this way. There is a systematic way that cuts short the analysis time to manageable proportions.

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It is accomplished by performing the following steps:

1. Calculate the time schedule for each activity. This represents the time by which an activity must begin and the time before which it must be completed. The time schedule data for each activity includes, the earliest start, earliest finish, the latest start, latest finish, and finally the float, which is the spare time associated with an activity.
2. Calculate the time schedule for the completion of entire project. This represents the

scheduled date for the completion of the entire project and the probability of completing the project on or before the deadline.

3. Identify the critical activities. These activities are the ones which must be started and completed on schedule, or else the project is likely to be delayed.

4. Determine the critical path for the network. This path represents the critical activities which must be closely followed in order to complete the project on time.

14.5.1 Scheduling computations: After preparing the network diagram, we wish to know how long it will take to complete the project and also to identify the activities in the network that are to be placed under strict control. The basic scheduling computations involve a forward and then a backward pass through the network. The process of tracing the network from START to END is called forward pass, and from END to START is called backward pass. Based on a specified occurrence time for the initial network event, the forward pass computations give the earliest start and finish times for each activity. By the specification of the latest allowable occurrence time for the terminal network event, the backward pass computation will give the latest allowable start and finish times for each activity.

14.5.2 Forward pass computations: As stated above, the purpose of the forward pass is to compute the earliest start (EST) and finish time (EFT) for each activity. The EST time indicates the earliest time that a given activity can be scheduled. Earliest finish time for an activity indicates the time by which the activity can be completed, at the earliest. To compute these time estimates, we will first of all compute the earliest allowable occurrence time for various events of the network.

It is a convention to keep the earliest allowable occurrence time of the START event as zero. To understand, how this time estimate for other events is computed, let us consider the following network diagram.

E3 = 14

3

8 6

6 16

1 2 0 5 6 E6 = 52

E1 = 0 10 20 E5 = 36

E2 = 6

4

E4 = 16

Figure 24

Critical Path Analysis 14.21

◆ Earliest allowable occurrence time of an event or Earliest Event Time :

In the network shown above, event 1 stands for the beginning of the activity 1 - 2 and we can say that it occurs at the time zero i.e. $E1 = 0$. Event 2 stands for the finish of the activity 1-2 thus event 2 can occur at the earliest time $E2$ which is computed as

$$E2 = 0 + D12 = 0 + 6 = 6$$

where $D12$ stands for the duration of activity 1 - 2

Event 3 stands for finish of the activity 2-3 and its earliest time is

$$E_3 = E_2 + D_{23} = 6 + 8 = 14$$

The event 4 can occur either at the end of the activity 3 - 4 or at the finish of activity 2 - 4. In this case, there will be two time estimates as follows:

$$E_4 = E_3 + D_{34} = 14 + 0 = 14$$

$$E_4 = E_2 + D_{24} = 6 + 10 = 16$$

In case two or more time estimates exist for a particular event, then the time estimate with maximum value is retained as the earliest event time and other values are discarded. This maximum value represents the completion of all the activities ending at the event under consideration. In the above example, the earliest event time for event 4 will be 16.

A general rule can also be given here for determining the earliest event time as below:

$$E_j = \text{Max} (E_i + D_{ij})$$

where E_j is the earliest time for event j , E_i is the earliest time for event i , and D_{ij} is the duration of the activity $i-j$.

♦ Earliest start and earliest finish times of an activity

After computing the earliest event time of various events, one can easily compute the earliest start and finish times of all the activities on the network. The earliest start time of an activity is given by the earliest allowable occurrence time of the tail event of that activity. Thus, in our example, the earliest start time of the activity 1-2 will be given by the earliest time of the event 1 i.e. it will be 0. The earliest start time for the activities 2 - 3 and 2 - 4 will be given by the earliest time of event 2 which is equal to 6. The earliest time for the activities 3-4 and 3-5 will be 14 which is the earliest time for the event 3 and so on.

The earliest finish time of an activity will be simply equal to the earliest start time of the activity plus the duration of that activity. Hence, in our example, earliest finish time of activity 1-2 will be $0+6 = 6$, for activity 2-3, it will be $6+8 = 14$ and for activity 2-4, it will be $6+10 = 16$ and so on.

The complete computations for all the activities are shown in columns (3) and (4) of table -1 given on the next page.

(*Note: In various networks we have omitted subscripts for E_i 's and L_j 's)

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14.5.3 Backward pass computations: The purpose of the backward pass is to compute the latest start and finish times for each activity. These computations are precisely a "mirror image" of the forward pass computations. The term "latest allowable occurrence time of an event" (denoted by L_i) is used in the sense that the project terminal event must occur on or before some arbitrary scheduled time. Thus, the backward pass computations are started rolling back by arbitrarily specifying the latest allowable occurrence time for the project terminal event. If no schedule date for the completion of the project is specified, then the convention of setting the latest allowable time for the terminal event equal to its earliest time, determined in the forward pass, is usually followed, i.e. $L = E^*$ for

the terminal event of the project. This convention is called the zero slack convention. Following this, one can also interpret the latest allowable activity finish time (LFT) as the time to which the completion of an activity can be delayed without directly causing any increase in the total time to complete the project.

To explain the computation, let us again consider the network diagram in figure-24. The terminal event is 6 so we set $L_6 = E_6 = 52$ and we start rolling back. The latest allowable occurrence time for the events 5 and 4 are $L_5 = 52 - 16 = 36$ and $L_4 = 36 - 20 = 16$ respectively. It may be noted here that we can roll back to event 3 via activity 3-5 as well as activity 4-3. So there are two latest allowable occurrence times for the event 3 as given below;

$$L_3 = L_4 - D_{34} = 16 - 0 = 16$$

$$L_3 = L_5 - D_{35} = 36 - 6 = 30$$

We retain the minimum value as the latest occurrence time for the event 3 and ignore other values. Therefore, the latest allowable occurrence time for the event 3 is 16. Similarly

$$L_2 = L_3 - D_{23} = 16 - 8 = 8$$

$$\text{and } L_2 = L_4 - D_{24} = 16 - 10 = 6$$

The latest occurrence time for the event 2 is thus 6 and the latest occurrence time for the event 1 is equal to its earliest time i.e. zero.

In general, the latest allowable occurrence time of an event can be calculated by selecting an appropriate formula among the following two:

$$L_i = L_j - D_{ij}$$

$$\text{or } L_i = \text{minimum } (L_j - D_{ij})$$

The second formula is used for the event having two or more latest allowable occurrence time estimates.

♦ Latest start and latest finish times of an activity

After computing latest allowable occurrence time for various events, one can compute the latest start and latest finish times of an activity. The latest finish time of an activity is equal to the latest allowable occurrence time of the head event of that activity.

$$\text{i.e. } LFT(i - j) = L_j$$

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The latest start time of an activity is equal to its latest finish time minus its duration.

$$\text{i.e. } LST(i - j) = LFT(i - j) - D_{ij}$$

These computations are shown in column (4) and column (6) of table-1 given below:

Table 1

START FINISH

Activity Duration Earliest time Latest time Earliest time Latest Time

(1) (2) (3) (4) (5) (6)

1-2 6 0 0 6 6

2-3 8 6 8 14 16

2-4 10 6 6 16 16

3-4 0 14 16 14 16

3-5 6 14 30 20 36

4-5 20 16 16 36 36

5-6 16 36 36 52 52

14.5.4 The critical path determination: After having computed various time estimates, we are now interested in finding the critical path of the network. A network will consist of a number of paths. A path is a continuous series of activities through the network that leads from the initial event (or node) of the network to its terminal event. For finding the critical path, we list out all possible paths through a network along with their duration. In the network under consideration, various paths have been listed below.

Path Length (in days)

1-2-3-5-6 36

1-2-4-5-6 52

1-2-3-4-5-6 50

◆ Critical path: A path in a project network is called critical if it is the longest path. The activities lying on the critical path are called the critical activities.

In the above example, the path 1-2-4-5-6 with the longest duration of 52 days is the critical path and the activities 1-2, 2-4, 4-5 and 5-6 are the critical activities.

14.5.5 Calculation of Floats: It may be observed that for every critical activity in a network, the earliest start and latest start time are the same. This is so since the critical activities cannot be scheduled later than their earliest schedule time without delaying the total project duration, they do not have any flexibility in scheduling. However, non-critical activities do have some flexibility i.e. these activities can be delayed for some time without affecting the project duration. This flexibility is termed as slack in case of an event and as float in case of an activity.

(*Note: In various networks we have omitted subscripts for E_i and L_j)

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Some people do not make any distinction between a slack and a float.

◆ Slack time for an event

The slack time or slack of an event in a network is the difference between the latest event time and the earliest event time. Mathematically it may be calculated using the formula $L_i - E_i$, where L_i is the latest allowable occurrence time and E_i is the earliest allowable occurrence time of an event i .

◆ Total float of an activity: The total activity float is equal to the difference between the earliest and latest allowable start or finish times for the activity in question. Thus, for an activity $(i-j)$, the total float is given by

$$TF_{ij} = LST - EST \text{ or } TF_{ij} = LFT - EFT$$

In other words, it is the difference between the maximum time available for the activity and the actual time it takes to complete. Thus, total float indicates the amount of time by which the actual completion of an activity can exceed its earliest expected completion time without causing any delay in the project duration.

◆ Free Float: It is defined as that portion of the total float within which an activity can be

manipulated without affecting the float of the succeeding activities. It can be determined by subtracting the head event slack from the total float of an activity.

$$\text{i.e } FF_{ij} = TF_{ij} - (\text{slack of event } j)$$

The free float indicates the value by which an activity in question can be delayed beyond the earliest starting point without affecting the earliest start, and therefore, the total float of the activities following it.

♦ Independent float : It is defined as that portion of the total float within which an activity can be delayed for start without affecting float of the preceding activities. It is computed by subtracting the tail event slack from the free float of an activity.

$$IF_{ij} = FF_{ij} - (\text{slack of event } i)$$

The independent float is always either equal to or less than the free float of an activity. If a negative value is obtained, the independent float is taken to be zero .

♦ Interfering float: Utilization of the float of an activity can affect the float of subsequent activities in the network. Thus, interfering float can be defined as that part of the total float which causes a reduction in the float of the successor activities. In other words, it can be defined as the difference between the latest finish time of the activity under consideration and the earliest start time of the following activity, or zero, whichever is larger. Thus, interfering float refers to that portion of the activity float which cannot be consumed without affecting adversely the float of the subsequent activity or activities.

Illustration

Activity Duration

1-2 4 days

1-3 12 days

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1-4 10 days

2-4 8 days

2-5 6 days

3-6 8 days

4-6 10 days

5-7 10 days

6-7 0 days

6-8 8 days

7-8 10 days

8-9 6 days

With the help of the activities given above draw a network. Determine its critical path, earliest start time, earliest finish time, latest start time, latest finish time, total float, free float and independent float.

Solution

The network based on the activities given in the example is as follows:

$$E=4$$

$$L=4 \quad E = 10$$

$6 L = 12$
 $2 5$
 $4 10$
 $8 77 E = 22$
 $L = 22$
 $10 10 0 10$
 $E=0 1 6$
 $4 16 8 9$
 $L=0 8 E = 38$
 $E = 12 E = 22 E = 32$
 $L = 32 L = 38$
 $12 L = 12 8 L = 22$
 $3 E = 12$
 $L = 14$

Figure - 25

The time estimates and floats are calculated as below:

Job or Duration EST EFT LST LFT Slack of
 activity of an (= E_i) ($E_i + D_{ij}$) ($L_j - D_{ij}$) (= L_j) tail head Total float Free float Independent
 activity event event (LST- (Total Float (Free float-
 D_{ij} ($L_i - E_i$) ($L_j - E_j$) EST) -Slack of slack of
 head event) tail event)

1-2 4 0 4 0 4 0 0 0 0 0

1-3 12 0 12 2 14 0 2 2 0 0

1-4 10 0 10 2 12 0 0 2 2 2

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2-4 8 4 12 4 12 0 0 0 0 0

2-5 6 4 10 6 12 0 2 2 0 0

3-6 8 12 20 14 22 2 0 2 2 0

4-6 10 12 22 12 22 0 0 0 0 0

5-7 10 10 20 12 22 2 0 2 2 0

6-7 0 22 22 22 22 0 0 0 0 0

6-8 8 22 30 24 32 0 0 2 2 2

7-8 10 22 32 22 32 0 0 0 0 0

8-9 6 32 38 32 38 0 0 0 0 0

Critical path is represented by 1-2-4-6-7-8-9.

(Note: In this table we have given detailed calculations merely for explanation.)

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Illustration

Analyse the network below for the critical path and for different floats

5

2 6

$E=3$ $E = 19$

$L=3$ $L = 19$

3 10 6 1

$E=0$ 1 3 15 $E = 17$

$L=0$ 4 4 $L = 18$

$E = 13$

$L = 13$

14 1

4 $E = 14$

$L = 17$

Figure 26

Network Analysis Table

Activity Duration Start Finish Float

E L E L Total FreeInd. Interfering

1-2 3 0 0 3 3 0 0 0 0

1-3 4 0 9 4 13 9 9 9 0

1-4 14 0 3 14 17 3 0 0 3

2-3 10 3 3 13 13 0 0 0 0

2-6 5 3 14 8 19 11 11 11 0

3-5 4 13 14 17 18 1 0 0 1

3-6 6 13 13 19 19 0 0 0 0

4-5 1 14 17 15 18 3 2 0 1

5-6 1 17 18 18 19 1 1 0 0

(N.B.: Activities must always be arranged in the i-j sequence.)

Explanation: Consider activity 1 - 3. The slack of head event 3 is $13 - 13 = 0$. Therefore, free float = $9 - 0 = 9$. Likewise slack of tail event 1 is $0 - 0 = 0$. Therefore independent float = 9.

Illustration

Analyse the network below for the critical path and calculate total float, free float and independent float for each activity.

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$E = 43$ $E = 63$ $E = 83$

$E = 36$ $L = 43$ $L = 63$ $L = 83$

$L = 41$

4 8 9 10

A/36 D/2 K/20 M/20

0 J/9

1 2 3 E/1 L/8

B/4 C/2

E=0 E=4 F/15 7

G/9

L=0 E=6 E = 34

L=4 L=6 L = 34

5

I/9

E = 21 H/4 Critical Path: 2-3-5-6-7-8-9-10

L = 21

6 E = 25

L = 25

Figure - 27

Solution

Network Analysis Table

i-j D Start Finish Floats

E L E L T F I

1- 2 4 0 0 4 4 0 0 0

1-4 36 0 5 36 41 5 0 0

2-3 2 4 4 6 6 0 0 0

3-5 15 6 6 21 21 0 0 0

3-8 10 6 33 16 43 27 27 27

4-8 2 36 41 38 43 5 5 0

5-6 4 21 21 25 25 0 0 0

5-8 9 21 34 30 43 13 13 13

6-7 9 25 25 34 34 0 0 0

7-8 9 34 34 43 43 0 0 0

7-9 8 34 55 42 63 21 21 21

8-9 20 43 43 63 63 0 0 0

9-10 20 63 63 83 83 0 0 0

Critical Path : BCHIJKM = 83 weeks.

Illustration

A small assembly plant assembles PCs through 9 interlinked stages according to following precedence / process:

Critical Path Analysis 14.29

Stage Duration

From to (Hours)

1-2 4

1-3 12

1-4 10

2-4 8

2-5 6

3-6 8
 4-6 10
 5-7 10
 6-7 0
 6-8 8
 7-8 10
 8-9 6

- (i) Draw an arrow diagram (network) representing the above assembly work.
 (ii) Tabulate earliest start, earliest finish, latest start and latest finish time for all the stages.
 (iii) Find the critical path and the assembly duration.
 (iv) Tabulate total float, free float and independent float.

Solution

- (i) The network diagram representing the given assembly work is as follows:

E=4 6 E = 10

> 5

L=4 2 L = 12

10 >

4 7

8

E=0 0 10

L=0

10 10 8 6

1 > 4 > 6 > 8 > 9

E = 12 E = 22 E = 32 E = 38

L = 12 L = 22 L = 32

12 L = 38

8

3 E = 12

L = 14

Figure 28

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- (ii) The earliest start, earliest finish, latest start and latest finish times for all the stages are calculated in the table given below:

Stage or Duration Earliest Earliest Latest Latest Floats

Activity (Hours) Start Finish Start Finish Total Free Independent

(EST) (EFT) (LST) (LFT) (LST-EST) (Total Float (Free Float)

or -Slack of - Slack of

(LFT-EFT) Head event) tail event)

(1) (2) (3) (4) (5) (6) (7) (8) (9)

1-2 4 0 4 0 4 0 0 0

1-3 12 0 12 2 14 2 0 0
 1-4 10 0 10 2 12 2 2 2
 2-4 8 4 12 4 12 0 0 0
 2-5 6 4 10 6 12 2 0 0
 3-6 8 12 20 14 22 2 2 0
 4-6 10 22 12 22 0 0 0
 5-7 10 12 20 12 22 2 2 0
 6-7 0 10 22 22 22 0 0 0
 6-8 8 22 30 24 32 2 2 2
 7-8 10 22 32 22 32 0 0 0
 8-9 6 22 38 32 38 0 0 0

(iii) The critical path for the above network is given by 1 – 2 – 4 – 6 – 7 – 8 – 9. The assembly duration is 38 hours.

(iv) The total float, free float and independent float are given by columns (7), (8) and (9) respectively of the above table.

14.6 Distinction between PERT & CPM

PERT CPM

PERT is used for non-repetitive Jobs like planning the CPM is used for non-Repetitive Job assembly of the space platform like building a house

It is a probabilistic Model It is a deterministic Model

It is event oriented It is activity oriented

It incorporates Statistical Analysis It does not incorporates Statistical Analysis

It serve as useful control device It is difficult to use CPM as a control device

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SUMMARY :

- Two most commonly used network Models
 - ü Critical Path Method(CPM)
 - ü Programme Evaluation Review Technique(PERT)
- Construction of Network diagrams
- Calculation of Critical Path and Project Completion time together with Floats, earliest start and finishing times
- Application of technique to crash cost problems
- Critical path should be longest route through the network
- Resource schedule problems and use of bar charts
- Identifying the project duration
- Cheapest critical activities to crash and not reduce project completion time by reducing non-critical activities

SELF-EXAMINATION QUESTIONS

1. Define a project and briefly explain the four common implications which characterise a project and state the five steps of the working methodology of critical path analysis.
2. Distinguish between PERT and CPM.
3. What kinds of decision making situation may be analysed by using CPM technique.
4. Draw a network corresponding to the following information. Obtain the early and late start and completion times, and determine the critical activities.

Activity Duration

- 1-2 4 days
- 1-3 6 days
- 2-6 8 days
- 3-4 7 days
- 3-5 4 days
- 4-6 6 days
- 5-6 5 days
- 5-7 19 days
- 6-7 10 days

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5. A work project consists of twelve activities labelled A through L. Upon being asked to specify the order in which the jobs had to be done, the managers answered as follows :

A, B and C are the first activities of the project and can start simultaneously and immediately; A and B precede D while B precedes E, F and H. Activities F and C precede G while E and H precede I and J. C, D, F and J precede K which in turn, precedes L. Further I, G and L are the terminal activities of the project.

The completion times of the various activities are listed here :

Activity Time (days)

- A 6
- B 4
- C 10
- D 1
- E 6
- F 3
- G 14
- H 1
- I 9
- J 2
- K 7
- L 5

- (a) Draw a network diagram corresponding to this project.
- (b) Obtain the lengths of all the paths and determine critical path.

ANSWER TO SELF-EXAMINATION QUESTIONS

5. The network diagram is shown in Figure below :

E 4

H

2 6

I

B F

C G

1 5 9

A J L

D K

3 7 8

Figure 29

Critical Path Analysis 14.33

The various paths and their lengths are as follows :

Path Length

1-3-7-8-9 19

1-2-3-7-8-9 17

1-2-4-5-7-8-9 19

1-2-5-7-8-9 24* Critical

1-2-5-9 19

1-2-4-5-9 14

1-2-6-7-8-9 19

1-2-6-9 21

1-6-7-8-9 22

1-6-9 24* Critical